A practical guide to GRASP2018
– A collection of Fortran 95 programs with parallel computing using MPI

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Preface

This is a practical guide to the GRASP2018 General Relativistic Atomic Structure Package, C. Froese Fischer, G. Gaigalas, P. Jönsson, J. Bieroń, Comput. Phys. Commun. (in press 2018), `https://doi.org/10.1016/j.cpc.2018.10.032`, which is an new version of GRASP2K, P. Jönsson, G. Gaigalas, J. Bieroń, C. Froese Fischer, I.P. Grant, Comput. Phys. Commun. 184, 2197 (2013), converted from Fortran 77 to Fortran 95. The revision makes more efficient use of memory and facilitates calculations with expansions of more than a few millions of configuration state functions that occur in calculations for complex systems. The guide assumes that GRASP2018 has been correctly installed and that the executables are on the path. All calculations are done with non-interacting blocks of given parity and J value. Operation of both the scalar programs and the parallel programs running under the Message Passing Interface (MPI) are discussed. Please note that the results from scalar and MPI calculations need not be identical since the codes are not entirely identical and there are some differences in convergence criteria etc. Results on different computer systems may not be identical due to differences in floating point arithmetic.

The guide can be read and used in several ways. For the ones that want to get started with practical calculations as soon as possible the best route would be to read relevant parts of chapters 1, 2, and 5 and then directly jump to chapter 7, which contains the first full sample calculations. After getting acquainted with the different programs and the steps needed to perform the calculations, it is probably a good thing to go back to the previous chapters and look more into details of the processing. Alternatively, just read through the guide in a more linear fashion.

Questions and comments regarding the guide and GRASP2018 can be put to the authors.

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The GRASP2018 software is provided under the MIT Licence.

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Part I

Overview of GRASP2018

Chapter 1

GRASP2018

1.1 Relativistic vs non-relativistic calculations

The methods implemented in the GRASP2018 General Relativistic Atomic Structure Package are based on the fully relativistic (four-component) multiconfiguration Dirac-Hartree-Fock (MCDHF) method and are suitable for medium to heavy atomic systems. For light and near neutral systems, where relativistic effects are often (though not always) comparatively small, the ATSP2K Atomic Structure Package [1] based on the non-relativistic multiconfiguration Hartree-Fock (MCHF) method with Breit-Pauli (BP) relativistic corrections may be the method of choice. The MCHF-BP method allows LS symmetries to be used, which often makes it possible to include more electron correlation. In addition semi-empirical fine tuning of the energies can be done, which leads to more accurate results, especially in cases with closely degenerate states. ATSP2K can be downloaded from `http://nlte.nist.gov/MCHF/`. The corresponding write-up is also available for download at the above site.

1.2 Features of the package

GRASP2018, like its predecessors, is based on the multiconfiguration Dirac-Hartree-Fock (MCDHF) method, see [2, 3] for an account of the general theory. The package consists of a number of application programs and tools to compute approximate relativistic wave functions, energy levels, hyperfine structures (HFS), isotope shifts (IS), Landé g_J-factors, interactions with external fields, angular couplings for labelling purposes, and transition energies and transition probabilities for many-electron atomic systems. There are also some graphical capabilities. The application programs and tools, along with the underlying theory, are described in the original write-ups [4, 5, 6, 7, 8, 9, 10, 11, 12, 13]. GRASP2018, whose use is described here, is similar to the Fortran 77 GRASP2K version 1.1 [5]. Using the package, research into highly accurate transition energies and transition rates becomes feasible for a wider range of systems.

The main features of these packages are:

1. The interaction matrix is considered to be a series of non-interacting blocks of given parity and J value, with selected eigenvalues determined from each. For a description of the sparse Davidson diagonalization library module see [14].

2. Angular integrations are based on second quantization in the coupled tensorial form, angular momentum theory in three spaces (orbital, spin and quasispin), and a generalized graphical technique. The theoretical background can be found in [15, 16, 17].

3. It is possible to transform the wave function in jj-coupling to a basis of LSJ-coupled configuration state functions, see [18, 19, 20]. Labels in LSJ-coupling are used by several programs

in the package.

4. It is possible to use separately optimized initial and final state wave functions when computing transition rates. The non-orthogonality between initial and final state radial orbitals is handled by a biorthonormal transformation technique [21].

5. MPI codes are available for the most time consuming parts of the package.

6. Zero- and first-order perturbative methods can be used to handle large configuration state function (CSF) expansions [2, 22].

1.3 Features of the present version

The present version maintains many of the computational procedures of earlier versions but is a complete revision that adheres to current Fortran 95 standards. The code has been translated from the earlier Fortran 77 standard to Fortran 95 standard. IMPLICIT statements pertaining to type have been replaced by IMPLICIT NONE so that the type of all variables is clear. The COMMON statements were replaced by MODULES. This greatly facilitates code maintenance in that it is easier to modify MODULES without side-effects than COMMON statements where long datatypes should precede shorter data types in the list. Interface modules are used to improve correctness of procedure calls when modifications are made during code maintenance. But the most important modification was the replacement of the non-standard Cray pointers by the pointer arrays that are part of Fortran 95. With the Cray pointers it was not possible to compile the code for array-bound checking, an important debugging tool. This modification was not part of the translation process, but involved a restructuring of the code. Together with long integers (kind=8), memory allocation for arrays exceeding a Gigabyte are now possible.

Many of these modification are important for future code development, but the more efficient use of memory opens the possibility of studying more complex systems. In the next section, we describe changes that are important for the user of the code.

1.4 Changes and updates from GRASP2K, version 1.1

The main changes relative to GRASP2K, version 1.1 [5] are:

1. A new environment file `make_environment_gfortran` making it easier to compile and set up the programs in the package.

2. A new and more logical naming convention for the programs has been adopted. Of course, a user can alias these to a preferred name.

3. Better information about what input and output files are needed for the different programs, making interactive use easier.

4. Input and output files now have more consistent names.

5. For the main programs there is a log-file showing the input data that was used for the run. By saving the log-file the user has a record of how the calculation was performed. The log-file can be edited and used as input file for future runs.

6. The diagonalization routines based on the Davidson method have been updated and now the eigenvector components from previous iterations of the `rmcdhf` program are used as starting values for the diagonalization in the current iteration.

7. The convergence criteria for diagonalization using the Davidson method have been tuned, improving convergence and stability especially for the `rmcdhf_mpi` program.

8. In the `rmcdhf` program, memory allocation for unneeded angular data have been removed, that considerably reduces the memory used in large cases.
9. The way Breit integrals are computed in the `rci` program has been changed resulting in a dramatic speed-up for CSF expansions built on large orbital sets.
10. A number of plotting tools using Matlab or GNU Octave are available.
11. There are several programs to automatically generate LaTeX tables.
12. There are several programs to create plots of properties along an iso-electronic sequence.
13. There are new programs for efficient and flexible generation of lists of CSFs.
14. There are new programs to support large scale calculations.
15. A number of small bugs have been corrected.

1.5 Backward compatibility

The configuration state function files, the mixing coefficient files and the radial orbitals files of GRASP2018 are compatible with the corresponding files in GRASP2K, version 1.1. Files from GRASP2018 can thus be used as input files for the `hfszeeman` program [11] that can be used for computing magnetically induced transitions [23] and the `ris3` and `ris4` programs [13, 12] that compute relativistically corrected normal and specific mass shift parameters and model independent field shift including effects from nuclear deformations. The three latter programs are applications in GRASP2K, but have not yet been translated and converted for inclusion into GRASP2018.

1.6 Quoting the package

Developing computational methods and programs is challenging, often requiring intensive effort. The work needs to be properly acknowledged and quoted in order to be continued.

Users of GRASP2018 should quote it as:

C. Froese Fischer, G. Gaigalas, P. Jönsson and J. Bieroń
GRASP2018 - A Fortran 95 version of the General Relativistic Atomic Structure Package
Comput. Phys. Commun. (in press 2018), `https://doi.org/10.1016/j.cpc.2018.10.032`
(the user should update the reference when volume, issue and pages are available).

1.7 Reporting errors

The programs have been used and tested extensively but as new calculations are tried errors may be encountered. If you, the user of the program package, have reasons to believe that there is an error somewhere in the package, please send an email to one of the authors specifying the case that resulted in the error. Also, if there are sections in this guide that are unclear, please let us know. Better yet, if you find the needed correction, let us know so that others may benefit as well.

Chapter 2

Installation

Our code has been thoroughly tested using the gfortran compiler. The code also compiles without error in the Intel environment, but a case has arisen where a run-time error occurred that could not be resolved. For this reason, the following discussion refers only to gfortran.

2.1 The gfortran compiler, MPI libraries and GNU Octave

GRASP2018 should be compiled with the `gfortran` compiler. The MPI programs should be compiled with `mpifort` compiler from Open MPI. The `mpifort` compiler is a wrapper compiler that manipulates the command line and adds in all the relevant compiler and linker flags and then invokes an underlying compiler/linker (hence, the name wrapper compiler). Normally the underlying compiler is `gfortran`, as required for GRASP2018, but dependent on how Open MPI has been set up there may be other underlying compilers. The user must pay attention to this.

To see if the `gfortran` compiler is installed on your computer system typing the following may be helpful:

```
gfortran --version
```

To see if the Open MPI libraries and the `mpifort` compiler are available type

```
mpirun --version
```

To see which underlying compiler is used by `mpifort` give the command

```
mpifort --showme
```

Please note that on some computer systems Open MPI, Lapack, and Blas are available as modules that need to be loaded. Often you can load Open MPI modules for different compilers. Please also note that `mpif77` and `mpif90` are deprecated as of Open MPI v1.7 and that `mpifort` should be used instead.

There are two ways to link to an MPI library: one is by adding the `INCLUDE 'mpif.h'` statement and the other is by adding the `USE mpi` statement, whereby the module `mpi.mod` is used. The use of `INCLUDE 'mpif.h'` is going to be deprecated in future MPI releases and is replaced by `USE mpi`, see `https://www.mpi-forum.org/docs/mpi-3.1/mpi31-report/node408.htm#Node408` for more information. In the current version of GRASP2018 we have `INCLUDE 'mpif.h'` in 9 places in the two files `/src/lib/mpi90/mpi_C.f90` and `/src/lib/mpi90/mpiu.f90`. If supported by the MPI release, the user is strongly encouraged to replace `INCLUDE 'mpif.h'` by `USE mpi` at all 9 occurrences in the two files.

Information about underlying compilers and how they can be switched can be found here `https://www.open-mpi.org/community/lists/users/2011/12/17972.php`.

Frequently asked questions about compiling MPI applications can be found at
`https://www.open-mpi.org/faq/?category=mpi-apps`

To install the `gfortran` compiler and the Open MPI libraries on Debian-like systems and its variants use the `apt-get` command

```
sudo apt-get install gfortran
```

and

```
sudo apt-get install openmpi
sudo apt-get install libopenmpi-dev
```

In case of problems with the installation of the compiler and the MPI-libraries contact your system manager. For the runs in this guide the code was compiled with `gfortran 4.8.2`. The MPI runs were based on `Open MPI 1.6.5` with `gfortran` as the underlying compiler.

The plotting tools utilize GNU Octave (alternatively Matlab). To install and use GNU Octave consult the home page `www.gnu.org/software/octave/` .

Unlike Grasp2K, version 1.1 or earlier, the present package includes only the new Fortran 95 programs. The source code for the needed `lapack` and `blas` are no longer provided. These can be downloaded from `http://www.netlib.org/lapack/lug/node14.html` when needed.

2.2 Download and unpack the package

Grasp2018 comes as a compressed tarball named `GRASP2018.tar.gz` that can be downloaded from the Computational Atomic Structure group home page `http://ddwap.mah.se/tsjoek/compas/` or from GitHub `https://www.github.com/compas/grasp2018`. To unpack the tarball issue the command

```
tar -zxvf GRASP2018.tar.gz
```

You should now get a directory named `GRASP2018`. Entering this directory there are environment script files where environment variables are set. There are also four sub-directories:

`bin`	here the executables reside after compilation
`lib`	here the libraries reside after compilation
`src`	source code in Fortran 95 in a number of sub-directories
`grasptest`	scripts for the examples described in this guide in a number of sub-directories

2.3 The environment scripts under bash

Compiling the package requires that the compiler and Open MPI software be on the system search `PATH` and that the lapack, blas, and Open MPI libraries be either on the `LD_LIBRARY_PATH` or `LIBRARY_PATH`, depending on the system. The Makefiles of the package assume that the location of the libraries can be found by the compiler. Thus no environment variable is needed for `LAPACK_DIR`, but possibly the names of the libraries themselves. How this is done depends on the system. At High-Performance Computing (HPC) computing centers, this often is done by loading modules that define variables and prepend directories to `PATH`. On Compute Canada (CC), the command

```
module show gcc/4.4.7/lapack
-------------------------------------------------------------------
/global/system/Modules/modulefiles/gcc/4.4.7/lapack:

module-whatis    loads the lapack/gcc (64 bit) library
```

```
prepend-path     LIBRARY_PATH /global/software/lib64/gcc/lapack
prepend-path     LD_LIBRARY_PATH /global/software/lib64/gcc/lapack
-------------------------------------------------------------------
```

displays the location of the lapack library, and shows that the module prepends this library to both `LIBRARY_PATH` and `LD_LIBRARY_PATH`.

When the numerical libraries are not available and the downloaded lapack and blas libraries have been compiled, the easiest solution is to place them in the same directory as other GRASP libraries, namely `$GRASP/lib`, where the environment variable `$GRASP` gives the location of the GRASP2018 root directory, see environment file below. If the libraries in `$LIB` (some other directory) are shared with other applications, then another simple solution is to create links from `$GRASP/lib` to the libraries:

```
 cd $GRASP/lib
 ln -s $LIB/lib   libblas.a
 ln -s $LIB/lib   liblapack.a
```

The environment script file below sets the various paths and then defines some variables needed by the Makefiles of the various applications. The script file `make_environment_gfortran_CC` for the gfortran compiler in the Compute Canada environment is shown below.

```
#!/bin/bash
# ---------------------------------------------------------------------------------------------
# GRASP2018 ENVIRONMENT FLAGS
# ---------------------------------------------------------------------------------------------
#
# Define the following global variables according to your environment and
# source this script or add these definitions to your terminal configuration
# file, eg. ~/.cshrc, ~/.bashrc or ~/.profile.
#
# Current version: Linux, gfortran gcc version 6.4.0
#
# Loads the modules (in module environment) of Compute Canada
module purge                         # clear all modules
module load gcc/6.4.0
module load openmpi/2.1.1
module load gcc/4.4.7/lapack
module load gcc/4.4.7/blas

# ---------------------------------------------------------------------------------------------
# Set up main flags
# ---------------------------------------------------------------------------------------------
export   FC=gfortran                                # Fortran compiler
export   FC_FLAGS="-O2 -fno-automatic "             # Serial code compiler flags
export   FC_LD=" "                                  # Serial linker flags
export   GRASP="${PWD}"                             # Location of the GRASP2018 root directory
export   LAPACK_LIBS="-llapack -lblas"              # Lapack libraries
# ---------------------------------------------------------------------------------------------
# Set up MPI related flags
# ---------------------------------------------------------------------------------------------
export   FC_MPI="mpifort"                           # MPI
export   FC_MPIFLAGS="${FC_FLAGS}"                  # Parallel code compiler flags
export   FC_MPILD=${FC_LD}                          # Parallel linker flags
# ---------------------------------------------------------------------------------------------
export MPI_TMP="/scratch"                           # Disk for storing temporary files
# ---------------------------------------------------------------------------------------------
```

First the `PATHs` are set, then the compiler is selected along with its various options. Because parts of the package consist of pre-F90 code, the `-fno-automatic` ("save") option is recommended. When compiler libraries are present, `LAPACK_LIBS=""` (blank) may be used but in other cases, the libraries need to be mentioned. Next the MPI flags are given. All the flags are for the `gfortran` compiler under Linux.

Finally, an optional `MPI_TMP` variable may be set defining the location of the disk on which temporary data should be stored. The location can also be defined through a "disks" file (see Section 7.4), but when no disks file is available, the temporary data are stored in `$MPI_TMP`. If neither is available the data will be stored on `/scratch/$USER`, provided the directory exists.

2.4 Installation of the package under bash

Make sure you are running under the bash shell. In the directory `GRASP2018` issue the following commands

```
source ./make_environment_gfortran
cd src
make clean
make
```

Upon successful compilation the executable programs and tools can be found in the `GRASP2018/bin` directory. The following 44 programs and tools should be produced. Programs running in parallel under MPI have the extension `mpi`.

```
hf                 rci            rhfs_lsj       rseqenergy   rtransition
jj2lsj             rci_mpi        rlevels        rseqhfs      rtransition_mpi
jjgen              rcsfblock      rlevelseV      rseqtrans    rwfnestimate
lscomp.pl          rcsfgenerate   rmcdhf         rsms         rwfnmchfmcdf
rangular           rcsfinteract   rmcdhf_mpi     rtabhfs      rwfnplot
rangular_mpi       rcsfmr         rmixaccumulate rtablevels   rwfnrelabel
rasfsplit          rcsfsplit      rmixextract    rtabtrans1   rwfnrotate
rbiotransform      rcsfzerofirst  rnucleus       rtabtrans2   wfnplot
rbiotransform_mpi  rhfs           rsave          rtabtransE1
```

2.5 Putting the executables on the path

To run the scripts in the test data set (see sections 2.2 and 3.5) the executable programs and tools must be on the path. Under bash (see `http://en.wikipedia.org/wiki/Bash_(Unix_shell)`) this can be done by editing the file `.profile` or the file `.bashrc`

Suppose, for example, that the executables are in `GRASP2018/bin` then we should add something like `PATH="$HOME/GRASP2018/bin:$PATH"` at the end of the `.profile` or `.bashrc` files.

2.6 Changing parameters in the package

The application programs are written in terms of some basic parameters. Most, but not all, are set in the directory `GRASP2018/src/lib/libmod` and can be changed by editing the file `parameter_def_M.f90`. These include parameters that define the grid. Often changes are with respect to the location of the first point away from the origin, defined in terms of a variable `RNT` that changes the number of points of the grid. The above installation sets the maximum number of grid points `NNNP` for representing the radial parts of the one-electron orbitals to the default value `NNNP=590`. This default value works fine in most cases. For heavy or super heavy elements it is sometimes necessary to extend the number of grid points. Another parameter defining the grid is the step-size `H`. Reducing this parameter would improve the numerical accuracy of the calculations but, at the same time, might require an increase of the number of grid-points. To install the program with an extended grid start by deleting the old executables and libraries in the `GRASP2018/bin` and `GRASP2018/lib` directories by issuing the `make clean` command in the `GRASP2018/src` directory and change the number of grid points from `NNNP=590` to a larger value, say `NNNP=1990`. At the same time set `NNN1=2000` (`NNN1 = NNNP + 10`). Recompile all the package.

After recompilation all programs and tools in the `GRASP2018/bin` directory will be based on the extended grid. Unless explicitly stated, all examples in this guide are based on programs with the default grid `NNNP=590`.

The `rci` programs (including the MPI version) have a parameter `NINCOR` that decides whether the eigenvalue problem stores the interaction matrix in memory or on disk in terms of the memory requirement for all the non-zero matrix elements. This parameter has been increased to the number of double precision matrix elements that can be stored in 2 Gigabytes of memory. For the MPI version, this is a memory requirement per CPU. Another parameter is `IOLPCK` that determines whether matrices are stored in a sparse format and solved by the Davidson method or are small enough to be stored in the dense, symmetric matrix format and eigenvalues computed using a Lapack routine. This parameter is set to 2000. Both parameters can be modified by the user.

Chapter 3

Program structure and file flow

3.1 Program naming conventions

In multiconfiguration calculations the wave function for an atomic state is approximated by an atomic state function (ASF). The ASF, in turn, is given as an expansion over configuration state functions (CSFs)

$$\Psi(\gamma PJ) = \sum_{i=1}^{N} c_i \Phi(\gamma_i PJ). \tag{3.1}$$

Here $\{\gamma_i\}$ denote the configurations together with the angular coupling trees, P is parity, J is the final angular quantum number, and $\{c_i\}$ are the expansion (mixing) coefficients. The CSFs are given as coupled anti-symmetric products of one-electron orbitals

$$\phi(\mathbf{x}) = \frac{1}{r} \left(\begin{array}{c} P(n\kappa; r)\chi_{\kappa m}(\theta, \varphi) \\ \mathrm{i}\, Q(n\kappa; r)\chi_{-\kappa m}(\theta, \varphi) \end{array} \right), \tag{3.2}$$

where the radial parts of the orbitals (the radial wave functions) $P(n\kappa; r), Q(n\kappa; r)$ are numerically represented on a grid.[1]

Given this description we identify three main concepts:

- lists of CSFs defining the ASF
- mixing coefficients
- radial parts (wave functions) of orbitals

These concepts are the basis for the program naming conventions: programs generating or manipulating lists of CSFs have names starting with `rcsf`, programs generating or manipulating mixing coefficients have names starting with `rmix`, programs generating or manipulating the radial parts of the orbitals (radial wave functions) have names starting with `rwfn`. Other programs are named according to the atomic properties they compute. There are also a number of programs that produce output tables in LaTeX format. These programs all have names starting with `rtab`. Finally, there are programs that create GNU Octave and Matlab M-files for plotting properties along iso-electronic sequences. These programs have names starting with `rseq`.

[1] In the guide the three terms radial orbital, radial part of the orbital, and radial wave function will be used intermixed meaning the same thing. Sometimes we will also loosely speak about the orbitals meaning the radial parts of the orbitals.

3.2 Application programs and tools

Below is a partial list of programs in the package:

1. `rnucleus` – define nuclear data.
2. Routines that generate and manipulate lists of CSFs:
   (a) `rcsfgenerate` – generate a list of CSFs using rules for excitations.
   (b) `rcsfinteract` – reduce a list of CSFs by retaining only CSFs that interact with CSFs of a reference list (former `jjreduce`).
   (c) `rcsfsplit` – split a list of CSFs into a number of lists with CSFs that can be formed from different sets of active orbitals.
   (d) `rcsfzerofirst` – rearrange a list of CSFs in such a way that the most important CSFs come at the beginning, defining the zero-order space, and the less important come at the end, defining the first-order space.
3. `rangular, rangular_mpi` – perform angular integration and compute angular coefficients (former `mcp`).
4. `rwfnestimate` – estimate the radial parts of the orbitals - radial wave functions (former `erwf`).
5. `rmcdhf, rmcdhf_mpi` – determine radial parts of the orbitals and mixing coefficients of the CSFs in a relativistic self-consistent-field (SCF) procedure.
6. `rci, rci_mpi` – perform relativistic configuration interaction (RCI) calculation with transverse photon (Breit) interaction and vacuum polarization and self-energy (QED) corrections.
7. `jj2lsj` – a program for converting a portion of the wave function expansion in jj-coupled CSFs to LSJ-coupled CSFs for labeling purposes. Includes a unique labeling feature.
8. Routines for computing transition probabilities:
   (a) `rbiotransform, rbiotransform_mpi` – perform biorthonormal transformations of wave functions.
   (b) `rtransition, rtransition_mpi` – compute transition properties from transformed wave functions. If the program `jj2lsj` has been run, the labels of the states in the output files are in LSJ-coupling.
9. `rhfs` – compute diagonal and off-diagonal hyperfine interaction constants and Landé g_J-factors.
10. `rsms` – compute isotope shift.

A number of generally short programs have been developed as tools to facilitate computational procedures.

1. `rmixaccumulate` – accumulate CSFs corresponding to a specified fraction of the total wave function.
2. `rmixextract` – extract and print the numerical values of the expansion coefficients above a cut-off value along with the corresponding CSFs, in descending order of magnitude, if requested (former `extmix`).
3. `rcsfmr` – analyse the wave function expansion in LSJ-coupled CSFs and determine a multireference.

4. `hf` – perform a non-relativistic Hartree-Fock (HF) calculation to produce a radial wave function file `wfn.out`. The file `wfn.out` should be copied to `wfn.inp` for further processing.

5. `rwfnmchfmcdf` – convert a non-relativistic Hartree-Fock radial wave function file `wfn.inp` file to a GRASP2018 radial wave function file `rwfn.out` that can be used with `rwfnestimate`.

6. `rwfnplot` – extract radial wave functions from a radial wave function file and generate a GNU Octave/Matlab M-file that plots the radial wave functions as functions of $\sqrt{r}$ or r.

7. `wfnplot` – extract radial wave functions from the non-relativistic radial wave function file as produced by the `hf` program and generate a GNU Octave/Matlab M-file that plots the radial wave functions as functions of $\sqrt{r}$ or r.

8. `rwfnrotate` – a routine that rotates radial orbitals, useful for debugging purposes (former `rotate_pair`).

9. `rlevels` – list the levels in a series of mixing files, in the order of increasing energy and report levels in cm^{-1} relative to the lowest. If the program `jj2lsj` has been run, the levels are given in LSJ-coupling notation.

10. `rlevelseV` – list the levels in a series of mixing files, in the order of increasing energy and report levels in eV relative to the lowest. If the program `jj2lsj` has been run, the levels are given in LSJ-coupling notation.

11. `rtablevels` – produce LaTeX and ASCII tables of energies from energy files produced by `rlevels`.

12. `lscomp.pl` – perl script to produce LaTeX tables with LSJ composition and energies from energy files `rlevels`.

13. `rtabtransE1` – produce LaTeX and ASCII tables of transition parameters from files produced by `rtransition` (E1 transitions only).

14. `rtabtrans1` and `rtabtrans2` – produce LaTeX tables of transition parameters and lifetimes from files produced by `rtransition`.

15. `rhfs_lsj` – give the output from the `rhfs` program and its variants in LSJ-coupling notation.

16. `rtabhfs` – produce LaTeX tables of hyperfine interaction constants.

17. `rseqenergy` – produce GNU Octave/Matlab M-files that plot energies as functions of Z along an iso-electronic sequence.

18. `rseqhfs` – produce GNU Octave/Matlab M-files that plot hyperfine interaction constants and Landé g_J-factors as functions of Z along an iso-electronic sequence.

19. `rseqtrans` – produce GNU Octave/Matlab M-files that plot transition parameters as functions of Z along an iso-electronic sequence.

20. `rsave` – a script file such that the command `rsave name` moves `rwfn.out` to `name.w`, `rmix.out` to `name.m`, `rcsf.inp` to `name.c`, `rmcdhf.sum` to `name.sum`, `rangular.log` to `name.alog` and `rmcdhf.log` to `name.log`.

21. `rasfsplit` – splits the files defining a number of ASFs of different symmetry blocks (J and parity) into groups of files, one for each symmetry block.

22. `jjgen` – generates a list of CSFs in non-block form.

23. `rcsfblock` – splits the list produced by `jjgen` into block-form.

3.3 File naming convention, program and data flow

Passing information between different programs is done through files. This process is greatly simplified by a file naming convention. GRASP2018 uses a convention similar to the one for ATSP2K [1]. A name is associated with the results from a calculation and an extension defines the content and format of a file. Thus the file name becomes `name.extension`. Common extensions are listed in Table 3.1. The tool `rsave` makes use of these default extensions to save the output files from an `rmcdhf` calculation.

Most programs produce a file that keeps a record of the input data. This file is called a log-file.

Table 3.1: Table of common extensions.

Extension	Type of file
`c`	List of CSFs.
`w`	Binary file of radial wave functions.
`m`	Binary file of expansion or mixing coefficients produced by `rmcdhf`.
`sum`	File containing information from an `rmcdhf` run.
`cm`	Binary file of mixing coefficients produced by `rci`.
`bw`	A `.w` file after biorthonormal transformation using `rbiotransform`.
`csum`	File containing information from an `rci` run.
`bm`	A `.m` file after biorthonormal transformation using `rbiotransform`.
`cbm`	A `.cm` file after biorthonormal transformation using `rbiotransform`.
`lsj.lbl`	File containing composition of wave functions in LSJ-coupling.
`uni.lsj.lbl`	File containing composition of wave functions in LSJ-coupling but arranged to give unique labels of all states.
`t`	Transition probability data from `rmcdhf` mixing coefficients.
`t.lsj`	Transition probability data from `rmcdhf` mixing coefficients. Labels in in LSJ-coupling.
`ct`	Transition probability data from `rci` mixing coefficients.
`ct.lsj`	Transition probability data from `rci` mixing coefficients. Labels in in LSJ-coupling.
`h`	Hyperfine structure data and Landé g_J-factors from `rmcdhf` mixing coefficients.
`ch`	Hyperfine structure data and Landé g_J-factors from `rci` mixing coefficients.
`hoffd`	Off-diagonal hyperfine structure data from `rmcdhf` mixing coefficients.
`choffd`	Off-diagonal hyperfine structure data from `rci` mixing coefficients.
`i`	Isotope shift data from `rmcdhf` mixing coefficients.
`ci`	Isotope shift data from `rci` mixing coefficients.
`log`	Log-file that keeps a record of program input.

To perform a calculation a number of programs needs to be run in a pre-determined sequence. Figure 3.1 displays a typical sequence of program calls to compute wave functions and different expectation values. The resulting flow of files is displayed in Figure 3.2.

Sequence of program calls to compute expectation values

`rnucleus`	Generation of nuclear data
`rcsfgenerate`	Generation of list of CSFs based on rules for excitations
`rcsfinteract`	Reduction of a list to CSFs interacting with the multireference
`rangular`	Angular integration
`rwfnestimate`	Initial estimates of radial orbitals
`rmcdhf`	Self-consistent field procedure
`rci`	Relativistic RCI with optional transverse photon (Breit) interaction and vacuum polarization and self-energy (QED) corrections
`jj2lsj`	Transform representation from jj- to LSJ-coupling

`rbiotransform`	Biorthonormal transf.	`rhfs`, `rsms`	Eval. of expect. values
`rtransition`	Eval. of expect. values		

Figure 3.1: Typical sequence of program calls to compute wave functions and different expectation values.

Flow of files between different programs

`rnucleus`	Output; `isodata`
↓	
`rcsfgenerate`	Output; `rcsf.out, rcsfgenerate.log`
↓	
`rcsfinteract`	Input; `rcsfmr.inp, rcsf.inp`
	Output; `rcsf.out`
`rangular`	Input; `rcsf.inp`
	Output; `rangular.log`, `mcp.30`, `mcp.31`, ...
↓	
`rwfnestimate`	Input; `isodata`, `rcsf.inp`, optional radial wave function file(s)
	Output; `rwfn.inp`
↓	
`rmcdhf`	Input; `isodata`, `rcsf.inp`, `rwfn.inp`, `mcp.30`, `mcp.31`, ...
	Output; `rmix.out`, `rwfn.out`, `rmcdhf.sum`, `rmcdhf.log`
↓	Use `rsave` to move output files to `name.c`, `name.m`, `name.w`, `name.sum`, `name.log`
`rci`	Input; `name.c`, `name.w`
	Output; `name.cm`, `name.csum`, `name.clog`, `rci.res` (restart file)
↓	
`jj2lsj`	Input; `name.c`, `name.(c)m`
	Output; `name.lsj.lbl`, `name.uni.lsj.lbl` (optional)
↓	
`rbiotransform`	Input; `name1.c`, `name1.(c)m`, `name1.w`, `name2.c`, `name2.(c)m`, `name2.w`
	Input; `name1.TB`, `name2.TB` (if available)
	Output; `name1.(c)bm`, `name1.(c)bw`, `name2.(c)bm`, `name2.(c)bw`
↓	Output; `name1.TB`, `name2.TB`
`rtransition`	Input; `name1.c`, `name1.(c)bm`, `name1.(c)bw`
	`name2.c`, `name2.(c)bm`, `name2.(c)bw`
	Input; `name1.name2.xT` (if available)
	Output; `name1.name2.(c)t, name1.name2.(c)t.lsj` (in LSJ-coupling)
	Output; `name1.name2.xT`

Figure 3.2: Flow of files for a normal sequence of program runs. Extensions `(c)` indicate data files based on `rci` mixing coefficients. For `rtransition` the extension `x` denotes the multipole.

3.4 Old and new program names

Below is a conversion table between old and new program names.

Table 3.2: Table of name conversions

GRASP2K old version	GRASP2018 version
bioscl	rtransition
bioscl_mpi	rtransition_mpi
biotra	rbiotransform
biotra_mpi	rbiotransform_mpi
erwf	rwfnestimate
extmix	rmixextract
iso	rnucleus
jj2lsj	jj2lsj
jjgen	jjgen
jjreduce	rcsfinteract
jsplit	rcsfblock obsolete, blocks handled automatically
mchfmcdf	rwfnmchfmcdf
mcp	rangular
mcp_mpi	rangular_mpi
plotmcdf	rwfnplot
rci	rci
rci_mpi	rci_mpi
rhfs	rhfs
rlevels	rlevels, rlevelseV
rotate_pair	rwfnrotate
rsave	rsave
rscf	rmcdhf
rscf_mpi	rmcdhf_mpi
sms	rsms

3.5 Test data set under bash

Along with GRASP2018 there is a test data set. Under the `grasptest` directory the following sub-directories reside

```
example1 : script, output
example2 : script, output
example3 : script, output
example4 : script, output, tmp_mpi
example5 : script, output
case1    : script
case1_mpi: script, tmp_mpi
case2    : script
case2_mpi: script, tmp_mpi
case3    : script
```

The `output` directories contain output files from the five examples in chapter 7. To validate program operations these output files can be used as references. Script files for the examples and the three case studies can be found in the `script` directories. In example 4 and in the first two cases studies the MPI codes are used and the temporary files from each process are written to the `tmp_mpi` sub-directories. Carefully look at the `README` file before running the scripts in these three cases. The scripts have been tested under bash.

Chapter 4

Important concepts and aspects of processing

4.1 Generating lists of CSFs

Wave functions are expanded in configuration state functions (CSFs), where effects beyond the single CSF approximation are referred to as correlation effects. Exploring different electron correlation models and generating lists of CSFs is thus a major task of the computation. To generate lists of CSFs based on the notion of excitations[1] from orbitals in a multireference (MR) to an active set of orbitals it is advantageous to use the `rcsfgenerate` program. Different restrictions can be put on the excitations, and it is possible to generate CSFs that describe valence-valence, core-valence and core-core correlation in different combinations. To make sure that the generated CSFs interact with the CSFs in the multireference the program `rcsfinteract` should be used.

The reader is advised to work through the examples in chapter 6 on how to use `rcsfgenerate`. The reader may also want to read the write-up of the `jjgen` program [10], the predecessor of `rcsfgenerate`. The write-up provides a number of examples on how to generate expansions capturing different correlation effects. The general theory, Z-dependent perturbation theory, for generating CSFs is described in [24], chapters 4 and 5. See also section 5.4 of this guide.

4.2 Lists of CSFs and symmetry blocks

A list of CSFs starts with a line that defines the core subshells (or orbitals). The core orbitals are fully occupied in all CSFs and need not be part of the specification of the CSFs. After the line with the core orbitals there is a line of the remaining subshells (peel subshells). The specification of the orbitals is followed by the list of CSFs, where each CSF comprise three lines. The CSFs are arranged into symmetry blocks, where the different blocks are separated by an asterisk. We take a concrete example.

```
Core subshells:
  1s   2s   2p-  2p
Peel subshells:
  3s   3p-  3p
CSF(s):
  3s ( 1)  3p ( 2)
      1/2        0
```

[1]We normally speak about excitations, but in fact it could also be de-excitations if we have configurations that are not the ground configuration. Instead of talking about excitations (de-excitations) we could talk about substitutions.

```
                1/2+
  3s ( 1)  3p-( 1)  3p ( 1)
      1/2      1/2       3/2
                    1    1/2+
  3s ( 1)  3p-( 2)
      1/2
                1/2+
 *
  3s ( 1)  3p ( 2)
      1/2        2
                3/2+
  3s ( 1)  3p-( 1)  3p ( 1)
      1/2      1/2       3/2
                    0    3/2+
  3s ( 1)  3p-( 1)  3p ( 1)
      1/2      1/2       3/2
                    1    3/2+
 *
  3s ( 1)  3p ( 2)
      1/2        2
                5/2+
  3s ( 1)  3p-( 1)  3p ( 1)
      1/2      1/2       3/2
                    1    5/2+
```

There are four core subshells `1s, 2s, 2p-, 2p` corresponding to a $1s^2 2s^2 2p^6$ closed core (in non-relativistic notation) that is common to all CSFs. After the line with core subshells there is the line with the peel subshells, `3s, 3p-, 3p`. The peel subshells (or orbitals) are the orbitals in the active set that are used in the construction of the CSFs in the list. The core subshells are not part of the active set. After the orbital specifications the list of CSFs appear. Each CSF is written on three lines. The first line gives the configuration. The second line gives the J quantum number of each subshell. The third line shows how the J quantum numbers of each subshell are coupled together from left to right. Looking at the first CSF in the list

```
  3s ( 1)  3p ( 2)
      1/2        0
                1/2+
```

the `3s ( 1)` subshell has $J = 1/2$ and the `3p ( 2)` subshell is coupled to $J = 0$. The third line defines how the J quantum numbers of the different shells are coupled from left to right to a final J quantum number $J = 1/2+$, where $+$ denotes positive (even) parity.

In the current version of the codes the CSFs are automatically arranged into symmetry blocks, where the different blocks are separated by an asterisk. In the example above there are three symmetry blocks $J = 1/2+$, $3/2+$, $5/2+$ separated by an asterisk $*$.

4.3 Providing initial estimates of the radial wave functions

The program application **`rwfnestimate`** generates initial estimates for radial wave functions. These estimates may be generated using a Thomas-Fermi potential. Alternatively, the initial radial wave functions can be taken as screened hydrogenic functions. Converted Hartree-Fock (HF) or multiconfiguration Hartree-Fock (MCHF) radial wave functions or radial wave functions from previous **`rmcdhf`** runs may also be used. It is the experience of the authors that the use of converted HF or MCHF radial wave functions generally gives very good starting values and that this may cut down on the number of needed iterations in the self-consistent-field procedure. The conversion of HF or MCHF radial wave functions to relativistic radial wave functions is done by `rwfnmchfmcdf`. In the present implementation, prior to normalization,

$$
\begin{aligned}
P(n\kappa;r) &= P^{\mathrm{HF}}(nl;r) \\
Q(n\kappa;r) &= \frac{\alpha}{2}\left(\frac{d}{dr}+\frac{\kappa}{r}\right)P(n\kappa;r),
\end{aligned}
$$

which means that the relativistic orbital pair is strictly kinetically matched [3].

Alternatively, the program DBSR_HF [25] can be used. This is a B-spline solution of the Dirac-Hartree-Fock equation in which orbitals are obtained from eigenvectors of a Dirac-Fock operator and orthogonality is achieved through the use of projection operators. Thus the node-counting used by differential equations methods is avoided. The code is easy to use. The command

```
dbsr_hf Li_092 atom=U ion=Li mbreit=1 out_w=1
```

will determine orbitals for Li-like Uranium, with orbitals output in GRASP format.

4.4 Spectroscopic orbitals and convergence

The orbitals building the reference CSFs of the targeted states are referred to as spectroscopic orbitals. They are required to have the same number of nodes as the corresponding hydrogen like orbitals [24]. The requirements of a specified number of nodes sometimes lead to convergence problems in the self-consistent field procedure, especially for near neutral systems where initial estimates from e.g. screened hydrogenic functions are inaccurate. If convergence problems occur our first recommendation is to use the DBSR_HF code and request orbital output in GRASP format. These generated orbitals can be directly used as input or alternatively combined with other orbitals using `rwfnestimate`. If the user does not want to install the DBSR_HF code, another way is to start from converted HF or MCHF radial wave functions, see section 7.2. If convergence problems remain for the spectroscopic orbitals, the user may increase the nuclear charge Z and perform the calculation for some more highly charged ion. The, hopefully, converged radial wave functions from this run can then be used for another calculation, where the nuclear charge has been slightly decreased. The radial wave functions from this run are, provided they are converged, used in a new calculation where the nuclear charge has been decreased further etc. If this still does not lead to convergence the user may override the default options in the self-consistent field procedure. There is a number of options to aid convergence such as increasing the orbital damping. Finally, it can help to recompile the program and use a finer grid. The steps to achieve convergence are summarized below. Start with step 1 and if this does not work try step 2 etc.

1. Start from converted HF or MCHF radial wave functions.
2. Increase nuclear charge, again starting from converted HF or MCHF radial wave functions. If convergence is achieved decrease the nuclear charge in small steps. Use the converged radial wave functions from the previous `rmcdhf` run as input for the new `rmcdhf` calculation.
3. Use the above strategies together with non-default options in `rmcdhf` allowing direct control of damping and orbital updates.
4. Recompile the program and use a finer grid. Again try the strategies above.
5. If nothing helps see if it is possible to start with a different multireference.

Convergence will be further discussed in Part IV of this document in connection to some practical examples on how convergence of spectroscopic orbitals can be achieved in problematic cases.

4.5 Correlation orbitals and layer-by-layer calculations

Orbitals introduced to build CSFs that correct the reference CSFs, accounting for electron correlation effects, are called correlation orbitals. Although desirable it is often not possible to optimize all radial orbitals, spectroscopic and correlation orbitals, simultaneously. Instead the calculations are done layer-by-layer in a procedure that can be described as follows:

1. Perform calculation for the multireference where the orbitals are required to be spectroscopic.
2. Use the active set approach to generate the list of CSFs. Increase the active set systematically by adding a layer of correlation orbitals (a layer is a set of correlation orbitals such that there are no two orbitals with the same symmetry). Optimize only the outermost layer and keep the remaining orbitals fixed from the previous calculation.
3. Monitor the convergence of the calculated properties such as energy differences, transition rates, hfs, isotope shift, as the active set is increased.
4. Stop the calculations when the properties are converged at some level and when it is not meaningful to extend the active set further.
5. Relax the rules for generating CSF, perform calculations using RCI and check if the calculated properties are converged also with respect to the type (valence-valence, core-valence and core-core etc) of included electron correlation, see [2] for a general discussion of systematic methodologies.

4.6 Simultaneous calculations for many levels

In GRASP2018 calculations can be done for many levels (states) simultaneously, sometimes referred to as 'all levels' calculations or, if both even and odd parity levels are targeted at the same time, spectrum calculations. Although the wave function for each individual level (state) may not be the most accurate, simultaneous calculations lead to a balanced description of the levels with accurate energy separations. Simultaneous calculations are often done by term, which determines all the levels of an LS-term, by configuration, which determines all the levels of a configuration, or by parity, which determines all the desired levels with the same parity. Simultaneous calculations can be done also in other ways and may include all desired levels of both parities. Studies have been performed where hundreds of levels in an atomic spectrum have been determined simultaneously [26, 27].

In `rmcdhf` simultaneous calculations on many levels are done in the so called extended optimal level (EOL) mode. Here a weighted energy functional of a selected set of levels is constructed, and by applying the variational principle both the radial wave functions and the corresponding expansion coefficients are determined. As an example we consider $1s^22s2p$. We want to do the calculation by parity and determine the four levels $1s^22s2p\ ^3P_{0,1,2}$ and $1s^22s2p\ ^1P_1$ simultaneously. The $J = 0$ and $J = 2$ levels are the lowest of their symmetry. The two $J = 1$ are the lowest and the second lowest of their symmetry. In the `rmcdhf` calculation we would specify this by saying that we want the serial number 1 of symmetry $J = 0$, the serial numbers 1 and 2 of symmetry $J = 1$, and the serial number 1 of symmetry $J = 2$. In previous studies levels entering the construction of the energy functional have been equally weighted [28] and also weighted by the statistical weight $2J + 1$ [26, 27]. Depending on the case other weights may be useful.

4.7 Transverse photon interaction and self-energy correction

Relativistic corrections beyond the Dirac-Coulomb approximation for a many-electron system are implemented using assumptions based on one-electron concepts. For example, in the transverse photon interaction

$$H_{\mathrm{T}} = -\sum_{i<j}^{N}\left[\boldsymbol{\alpha}_i\cdot\boldsymbol{\alpha}_j\frac{\cos(\omega_{ij}r_{ij}/c)}{r_{ij}} \qquad + (\boldsymbol{\alpha}_i\cdot\boldsymbol{\nabla}_i)(\boldsymbol{\alpha}_j\cdot\boldsymbol{\nabla}_j)\frac{\cos(\omega_{ij}r_{ij}/c)-1}{\omega_{ij}^2 r_{ij}/c^2}\right], \tag{4.1}$$

which is the leading correction to the electron-electron Coulomb interaction, the frequency ω_{ij} is assumed to be the difference between the diagonal orbital energy parameters. This may be an appropriate assumption for singly occupied orbitals, but is not correct for multiply occupied ones and certainly is not true for correlation orbitals. For these reasons transverse photon interaction is often computed in the low-frequency limit by multiplying the frequency with a scale factor. The scale factor is often set to 10^{-6}. The transverse photon interaction with scaled frequencies is sometimes referred to as the Breit interaction.

Similarly, the self-energy correction is computed from a screened-hydrogenic approximation, a model that does not apply well to correlation orbitals that are far from hydrogenic. The `rci` code allows the user to specify the largest principal quantum number for which CSFs are to be considered in the self-energy corrections. For small calculations with a few correlation orbitals this cut-off is set to the largest principal quantum number of the included orbitals. In large calculations with many correlation orbitals the cut-off is typically set to a number somewhat larger than the highest principal quantum number of the spectroscopic orbitals. In many research articles the vacuum polarization and the self-energy correction are referred to as the leading quantum electrodynamic (QED) corrections.

4.8 Biorthonormal transformations for transition calculations

Transition parameters, such as rate and weighted oscillator strength, for a multipole transition of rank K from γPJ to $\gamma' P' J'$ are related to the reduced transition matrix element

$$\langle \Psi(\gamma' P' J') \| \mathbf{O}^K \| \Psi(\gamma PJ) \rangle, \tag{4.2}$$

where $\mathbf{O}^K$ is the transition operator. This matrix element is very time consuming to evaluate between separately determined initial and final state wave functions since the non-orthogonalities of the initial and final state orbital sets prevents Racah-algebra to be used. Provided the CSF expansions for the initial and final states are closed under de-excitation[2] it is possible to change the wave function representation of the two states in such a way that Racah-algebra can be used for the matrix elements in the new representation [21].

The procedure for calculating the oscillator strength can be summarized as follows:

1. Perform separate `rmcdhf` or `rci` calculations for the initial and the final states.
2. Change the initial and final state wave function representations by transforming the radial orbital sets to a biorthonormal orbital set. This is followed by a counter-transformation of the initial and final state expansion coefficients so as to leave the total wave functions invariant.
3. Calculate the transition matrix element with the transformed wave functions for which now the Racah-algebra can be used.

The biorthonormal transformation is very fast and is performed with the program `rbiotransform`. The evaluation of the transition parameters from the transformed initial and final wave functions is then performed with `rtransition`.

4.9 Angular data from rbiotransform and rtransition

The `rbiotransform` and `rtransition` programs and their MPI variants save angular data on file to speed up calculations for an iso-electronic sequence. If angular files are available the programs

[2] If a CSF based on a configuration is part of the list, then the CSFs based on the configurations where the orbitals are de-excited to orbitals with lower principal quantum numbers should also be part of the list. An expansion based on the active set approach is closed under de-excitation if the MR is closed under de-excitation. Please note that CSF lists based on the active set approach from a single core-excited configurations may not be closed under de-excitation although additional CSFs can be introduced for closure.

read these files and the execution time is reduced considerably. If, for some reason, there are incomplete files with angular coefficients these programs will end with some error message when trying to process the angular data files. In these cases the user should remove the angular files (they all have a capital `T` in the extension) and rerun the case again.

4.10 Managing large expansions - zero- and first order calculations

Often the CSFs expansions grow so large that they can not be handled with the available computational resources. In these cases an approximate computational scheme can be employed in which the CSF list is rearranged into zero- and a first-order spaces:

$$\underbrace{\Phi(\gamma_1^0 PJ), \Phi(\gamma_2^0 PJ), \ldots, \Phi(\gamma_M^0 PJ)}_{\text{zero-order space}}, \underbrace{\Phi(\gamma_1^1 PJ), \Phi(\gamma_2^1 PJ), \ldots, \Phi(\gamma_N^1 PJ)}_{\text{first-order space}}$$

where $M + N$ is the total number of CSFs in the original list. The zero-order space contains the most important CSFs while the first-order space contain less important CSFs that can be regarded as minor corrections. Normally $M \ll N$. Associated with the rearrangement of the CSFs is a decomposition of the Hamiltonian interaction matrix in submatrices

$$\begin{pmatrix} H^{00} & H^{01} \\ H^{10} & H^{11} \end{pmatrix}.$$

The energy expression, on which to optimize, is now obtained from the limited interaction matrix where the full H^{00}, H^{01}, H^{10} submatrices are included (interactions within the zero-order space and between the zero- and first-order spaces) but only the diagonal part of H^{11}. The rearrangement of the list of CSFs in zero- and first-order spaces is done by the program `rcsfzerofirst`. In the programs `rangular` and `rci`, which sets up expressions for the Hamiltonian, there is a question if full interaction should be considered or not. If not full interaction the user can specify the size of the zero-order space for each symmetry block. See [22] for recent applications of this methodology. The handling of large expansions is discussed and exemplified in chapter 14.

4.11 Running parallel programs using MPI

Some of the more time-consuming programs in GRASP2018 have been converted to run in parallel under MPI, a language-independent communication protocol used to program parallel computers. In order to compile the programs MPI libraries need to be installed. For cases where the MPI codes can be used the increase of speed is often substantial. In section 7.4 we show in detail how to set up the computational environment and use the MPI codes.

4.12 Restarting rci

`rci` and `rci_mpi` produce a file `rci.res` containing, in sparse representation, the matrix elements of the Hamiltonina. If, for some reason, an `rci` or `rci_mpi` run stalls then the programs can be restarted. During a restart the `rci.res` file is read, and the computation continues at the place where the original computation stalled. The restart option is described in section 7.7.

Part II

Generating lists of CSFs

Chapter 5

Lists of CSFs

5.1 Configurations, configuration state functions

A configuration is a number of orbitals with occupation numbers, e.g.

$$1s^2 2s^2 (2p\text{-})^2, \quad 1s^2 2s^2 (2p\text{-}) 2p, \quad 1s^2 2s^2 2p^2,$$

where we use the notation $1s, 2s, 2p\text{-}, 2p$ for $1s_{1/2}, 2s_{1/2}, 2p_{1/2}, 2p_{3/2}$. Frequently, the non-relativistic notation is used instead

$$1s^2 2s^2 2p^2.$$

Configuration state functions (CSFs) are formed by angular couplings of the orbitals in a configuration. Depending on the structure of the configuration, i.e. number of open shells, there may be many angular couplings and thus CSFs for each configuration. An angular coupling is sometimes referred to as a coupling tree.

In GRASP2018 the CSFs are given in `rcsf.inp`. The CSFs comprise three lines in the file. The first line gives the configuration and lines two and three define the coupling tree. The CSFs are ordered in blocks specified by parity and J symmetry, the blocks being separated by an asterisk $*$. Below are all the CSFs of even parity belonging to the configuration $1s^2 2s^2 2p^2$.

```
  1s ( 2)  2s ( 2)  2p ( 2)
                         0
                          0+
  1s ( 2)  2s ( 2)  2p-( 2)

                          0+
 *
  1s ( 2)  2s ( 2)  2p-( 1)  2p ( 1)
                      1/2      3/2
                                  1+
 *
  1s ( 2)  2s ( 2)  2p ( 2)
                         2
                          2+
  1s ( 2)  2s ( 2)  2p-( 1)  2p ( 1)
                      1/2      3/2
                                  2+
```

In the case above we have three symmetry blocks with even parity corresponding to $J = 0, 1, 2$.

GRASP2018 handles expansions with hundreds of thousands of CSFs even on a scalar computer. On a cluster expansions with millions of CSFs can be used. The success of a calculation depends on judiciously chosen CSFs.

5.2 Multireference

The starting point for a study is normally a calculation for a number of important CSFs that define the multireference (MR). The CSFs in the MR are those that can be formed from nearly degenerate configurations, see [24] chapter 4.[1] The wave function based on the CSFs in the MR is the first approximation and it is the starting point for further refinements. The concept of an MR is best illustrated by some examples.

Suppose we want to compute the wave function for the ground state $3s^2\ {}^1S_0$ of Mg I. The $3s^2$, $3p^2$ and $3d^2$ configurations are formed by orbitals with the same principal quantum numbers and the configurations are closely degenerate. An MR in this case could consist of the CSFs that can be formed from these configurations.

Suppose we want to compute the wave functions for the $3s3p\ {}^3P_{0,1,2},\ {}^1P_1$ excited states of Mg I. The $3s3p$ and $3p3d$ configurations are formed by orbitals with the same principal quantum numbers and these configurations are closely degenerate. An MR in this case could consist of the CSFs that can be formed from these configurations. However, it turns out that $3s4p$ is important and thus a more suitable MR should consist of CSFs also from the latter configuration.

We want to compute the wave functions for the states of $2s^22p^2$ and $2s2p^3$. The $2s^22p^2$ and $2p^4$ configurations are formed by orbitals with the same principal quantum numbers and the configurations are closely degenerate. Thus the MR for the even states would consist of the CSFs that can be formed from these two configurations. It turns out that the $2s2p^23d$ and $2s^23d^2$ configurations are important and a better MR includes CSFs also from these configurations. Looking at $2s2p^3$ there is no other configuration of the same parity that can be formed by orbitals with the same principal quantum numbers. In this case the MR would consist of CSFs formed from this single configuration. But, also $2p^33d$, $2s^22p3d$ and $2s2p3d^2$ are important and the MR should consist of CSFs also from these configurations.

We see that the selection of the MR is far from trivial and it often requires a number of exploratory calculations to find a good MR. The program `rcsfmr`, described in section 7.6, is designed to support the exploratory process.

For 'all levels' calculations or spectrum calculations, see section 4.7, where wave functions are determined for a number of states belonging to several configurations the MR is often taken as the set of CSFs that can be formed from these configurations. Suppose that we want to determine the wave functions for states of the $3l3l'$, $3l4l'$ and $3l5s$ configurations in Mg-like ions. The MR in this case would just be the CSFs that can be formed from these configurations. If we do the calculations by parity, the MR for the even parity states would be the CSFs formed from even parity configurations and the MR for the odd parity states would be the CSFs formed from odd parity configurations.

5.3 Active set approach

CSFs are often generated using the active set approach. In the active set approach CSFs of a specified parity and J symmetry are obtained from angular couplings of configurations generated by excitations from orbitals of one or more configurations in the MR to orbitals in an active set (AS). Orbitals of a reference configuration are classified as closed (c), inactive (i), active (*), or

[1] When talking about the MR we will, somewhat loosely, refer to both the set of CSFs and the set of configurations from which the CSFs are formed

active having minimal occupation (m). The active set consists of the active orbitals in the reference configuration together with orbitals up to a given limit specified by the highest principal quantum number of each orbital symmetry. Closed orbitals are fully occupied and make up the core. No excitations are allowed from inactive orbitals of the reference configuration. Excitations are allowed from the active orbitals of the reference configuration to orbitals in the active set. Excitations from active orbitals having minimal occupations are such that the occupations after the excitations are always larger or equal to the specified minimal occupation.

Based on perturbation theory one can show that the major electron correlation effects are captured by including, in the ASF, the CSFs that can be formed from configurations obtained by allowing single (S) and double (D) excitations from the most important configurations, defining the multireference (MR), to an extended active set of orbitals [24].

For small systems, e.g. nominal three and four electron systems, it is sometimes advantageous to include CSFs that can be formed from all possible excitations: single (S), double (D), triple (T), (Q) quadruple, etc. This expansion is referred to as the complete active space (CAS).

5.4 Different types of correlation effects

For complex systems it may not be possible, or even desirable, to allow excitations from all orbitals of the MR. Often excitations are done only from outer orbitals and the corresponding CSFs are said to describe valence-valence correlation. If one excitation is from a core orbital and one from an outer orbital then the corresponding CSFs are said to describe core-valence correlation. If both excitations are from the core the corresponding CSFs are said to describe core-core correlation.

As an example of correlation effects we look at the ground state of Mg $1s^22s^22p^63s^2\ ^1S_0$. To make things simple we consider only a single reference.

Valence-valence correlation
CSFs based on configurations of the type $1s^22s^22p^6nln'l'$ represent valence-valence correlation. In the active set approach these configurations can be formed by starting from $1s^22s^22p^63s^2$ and classifying the $1s, 2s, 2p$ orbitals as inactive (i) and the $3s$ orbital as active (*). In our notation we have

```
1s(2,i)2s(2,i)2p(6,i)3s(2,*)
```

SD-excitations are then allowed to orbitals in the active set.

Core-valence correlation
CSFs based on configurations of the type $1s^22s^22p^5nl3sn'l'$, $1s^22snl2p^63sn'l'$ represent core-valence correlation involving the $2s^22p^6$ core. In the active set approach the configurations of the first type can be formed by starting from $1s^22s^22p^63s^2$ and classifying the $1s, 2s$ orbitals as inactive (i), the $2p$ orbital as active having a minimal occupation 5 and the $3s$ orbital as active having a minimal occupation 1. In our notation we have

```
1s(2,i)2s(2,i)2p(6,5)3s(2,1)
```

SD-excitations are then allowed to orbitals in the active set. Configurations of the second type can be formed by starting from $1s^22s^22p^63s^2$ and classifying the $1s, 2p$ orbitals as inactive (i), the $2s$ orbital as active having a minimal occupation 1 and the $3s$ orbital as active having a minimal occupation 1. In our notation this is

```
1s(2,i)2s(2,1)2p(6,i)3s(2,1)
```

SD-excitations are then allowed to orbitals in the active set. In practical applications one most often treats valence-valence and core-valence correlation together and this is achieved by classifying the $3s$ orbital as active instead of active with minimal occupation 1. This corresponds to

```
1s(2,i)2s(2,i)2p(6,5)3s(2,*)
```

and

```
1s(2,i)2s(2,1)2p(6,i)3s(2,*)
```

Core-core correlation
CSFs based on configurations of the type $1s^2 2s^2 2p^4 nln'l'3s^2$, $1s^2 2snl 2p^5 n'l'3s^2$, $1s^2 nln'l'2p^6 3s^2$ represent core-core correlation in the $2s^2 2p^6$ core. In the active set approach these configurations can be formed by starting from $1s^2 2s^2 2p^6 3s^2$ and classifying the $1s$ orbital as inactive (i), the $2s, 2p$ orbitals as active (*) and the $3s$ orbital as inactive (i). In our notation

```
1s(2,i)2s(2,*)2p(6,*)3s(2,i)
```

SD-excitations are then allowed to orbitals in the active set. In practical applications we very seldom treat core-core correlation alone. Instead we treat valence-valence, core-valence and core-core correlation together and this is achieved by classifying the $2s, 2p, 3s$ orbitals active (*)

```
1s(2,i)2s(2,*)2p(6,*)3s(2,*)
```

and allowing SD-excitations to orbitals in the active set.

Atomic properties depend in various ways on electron correlation effects. For transition rate calculations it is important to include valence-valence and core-valence correlation. For calculations of hyperfine structure and isotope shift it is important to include also deep core-valence correlation effects.

5.5 Doubly occupied correlation orbitals

Accounting for electron correlation effects including core-core often leads to very large expansions. Imposing the restriction that correlation orbitals are doubly occupied reduces the expansion size. For example, if $3s$, $4s$, $5s$, $3p$-, $3p$, $4p$-, $4p$, $5p$-, $5p$, $3d$-, $3d$, $4d$-, $4d$, $5d$-, $5d$ are correlation orbitals in relativistic notation only excitation pairs $3s^2$, $4s^2$, $5s^2$, $(3p\text{-})^2$, $(3p\text{-})3p$, $3p^2$, $(4p\text{-})^2$, $(4p\text{-})4p$, $4p^2$ etc are allowed. Such an expansion still describes a fair part of the correlation. A practical example of how to use the restriction that correlation orbitals are doubly occupied is given in the sixth example of the next chapter.

5.6 CSFs interacting with the MR

It is important to realize that the active set approach, as we have described it above, is based on generation of configurations that are then coupled to form CSFs. However, not all CSFs generated in this way have non-zero Hamiltonian matrix elements (interact) with CSFs in the MR. Generated CSFs not interacting with the CSFs of the MR can, without major loss of accuracy, be removed from the list of CSFs [24]. This is done by the program `rcsfinteract`. The reduction of CSFs is important mainly for complex systems, where the list of CSFs grow very rapidly with the increasing active set of orbitals.

Chapter 6

Running the CSFs generation programs

6.1 First example: valence-valence, core-valence and core-core for $1s^2 2s^2\ {}^1S_0$

We want to generate an expansion for the $1s^2 2s^2\ {}^1S_0$ state. In this example the CSFs are generated by SD-excitations from the $\{1s^2 2s^2, 1s^2 2p^2\}$ multireference set to an active set characterized by a maximal principal quantum number $n = 4$. The expansion accounts for valence-valence, core-valence and core-core correlation.

```
******************************************************************************
*          RUN RCSFGENERATE                                                  *
*          OUTPUT FILES: rcsf.out, rcsfgenerate.log                          *
******************************************************************************

>>rcsfgenerate

 RCSFGENERATE
 This program generates a list of CSFs

 Configurations should be entered in spectroscopic notation
 with occupation numbers and indications if orbitals are
 closed (c), inactive (i), active (*) or has a minimal
 occupation e.g. 1s(2,1)2s(2,*)
 Outputfiles: rcsf.out, rcsfgenerate.log

 Default, reverse, symmetry or user specified ordering? (*/r/s/u)
>>*

 Select core
        0: No core
        1: He (       1s(2)                  =  2 electrons)
        2: Ne ([He] + 2s(2)2p(6)             = 10 electrons)
        3: Ar ([Ne] + 3s(2)3p(6)             = 18 electrons)
        4: Kr ([Ar] + 3d(10)4s(2)4p(6)       = 36 electrons)
        5: Xe ([Kr] + 4d(10)5s(2)5p(6)       = 54 electrons)
        6: Rn ([Xe] + 4f(14)5d(10)6s(2)6p(6) = 86 electrons)
>>0
```

```
 Enter list of (maximum 100) configurations. End list with a blank line or an asterisk (*)

 Give configuration  1
>>1s(2,*)2s(2,*)
 Give configuration  2
>>1s(2,*)2p(2,*)
 Give configuration  3
>>
 Give set of active orbitals, as defined by the highest principal quantum number
 per l-symmetry, in a comma delimited list in s,p,d etc order, e.g. 5s,4p,3d
>>4s,4p,4d,4f
 Resulting 2*J-number? lower, higher (J=1 -> 2*J=2 etc.)
>>0,0
 Number of excitations (if negative number e.g. -2, correlation
 orbitals will always be doubly occupied)
>>2
 Generate more lists ? (y/n)
>>n
    ......

  1 blocks were created

      block  J/P            NCSF
          1    0+            361
```

Note that by answering 2 for the the number of excitations we will include both single (S) and double (D) excitations. By default the orbitals will be in the order $1s, 2s, 2p, 3s, 3p, 3d$ etc. There is also the possibility to have a reverse orbital order $\ldots 3d, 3p, 3s, 2p, 2s, 1s$, a symmetry order $1s, 2s, 3s, \ldots, 2p, 3p, \ldots, 3d, 4d, \ldots$ or a user defined order. We will look at these options in section 6.9.

6.2 Second example: valence-valence, core-valence for $1s^22s^22p^63s3p\ ^3P_{0,1,2}, {}^1P_1$

We want to generate expansions for $1s^22s^22p^63s3p\ ^3P_{0,1,2},\ ^1P_1$. In this example the CSFs are generated by SD-excitations from $\{1s^22s^22p^63s3p, 1s^22s^22p^63p3d\}$ to an active set $n = 5$ with the restrictions that $1s$ is closed and that there is at most one excitation from orbitals with $n = 2$. The expansions account for valence-valence and core-valence correlation.

```
*******************************************************************************
*          RUN RCSFGENERATE                                                   *
*          OUTPUT FILES: rcsf.out, rcsfgenerate.log                           *
*******************************************************************************

>>rcsfgenerate

 RCSFGENERATE
 This program creates a list of CSFs
 Configurations should be entered in spectroscopic notation
 with occupation numbers and indications if orbitals are
 closed (c), inactive (i), active (*) or has a minimal
 occupation e.g. 1s(2,1)2s(2,*)
 Outputfiles: rcsf.out, rcsfgenerate.log
```

```
 Default, reverse, symmetry or user specified ordering? (*/r/s/u)
>>*

 Select core
        0: No core
        1: He (       1s(2)                  =  2 electrons)
        2: Ne ([He] + 2s(2)2p(6)             = 10 electrons)
        3: Ar ([Ne] + 3s(2)3p(6)             = 18 electrons)
        4: Kr ([Ar] + 3d(10)4s(2)4p(6)       = 36 electrons)
        5: Xe ([Kr] + 4d(10)5s(2)5p(6)       = 54 electrons)
        6: Rn ([Xe] + 4f(14)5d(10)6s(2)6p(6) = 86 electrons)
>>1
 Enter list of (maximum 100) configurations. End list with a blank line or an asterisk (*)

 Give configuration  1
>>2s(2,1)2p(6,i)3s(1,*)3p(1,*)
 Give configuration  2
>>2s(2,i)2p(6,5)3s(1,*)3p(1,*)
 Give configuration  3
>>2s(2,1)2p(6,i)3p(1,*)3d(1,*)
 Give configuration  4
>>2s(2,i)2p(6,5)3p(1,*)3d(1,*)
 Give configuration  5
>>
 Give set of active orbitals, as defined by the highest principal quantum number
 per l-symmetry, in a comma delimited list in s,p,d etc order, e.g. 5s,4p,3d
>>5s,5p,5d,5f,5g
 Resulting 2*J-number? lower, higher (J=1 -> 2*J=2 etc.)
>>0,4
 Number of excitations (if negative number e.g. -2, correlation
 orbitals will always be doubly occupied)
>>2
 Generate more lists ? (y/n)
>>n
   ......

  3 blocks were created

      block  J/P            NCSF
          1    0-           1912
          2    1-           5210
          3    2-           7122
```

6.3 Third example: valence-valence, core-valence and intercore for $1s^22s^22p^63s3p\ ^3P_{0,1,2}, {}^1P_1$

We want to generate expansions for $1s^22s^22p^63s3p\ ^3P_{0,1,2},\ ^1P_1$. In this example the CSFs are generated by SD-excitations from $\{1s^22s^22p^63s3p, 1s^22s^22p^63p3d\}$ to an active set $n = 5$ with the restrictions that $1s$ is closed (and hence inactive) and that there is at most one excitation from $2s$ and $2p$, respectively. In this case, in addition to valence-valence and core-valence correlation, also inter-core correlation are accounted for through configurations of the form $1s^22snl2p^5n'l'3s3p$ $1s^22snl2p^5n'l'3p3d$, where $1s^2$ is inactive. Note how much the number of CSFs has increased.

```
******************************************************************************
*          RUN RCSFGENERATE                                                  *
*          OUTPUT FILES: rcsf.out, rcsfgenerate.log                          *
******************************************************************************

>>rcsfgenerate

 RCSFGENERATE
 This program creates a list of CSFs
 Configurations should be entered in spectroscopic notation
 with occupation numbers and indications if orbitals are
 closed (c), inactive (i), active (*) or has a minimal
 occupation e.g. 1s(2,1)2s(2,*)
 Outputfiles: rcsf.out, rcsfgenerate.log

 Default, reverse, symmetry or user specified ordering? (*/r/s/u)
>>*

 Select core
        0: No core
        1: He (       1s(2)                  =  2 electrons)
        2: Ne ([He] + 2s(2)2p(6)             = 10 electrons)
        3: Ar ([Ne] + 3s(2)3p(6)             = 18 electrons)
        4: Kr ([Ar] + 3d(10)4s(2)4p(6)       = 36 electrons)
        5: Xe ([Kr] + 4d(10)5s(2)5p(6)       = 54 electrons)
        6: Rn ([Xe] + 4f(14)5d(10)6s(2)6p(6) = 86 electrons)
>>1
 Enter list of (maximum 100) configurations. End list with a blank line or an asterisk (*)

 Give configuration  1
>>2s(2,1)2p(6,5)3s(1,*)3p(1,*)
 Give configuration  2
>>2s(2,1)2p(6,5)3p(1,*)3d(1,*)
 Give configuration  3
>>
 Give set of active orbitals, as defined by the highest principal quantum number
 per l-symmetry, in a comma delimited list in s,p,d etc order, e.g. 5s,4p,3d
>>5s,5p,5d,5f,5g
 Resulting 2*J-number? lower, higher (J=1 -> 2*J=2 etc.)
>>0,4
 Number of excitations (if negative number e.g. -2, correlation
 orbitals will always be doubly occupied)
>>2
 Generate more lists ? (y/n)
>>n
   ...........

 3 blocks were created

      block  J/P            NCSF
          1    0-          10743
```

```
    2    1-          29589
    3    2-          41500
```

6.4 Fourth example: valence-valence and core-valence and large multireference

We want to generate CSF expansions that describe all 92 states with symmetries $J = 0, 1, 2, 3, 4, 5$ of the configurations $\{2s^2 2p^2, 2p^4, 2s^2 2p3p, 2s2p^2 3s, 2s2p^2 3d\}$. In this example the CSFs are generated by SD-excitations from $\{2s^2 2p^2, 2p^4, 2s^2 2p3p, 2s2p^2 3s, 2s2p^2 3d\}$ to an active set $n = 5$ with the restriction that there is at most one excitation from $1s$. The expansions account for valence-valence and core-valence correlation.

```
*******************************************************************************
*          RUN RCSFGENERATE                                                   *
*          OUTPUT FILES: rcsf.out, rcsfgenerate.log                           *
*******************************************************************************

 RCSFGENERATE
 This program creates a list of CSFs
 Configurations should be entered in spectroscopic notation
 with occupation numbers and indications if orbitals are
 closed (c), inactive (i), active (*) or has a minimal
 occupation e.g. 1s(2,1)2s(2,*)
 Outputfiles: rcsf.out, rcsfgenerate.log

 Default, reverse, symmetry or user specified ordering? (*/r/s/u)
>>*

 Select core
        0: No core
        1: He (       1s(2)                  =  2 electrons)
        2: Ne ([He] + 2s(2)2p(6)             = 10 electrons)
        3: Ar ([Ne] + 3s(2)3p(6)             = 18 electrons)
        4: Kr ([Ar] + 3d(10)4s(2)4p(6)       = 36 electrons)
        5: Xe ([Kr] + 4d(10)5s(2)5p(6)       = 54 electrons)
        6: Rn ([Xe] + 4f(14)5d(10)6s(2)6p(6) = 86 electrons)
>>0
 Enter list of (maximum 100) configurations. End list with a blank line or an asterisk (*)

 Give configuration  1
>>1s(2,1)2s(2,*)2p(2,*)
 Give configuration  2
>>1s(2,1)2p(4,*)
 Give configuration  3
>>1s(2,1)2s(2,*)2p(1,*)3p(1,*)
 Give configuration  4
>>1s(2,1)2s(1,*)2p(2,*)3s(1,*)
 Give configuration  5
>>1s(2,1)2s(1,*)2p(2,*)3d(1,*)
 Give configuration  6
>>
 Give set of active orbitals, as defined by the highest principal quantum number
 per l-symmetry, in a comma delimited list in s,p,d etc order, e.g. 5s,4p,3d
```

```
>>5s,5p,5d,5f,5g
 Resulting 2*J-number? lower, higher (J=1 -> 2*J=2 etc.)
>>0,10
 Number of excitations (if negative number e.g. -2, correlation
 orbitals will always be doubly occupied)
>>2
 Generate more lists ? (y/n)
>>n
    ......

  6 blocks were created

       block  J/P            NCSF
           1    0+          14351
           2    1+          38928
           3    2+          53645
           4    3+          56147
           5    4+          48973
           6    5+          36562
```

6.5 Fifth example: CSFs interacting with CSFs in the MR

In this example we show how to reduce the number of CSFs in the previous list by retaining only the CSFs that interact with the CSFs of the multireference through the Dirac-Coulomb or Dirac-Coulomb-Breit Hamiltonian. We start by copying `rcsf.out` from the previous run to `rcsf.inp`. After that we generate the list of CSFs for the MR.

```
******************************************************************************
*          COPY FILE                                                         *
******************************************************************************

>>cp rcsf.out rcsf.inp

******************************************************************************
*          RUN RCSFGENERATE                                                  *
*          OUTPUT FILES: rcsf.out, rcsfgenerate.log                          *
******************************************************************************

>>rcsfgenerate

 RCSFGENERATE
 This program creates a list of CSFs
 Configurations should be entered in spectroscopic notation
 with occupation numbers and indications if orbitals are
 closed (c), inactive (i), active (*) or has a minimal
 occupation e.g. 1s(2,1)2s(2,*)
 Outputfiles: rcsf.out, rcsfgenerate.log

 Default, reverse, symmetry or user specified ordering? (*/r/s/u)
>>*

 Select core
        0: No core
        1: He (       1s(2)                  =  2 electrons)
```

```
        2: Ne ([He] + 2s(2)2p(6)             = 10 electrons)
        3: Ar ([Ne] + 3s(2)3p(6)             = 18 electrons)
        4: Kr ([Ar] + 3d(10)4s(2)4p(6)       = 36 electrons)
        5: Xe ([Kr] + 4d(10)5s(2)5p(6)       = 54 electrons)
        6: Rn ([Xe] + 4f(14)5d(10)6s(2)6p(6) = 86 electrons)
>>0
 Enter list of (maximum 100) configurations. End list with a blank line or an asterisk (*)

 Give configuration  1
>>1s(2,i)2s(2,i)2p(2,i)
 Give configuration  2
>>1s(2,i)2p(4,i)
 Give configuration  3
>>1s(2,i)2s(2,i)2p(1,i)3p(1,i)
 Give configuration  4
>>1s(2,i)2s(1,i)2p(2,i)3s(1,i)
 Give configuration  5
>>1s(2,i)2s(1,i)2p(2,i)3d(1,i)
 Give configuration  6
>>
 Give set of active orbitals, as defined by the highest principal quantum number
 per l-symmetry, in a comma delimited list in s,p,d etc order, e.g. 5s,4p,3d
>>3s,3p,3d
 Resulting 2*J-number? lower, higher (J=1 -> 2*J=2 etc.)
>>0,10
 Number of excitations (if negative number e.g. -2, correlation
 orbitals will always be doubly occupied)
>>0
 Generate more lists ? (y/n)
>>n
    ......

  6 blocks were created

      block  J/P           NCSF
          1    0+            14
          2    1+            25
          3    2+            28
          4    3+            16
          5    4+             7
          6    5+             2

*******************************************************************************
*          COPY RCSF.OUT TO RCSFMR.INP                                        *
*******************************************************************************

>>cp rcsf.out rcsfmr.inp

*******************************************************************************
*          RUN RCSFINTERACT                                                   *
*          INPUT FILES: rcsf.inp, rcsfmr.inp                                  *
*          OUTPUT FILE: rcsf.out                                              *
*******************************************************************************
```

```
>>rcsfinteract

RCSFinteract: Determines all the CSFs (rcsf.inp) that interact
              with the CSFs in the multireference (rcsfmr.inp)
              (C)  Copyright by G. Gaigalas and Ch. F. Fischer
              (Fortran 95 version)              NIST  (2017).
              Input files: rcsfmr.inp, rcsf.inp
              Output file: rcsf.out

 Reduction based on Dirac-Coulomb (1) or
 Dirac-Coulomb-Breit (2) Hamiltonian?
>>1

  .....

 There are 25 relativistic subshells;
  Block    MR NCSF   Before NCSF   After NCSF
    1          14        14351         7765
    2          25        38928        24492
    3          28        53645        33925
    4          16        56147        29299
    5           7        48973        17134
    6           2        36562         7542
 RCSFINTERACT: Execution complete
```

Comparing with what we had before we see that there is a great reduction in the number of CSFs where the removed CSFs are relatively unimportant. The reduction based on the Dirac-Coulomb-Breit Hamiltonian gives somewhat more CSFs compared to the reduction based on the Dirac-Coulomb Hamiltonian. There is however not a big difference.

6.6 Sixth example: core-core and doubly occupied orbitals

Allowing SD-excitations from all shells of a MR without restrictions leads to large expansions. We may impose different restrictions allowing, for example, at most one excitation from the core. The resulting expansion accounts for valence-valence and core-valence electron correlation. Another restriction is to require that all correlation orbitals are doubly occupied in the generated CSFs. This cuts down the expansion size quite substantially, but still efficiently accounts for much of the correlation.

We generate a CSF expansion that describes the states with symmetries $J = 0, 1, 2$ of the configuration $2s^2 2p^6 3s3p$. CSFs are generated by SD-excitations from $\{2s^2 2p^6 3s3p, 2s^2 2p^6 3p3d\}$ to an active set $n = 8$ and symmetry $l = h$ with the restriction that there is at most one excitation from $2s^2 2p^6$. The expansion accounts for valence-valence and core-valence correlation. In addition there are SD-excitations from $\{2s^2 2p^6 3s3p, 2s^2 2p^6 3p3d\}$ to an active set $n = 8$ and symmetry $l = h$ with the restriction that the correlation orbitals are doubly occupied (see section 5.5). This part of the expansion accounts for part of the core-core correlation.

```
******************************************************************************
*          RUN RCSFGENERATE                                                  *
*          OUTPUT FILES: rcsf.out, rcsfgenerate.log                          *
******************************************************************************

 RCSFGENERATE
```

```
 This program creates a list of CSFs
 Configurations should be entered in spectroscopic notation
 with occupation numbers and indications if orbitals are
 closed (c), inactive (i), active (*) or has a minimal
 occupation e.g. 1s(2,1)2s(2,*)
 Outputfiles: rcsf.out, rcsfgenerate.log

 Default, reverse, symmetry or user specified ordering? (*/r/s/u)
>>*

 Select core
        0: No core
        1: He (       1s(2)                  =  2 electrons)
        2: Ne ([He] + 2s(2)2p(6)             = 10 electrons)
        3: Ar ([Ne] + 3s(2)3p(6)             = 18 electrons)
        4: Kr ([Ar] + 3d(10)4s(2)4p(6)       = 36 electrons)
        5: Xe ([Kr] + 4d(10)5s(2)5p(6)       = 54 electrons)
        6: Rn ([Xe] + 4f(14)5d(10)6s(2)6p(6) = 86 electrons)
>>1
 Enter list of (maximum 100) configurations. End list with a blank line or an asterisk (*)

 Give configuration           1
>>2s(2,i)2p(6,5)3s(1,*)3p(1,*)
 Give configuration           2
>>2s(2,1)2p(6,i)3s(1,*)3p(1,*)
 Give configuration           3
>>2s(2,i)2p(6,5)3p(1,*)3d(1,*)
 Give configuration           4
>>2s(2,1)2p(6,i)3p(1,*)3d(1,*)
 Give configuration           5
>>
 Give set of active orbitals, as defined by the highest principal quantum number
 per l-symmetry, in a comma delimited list in s,p,d etc order, e.g. 5s,4p,3d
>>8s,8p,8d,8f,8g,8h
 Resulting 2*J-number? lower, higher (J=1 -> 2*J=2 etc.)
>>0,4
 Number of excitations (if negative number e.g. -2, correlation
 orbitals will always be doubly occupied)
>>2
 Generate more lists ? (y/n)
>>y
 Enter list of (maximum 100) configurations. End list with a blank line or an asterisk (*)

 Give configuration           1
>>2s(2,*)2p(6,*)3s(1,*)3p(1,*)
 Give configuration           2
>>2s(2,*)2p(6,*)3p(1,*)3d(1,*)
 Give configuration           3
>>
 Give set of active orbitals, as defined by the highest principal quantum number
 per l-symmetry, in a comma delimited list in s,p,d etc order, e.g. 5s,4p,3d
>>8s,8p,8d,8f,8g,8h
 Resulting 2*J-number? lower, higher (J=1 -> 2*J=2 etc.)
>>0,4
```

```
 Number of excitations (if negative number e.g. -2, correlation
 orbitals will always be doubly occupied)
>>-2
 Generate more lists ? (y/n)
>>n
    ......

 3 blocks were created

      block  J/P           NCSF
          1    0-         21399
          2    1-         59512
          3    2-         85284
```

6.7 Running rcsfgenerate more than once

We may merge CSF expansions by running `rcsfgenerate` twice in the following way:

```
******************************************************************************
*          RUN RCSFGENERATE                                                  *
*          OUTPUT FILES: rcsf.out, rcsfgenerate.log                          *
******************************************************************************

>>rcsfgenerate

 RCSFGENERATE
 This program creates a list of CSFs
 Configurations should be entered in spectroscopic notation
 with occupation numbers and indications if orbitals are
 closed (c), inactive (i), active (*) or has a minimal
 occupation e.g. 1s(2,1)2s(2,*)
 Outputfiles: rcsf.out, rcsfgenerate.log

 Default, reverse, symmetry or user specified ordering? (*/r/s/u)
>>*

 Select core

        0: No core
        1: He (       1s(2)                  =  2 electrons)
        2: Ne ([He] + 2s(2)2p(6)             = 10 electrons)
        3: Ar ([Ne] + 3s(2)3p(6)             = 18 electrons)
        4: Kr ([Ar] + 3d(10)4s(2)4p(6)       = 36 electrons)
        5: Xe ([Kr] + 4d(10)5s(2)5p(6)       = 54 electrons)
        6: Rn ([Xe] + 4f(14)5d(10)6s(2)6p(6) = 86 electrons)
>>0
 Enter list of (maximum 100) configurations. End list with a blank line or an asterisk (*)

 Give configuration           1
>>1s(2,*)2p(1,*)
 Give configuration           2
>>
 Give set of active orbitals, as defined by the highest principal quantum number
 per l-symmetry, in a comma delimited list in s,p,d etc order, e.g. 5s,4p,3d
```

```
>>5s,5p,5d,5f,5g
 Resulting 2*J-number? lower, higher (J=1 -> 2*J=2 etc.)
>>1,3
 Number of excitations (if negative number e.g. -2, correlation
 orbitals will always be doubly occupied)
>>3
 Generate more lists ? (y/n)
>>y
 Enter list of (maximum 100) configurations. End list with a blank line or an asterisk (*)

 Give configuration           1
>>1s(2,*)2p(1,*)
 Give configuration           2
>>
 Give set of active orbitals, as defined by the highest principal quantum number
 per l-symmetry, in a comma delimited list in s,p,d etc order, e.g. 5s,4p,3d
>>7s,7p,7d,7f,7g,7h,7i
 Resulting 2*J-number? lower, higher (J=1 -> 2*J=2 etc.)
>>1,3
 Number of excitations (if negative number e.g. -2, correlation
 orbitals will always be doubly occupied)
>>2
 Generate more lists ? (y/n)
>>n
.........

 2 blocks were created

      block  J/P           NCSF
          1  1/2-          2408
          2  3/2-          4174
```

As expected we get the same number of CSFs in the two runs. Please note that the resulting J number needs to be the same when running `rcsfgenerate` several times for the same parity.

6.8 Running rcsfgenerate for even and odd parity

We want to generate CSFs for odd states with $J = 1/2, 3/2$ by allowing all SDT-excitations from $1s^2 2p$ and for even states with $J = 1/2$ by allowing all SDT-excitations from $1s^2 2s$. In both cases the excitations are to an active set with $n = 5$.

```
******************************************************************************
*          RUN RCSFGENERATE FOR ODD AND EVEN PARITY                          *
*          OUTPUT FILES: rcsf.out, rcsfgenerate.log                          *
******************************************************************************

>>rcsfgenerate

 RCSFGENERATE
 This program creates a list of CSFs
 Configurations should be entered in spectroscopic notation
 with occupation numbers and indications if orbitals are
 closed (c), inactive (i), active (*) or has a minimal
 occupation e.g. 1s(2,1)2s(2,*)
```

```
 Outputfiles: rcsf.out, rcsfgenerate.log

 Default, reverse, symmetry or user specified ordering? (*/r/s/u)
>>*

 Select core
        0: No core
        1: He (       1s(2)                  =  2 electrons)
        2: Ne ([He] + 2s(2)2p(6)             = 10 electrons)
        3: Ar ([Ne] + 3s(2)3p(6)             = 18 electrons)
        4: Kr ([Ar] + 3d(10)4s(2)4p(6)       = 36 electrons)
        5: Xe ([Kr] + 4d(10)5s(2)5p(6)       = 54 electrons)
        6: Rn ([Xe] + 4f(14)5d(10)6s(2)6p(6) = 86 electrons)
>>0
 Enter list of (maximum 100) configurations. End list with a blank line or an asterisk (*)

 Give configuration           1
>>1s(2,*)2p(1,*)
 Give configuration           1
>>
 Give set of active orbitals, as defined by the highest principal quantum number
 per l-symmetry, in a comma delimited list in s,p,d etc order, e.g. 5s,4p,3d
>>5s,5p,5d,5f,5g
 Resulting 2*J-number? lower, higher (J=1 -> 2*J=2 etc.)
>>1,3
 Number of excitations (if negative number e.g. -2, correlation
 orbitals will always be doubly occupied)
>>3
 Generate more lists ? (y/n)
>>y
 Enter list of (maximum 100) configurations. End list with a blank line or an asterisk (*)

 Give configuration           1
>>1s(2,*)2s(1,*)
 Give configuration           2
>>
 Give set of active orbitals, as defined by the highest principal quantum number
 per l-symmetry, in a comma delimited list in s,p,d etc order, e.g. 5s,4p,3d
>>5s,5p,5d,5f,5g
 Resulting 2*J-number? lower, higher (J=1 -> 2*J=2 etc.)
>>1,1
 Number of excitations (if negative number e.g. -2, correlation
 orbitals will always be doubly occupied)
>>3
 Generate more lists ? (y/n)
>>n

.........

  3 blocks were created

       block  J/P            NCSF
           1  1/2+           1463
           1  1/2-           1454
```

```
          2  3/2-          2478
```

6.9 User defined orbital ordering

In Ce III the ground configuration is $5s^25p^64f^2$, where $4f$ is to the right of the $5s$ and $5p$ orbitals and a user defined orbital order is needed. To illustrate the user defined orbital ordering we generate a list of CSFs by allowing SD-excitations from $4s^24p^64d^{10}5s^25p^64f^2$ to an active orbital set $\{1s, 2s, 3s, 4s, 5s, 6s, 2p, 3p, 4p, 5p, 6p, 3d, 4d, 5d, 4f, 5f\}$ (or `6s,6p,5d,5f` in the notation of the `rcsfgenerate` program).

To generate a list of CSFs where, in the configurations, $4f$ is to the right of the $5s$ and $5p$ orbitals start by creating a file `clist.ref` with the desired orbital order; one orbital per line, left justified and with a non-relativistic notation.

```
1s
2s
2p
3s
3p
3d
4s
4p
4d
5s
5p
4f
5d
5f
6s
6p
```

Then run `rcsfgenerate` as usual, but select the user defined orbital order.

```
*******************************************************************************
*          RUN RCSFGENERATE USING USER DEFINED ORBITAL ORDERING               *
*          INPUT FILE: clist.ref                                              *
*          OUTPUT FILES: rcsf.out, rcsfgenerate.log                           *
*******************************************************************************

>>rcsfgenerate

 RCSFGENERATE
 This program creates a list of CSFs
 Configurations should be entered in spectroscopic notation
 with occupation numbers and indications if orbitals are
 closed (c), inactive (i), active (*) or has a minimal
 occupation e.g. 1s(2,1)2s(2,*)
 Outputfiles: rcsf.out, rcsfgenerate.log

 Default, reverse, symmetry or user specified ordering? (*/r/s/u)
>>u

 Select core
       0: No core
       1: He (       1s(2)                  =  2 electrons)
```

```
        2: Ne ([He] + 2s(2)2p(6)             = 10 electrons)
        3: Ar ([Ne] + 3s(2)3p(6)             = 18 electrons)
        4: Kr ([Ar] + 3d(10)4s(2)4p(6)       = 36 electrons)
        5: Xe ([Kr] + 4d(10)5s(2)5p(6)       = 54 electrons)
        6: Rn ([Xe] + 4f(14)5d(10)6s(2)6p(6) = 86 electrons)
>>3
 Enter list of (maximum 100) configurations. End list with a blank line or an asterisk (*)

 Give configuration           1
>>3d(10,c)4s(2,*)4p(6,*)4d(10,*)5s(2,*)5p(6,*)4f(2,*)
 Give configuration           2
>>
 Give set of active orbitals, as defined by the highest principal quantum number
 per l-symmetry, in a comma delimited list in s,p,d etc order, e.g. 5s,4p,3d
>>6s,6p,5d,5f
 Resulting 2*J-number? lower, higher (J=1 -> 2*J=2 etc.)
>>0,12
 Number of excitations (if negative number e.g. -2, correlation
 orbitals will always be doubly occupied)
>>2
 Generate more lists ? (y/n)
>>n

.........

 7 blocks were created

        block  J/P            NCSF
           1    0+          26477
           2    1+          74434
           3    2+         112054
           4    3+         133012
           5    4+         137871
           6    5+         127297
           7    6+         107194
```

The produced output file `rcsf.out` looks like this

```
Core subshells:
  1s   2s   2p-  2p   3s   3p-  3p   3d-  3d
Peel subshells:
  4s   4p-  4p   4d-  4d   5s   5p-  5p   4f-  4f   5d-  5d   5f-  5f   6s   6p-  6p
CSF(s):
  4s ( 2)  4p-( 2)  4p ( 4)  4d-( 4)  4d ( 6)  5s ( 2)  5p ( 4)  4f ( 4)
                                                                      0
                                                                       0+
  4s ( 2)  4p-( 2)  4p ( 4)  4d-( 4)  4d ( 6)  5s ( 2)  5p-( 1)  5p ( 3)  4f ( 4)
                                                             1/2      3/2   2;   2
                                                                               2      0+
  4s ( 2)  4p-( 2)  4p ( 4)  4d-( 4)  4d ( 6)  5s ( 2)  5p-( 1)  5p ( 3)  4f ( 4)
                                                             1/2      3/2   4;   2
                                                                               2      0+
  4s ( 2)  4p-( 2)  4p ( 4)  4d-( 4)  4d ( 6)  5s ( 2)  5p-( 2)  5p ( 2)  4f ( 4)
                                                                      0        0
                                                                                0+
```

```
 4s ( 2)  4p-( 2)  4p ( 4)  4d-( 4)  4d ( 6)  5s ( 2)  5p-( 2)  5p ( 2)  4f ( 4)
                                                                       2   2;   2
                                                                                 0+
 4s ( 2)  4p-( 2)  4p ( 4)  4d-( 4)  4d ( 6)  5s ( 2)  5p-( 2)  5p ( 2)  4f ( 4)
                                                                       2   4;   2
                                                                                 0+
 4s ( 2)  4p-( 2)  4p ( 4)  4d-( 4)  4d ( 6)  5p-( 2)  5p ( 4)  4f ( 4)
                                                                 0
                                                                   0+
 4s ( 2)  4p-( 2)  4p ( 4)  4d-( 4)  4d ( 6)  5s ( 2)  5p ( 4)  4f-( 1)  4f ( 3)
                                                                     5/2      5/2
                                                                                 0+
...............
```

Comment: when using `rcsfinteract` make sure that you have the same orbital order (and core) for both `rcsf.inp` and `rcsfmr.inp`. The additional quantum numbers `2;` and `4;` for the `4f ( 4)` subshell are the seniority quantum numbers.

6.10 Running jjgen

The `jjgen` program is a more flexible generation program than `rcsfgenerate`. It has several useful properties, but the input is somewhat longer. The use of `jjgen` is described in detail in the original write-up [10]. Note that after generating a CSF list with `jjgen` the list needs to be put on block form by `rcsfblock`.

Part III

Sample runs

Chapter 7

Running the application programs

In this chapter we demonstrate the use of the application programs of GRASP2018 in six cases described below. The use of the tools of GRASP2018 is described in the next chapter. All data written to the output files are shown, explained and discussed in detail in chapter 9. Output files from the runs are available in `grasptest` in the `output` directories of `example1`, `example2`, `example3`, `example4`, `example5`. Scripts for running the examples can be found in the `script` directories (see section 3.5). Please note that the executables must be on the path! When running the application programs and the tools the user is encouraged to look at all the output files and use the information in chapter 9 to correctly interpret the output data.

7.1 First example: $1s^2 2s\ ^2S$ and $1s^2 2p\ ^2P$ in Li I

The first example is for $1s^2 2s\ ^2S_{1/2}$ and $1s^2 2p\ ^2P_{1/2,3/2}$ in Li.

Overview

1. Define nuclear data.
2. Obtain common spectroscopic orbitals for the MR set.
   (a) Generate configuration state list containing three CSFs: $1s^2 2s\ ^2S_{1/2}, 1s^2 2p\ ^2P_{1/2,3/2}$.
   (b) Perform angular integration.
   (c) Generate initial estimates of radial orbitals.
   (d) Perform self-consistent field calculation on the weighted average of $1s^2 2s\ ^2S_{1/2}, 1s^2 2p\ ^2P_{1/2,3/2}$.
   (e) Save output to `2s_2p_DF`.
3. Improve even states
   (a) Generate $n = 3$ complete active space (CAS) expansion for $1s^2 2s\ ^2S_{1/2}$.
   (b) Perform angular integration.
   (c) Generate initial estimates of radial orbitals.
   (d) Perform self-consistent field calculation on $1s^2 2s\ ^2S_{1/2}$.
   (e) Save output to `2s_3`.
   (f) Perform RCI calculation in which the transverse photon interaction (Breit) and vacuum polarization and self-energy (QED) corrections are added.
4. Transform from jj- to LSJ-coupling
5. Improve odd states

(a) Generate $n = 3$ complete active space (CAS) expansion for $1s^22p\ ^2P_{1/2,3/2}$.

(b) Perform angular integration.

(c) Generate initial estimates of radial orbitals.

(d) Perform self-consistent field calculation on the weighted average of $1s^22p\ ^2P_{1/2,3/2}$.

(e) Save output to `2p_3`.

(f) Perform RCI calculation in which the transverse photon interaction (Breit) and vacuum polarization and self-energy (QED) corrections are added.

6. Transform from jj- to LSJ-coupling

7. Run `rlevels` to view energy separations.

8. Calculate properties

   (a) Calculate hyperfine structure using the RCI wave functions.

   (b) Compute the transition rates from the RCI wave functions. Calculation in two steps: biorthonormal transformation and evaluation of transition matrix elements using standard Racah algebra methods.

Program input

In the test-runs input marked by >> or >>3, for example, indicates that the user should input 3 and then strike the return key. When >> is followed by blanks just strike the return key.

```
*****************************************************************************
* RUN RNUCLEUS TO GENERATE NUCLEAR DATA AND DEFINE RADIAL GRID               *
* OUTPUT FILE: isodata                                                      *
*****************************************************************************

>>rnucleus

 Enter the atomic number:
>>3
 Enter the mass number (0 if the nucleus is to be modelled as a point source:
>>7
 The default root mean squared radius is    2.1692104687977172      fm;
  the default nuclear skin thickness is    2.2999999999999998      fm;
 Revise these values?
>>n
 Enter the mass of the neutral atom (in amu) (0 if the nucleus is to be static):
>>6.941
 Enter the nuclear spin quantum number (I) (in units of h / 2 pi):
>>1.5
 Enter the nuclear dipole moment (in nuclear magnetons):
>>3.2564268
 Enter the nuclear quadrupole moment (in barns):
>>-0.040


*****************************************************************************
*        RUN RCSFGENERATE TO GENERATE LIST OF CSFs FOR 2S                   *
*        AND 2P WITH THREE CSFs: 1s(2)2s J=1/2, 1s(2)2p- J=1/2,             *
*                                                  1s(2)2p J=3/2            *
*        OUTPUT FILES: rcsfgenerate.log, rcsf.out                           *
```

```
*****************************************************************************

>>rcsfgenerate

 RCSFGENERATE
 This program generates a list of CSFs
 Configurations should be entered in spectroscopic notation
 with occupation numbers and indications if orbitals are
 closed (c), inactive (i), active (*) or has a minimal
 occupation e.g. 1s(2,1)2s(2,*)
 OUTPUT FILES: rcsf.out, rcsfgenerate.log

 Default, reverse, symmetry or user specified ordering? (*/r/s/u)
>>*

 Select core
        0: No core
        1: He (       1s(2)                  =  2 electrons)
        2: Ne ([He] + 2s(2)2p(6)             = 10 electrons)
        3: Ar ([Ne] + 3s(2)3p(6)             = 18 electrons)
        4: Kr ([Ar] + 3d(10)4s(2)4p(6)       = 36 electrons)
        5: Xe ([Kr] + 4d(10)5s(2)5p(6)       = 54 electrons)
        6: Rn ([Xe] + 4f(14)5d(10)6s(2)6p(6) = 86 electrons)
>>0
 Enter list of (maximum 100) configurations. End list with a blank line or an asterisk (*)

 Give configuration  1
>>1s(2,i)2s(1,i)
 Give configuration  2
>>
 Give set of active orbitals, as defined by the highest principal quantum number
 per l-symmetry, in a comma delimited list in s,p,d etc order, e.g. 5s,4p,3d
>>2s
 Resulting 2*J-number? lower, higher (J=1 -> 2*J=2 etc.)
>>1,1
 Number of excitations (if negative number e.g. -2, correlation
 orbitals will always be doubly occupied)
>>0
 Generate more lists ? (y/n)
>>y
 Enter list of (maximum 100) configurations. End list with a blank line or an asterisk (*).

 Give configuration  1
>>1s(2,i)2p(1,i)
 Give configuration  2
>>
 Give set of active orbitals in a comma delimited list ordered by l-symmetry, e.g. 5s,4p,3d
>>1s,2p
 Resulting 2*J-number? lower, higher (J=1 -> 2*J=2 etc.)
>>1,3
 Number of excitations (if negative number e.g. -2, correlation
 orbitals will always be doubly occupied)
>>0
 Generate more lists ? (y/n)
```

```
>>n

        .........

 3 blocks were created

      block  J/P           NCSF
          1  1/2+             1
          2  1/2-             1
          3  3/2-             1

******************************************************************************
*        COPY FILES                                                          *
*        IT IS ADVISABLE TO SAVE THE rcsfgenerate.log FILE TO HAVE A         *
*        RECORD ON HOW THE LIST OF CSFs WAS CREATED                          *
******************************************************************************

>>cp rcsfgenerate.log 2s_2p_DF.exc
>>cp rcsf.out rcsf.inp


******************************************************************************
*        RUN RANGULAR TO GENERATE ENERGY EXPRESSION                          *
*        INPUT FILE  : rcsf.inp                                              *
*        OUTPUT FILES: rangular.log, mcp.30, mcp.31,....                     *
******************************************************************************

>>rangular

 RANGULAR
 This program performs angular integration
 Input file:  rcsf.inp
 Outputfiles: mcp.30, mcp.31, ....
              rangular.log

 Full interaction?  (y/n)
>>y

  .....

 RANGULAR: Execution complete.

******************************************************************************
*        RUN RWFNESTIMATE TO GENERATE INITIAL ESTIMATES FOR RADIAL ORBITALS  *
*        WE CAN USE WILD CARDS * FOR SPECIFYING ORBITALS                     *
*        * MEANS ALL ORBITALS                                                *
*        INPUT FILES: isodata, rcsf.inp, previous rwfn files                 *
*        OUTPUT FILE: rwfn.inp, rwfnestimate.log                             *
******************************************************************************

>>rwfnestimate

 RWFNESTIMATE
 This program estimates radial wave functions
```

```
 for orbitals
 Input files: isodata, rcsf.inp, optional rwfn file
 Output file: rwfn.inp

 Default settings ?
>>y
 Loading CSF file ... Header only
 There are/is            4  relativistic subshells;

 The following subshell radial wavefunctions remain to be estimated:
 1s 2s 2p- 2p

 Read subshell radial wavefunctions. Choose one below
     1 -- GRASP2K File
     2 -- Thomas-Fermi
     3 -- Screened Hydrogenic
>>2
 Enter the list of relativistic subshells:
>>*

 All required subshell radial wavefunctions  have been estimated:
Shell      e           p0        gamma        P(2)       Q(2)      MTP  SRC

  1s   0.2476D+01  0.9246D+01  0.1000D+01  0.3160D-06 -0.4186D-11  332  T-F
  2s   0.2895D+00  0.2308D+01  0.1000D+01  0.7888D-07 -0.1045D-11  355  T-F
  2p-  0.2173D+00  0.1444D-03  0.1000D+01  0.8084D-14  0.4509D-09  358  T-F
  2p   0.2173D+00  0.1204D+01  0.2000D+01  0.1406D-14 -0.1863D-19  358  T-F

 RWNESTIMATE: Execution complete.

*******************************************************************************
*         RUN RMCDHF TO OBTAIN SELF CONSISTENT SOLUTIONS                      *
*         INPUT FILES: isodata, rcsf.inp, rwfn.inp, mcp.30, mcp.31,...        *
*         OUTPUT FILES: rwfn.out, rmix.out, rmcdhf.sum, rmcdhf.log            *
*                                                                             *
*         NOTE: ORBITALS BUILDING REFERENCE STATES ARE REQUIRED TO HAVE       *
*         THE CORRECT NUMBER OF NODES. THEY ARE REFERRED TO AS SPECTROSCOPIC  *
*         ORBITALS. IN THIS RUN WE VARY 1s, 2s, 2p AND THEY ARE ALL           *
*         SPECTROSCOPIC. WE CAN USE WILD CARDS * FOR SPECIFYING ORBITALS      *
*         * MEANS ALL ORBITALS                                                *
*******************************************************************************

>>rmcdhf

 RMCDHF
 This program determines the radial orbitals
 and the expansion coefficients of the CSFs
 in a self-onsistent field proceedure
 Input file:  isodata, rcsf.inp, rwfn.inp, mcp.30, ...
 Outputfiles: rwfn.out, rmix.out, rmcdhf.sum, rmcdhf.log


 Default settings?  (y/n)
>>y
```

```
 Loading CSF file ... Header only
 There are/is           4  relativistic subshells;
 Loading CSF File for ALL blocks
 There are            3  relativistic CSFs... load complete;

 Loading Radial WaveFunction File ...
 There are            3  blocks  (block   J/Parity   NCF):
  1  1/2+     1       2  1/2-     1       3  3/2-     1

 Enter ASF serial numbers for each block
 Block           1    ncf =           1  id =  1/2+
>>1
 Block           2    ncf =           1  id =  1/2-
>>1
 Block           3    ncf =           1  id =  3/2-
>>1
 level weights (1 equal;  5 standard;  9 user)
>>5
 Radial functions
 1s 2s 2p- 2p
 Enter orbitals to be varied (Updating order)
>>*
 Which of these are spectroscopic orbitals?
>>*
 Enter the maximum number of SCF cycles:
>>100

        ..........


 RMCDHF: Execution complete.

*******************************************************************************
*         RUN RSAVE TO SAVE OUTPUT FILES: name.c, name.w, name.m, name.sum    *
*                                         name.alog, name.log                 *
*******************************************************************************

>>rsave 2s_2p_DF
 Created 2s_2p_DF.w, 2s_2p_DF.c, 2s_2p_DF.m, 2s_2p_DF.sum, 2s_2p_DF.alog and 2s_2p_DF.log

*******************************************************************************
*         RUN RCSFGENERATE TO GENERATE n = 3 CAS LIST                         *
*         OF CSFs FOR 2S                                                      *
*         OUTPUT FILES: rcsfgenerate.log, rcsf.out                            *
*******************************************************************************

>>rcsfgenerate

 RCSFGENERATE
 This program creates a list of CSFs
 Configurations should be entered in spectroscopic notation
 with occupation numbers and indications if orbitals are
 closed (c), inactive (i), active (*) or has a minimal
 occupation e.g. 1s(2,1)2s(2,*)
```

```
 Outputfile: rcsf.out, rcsfgenerate.log

 Default, reverse, symmetry or user specified ordering? (*/r/s/u)
>>*

 Select core
        0: No core
        1: He (       1s(2)                  =  2 electrons)
        2: Ne ([He] + 2s(2)2p(6)             = 10 electrons)
        3: Ar ([Ne] + 3s(2)3p(6)             = 18 electrons)
        4: Kr ([Ar] + 3d(10)4s(2)4p(6)       = 36 electrons)
        5: Xe ([Kr] + 4d(10)5s(2)5p(6)       = 54 electrons)
        6: Rn ([Xe] + 4f(14)5d(10)6s(2)6p(6) = 86 electrons)
>>0
 Enter list of (maximum 100) configurations. End list with a blank line or an asterisk (*)

 Give configuration  1
>>1s(2,*)2s(1,*)
 Give configuration  2
>>
 Give set of active orbitals, as defined by the highest principal quantum number
 per l-symmetry, in a comma delimited list in s,p,d etc order, e.g. 5s,4p,3d
>>3s,3p,3d
 Resulting 2*J-number? lower, higher (J=1 -> 2*J=2 etc.)
>>1,1
 Number of excitations (if negative number e.g. -2, correlation
 orbitals will always be doubly occupied)
>>3
 Generate more lists ? (y/n)
>>n

        .........

  1  blocks were created

      block  J/P           NCSF
          1  1/2+            79

******************************************************************************
*         COPY FILES                                                         *
*         IT IS ADVISABLE TO SAVE THE rcsfgenerate.log FILE TO HAVE A        *
*         RECORD ON HOW THE LIST OF CSFs WAS CREATED                         *
******************************************************************************

>>cp rcsfgenerate.log 2s_3.exc
>>cp rcsf.out rcsf.inp


******************************************************************************
*         RUN RANGULAR TO GENERATE ENERGY EXPRESSION                         *
*         INPUT FILE  : rcsf.inp                                             *
*         OUTPUT FILES: rangular.log, mcp.30, mcp.31,....                    *
******************************************************************************
```

```
>>rangular

 RANGULAR
 This program performs angular integration
 Input file:  rcsf.inp
 Outputfiles: mcp.30, mcp.31, ....
              rangular.log

 Full interaction?  (y/n)
>>y

  ...........

 RANGULAR: Execution complete.

*******************************************************************************
*         RUN RWFNESTIMATE TO GENERATE INITIAL ESTIMATES FOR RADIAL ORBITALS  *
*         INPUT FILES: isodata, rcsf.inp, previous rwfn files                 *
*         OUTPUT FILE: rwfn.inp                                               *
*******************************************************************************

>>rwfnestimate

 RWFNESTIMATE
 This program estimates radial wave functions
 for orbitals
 Input files: isodata, rcsf.inp, optional rwfn file
 Output file: rwfn.inp

 Default settings ?
>>y
 Loading CSF file ... Header only
 There are/is            9  relativistic subshells;

 The following subshell radial wavefunctions remain to be estimated:
 1s 2s 2p- 2p 3s 3p- 3p 3d- 3d

 Read subshell radial wavefunctions. Choose one below
     1 -- GRASP2K File
     2 -- Thomas-Fermi
     3 -- Screened Hydrogenic
>>1
 Enter the file name (Null then "rwfn.out")
>>
 Enter the list of relativistic subshells:
>>*
 The following subshell radial wavefunctions remain to be estimated:
 3s 3p- 3p 3d- 3d

 Read subshell radial wavefunctions. Choose one below
     1 -- GRASP2K File
     2 -- Thomas-Fermi
     3 -- Screened Hydrogenic
>>2
```

```
 Enter the list of relativistic subshells:
>>*
 All required subshell radial wavefunctions  have been estimated:
Shell      e           p0        gamma        P(2)       Q(2)      MTP  SRC

  1s   0.2518D+01  0.9281D+01  0.1000D+01  0.3172D-06 -0.4202D-11  355  rwf
  2s   0.1963D+00  0.1453D+01  0.1000D+01  0.4965D-07 -0.6576D-12  361  rwf
  2p-  0.1287D+00  0.5116D-04  0.1000D+01  0.2864D-14  0.1598D-09  366  rwf
  2p   0.1287D+00  0.4265D+00  0.2000D+01  0.4983D-15 -0.6601D-20  366  rwf
  3s   0.9128D-01  0.9784D+00  0.1000D+01  0.3344D-07 -0.4429D-12  369  T-F
  3p-  0.7531D-01  0.6592D-04  0.1000D+01  0.3691D-14  0.2058D-09  371  T-F
  3p   0.7531D-01  0.5495D+00  0.2000D+01  0.6419D-15 -0.8503D-20  371  T-F
  3d-  0.6228D-01  0.3234D-05  0.2000D+01  0.1238D-22  0.6903D-18  373  T-F
  3d   0.6228D-01  0.3237D-01  0.3000D+01  0.1293D-23 -0.1712D-28  373  T-F

 RWFNESTIMATE: Execution complete.
```

Comment: please note how we used the wild card * twice. We start by reading the orbitals from a GRASP2018 file (previous run `rwfn.out`). Using the wild card * the program reads as many orbitals as possible i.e. $1s, 2s, 2p-, 2p$. The orbitals $3s, 3p-, 3p, 3d-, 3d$ then remain to be estimated and we use Thomas-Fermi estimates. By again using the wild card * all the remaining orbitals will be Thomas-Fermi estimates.

```
*****************************************************************************
*         RUN RMCDHF TO OBTAIN SELF CONSISTENT SOLUTIONS                    *
*         INPUT FILES: isodata, rcsf.inp, rwfn.inp, mcp.30, mcp.31,...      *
*         OUTPUT FILES: rwfn.out, rmix.out, rmcdhf.sum, rmcdhf.log          *
*                                                                           *
*         NOTE: FOR CORRELATION ORBITALS THERE ARE NO RESTRICTIONS ON THE   *
*         NUMBER OF NODES, I.E. THEY ARE NOT SPECTROSCOPIC. IN THIS RUN WE  *
*         VARY THE CORRELATION ORBITALS 3s,3p, 3d. NONE OF THESE ARE        *
*         SPECTROSCOPIC. WE CAN USE WILD CARDS * FOR SPECIFYING ORBITALS    *
*         3* MEANS 3s, 3p-, 3p, 3d-, 3d                                     *
*****************************************************************************

>>rmcdhf

 RMCDHF
 This program determines the radial orbitals
 and the expansion coefficients of the CSFs
 in a self-consistent field proceedure
 Input file:  isodata, rcsf.inp, rwfn.inp, mcp.30, ...
 Outputfiles: rwfn.out, rmix.out, rmcdhf.sum, rmcdhf.log

 Default settings?  (y/n)
>>y
 Loading CSF file ... Header only
 There are/is            9  relativistic subshells;
 Loading CSF File for ALL blocks
 There are           79  relativistic CSFs... load complete;

 Loading Radial WaveFunction File ...
 There are            1  blocks  (block   J/Parity   NCF):
  1  1/2+    79
```

```
 Enter ASF serial numbers for each block
 Block            1    ncf =          79  id =  1/2+
>>1
 Radial functions
 1s 2s 2p- 2p 3s 3p- 3p 3d- 3d
 Enter orbitals to be varied (Updating order)
>>3*
 Which of these are spectroscopic orbitals?
>>
 Enter the maximum number of SCF cycles:
>>100

    ..........

 RMCDHF: Execution complete.

******************************************************************************
*         RUN RSAVE TO SAVE OUTPUT FILES: name.c, name.w, name.m, name.sum    *
*                                         name.alog, name.log                 *
******************************************************************************

>>rsave 2s_3
 Created 2s_3.w, 2s_3.c, 2s_3.m, 2s_3.sum, 2s_3.alog and 2s_3.log

******************************************************************************
*         RUN RCI TO INCLUDE TRANSVERSE PHOTON INTERACTION AND QED EFFECTS    *
*         INPUT FILES : isodata, 2s_3.c, 2s_3.w                               *
*         OUTPUT FILES: 2s_3.cm, 2s_3.csum, 2s_3.clog, rci.res                *
*                                                                             *
*         THE TRANSVERSE PHOTON FREQUENCIES CAN BE SET TO THE LOW FREQUENCY   *
*         LIMIT. RECOMMENDED IN CASES WHERE YOU HAVE CORRELATION ORBITALS     *
*         THE SELF ENERGY CORRECTION MAY FAIL FOR CORRELATION ORBITALS WITH   *
*         HIGH N.                                                             *
******************************************************************************

>>rci

 RCI
 This is the configuration interaction program
 Input file:  isodata, name.c, name.w
 Outputfiles: name.cm, name.csum, name.clog, rci.res

 Default settings?
>>y
 Name of state:
>>2s_3
 Block            1 ,  ncf =          79
 Loading CSF file ... Header only
 There are/is            9  relativistic subshells;
 Include contribution of H (Transverse)?
>>y
 Modify all transverse photon frequencies?
>>y
```

```
 Enter the scale factor:
>>1.d-6
 Include H (Vacuum Polarisation)?
>>y
 Include H (Normal Mass Shift)?
>>n
 Include H (Specific Mass Shift)?
>>n
 Estimate self-energy?
>>y
 Largest n quantum number for including self-energy for orbital
 n should be less or equal 8
>>3

 Loading Radial WaveFunction File ...
 There are           1  blocks  (block   J/Parity   NCF):
  1  1/2+    79

 Enter ASF serial numbers for each block
 Block           1    ncf =          79  id =  1/2+
>>1

   ......

 RCI: Execution complete.

*******************************************************************************
*         RUN JJ2LSJ TO TRANSFORM FROM JJ- TO LSJ-COUPLING                    *
*         INPUT FILES: 2s_3.c, 2s_3.cm                                        *
*         OUTPUT FILE: 2s_3.lsj.lbl, 2s_3.uni.lsj.lbl                         *
*******************************************************************************

>>jj2lsj

 jj2lsj: Transformation of ASFs from a jj-coupled CSF basis
         into an LSJ-coupled CSF basis  (Fortran 95 version)
         (C) Copyright by   G. Gaigalas and Ch. F. Fischer,
         (2017).
 Input files: name.c, name.(c)m
 Output files: name.lsj.lbl
   (optional)  name.lsj.c, name.lsj.j,
               name.uni.lsj.lbl, name.uni.lsj.sum


 Name of state
>>2s_3
 Loading Configuration Symmetry List File ...
 There are 9 relativistic subshells;
 There are 79 relativistic CSFs;
  ... load complete;

 Mixing coefficients from a CI calc.?
>>y
 Do you need a unique labeling? (y/n)
```

```
>>y
    nelec  =           3
    ncftot =          79
    nw     =           9
    nblock =           1

   block     ncf     nev    2j+1  parity
       1      79       1       2       1
 Default settings?  (y/n)
>>y

....

 jj2lsj: Execution Complete

*******************************************************************************
*         RUN RCSFGENERATE TO GENERATE n = 3 CAS LIST                          *
*         OF CSFs FOR 2P                                                       *
*         OUTPUT FILES: rcsfgenerate.log, rcsf.out                             *
*******************************************************************************

>>rcsfgenerate

 RCSFGENERATE
 This program creates a list of CSFs
 Configurations should be entered in spectroscopic notation
 with occupation numbers and indications if orbitals are
 closed (c), inactive (i), active (*) or has a minimal
 occupation e.g. 1s(2,1)2s(2,*)
 Outputfiles: rcsf.out, rcsfgenerate.log

 Default, reverse, symmetry or user specified ordering? (*/r/s/u)
>>*

 Select core
        0: No core
        1: He (       1s(2)                  =  2 electrons)
        2: Ne ([He] + 2s(2)2p(6)             = 10 electrons)
        3: Ar ([Ne] + 3s(2)3p(6)             = 18 electrons)
        4: Kr ([Ar] + 3d(10)4s(2)4p(6)       = 36 electrons)
        5: Xe ([Kr] + 4d(10)5s(2)5p(6)       = 54 electrons)
        6: Rn ([Xe] + 4f(14)5d(10)6s(2)6p(6) = 86 electrons)
>>0
 Enter list of (maximum 100) configurations. End list with a blank line or an asterisk (*)

 Give configuration  1
>>1s(2,*)2p(1,*)
 Give configuration  2
>>
 Give set of active orbitals, as defined by the highest principal quantum number
 per l-symmetry, in a comma delimited list in s,p,d etc order, e.g. 5s,4p,3d
>>3s,3p,3d
 Resulting 2*J-number? lower, higher (J=1 -> 2*J=2 etc.)
>>1,3
```

```
 Number of excitations (if negative number e.g. -2, correlation
 orbitals will always be doubly occupied)
>>3
 Generate more lists ? (y/n)
>>n

  ....

  2  blocks were created

      block  J/P            NCSF
          1  1/2-             76
          2  3/2-            110


*******************************************************************************
*         COPY FILES                                                          *
*******************************************************************************

>>cp rcsfgenerate.log 2p_3.exc
>>cp rcsf.out rcsf.inp

*******************************************************************************
*         RUN RANGULAR TO GENERATE ENERGY EXPRESSION                          *
*         INPUT FILE  : rcsf.inp                                              *
*         OUTPUT FILES: rangular.log, mcp.30, mcp.31,....                     *
*******************************************************************************

>>rangular

 RANGULAR
 This program performs angular integration
 Input file:  rcsf.inp
 Outputfiles: mcp.30, mcp.31, ....
              rangular.log

 Full interaction?  (y/n)
>>y

   ....

 RANGULAR: Execution complete.

*******************************************************************************
*         RUN RWFNESTIMATE TO GENERATE INITIAL ESTIMATES FOR RADIAL ORBITALS  *
*         WE CAN USE WILD CARDS * TO SPECIFY ORBITALS                         *
*         * MEANS ALL ORBITALS                                                *
*         INPUT FILES: isodata, rcsf.inp, previous rwfn files                 *
*         OUTPUT FILE: rwfn.inp                                               *
*******************************************************************************

>>rwfnestimate

 RWFNESTIMATE
```

```
 This program estimates radial wave functions
 for orbitals
 Input files: isodata, rcsf.inp, optional rwfn file
 Output file: rwfn.inp

 Default settings ?
>>y
 Loading CSF file ... Header only
 There are/is            9  relativistic subshells;

 The following subshell radial wavefunctions remain to be estimated:
 1s 2s 2p- 2p 3s 3p- 3p 3d- 3d

 Read subshell radial wavefunctions. Choose one below
     1 -- GRASP2K File
     2 -- Thomas-Fermi
     3 -- Screened Hydrogenic
>>1
 Enter the file name (Null then "rwfn.out")
>>2s_2p_DF.w
 Enter the list of relativistic subshells:
>>*
 The following subshell radial wavefunctions remain to be estimated:
 3s 3p- 3p 3d- 3d

 Read subshell radial wavefunctions. Choose one below
     1 -- GRASP2K File
     2 -- Thomas-Fermi
     3 -- Screened Hydrogenic
>>2
 Enter the list of relativistic subshells:
>>*
 All required subshell radial wavefunctions  have been estimated:
Shell      e           p0        gamma        P(2)       Q(2)      MTP  SRC

  1s   0.2518D+01  0.9281D+01  0.1000D+01  0.3172D-06 -0.4202D-11  355  2s_
  2s   0.1963D+00  0.1453D+01  0.1000D+01  0.4965D-07 -0.6576D-12  361  2s_
  2p-  0.1287D+00  0.5116D-04  0.1000D+01  0.2864D-14  0.1598D-09  366  2s_
  2p   0.1287D+00  0.4265D+00  0.2000D+01  0.4983D-15 -0.6601D-20  366  2s_
  3s   0.9128D-01  0.9784D+00  0.1000D+01  0.3344D-07 -0.4429D-12  369  T-F
  3p-  0.7531D-01  0.6592D-04  0.1000D+01  0.3691D-14  0.2058D-09  371  T-F
  3p   0.7531D-01  0.5495D+00  0.2000D+01  0.6419D-15 -0.8503D-20  371  T-F
  3d-  0.6228D-01  0.3234D-05  0.2000D+01  0.1238D-22  0.6903D-18  373  T-F
  3d   0.6228D-01  0.3237D-01  0.3000D+01  0.1293D-23 -0.1712D-28  373  T-F

 RWFNESTIMATE: Execution complete.

***************************************************************************
*         RUN RMCDHF TO OBTAIN SELF CONSISTENT SOLUTIONS                  *
*         INPUT FILES: isodata, rcsf.inp, rwfn.inp, mcp.30, mcp.31,...    *
*         OUTPUT FILES: rwfn.out, rmix.out, rmcdhf.sum, rmcdhf.log        *
*                                                                         *
*         NOTE: FOR CORRELATION ORBITALS THERE ARE NO RESTRICTIONS ON THE *
*         NUMBER OF NODES, I.E. THEY ARE NOT SPECTROSCOPIC. IN THIS RUN WE *
```

```
*         VARY THE CORRELATION ORBITALS 3s,3p, 3d. NONE OF THESE ARE          *
*         SPECTROSCOPIC. WE CAN USE WILD CARDS * FOR SPECIFYING ORBITALS      *
*         3* MEANS 3s, 3p-, 3p, 3d-, 3d                                       *
******************************************************************************

>>rmcdhf

 RMCDHF
 This program determines the radial orbitals
 and the expansion coefficients of the CSFs
 in a self-onsistent field proceedure
 Input file:  isodata, rcsf.inp, rwfn.inp, mcp.30, ...
 Outputfiles: rwfn.out, rmix.out, rmcdhf.sum, rmcdhf.log

 Default settings?  (y/n)
>>y
 Loading CSF file ... Header only
 There are/is            9  relativistic subshells;
 Loading CSF File for ALL blocks
 There are          186  relativistic CSFs... load complete;

 Loading Radial WaveFunction File ...

 There are            2  blocks  (block   J/Parity   NCF):
  1  1/2-    76       2  3/2-   110

 Enter ASF serial numbers for each block
 Block            1    ncf =           76  id =  1/2-
>>1
 Block            2    ncf =          110  id =  3/2-
>>1
 level weights (1 equal;  5 standard;  9 user)
>>5
 Radial functions
 1s 2s 2p- 2p 3s 3p- 3p 3d- 3d
 Enter orbitals to be varied (Updating order)
>>3*
 Which of these are spectroscopic orbitals?
>>
 Enter the maximum number of SCF cycles:
>>100

  ......

 RMCDHF: Execution complete.

******************************************************************************
*         RUN RSAVE TO SAVE OUTPUT FILES                                      *
******************************************************************************

>>rsave 2p_3
 Created 2p_3.w, 2p_3.c, 2p_3.m, 2p_3.sum, 2p_3.alog and 2p_3.log

******************************************************************************
```

```
*         RUN RCI TO INCLUDE TRANSVERSE PHOTON INTERACTION AND QED EFFECTS    *
*         INPUT FILES : isodata, 2p_3.c, 2p_3.w                               *
*         OUTPUT FILES: 2p_3.cm, 2p_3.csum, 2p_3.clog, rci.res                *
*                                                                             *
*         THE TRANSVERSE PHOTON FREQUENCIES CAN BE SET TO THE LOW FREQUENCY   *
*         LIMIT. RECOMMENDED IN CASES WHERE YOU HAVE CORRELATION ORBITALS     *
*         THE SELF ENERGY CORRECTION MAY FAIL FOR CORRELATION ORBITALS WITH   *
*         HIGH N.                                                             *
*******************************************************************************

>>rci

 RCI
 This is the configuration interaction program
 Input file:  isodata, name.c, name.w
 Outputfiles: name.cm, name.csum, name.clog, rci.res

 Default settings?
>>y
 Name of state:
>>2p_3

 Block            1 ,  ncf =           76
 Block            2 ,  ncf =          110
 Loading CSF file ... Header only
 There are/is            9  relativistic subshells;
 Include contribution of H (Transverse)?
>>y
 Modify all transverse photon frequencies?
>>y
 Enter the scale factor:
>>1.d-6
 Include H (Vacuum Polarisation)?
>>y
 Include H (Normal Mass Shift)?
>>n
 Include H (Specific Mass Shift)?
>>n
 Estimate self-energy?
>>y
 Largest n quantum number for including self-energy for orbital
 n should be less or equal 8
>>3

 Loading Radial WaveFunction File ...
 There are            2  blocks  (block   J/Parity   NCF):
  1  1/2-    76       2  3/2-   110

 Enter ASF serial numbers for each block
 Block            1    ncf =           76  id =  1/2-
>>1
 Block            2    ncf =          110  id =  3/2-
>>1
```

```
   ....

 RCI: Execution complete.

*****************************************************************************
*         RUN JJ2LSJ TO TRANSFORM FROM JJ- TO LSJ-COUPLING                  *
*         INPUT FILES: 2p_3.c, 2p_3.cm                                      *
*         OUTPUT FILE: 2p_3.lsj.lbl, 2p_3.uni.lsj.lbl                       *
*****************************************************************************

>>jj2lsj

 jj2lsj: Transformation of ASFs from a jj-coupled CSF basis
         into an LSJ-coupled CSF basis  (Fortran 95 version)
         (C) Copyright by   G. Gaigalas and Ch. F. Fischer,
         (2017).
 Input files: name.c, name.(c)m
 Output files: name.lsj.lbl
   (optional)  name.lsj.c, name.lsj.j,
               name.uni.lsj.lbl, name.uni.lsj.sum

 Name of state
>>2p_3
 Loading Configuration Symmetry List File ...
 There are 9 relativistic subshells;
 There are 186 relativistic CSFs;
  ... load complete;

 Mixing coefficients from a CI calc.?
>>y
 Do you need a unique labeling? (y/n)
>>y
    nelec  =           3
    ncftot =         186
    nw     =           9
    nblock =           2

   block     ncf     nev    2j+1  parity
       1      76       1       2      -1
       2     110       1       4      -1
 Default settings?  (y/n)
>>y

  ...

jj2lsj: Execution Complete

*****************************************************************************
*         RUN RLEVELS TO VIEW ENERGIES AND ENERGY SEPARATIONS.              *
*         IF DESIRED WE CAN INSTEAD RUN RLEVELSEV TO GET THE SEPARATION IN EV *
*****************************************************************************

>> rlevels 2s_3.cm 2p_3.cm
```

```
 nblock =            1   ncftot =          79   nw =            9   nelec =            3
 nblock =            2   ncftot =         186   nw =            9   nelec =            3

 Energy levels for ...
 Rydberg constant is   109737.31569
 Splitting is the energy difference with the lower neighbour
-----------------------------------------------------------------------------------------
 No Pos  J Parity Energy Total    Levels     Splitting     Configuration
                    (a.u.)      (cm^-1)     (cm^-1)
-----------------------------------------------------------------------------------------
  1  1  1/2 +      -7.4719740        0.00        0.00  1s(2).2s_2S
  2  1  1/2 -      -7.4042610    14861.28    14861.28  1s(2).2p_2P
  3  1  3/2 -      -7.4042597    14861.57        0.29  1s(2).2p_2P
-----------------------------------------------------------------------------------------


*******************************************************************************
*        RUN RHFS FOR 2s_3                                                    *
*        INPUT FILES: isodata, 2s_3.c, 2s_3.w, 2s_3.cm                        *
*        OUTPUT FILE: 2s_3.ch, 2s_3.choffd                                    *
*******************************************************************************

>>rhfs

 RHFS
 This is the hyperfine structure program
 Input files:  isodata, name.c, name.(c)m, name.w
 Output files: name.(c)h, name.(c)hoffd

 Default settings?
>>y
 Name of state
>>2s_3
 Mixing coefficients from a CI calc.?
>>y

   ....

 RHFS: Execution complete.

*******************************************************************************
*        RUN RHFS FOR 2p_3                                                    *
*        INPUT FILES: isodata, 2p_3.c, 2p_3.w, 2p_3.cm                        *
*        OUTPUT FILE: 2p_3.ch, 2p_3.choffd                                    *
*******************************************************************************

>>rhfs

 RHFS
 This is the hyperfine structure program
 Input files:  isodata, name.c, name.(c)m, name.w
 Output files: name.(c)h, name.(c)hoffd

 Default settings?
```

```
>>y
 Name of state
>>2p_3
 Mixing coefficients from a CI calc.?
>>y

   .....

 RHFS: Execution complete.
```

Please note that `rhfs` computes both diagonal and off-diagonal hyperfine interaction constants. The off-diagonal parameters are sometimes available from experiment. For Li I the $A_{3/2,1/2}$ interaction constant is for example measured from level-crossing spectroscopy [H. Orth, H. Ackermann and E.W. Otten, Z. Phys. A 273, 221 (1975)]. For systems with small fine-structure separations the off-diagonal hyperfine parameters are of crucial importance in order to model the observed hyperfine line profiles [M. Andersson, P. Jönsson and H. Sabel, Journal of Physics B 39, 4239 (2006)]. For systems with large fine structure separations the off-diagonal hyperfine constants may be neglected.

```
******************************************************************************
*         RUN RBIOTRANSFORM FOR 2s_3 AND 2p_3 TO TRANSFORM WAVE FUNCTIONS    *
*         INPUT FILES:  isodata, 2s_3.c, 2s_3.w, 2s_3.cm,                    *
*                                2p_3.c, 2p_3.w, 2p_3.cm                     *
*         OUTPUT FILES: 2s_3.cbm, 2s_3.bw, 2p_3.cbm, 2p_3.bw                 *
*                       2s_3.TB, 2p_3.TB (angular files)                     *
******************************************************************************

>>rbiotransform

 RBIOTRANSFORM
 This program transforms the initial and final wave
 functions so that standard tensor albegra can be
 used in evaluation of the transition parameters
 Input files:  isodata, name1.c, name1.w, name1.(c)m
               name2.c, name2.w, name2.(c)m
               name1.TB, name2.TB (optional angular files)
 Output files: name1.bw, name1.(c)bm,
               name2.bw, name2.(c)bm
               name1.TB, name2.TB (angular files)
 Default settings?
>>y
 Input from a CI calculation?
>>y
  Name of the Initial state
>>2s_3
  Name of the Final state
>>2p_3
  Transformation of all J symmetries?
>>y

   ....

 BIOTRANSFORM: Execution complete.

******************************************************************************
```

```
*         RUN RTRANSITION FOR 2s_3 and 2p_3 TO COMPUTE TRANSITION PARAMETERS  *
*         INPUT FILES: isodata, 2s_3.c, 2s_3.bw, 2s_3.cbm                     *
*                      2p_3.c, 2p_3.bw, 2p_3.cbm                              *
*         OUTPUT FILES: 2s_3.2p_3.ct                                          *
*                       2s_3.2p_3.-1T (angular file)                          *
*****************************************************************************

>>rtransition

 RTRANSITION
 This program computes transition parameters from
 transformed wave functions
 Input files:  isodata, name1.c, name1.bw, name1.(c)bm
               name2.c, name2.bw, name2.(c)bm
               optional, name1.lsj.lbl, name2.lsj.lbl
               name1.name2.KT (optional angular files)
 Output files: name1.name2.(c)t
               optional, name1.name2.(c)t.lsj
               name1.name2.KT (angular files)
 Here K is parity and rank of transition: -1,+1 etc

 Default settings?
>>y
 Input from a CI calculation?
>>y
 Name of the Initial state
>>2s_3
 Name of the Final state
>>2p_3

 MRGCSL: Execution begins ...
 Loading Configuration Symmetry List File ...
 There are 9 relativistic subshells;
 There are 79 relativistic CSFs;
  ... load complete;
 Loading Configuration Symmetry List File ...
 There are 9 relativistic subshells;
 There are 186 relativistic CSFs;
  ... load complete;
           1 s
           2 s
           2 p-
           2 p
           3 s
           3 p-
           3 p
           3 d-
           3 d
           1
          79
           2
          76          186
 Loading Configuration Symmetry List File ...
  there are 9 relativistic subshells;
```

```
  there are 265 relativistic CSFs;
  ... load complete;
 Enter the list of transition specifications
  e.g.,  E1,M2  or  E1 M2  or  E1;M2 :
>>E1

   .....

 RTRANSITION: Execution complete.
```

Comment: it does not matter in which order the two files `2s_3` and `2p_3` are specified.

7.2 Second example: $1s^22s2p\ ^3P_{0,1,2},\ ^1P_1$ in B II

The second example is $1s^22s2p\ ^3P_{0,1,2},\ ^1P_1$ in B II.

Overview

1. Define nuclear data
2. Obtain common spectroscopic orbitals for the MR set
   (a) Generate configuration list containing 4 CSFs belonging to $1s^22s2p\ ^{1,3}P$
   (b) Perform angular integration
   (c) Perform HF calculation
   (d) Convert HF orbitals to relativistic orbitals. We do not need to run `rwfnestimate` since all orbitals have been estimated
   (e) Perform self-consistent field calculation on the weighted average on the state belonging to $1s^22s2p\ ^{1,3}P$
   (f) Save output to `2s2p_DF`
3. Transform from jj- to LSJ-coupling
4. Run `rlevels` to view energy separations.

Program input

In the test-runs input marked by >> or >>3, for example, indicates that the user should input 3 and then strike the return key. When >> is followed by blanks just strike the return key.

```
*****************************************************************************
* RUN RNUCLEUS TO GENERATE NUCLEAR DATA AND DEFINE RADIAL GRID              *
* OUTPUT FILE: isodata                                                      *
*****************************************************************************

>>rnucleus

 RNUCLEUS
 This program defines nuclear data and the radial grid
 Outputfile: isodata

 Enter the atomic number:
>>5
 Enter the mass number (0 if the nucleus is to be modelled as a point source:
```

```
>>11
 The default root mean squared radius is    2.4292473557159475       fm;
 the default nuclear skin thickness is    2.2999999999999998       fm;
 Revise these values?
>>n
 Enter the mass of the neutral atom (in amu) (0 if the nucleus is to be static):
>>10.81
 Enter the nuclear spin quantum number (I) (in units of h / 2 pi):
>>1.5
 Enter the nuclear dipole moment (in nuclear magnetons):
>>2.6886489
 Enter the nuclear quadrupole moment (in barns):
>>1

******************************************************************************
*         RUN RCSFGENERATE TO GENERATE LIST FOR                              *
*         1P_1 AND 3P_0,1,2  WITH FOUR CSFs: 2s2p- J=0, 2s2p- J=1,           *
*                                            2s2p J=1, 2s2p J = 2            *
*         OUTPUT FILES: rcsfgenerate.log, rcsf.out                           *
******************************************************************************

>>rcsfgenerate

 RCSFGENERATE
 This program creates a list of CSFs
 Configurations should be entered in spectroscopic notation
 with occupation numbers and indications if orbitals are
 closed (c), inactive (i), active (*) or has a minimal
 occupation e.g. 1s(2,1)2s(2,*)
 Outputfiles: rcsf.out, rcsfgenerate.log

 Default, reverse, symmetry or user specified ordering? (*/r/s/u)
>>*

 Select core
       0: No core
       1: He (       1s(2)                  =  2 electrons)
       2: Ne ([He] + 2s(2)2p(6)             = 10 electrons)
       3: Ar ([Ne] + 3s(2)3p(6)             = 18 electrons)
       4: Kr ([Ar] + 3d(10)4s(2)4p(6)       = 36 electrons)
       5: Xe ([Kr] + 4d(10)5s(2)5p(6)       = 54 electrons)
       6: Rn ([Xe] + 4f(14)5d(10)6s(2)6p(6) = 86 electrons)
>>0
 Enter list of (maximum 100) configurations. End list with a blank line or an asterisk (*)

 Give configuration  1
>>1s(2,i)2s(1,i)2p(1,i)
 Give configuration  2
>>
 Give set of active orbitals, as defined by the highest principal quantum number
 per l-symmetry, in a comma delimited list in s,p,d etc order, e.g. 5s,4p,3d
>>2s,2p
 Resulting 2*J-number? lower, higher (J=1 -> 2*J=2 etc.)
>>0,4
```

```
 Number of excitations (if negative number e.g. -2, correlation
 orbitals will always be doubly occupied)
>>0
 Generate more lists ? (y/n)
>>n
        .........

 3  blocks were created

       block  J/P           NCSF
          1    0-             1
          2    1-             2
          3    2-             1


*******************************************************************************
*         COPY FILES                                                          *
*         IT IS ADVISABLE TO SAVE THE rcsfgenerate.log FILE TO HAVE A         *
*         RECORD ON HOW THE LIST OF CSFs WAS CREATED                          *
*******************************************************************************

>>cp rcsfgenerate.log 2s2p_DF.exc
>>cp rcsf.out rcsf.inp

*******************************************************************************
*         RUN RANGULAR TO GENERATE ENERGY EXPRESSION                          *
*         INPUT FILE  : rcsf.inp                                              *
*         OUTPUT FILES: rangular.alog, mcp.30, mcp.31,....                    *
*******************************************************************************

>>rangular

 RANGULAR
 This program performs angular integration
 Input file:  rcsf.inp
 Outputfiles: mcp.30, mcp.31, ....
              rangular.log

 Full interaction?  (y/n)
>>y

  ....

 RANGULAR: Execution complete.

*******************************************************************************
*         RUN HF PROGRAM TO GENERATE NON-RELATIVISTIC RADIAL ORBITALS         *
*         THAT CAN BE CONVERTED TO RELATIVISTIC ORBITALS                      *
*         OUTPUT FILE: wfn.out                                                *
*******************************************************************************

>>hf

                      =============================
```

```
                    H A R T R E E - F O C K . 96
                   ============================


             THE DIMENSIONS FOR THE CURRENT VERSION ARE:
                        NWF= 20       NO=220

  START OF CASE
  =============

  Enter ATOM,TERM,Z
  Examples: O,3P,8. or Oxygen,AV,8.
>>B,AV,5.

  List the CLOSED shells in the fields indicated (blank line if none)
  ... ... ... ... ... ... ... ... etc.
>> 1s   (Note that the closed shells should be entered right-justified with
         respect to the dots on the line above!!!)

  Enter electrons outside CLOSED shells (blank line if none)
  Example: 2s(1)2p(3)
>>2s(1)2p(1)

  There are   3 orbitals as follows:
     1s  2s  2p

  Orbitals to be varied: ALL/NONE/=i (last i)/comma delimited list/H
>>all

  Default electron parameters ? (Y/N/H)
>>y

  Default values for remaining parameters? (Y/N/H)
>>y


          WEAK ORTHOGONALIZATION DURING THE SCF CYCLE=   T
          SCF CONVERGENCE TOLERANCE (FUNCTIONS)      = 1.00D-08
          NUMBER OF POINTS IN THE MAXIMUM RANGE      = 220


          ITERATION NUMBER  1
          ----------------

     ................


          ITERATION NUMBER  6
          ----------------

          SCF CONVERGENCE CRITERIA (SCFTOL*SQRT(Z*NWF)) =   1.2D-06

          C( 1s 2s) =     0.00000   V( 1s 2s) =    -7.06535   EPS = 0.000000
          E( 2s 1s) =     0.02654   E( 1s 2s) =     0.01327
```

```
                    EL         ED              AZ           NORM        DPM
                    1s     16.3418222     20.8332819   1.0000000    1.93D-08
                    2s      1.8579695      4.7336947   1.0000000    1.38D-08
                    2p      1.4015370      4.0799511   1.0000000    1.74D-08


      < 1s| 2s>= 8.0D-09


     TOTAL ENERGY (a.u.)
     ----- ------
           Non-Relativistic      -24.06678870    Kinetic       24.06678852
           Relativistic Shift     -0.00587815    Potential    -48.13357722
           Relativistic          -24.07266685    Ratio        -2.000000008

  Additional parameters ? (Y/N/H)
>>n

  Do you wish to continue along the sequence ?
>>n


 END OF CASE
 ===========

*******************************************************************************
*         COPY FILES                                                          *
*******************************************************************************

>>cp wfn.out wfn.inp

*******************************************************************************
*         RUN RWFNMCHFMCDF TO CONVERT NON-RELATIVISTIC RADIAL ORBITALS TO     *
*         RELATIVISTIC ONES                                                   *
*         INPUT FILE:  wfn.inp                                                *
*         OUTPUT FILE: rwfn.out                                               *
*******************************************************************************

>>rwfnmchmcdf

 RWFNMCHFMCDF
 This program converts non-relativistic radial
 orbitals to relativistic ones in GRASP format
 Input file: wfn.inp
 Output file: rwfn.out

*******************************************************************************
*         COPY FILES                                                          *
*         WE DONT NEED TO INVOKE RWFNESTIMATE SINCE ALL ORBITALS HAVE         *
*         BEEN ESTIMATED THROUGH THE MCHF MCDF CONVERSION                     *
*******************************************************************************

>>cp rwfn.out rwfn.inp
```

```
***************************************************************************
*         RUN RMCDHF TO OBTAIN SELF CONSISTENT SOLUTIONS                  *
*         INPUT FILES: isodata, rcsf.inp, rwfn.inp, mcp.30, mcp.31,...    *
*         OUTPUT FILES: rwfn.out, rmix.out, rmcdhf.sum, rmcdhf.log        *
*                                                                         *
*         NOTE: ORBITALS BUILDING REFERENCE STATES ARE REQUIRED TO HAVE   *
*         THE CORRECT NUMBER OF NODES. THEY ARE REFERRED TO AS SPECTROSCOPIC  *
*         ORBITALS. IN THIS RUN WE VARY 1s, 2s, 2p AND THEY ARE ALL      *
*         SPECTROSCOPIC. WE CAN USE WILD CARDS FOR SPECIFYING ORBITALS    *
***************************************************************************

>>rmcdhf

 RMCDHF
 This program determines the radial orbitals
 and the expansion coefficients of the CSFs
 in a self-onsistent field proceedure
 Input file:  isodata, rcsf.inp, rwfn.inp, mcp.30, ...
 Outputfiles: rwfn.out, rmix.out, rmcdhf.sum, rmcdhf.log

 Default settings?  (y/n)
>>y
 Loading CSF file ... Header only
 There are/is            4  relativistic subshells;
 Loading CSF File for ALL blocks
 There are            4  relativistic CSFs... load complete;

 Loading Radial WaveFunction File ...
 There are            3  blocks  (block   J/Parity   NCF):
  1    0-     1       2    1-     2       3    2-     1

 Enter ASF serial numbers for each block
 Block            1    ncf =           1  id =    0-
>>1
 Block            2    ncf =           2  id =    1-
>>1,2
 Block            3    ncf =           1  id =    2-
>>1
 level weights (1 equal;  5 standard;  9 user)
>>5
 Radial functions
 1s 2s 2p- 2p
 Enter orbitals to be varied (Updating order)
>>*
 Which of these are spectroscopic orbitals?
>>*
 Enter the maximum number of SCF cycles:
>>100

 .....

 RMCDHF: Execution complete.
```

```
******************************************************************************
*         RUN RSAVE TO SAVE OUTPUT FILES                                     *
******************************************************************************

>>rsave 2s2p_DF
 Created 2s2p_DF.w, 2s2p_DF.c, 2s2p_DF.m, 2s2p_DF.sum, 2s2p_DF.alog and 2s2p_DF.log

******************************************************************************
*         RUN JJ2LSJ TO GET THE LSJ-COMPOSITION                              *
*         INPUT FILE: 2s2p_DF.c, 2s2p_DF.m                                   *
*         OUTPUT FILE: 2s2p_DF.lsj.lbl, 2s2p_DF.uni.lsj.lbl                  *
******************************************************************************

>>jj2lsj

 jj2lsj: Transformation of ASFs from a jj-coupled CSF basis
         into an LSJ-coupled CSF basis  (Fortran 95 version)
         (C) Copyright by   G. Gaigalas and Ch. F. Fischer,
         (2017).
 Input files: name.c, name.(c)m
 Output files: name.lsj.lbl
   (optional)  name.lsj.c, name.lsj.j,
               name.uni.lsj.lbl, name.uni.lsj.sum

 Name of state
>>2s2p_DF
 Loading Configuration Symmetry List File ...
 There are 4 relativistic subshells;
 There are 4 relativistic CSFs;
  ... load complete;

 Mixing coefficients from a CI calc.?
>>n
  Do you need a unique labeling? (y/n)
>>y
    nelec  =            4
    ncftot =            4
    nw     =            4
    nblock =            3

   block     ncf     nev    2j+1  parity
       1       1       1       1      -1
       2       2       2       3      -1
       3       1       1       5      -1
 Default settings?  (y/n)
>>y

  ....

 jj2lsj: Execution complete.

******************************************************************************
*         RUN RLEVELS TO VIEW ENERGIES AND ENERGY SEPARATIONS                *
*         NOTE: SINCE LSJ-INFORMATION NOW IS AVAILABLE OUTPUT LABELS         *
```

```
*        WILL BE IN LSJ-COUPLING                                         *
*        IF DESIRED WE CAN INSTEAD RUN RLEVELSEV TO GET THE SEPARATION IN EV *
**************************************************************************

>> rlevels 2s2p_DF.m
 2s2p_DF.m
 nblock =           3   ncftot =           4   nw =           4   nelec =           4

 Energy levels for ...
 Rydberg constant is   109737.31569
 Splitting is the energy difference with the lower neighbor
-----------------------------------------------------------------------------------------
 No Pos  J Parity Energy Total    Levels     Splitting     Configuration
                     (a.u.)      (cm^-1)     (cm^-1)
-----------------------------------------------------------------------------------------
  1  1   0  -     -24.1270877        0.00        0.00  1s(2).2s_2S.2p_3P
  2  1   1  -     -24.1270404       10.39       10.39  1s(2).2s_2S.2p_3P
  3  1   2  -     -24.1269457       31.17       20.79  1s(2).2s_2S.2p_3P
  4  2   1  -     -23.9154061    46458.75    46427.58  1s(2).2s_2S.2p_1P
-----------------------------------------------------------------------------------------
```

For interpretation of LSJ-coupling notation produced by `jj2lsj` see section 9.2.

7.3 Third example: $2s^2 2p^3$ and $2p^5$ in Si VIII

The third example is $2s^2 2p^3$ and $2p^5$ in Si VIII.

Overview

1. Define nuclear data
2. Obtain common spectroscopic orbitals for the MR set
   (a) Generate configuration list belonging to $2s^2 2p^3$ and $2p^5$
   (b) Perform angular integration
   (c) Generate initial estimates of radial orbitals
   (d) Perform self-consistent field calculation on the weighted average of all states belonging to $2s^2 2p^3$ and $2p^5$ (there are two states with $J = 1/2$, four states with $J = 3/2$ and one state with $J = 5/2$, see NIST Tables)
   (e) Save output to `2s22p3_2p5_DF`
3. Improve states
   (a) Generate CSF list from SD-excitations from $2s^2 2p^3$ and $2p^5$ to $n = 3$
   (b) Run `rcsfinteract` to extract CSFs that interact with CSFs belonging to $2s^2 2p^3$ or $2p^5$
   (c) Perform angular integration
   (d) Generate initial estimates of radial orbitals
   (e) Perform self-consistent field calculation on the weighted average of all states belonging to $2s^2 2p^3$ and $2p^5$
   (f) Save output to `2s22p3_2p5_3`
   (g) Perform RCI calculation in which Breit and QED effects are added.

4. Transform from jj- to LSJ-coupling
5. Run `rlevels` to view energy separations.
6. Calculate properties
   (a) Compute the M1 transition rates from the RCI wave functions. biorthonormal transformation not needed in this case since the states are described using the same orthogonal orbital set. Copy files and run the transition program.

Program input

In the test-runs input marked by >> or >>3, for example, indicates that the user should input 3 and then strike the return key. When >> is followed by blanks just strike the return key.

```
******************************************************************************
* RUN RNUCLEUS TO GENERATE NUCLEAR DATA AND DEFINE RADIAL GRID               *
* OUTPUT FILE: isodata                                                       *
******************************************************************************

>>rnucleus

 RNUCLEUS
 This program defines nuclear data and the radial grid
 Outputfile: isodata

 Enter the atomic number:
>>14
 Enter the mass number (0 if the nucleus is to be modelled as a point source:
>>28
 The default root mean squared radius is    3.1085883804880532       fm;
 the default nuclear skin thickness is    2.2999999999999998       fm;
 Revise these values?
>>n
 Enter the mass of the neutral atom (in amu) (0 if the nucleus is to be static):
>>27.9769271
 Enter the nuclear spin quantum number (I) (in units of h / 2 pi):
>>1
 Enter the nuclear dipole moment (in nuclear magnetons):
>>1
 Enter the nuclear quadrupole moment (in barns):
>>1
```

Comment: if we are not interested in the hyperfine structure constants we may just set nuclear spin and electromagnetic moments (magnetic dipole and electric quadrupole) to 1.

```
******************************************************************************
*         RUN RCSFGENERATE TO GENERATE LIST FOR ALL                          *
*         STATES OF 2s(2)2p(3) + 2p(5)                                       *
*         OUTPUT FILES: rcsfgenerate.log, rcsf.out                           *
******************************************************************************

>>rcsfgenerate

 RCSFGENERATE
```

```
 This program creates a list of CSFs
 Configurations should be entered in spectroscopic notation
 with occupation numbers and indications if orbitals are
 closed (c), inactive (i), active (*) or has a minimal
 occupation e.g. 1s(2,1)2s(2,*)
 Outputfiles: rcsf.out, rcsfgenerate.log

 Default, reverse, symmetry or user specified ordering? (*/r/s/u)
>>*

 Select core
        0: No core
        1: He (       1s(2)                  =  2 electrons)
        2: Ne ([He] + 2s(2)2p(6)             = 10 electrons)
        3: Ar ([Ne] + 3s(2)3p(6)             = 18 electrons)
        4: Kr ([Ar] + 3d(10)4s(2)4p(6)       = 36 electrons)
        5: Xe ([Kr] + 4d(10)5s(2)5p(6)       = 54 electrons)
        6: Rn ([Xe] + 4f(14)5d(10)6s(2)6p(6) = 86 electrons)
>>0
 Enter list of (maximum 100) configurations. End list with a blank line or an asterisk (*)

 Give configuration  1
>>1s(2,i)2s(2,i)2p(3,i)
 Give configuration  2
>>1s(2,i)2p(5,i)
 Give configuration  3
>>
 Give set of active orbitals, as defined by the highest principal quantum number
 per l-symmetry, in a comma delimited list in s,p,d etc order, e.g. 5s,4p,3d
>>2s,2p
 Resulting 2*J-number? lower, higher (J=1 -> 2*J=2 etc.)
>>1,5
 Number of excitations (if negative number e.g. -2, correlation
 orbitals will always be doubly occupied)
>>0
 Generate more lists ? (y/n)
>>n

        .........

  3  blocks were created

      block  J/P            NCSF
          1  1/2-              2
          2  3/2-              4
          3  5/2-              1

******************************************************************************
*         COPY FILES                                                         *
*         NOTE THAT WE COPY THE FILE TO RCSFMR.INP FOR FUTURE USE            *
*         TOGETHER WITH RCSFINTERACT                                         *
******************************************************************************

>>cp rcsfgenerate.log 2s22p3_2p5_DF.exc
```

```
>>cp rcsf.out rcsf.inp
>>cp rcsf.out rcsfmr.inp

******************************************************************************
*         RUN RANGULAR TO GENERATE ENERGY EXPRESSION                         *
*         INPUT FILE  : rcsf.inp                                             *
*         OUTPUT FILES: rangular.alog, mcp.30, mcp.31,....                   *
******************************************************************************

>>rangular

 RANGULAR
 This program performs angular integration
 Input file:  rcsf.inp
 Outputfiles: mcp.30, mcp.31, ....
              rangular.log

 Full interaction?  (y/n)
>>y

  ........

 RANGULAR: Execution complete.

******************************************************************************
*         RUN RWFNESTIMATE TO GENERATE INITIAL ESTIMATES FOR RADIAL ORBITALS  *
*         INPUT FILES: isodata, rcsf.inp, previous rwfn files                *
*         OUTPUT FILE: rwfn.inp, rwfnestimate.log                            *
******************************************************************************

>>rwfnestimate

 RWFNESTIMATE
 This program estimates radial wave functions
 for orbitals
 Input files: isodata, rcsf.inp, optional rwfn file
 Output file: rwfn.inp

 Default settings ?
>>y
 Loading CSF file ... Header only
 There are/is            4  relativistic subshells;

 The following subshell radial wavefunctions remain to be estimated:
 1s 2s 2p- 2p

 Read subshell radial wavefunctions. Choose one below
     1 -- GRASP2K File
     2 -- Thomas-Fermi
     3 -- Screened Hydrogenic
>>3
 Enter the list of relativistic subshells:
>>*
```

```
 ***** Screening parameters ******
 1s      0.00
 2s      0.00
 2p-     0.00
 2p      0.00
 All required subshell radial wavefunctions  have been estimated:
Shell      e           p0        gamma        P(2)       Q(2)      MTP  SRC

  1s   0.9826D+02  0.1033D+03  0.1000D+01  0.8347D-06 -0.4275D-07  328  Hyd
  2s   0.2458D+02  0.3670D+02  0.1000D+01  0.2965D-06 -0.1518D-07  344  Hyd
  2p-  0.2458D+02  0.8338D-01  0.1000D+01  0.6736D-09  0.1315D-07  343  Hyd
  2p   0.2452D+02  0.1492D+03  0.2000D+01  0.8406D-14 -0.2148D-15  343  Hyd

 RWFNESTIMATE: Execution complete.

*******************************************************************************
*         RUN RMCDHF TO OBTAIN SELF CONSISTENT SOLUTIONS                      *
*         INPUT FILES: isodata, rcsf.inp, rwfn.inp, mcp.30, mcp.31,...        *
*         OUTPUT FILES: rwfn.out, rmix.out, rmcdhf.sum, rmcdhf.log            *
*                                                                             *
*         NOTE: ORBITALS BUILDING REFERENCE STATES ARE REQUIRED TO HAVE       *
*         THE CORRECT NUMBER OF NODES. THEY ARE REFERRED TO AS SPECTROSCOPIC  *
*         ORBITALS. IN THIS RUN WE VARY 1s, 2s, 2p AND THEY ARE ALL           *
*         SPECTROSCOPIC. WE CAN USE WILD CARDS * FOR SPECIFYING ORBITALS      *
*                                                                             *
*         NOTE: INSTEAD OF SAYING THAT WE WILL OPTIMIZE ON, FOR EXAMPLE,      *
*         STATES 1,2,3,4 WE CAN WRITE 1-4 MEANING THE SAME THING              *
*******************************************************************************

>>rmcdhf

 RMCDHF
 This program determines the radial orbitals
 and the expansion coefficients of the CSFs
 in a self-onsistent field proceedure
 Input file:  isodata, rcsf.inp, rwfn.inp, mcp.30, ...
 Outputfiles: rwfn.out, rmix.out, rmcdhf.sum, rmcdhf.log

 Default settings?  (y/n)
>>y
 Loading CSF file ... Header only
 There are/is            4  relativistic subshells;
 Loading CSF File for ALL blocks
 There are            7  relativistic CSFs... load complete;

 Loading Radial WaveFunction File ...

 There are            3  blocks  (block   J/Parity   NCF):
  1  1/2-     2       2  3/2-     4       3  5/2-     1

 Enter ASF serial numbers for each block
 Block           1    ncf =           2  id =  1/2-
>>1-2
 Block           2    ncf =           4  id =  3/2-
```

```
>>1-4
 Block            3    ncf =           1  id =  5/2-
>>1
 level weights (1 equal;  5 standard;  9 user)
>>5
 Radial functions
 1s 2s 2p- 2p
 Enter orbitals to be varied (Updating order)
>>*
 Which of these are spectroscopic orbitals?
>>*
 Enter the maximum number of SCF cycles:
>>100

 ......

 RMCDHF: Execution complete.

*******************************************************************************
*         RUN RSAVE TO SAVE OUTPUT FILES                                      *
*******************************************************************************

>>rsave 2s22p3_2p5_DF
 Created 2s22p3_2p5_DF.w, 2s22p3_2p5_DF.c, 2s22p3_2p5_DF.m, 2s2p3_2p5_DF.sum,
         2s2p3_2p5_DF.alog and 2s22p3_2p5_DF.log

*******************************************************************************
*         RUN RCSFGENERATE TO GENERATE LIST OBTAINED BY                       *
*         SD-EXCITATIONS FROM 1s(2)2s(2)2p(3) + 1s(2)2p(5) TO n = 3           *
*         OUTPUT FILES: rcsfgenerate.log, rcsf.out                            *
*******************************************************************************

>>rcsfgenerate

 RCSFGENERATE
 This program creates a list of CSFs
 Configurations should be entered in spectroscopic notation
 with occupation numbers and indications if orbitals are
 closed (c), inactive (i), active (*) or has a minimal
 occupation e.g. 1s(2,1)2s(2,*)
 Outputfiles: rcsf.out, rcsfgenerate.log

 Default, reverse, symmetry or user specified ordering? (*/r/s/u)
>>*

 Select core
        0: No core
        1: He (       1s(2)                  =  2 electrons)
        2: Ne ([He] + 2s(2)2p(6)             = 10 electrons)
        3: Ar ([Ne] + 3s(2)3p(6)             = 18 electrons)
        4: Kr ([Ar] + 3d(10)4s(2)4p(6)       = 36 electrons)
        5: Xe ([Kr] + 4d(10)5s(2)5p(6)       = 54 electrons)
        6: Rn ([Xe] + 4f(14)5d(10)6s(2)6p(6) = 86 electrons)
>>0
```

```
 Enter list of (maximum 100) configurations. End list with a blank line or an asterisk (*)

 Give configuration  1
>>1s(2,*)2s(2,*)2p(3,*)
 Give configuration  2
>>1s(2,*)2p(5,*)
 Give configuration  3
>>
 Give set of active orbitals, as defined by the highest principal quantum number
 per l-symmetry, in a comma delimited list in s,p,d etc order, e.g. 5s,4p,3d
>>3s,3p,3d
 Resulting 2*J-number? lower, higher (J=1 -> 2*J=2 etc.)
>>1,5
 Number of excitations (if negative number e.g. -2, correlation
 orbitals will always be doubly occupied)
>>2
 Generate more lists ? (y/n)
>>n

        .........

 3 blocks were created

      block  J/P           NCSF
          1  1/2-           595
          2  3/2-           914
          3  5/2-           847

*****************************************************************************
*         COPY FILES                                                        *
*****************************************************************************

>>cp rcsfgenerate.log 2s22p3_2p5_3.exc
>>cp rcsf.out rcsf.inp

*****************************************************************************
*         RUN RCSFINTERACT PROGRAM TO DETERMINE WHICH OF THE CSFs IN THE    *
*         rcsf.inp LIST INTERACTS WITH THE CSFs IN rcsfmr.inp               *
*         THE INTERACTING CSFs ARE WRITTEN TO rcsf.out                      *
*         INPUT FILES: rcsfmr.inp, rcsf.inp                                 *
*         OUTPUT FILE: rcsf.out                                             *
*****************************************************************************

>>rcsfinteract

 RCSFinteract: Determines all the CSFs (rcsf.inp) that interact
              with the CSFs in the multireference (rcsfmr.inp)
              (C)  Copyright by G. Gaigalas and Ch. F. Fischer
              (Fortran 95 version)              NIST  (2017).
              Input files: rcsfmr.inp, rcsf.inp
              Output file: rcsf.out

 Reduction based on Dirac-Coulomb (1) or
 Dirac-Coulomb-Breit (2) Hamiltonian?
```

```
>>1

   ....

 There are 9 relativistic subshells;
  Block    MR NCSF   Before NCSF   After NCSF
    1          2         595          274
    2          4         914          591
    3          1         847          300

 RCSFINTERACT: Execution complete

******************************************************************************
*         COPY FILES                                                         *
******************************************************************************

>>cp rcsf.out rcsf.inp

******************************************************************************
*         RUN RANGULAR TO GENERATE ENERGY EXPRESSION                         *
*         INPUT FILE  : rcsf.inp                                             *
*         OUTPUT FILES: rangular.alog, mcp.30, mcp.31,....                   *
******************************************************************************

>>rangular

 RANGULAR
 This program performs angular integration
 Input file:  rcsf.inp
 Outputfiles: mcp.30, mcp.31, ....
              rangular.log

 Full interaction?  (y/n)
>>y

 ....

 RANGULAR: Execution complete.

******************************************************************************
*         RUN RWFNESTIMATE TO GENERATE INITIAL ESTIMATES FOR RADIAL ORBITALS  *
*         INPUT FILES: isodata, rcsf.inp, previous rwfn files                *
*         OUTPUT FILE: rwfn.inp, rwfnestimate.log                            *
******************************************************************************

>>rwfnestimate

 RWFNESTIMATE
 This program estimates radial wave functions
 for orbitals
 Input files: isodata, rcsf.inp, optional rwfn file
 Output file: rwfn.inp

 Default settings ?
```

```
>>y
 Loading CSF file ... Header only
 There are/is            9  relativistic subshells;

 The following subshell radial wavefunctions remain to be estimated:
 1s 2s 2p- 2p 3s 3p- 3p 3d- 3d

 Read subshell radial wavefunctions. Choose one below
     1 -- GRASP2K File
     2 -- Thomas-Fermi
     3 -- Screened Hydrogenic
>>1
 Enter the file name (Null then "rwfn.out")
>>
 Enter the list of relativistic subshells:
>>*
 The following subshell radial wavefunctions remain to be estimated:
 3s 3p- 3p 3d- 3d

 Read subshell radial wavefunctions. Choose one below
     1 -- GRASP2K File
     2 -- Thomas-Fermi
     3 -- Screened Hydrogenic
>>3
 Enter the list of relativistic subshells:
>>*

 ***** Screening parameters ******
 3s      0.00
 3p-     0.00
 3p      0.00
 3d-     0.00
 3d      0.00
 All required subshell radial wavefunctions  have been estimated:
Shell      e           p0        gamma        P(2)       Q(2)      MTP  SRC

  1s   0.7698D+02  0.1056D+03  0.1000D+01  0.7732D-06 -0.6275D-11  347  rwf
  2s   0.1236D+02  0.3088D+02  0.1000D+01  0.2262D-06 -0.1836D-11  351  rwf
  2p-  0.1089D+02  0.5761D-01  0.1000D+01  0.7535D-13  0.8261D-08  352  rwf
  2p   0.1086D+02  0.1007D+03  0.2000D+01  0.5403D-14 -0.4386D-19  352  rwf
  3s   0.1092D+02  0.1998D+02  0.1000D+01  0.1614D-06 -0.8267D-08  354  Hyd
  3p-  0.1092D+02  0.4942D-01  0.1000D+01  0.3992D-09  0.7795D-08  354  Hyd
  3p   0.1090D+02  0.8855D+02  0.2000D+01  0.4988D-14 -0.1275D-15  354  Hyd
  3d-  0.1090D+02  0.4311D-01  0.2000D+01  0.2429D-17  0.9503D-16  353  Hyd
  3d   0.1089D+02  0.9250D+02  0.3000D+01  0.3755D-22 -0.6395D-24  353  Hyd

 RWFNESTIMATE: Execution complete.

*******************************************************************************
*         RUN RMCDHF TO OBTAIN SELF CONSISTENT SOLUTIONS                      *
*         INPUT FILES: isodata, rcsf.inp, rwfn.inp, mcp.30, mcp.31,...        *
*         OUTPUT FILES: rwfn.out, rmix.out, rmcdhf.sum, rmcdhf.log            *
*                                                                             *
*         NOTE: FOR CORRELATION ORBITALS THERE ARE NO RESTRICTIONS ON THE     *
```

```
*        NUMBER OF NODES, I.E. THEY ARE NOT SPECTROSCOPIC. IN THIS RUN WE   *
*        VARY THE CORRELATION ORBITALS 3s,3p, 3d. NONE OF THESE ARE         *
*        SPECTROSCOPIC. WE CAN USE WILD CARDS * FOR SPECIFYING ORBITALS     *
*                                                                           *
*        NOTE: INSTEAD OF SAYING THAT WE WILL OPTIMIZE ON, FOR EXAMPLE,     *
*        STATES 1,2,3,4 WE CAN WRITE 1-4 MEANING THE SAME THING             *
*****************************************************************************

>>rmcdhf

 RMCDHF
 This program determines the radial orbitals
 and the expansion coefficients of the CSFs
 in a self-onsistent field proceedure
 Input file:  isodata, rcsf.inp, rwfn.inp, mcp.30, ...
 Outputfiles: rwfn.out, rmix.out, rmcdhf.sum, rmcdhf.log

 Default settings?  (y/n)
>>y
 Loading CSF file ... Header only
 There are/is            9  relativistic subshells;
 Loading CSF File for ALL blocks
 There are         1164  relativistic CSFs... load complete;

 Loading Radial WaveFunction File ...
 There are            3  blocks  (block   J/Parity   NCF):
  1  1/2-   274       2  3/2-   590       3  5/2-   300

 Enter ASF serial numbers for each block
 Block           1    ncf =         274  id =  1/2-
>>1-2
 Block           2    ncf =         590  id =  3/2-
>>1-4
 Block           3    ncf =         300  id =  5/2-
>>1
 level weights (1 equal;  5 standard;  9 user)
>>5
 Radial functions
 1s 2s 2p- 2p 3s 3p- 3p 3d- 3d
 Enter orbitals to be varied (Updating order)
>>3*
 Which of these are spectroscopic orbitals?
>>
 Enter the maximum number of SCF cycles:
>>100

 .....

 RMCDHF: Execution complete.

*****************************************************************************
*        RUN RSAVE TO SAVE OUTPUT FILES                                     *
*****************************************************************************
```

```
>>rsave 2s22p3_2p5_3
 Created 2s22p3_2p5_3.w, 2s22p3_2p5_3.c, 2s22p3_2p5_3.m, 2s22p3_2p5_3.sum,
         2s22p3_2p5_3.alog and 2s22p3_2p5_3.log

******************************************************************************
*         RUN RCI TO INCLUDE TRANSVERSE PHOTON INTERACTION AND QED EFFECTS   *
*         INPUT FILES: isodata, 2s22p3_2p5_3.c, 2s22p3_2p5_3.w               *
*         OUTPUT FILES: 2s22p3_2p5_3.cm, 2s22p3_2p5_3.csum, 2s22p3_2p5_3.clog *
*                       rci.res                                              *
*                                                                            *
*         THE TRANSVERSE PHOTON FREQUENCIES CAN BE SET TO THE LOW FREQUENCY  *
*         LIMIT. RECOMMENDED IN CASES WHERE YOU HAVE CORRELATION ORBITALS    *
*         THE SELF ENERGY CORRECTION MAY FAIL FOR CORRELATION ORBITALS WITH  *
*         HIGH N.                                                            *
*                                                                            *
*         NOTE: INSTEAD OF SAYING THAT WE WILL COMPUTE EIGENVALUES FOR       *
*         STATES 1,2,3,4 WE CAN WRITE 1-4 MEANING THE SAME THING             *
******************************************************************************

>>rci

 RCI
 This is the configuration interaction program
 Input file:  isodata, name.c, name.w
 Outputfiles: name.cm, name.csum, name.clog. rci.res

 Default settings?
>>y
 Name of state:
>>2s22p3_2p5_3
 Full interaction?
>>y
 Block            1 ,  ncf =          274
 Block            2 ,  ncf =          590
 Block            3 ,  ncf =          300
 Loading CSF file ... Header only
 There are/is            9  relativistic subshells;
 Include contribution of H (Transverse)?
>>y
 Modify all transverse photon frequencies?
>>y
 Enter the scale factor:
>>1.d-6
 Include H (Vacuum Polarisation)?
>>y
 Include H (Normal Mass Shift)?
>>n
 Include H (Specific Mass Shift)?
>>n
 Estimate self-energy?
>>y
 Largest n quantum number for including self-energy for orbital
 n should be less or equal 8
>>3
```

```
 Loading Radial WaveFunction File ...
 There are           3  blocks  (block   J/Parity   NCF):
  1  1/2-   274       2  3/2-   590       3  5/2-   300

 Enter ASF serial numbers for each block
 Block           1    ncf =         274  id =  1/2-
>>1-2
 Block           2    ncf =         590  id =  3/2-
>>1-4
 Block           3    ncf =         300  id =  5/2-
>>1


 ....

 RCI: Execution complete.

******************************************************************************
*        RUN JJ2LSJ TO GET THE LSJ-COMPOSITION                              *
*        INPUT FILE: 2s22p3_2p5_3.c, 2s22p3_2p5_3.cm                         *
*        OUTPUT FILE: 2s22p3_2p5_3.lsj.lbl, 2s22p3_2p5_3.uni.lsj.lbl        *
******************************************************************************

>>jj2lsj

 jj2lsj: Transformation of ASFs from a jj-coupled CSF basis
         into an LSJ-coupled CSF basis  (Fortran 95 version)
         (C) Copyright by   G. Gaigalas and Ch. F. Fischer,
         (2017).
 Input files: name.c, name.(c)m
 Output files: name.lsj.lbl
   (optional)  name.lsj.c, name.lsj.j,
               name.uni.lsj.lbl, name.uni.lsj.sum

 Name of state
>>2s22p3_2p5_3
 Loading Configuration Symmetry List File ...
 There are 9 relativistic subshells;
 There are 1164 relativistic CSFs;
  ... load complete;

 Mixing coefficients from a CI calc.?
>>y
 Do you need a unique labeling? (y/n)
>>y
    nelec  =           7
    ncftot =        1164
    nw     =           9
    nblock =           3

   block     ncf     nev    2j+1  parity
       1     274       2       2      -1
       2     591       4       4      -1
       3     300       1       6      -1
```

```
 Default settings?  (y/n)
>>y

      ..........

 jj2lsj: Execution complete.

*******************************************************************************
*         RUN RLEVELS TO VIEW ENERGIES AND ENERGY SEPARATIONS                  *
*         NOTE: SINCE LSJ-INFORMATION NOW IS AVAILABLE OUTPUT LABELS           *
*         WILL BE IN LSJ-COUPLING                                              *
*         IF DESIRED WE CAN INSTEAD RUN RLEVELSEV TO GET THE SEPARATION IN EV *
*******************************************************************************

>>rlevels 2s22p3_2p5_3.cm

 nblock =            3   ncftot =         1165   nw =            9   nelec =            7

 Energy levels for ...
 Rydberg constant is   109737.31569
 Splitting is the energy difference with the lower neighbor
------------------------------------------------------------------------------------------
 No Pos  J Parity Energy Total    Levels     Splitting     Configuration
                    (a.u.)      (cm^-1)     (cm^-1)
------------------------------------------------------------------------------------------
  1  1  3/2 -    -263.2797858        0.00        0.00  1s(2).2s(2).2p(3)4S3_4S
  2  2  3/2 -    -262.9550573    71269.67    71269.67  1s(2).2s(2).2p(3)2D3_2D
  3  1  5/2 -    -262.9538223    71540.71      271.04  1s(2).2s(2).2p(3)2D3_2D
  4  1  1/2 -    -262.7906356   107356.06    35815.35  1s(2).2s(2).2p(3)2P1_2P
  5  3  3/2 -    -262.7882760   107873.94      517.88  1s(2).2s(2).2p(3)2P1_2P
  6  4  3/2 -    -259.5241195   824273.48   716399.54  1s(2).2p(5)_2P
  7  2  1/2 -    -259.4979415   830018.89     5745.41  1s(2).2p(5)_2P
------------------------------------------------------------------------------------------
```

To interpret the LSJ-coupling notation produced by `jj2lsj`, see section 9.2

```
*******************************************************************************
*         WE WILL NOW COMPUTE THE M1 TRANSITION RATES                          *
*         IN THIS CASE THE INITIAL AND FINAL STATE FILES ARE THE SAME          *
*         AND WE DO NOT NEED TO PERFORM A biorthonormal TRANSFORMATION          *
*         USING RBIOTRANSFORM. JUST COPY FILES TO name.bw AND name.cbm         *
*******************************************************************************

>>cp 2s22p3_2p5_3.w 2s22p3_2p5_3.bw
>>cp 2s22p3_2p5_3.cm 2s22p3_2p5_3.cbm

*******************************************************************************
*         RUN RTRANSITION FOR 2s22p3_2p5_3 TO COMPUTE M1 TRANSITION PARAMETERS*
*         INPUT FILES: isodata, 2s22p3_2p5_3.c, 2s22p3_2p5_3.bw,               *
*                               2s22p3_2p5_3.cbm,                              *
*         OUTPUT FILE: 2s22p3_2p5_3.2s22p3_2p5_3.ct,                           *
*                      2s22p3_2p5_3.2s22p3_2p5_3.ct.lsj                        *
*                      2s22p3_2p5_3.2s22p3_2p5_3.+1T (angular file)            *
*                                                                              *
*         NOTE THAT THE LATTER OUTPUT FILE HAS ALL THE LABELS IN LSJ-          *
```

```
*        COUPLING WHICH IS VERY CONVENIENT                                 *
*                                                                          *
*        PLEASE OBSERVE!! IF WE ARE GOING TO RUN RTRANSITION FOR AN RCI WAVE *
*        FUNCTIONS THEN THE LSJ-INFORMATION SHOULD BE AVAILABLE FOR THE SAME *
*        WAVE FUNCTION. IF FOR EXAMPLE THE LSJ-INFORMATION FROM JJ2LSJ IS   *
*        IS AVAILABLE FROM AN RMCDHF RUN AND WE RUN RTRANSITION ON THE RCI  *
*        WAVE FUNCTION THEN RTRANSITION WILL STOP. IN THIS CASE JUST RERUN  *
*        JJ2LSJ FOR THE RCI WAVE FUNCTION AND START RTRANSITION AGAIN FOR   *
*        THE SAME WAVE FUNCTION. IN OUR EXAMPLE JJ2LJS AND RTRANSITION ARE  *
*        RUN FOR RCI WAVE FUNCTIONS AND EVERYTHING IS OK.                   *
****************************************************************************

>>rtransition

 RTRANSITION
 This program computes transition parameters from
 transformed wave functions
 Input files:  isodata, name1.c, name1.bw, name1.(c)bm
               name2.c, name2.bw, name2.(c)bm
               optional, name1.lsj.lbl, name2.lsj.lbl
               name1.name2.KT (optional angular files)
 Output files: name1.name2.(c)t
               optional, name1.name2.(c)t.lsj
               name1.name2.KT (angular files)
 Here K is parity and rank of transition: -1,+1 etc

  Default settings?
>>y
  Input from a CI calculation?
>>y

  Name of the Initial state
>>2s22p3_2p5_3
  Name of the Final state
>>2s22p3_2p5_3

 MRGCSL: Execution begins ...
 Loading Configuration Symmetry List File ...
 There are 9 relativistic subshells;
 There are 1164 relativistic CSFs;
  ... load complete;
 Loading Configuration Symmetry List File ...
 There are 9 relativistic subshells;
 There are 1164 relativistic CSFs;
  ... load complete;
          1 s
          2 s
          2 p-
          2 p
          3 s
          3 p-
          3 p
          3 d-
          3 d
```

```
          3
        274         864        1164
          3
        274         864        1164
 Loading Configuration Symmetry List File ...
  there are 9 relativistic subshells;
          1
          2           2
  there are 2328 relativistic CSFs;
  ... load complete;
 Enter the list of transition specifications
  e.g.,  E1,M2  or  E1 M2  or  E1;M2 :
>>M1
 M1 transitions only between levels with different J?
>>n

  .....

 RTRANSITION: Execution complete.
```

7.4 Fourth example: $3l3l'$ states in Fe XV using MPI

The fourth example is to determine the energies for the 10 states belonging to the three even configurations $3s^2, 3p^2, 3s3d$ and the 16 states belonging to the two odd configurations $3s3p, 3p3d$. In addition the E1, M2 and M1 transition data should be computed. The NIST table for these states is shown below.

```
------------------------------------------------------------
Configuration       | Term    |  J  |           Level       |
--------------------|---------|-----|-----------------------|
                    |         |     |                       |
2p6.3s2             | 1S      |   0 |                0      |
                    |         |     |                       |
3s.3p               | 3P*     |   0 |           233842      |
                    |         |   1 |           239660      |
                    |         |   2 |           253820      |
                    |         |     |                       |
3s.3p               | 1P*     |   1 |           351911      |
                    |         |     |                       |
3p2                 | 3P      |   0 |           554524      |
                    |         |     |                       |
3p2                 | 1D      |   2 |           559600      |
                    |         |     |                       |
3p2                 | 3P      |   1 |           564602      |
                    |         |   2 |           581803      |
                    |         |     |                       |
3p2                 | 1S      |   0 |           659627      |
                    |         |     |                       |
3s.3d               | 3D      |   1 |           678772      |
                    |         |   2 |           679785      |
                    |         |   3 |           681416      |
                    |         |     |                       |
3s.3d               | 1D      |   2 |           762093      |
                    |         |     |                       |
```

```
3p.3d                  | 3F*    |   2 |         928241      |
                       |        |   3 |         938126      |
                       |        |     |                     |
3p.3d                  | 1D*    |   2 |         948513      |
                       |        |     |                     |
3p.3d                  | 3F*    |   4 |         949658      |
                       |        |     |                     |
3p.3d                  | 3D*    |   1 |         982868      |
                       |        |     |                     |
3p.3d                  | 3P*    |   2 |         983514      |
                       |        |     |                     |
3p.3d                  | 3D*    |   3 |         994852      |
                       |        |     |                     |
3p.3d                  | 3P*    |   0 |         995889      |
                       |        |   1 |         996243      |
                       |        |     |                     |
3p.3d                  | 3D*    |   2 |         996623      |
                       |        |     |                     |
3p.3d                  | 1F*    |   3 |        1062515      |
                       |        |     |                     |
3p.3d                  | 1P*    |   1 |        1074887      |
-------------------------------------------------------------
```

The starting point is two separate `rmcdhf` calculations for the even and odd reference states, respectively. Then one layer of correlation orbitals is included describing valence-valence and core-valence correlation.

Overview

1. Define nuclear data.
2. Obtain common spectroscopic orbitals for the even parity MR set
   (a) Generate list of CSFs describing the even states belonging to $3s^2, 3p^2, 3s3d$
   (b) Perform angular integration.
   (c) Generate initial estimates of radial orbitals.
   (d) Perform self-consistent field calculation on the weighted average of the even states.
   (e) Save output to `evenmr`.
3. Improve even states
   (a) Generate $n = 4$ valence-valence and core-valence CSF expansions
   (b) Perform angular integration using MPI.
   (c) Generate initial estimates of radial orbitals.
   (d) Perform self-consistent field MPI calculation on the weighted average of the even states.
   (e) Save output to `even4`.
   (f) Perform RCI MPI calculation in which Breit and QED effects are added.
4. Run `jj2lsj` to transform to LSJ-coupling.
5. Obtain common spectroscopic orbitals for the odd parity MR set
   (a) Generate list of CSFs describing the even states belonging to $3s3p, 3p3d$

   (b) Perform angular integration.
   (c) Generate initial estimates of radial orbitals.
   (d) Perform self-consistent field calculation on the weighted average of the odd states.
   (e) Save output to `oddmr`.

6. Improve odd states.
   (a) Generate $n = 4$ valence-valence and core-valence CSF expansions.
   (b) Perform angular integration using MPI.
   (c) Generate initial estimates of radial orbitals.
   (d) Perform self-consistent field MPI calculation on the weighted average of the odd states.
   (e) Save output to `odd4`.
   (f) Perform RCI MPI calculation in which Breit and QED effects are added.

7. Run `jj2lsj` to transform to LSJ-coupling.

8. Run `rlevels`, `rlevelseV` to view energy separations.

9. Compute properties.
   (a) Compute transition rates from the RCI wave functions. Computation in two steps: biorthonormal transformation and then evaluation of transition matrix elements using standard Racah algebra methods. Both steps use MPI code.

Preparation for the MPI run

We intend to run `rangular_mpi`, `rmcdhf_mpi`, and `rci_mpi` using four processors and a `disks` file defining the location of the directory (on the the disk) in which temporary data should be stored. On our computer we will run the MPI jobs on four processors from a directory called

```
/home/tspejo/GRASP2018/grasptest/example4/script
```

and store the temporary data in

```
/home/tspejo/GRASP2018/grasptest/example4/tmp_mpi
```

The `disks` file corresponding to this case is shown below.

```
'/home/tspejo/GRASP2018/grasptest/example4/script'
'/home/tspejo/GRASP2018/grasptest/example4/tmp_mpi'
'/home/tspejo/GRASP2018/grasptest/example4/tmp_mpi'
'/home/tspejo/GRASP2018/grasptest/example4/tmp_mpi'
'/home/tspejo/GRASP2018/grasptest/example4/tmp_mpi'
```

If we use four processors for the MPI run the full path to the directory storing temporary data should be given four times in the `disks` file. If we use eight processors for the MPI run the full path to the directory storing temporary data should be given eight times etc. The directory storing temporary data can be anywhere in the file system and need not be on the same level in the file system as the working directory.

Provided the `disks` file is set up correctly according to the file structure of the local computer, the `cpath.f90` routine of the GRASP `mpi90` library automatically creates the directory in which temporary data are stored along with sub-directories `000`, `001`, `002` etc. named after the processors, starting with 0. On our system `cpath.f90` creates

```
/home/tspejo/GRASP2018/grasptest/example4/tmp_mpi
```

along with four sub-directories `000`, `001`, `002`, `003` named after the four processors, starting with 0. If `cpath.f90` fails to create the directories specified in the `disks` file then temporary data are stored in the directory specified by the `MPI_TMP` environment variable, see section 2.3.

On some computer systems the MPI libraries need to be loaded before the calculation starts. The commands for this depend on the system but could look like

```
module add openmpi
```

Make sure you load MPI libraries for `gfortran`.

Program input

In the test-runs input marked by >> or >>3, for example, indicates that the user should input 3 and then strike the return key. When >> is followed by blanks just strike the return key.

```
******************************************************************************
* RUN RNUCLEUS TO GENERATE NUCLEAR DATA AND DEFINE RADIAL GRID               *
* OUTPUT FILE: isodata                                                       *
******************************************************************************

>>rnucleus

 RNUCLEUS
 This program defines nuclear data and the radial grid
 Outputfile: isodata

 Enter the atomic number:
>>26
 Enter the mass number (0 if the nucleus is to be modelled as a point source:
>>56
 The default root mean squared radius is    3.7684209375954341       fm;
 the default nuclear skin thickness is    2.2999999999999998       fm;
 Revise these values?
>>n
 Enter the mass of the neutral atom (in amu) (0 if the nucleus is to be static):
>>55.845
 Enter the nuclear spin quantum number (I) (in units of h / 2 pi):
>>1
 Enter the nuclear dipole moment (in nuclear magnetons):
>>1
 Enter the nuclear quadrupole moment (in barns):
>>1

******************************************************************************
*         RUN RCSFGENERATE TO GENERATE LIST OF CSFs FOR                      *
*         CONFIGURATIONS 3s(2), 3p(2), 3s3p                                  *
*         OUTPUT FILES: rcsfgenerate.log, rcsf.out                           *
******************************************************************************

>>rcsfgenerate

 RCSFGENERATE
 This program creates a list of CSFs
 Configurations should be entered in spectroscopic notation
 with occupation numbers and indications if orbitals are
```

```
 closed (c), inactive (i), active (*) or has a minimal
 occupation e.g. 1s(2,1)2s(2,*)
 Outputfiles: rcsf.out, rcsfgenerate.log

 Default, reverse, symmetry or user specified ordering? (*/r/s/u)
>>*

 Select core
       0: No core
       1: He (      1s(2)                  =  2 electrons)
       2: Ne ([He] + 2s(2)2p(6)            = 10 electrons)
       3: Ar ([Ne] + 3s(2)3p(6)            = 18 electrons)
       4: Kr ([Ar] + 3d(10)4s(2)4p(6)      = 36 electrons)
       5: Xe ([Kr] + 4d(10)5s(2)5p(6)      = 54 electrons)
       6: Rn ([Xe] + 4f(14)5d(10)6s(2)6p(6) = 86 electrons)
>>1
 Enter list of (maximum 100) configurations. End list with a blank line or an asterisk (*)

 Give configuration  1
>>2s(2,i)2p(6,i)3s(2,i)
 Give configuration  2
>>2s(2,i)2p(6,i)3p(2,i)
 Give configuration  3
>>2s(2,i)2p(6,i)3s(1,i)3d(1,i)
 Give configuration  4
>>
 Give set of active orbitals, as defined by the highest principal quantum number
 per l-symmetry, in a comma delimited list in s,p,d etc order, e.g. 5s,4p,3d
>>3s,3p,3d
 Resulting 2*J-number? lower, higher (J=1 -> 2*J=2 etc.)
>>0,6
 Number of excitations (if negative number e.g. -2, correlation
 orbitals will always be doubly occupied)
>>0
 Generate more lists ? (y/n)
>>n

       .........

  4  blocks were created

block  J/P           NCSF
          1    0+            3
          2    1+            2
          3    2+            4
          4    3+            1

******************************************************************************
*         COPY FILES                                                         *
*         IT IS ADVISABLE TO SAVE THE rcsfgenerate.log FILE TO HAVE A        *
*         RECORD ON HOW THE LIST OF CSFs WAS CREATED                         *
******************************************************************************

>>cp rcsfgenerate.log evenmr.exc
```

```
>>cp rcsf.out rcsf.inp

*****************************************************************************
*         RUN RANGULAR TO GENERATE ENERGY EXPRESSION                        *
*         INPUT FILE  : rcsf.inp                                            *
*         OUTPUT FILES: rangular.alog, mcp.30, mcp.31,....                  *
*****************************************************************************

>>rangular

 RANGULAR
 This program performs angular integration
 Input file:  rcsf.inp
 Outputfiles: mcp.30, mcp.31, ....
              rangular.log

 Full interaction?  (y/n)
>>y

  .....

 RANGULAR: Execution complete.

*****************************************************************************
*         RUN RWFNESTIMATE TO GENERATE INITIAL ESTIMATES FOR RADIAL ORBITALS  *
*         INPUT FILES: isodata, rcsf.inp, previous rwfn files               *
*         OUTPUT FILE: rwfn.inp, rwfnestimate.log                           *
*****************************************************************************

>>rwfnestimate

 RWFNESTIMATE
 This program estimates radial wave functions
 for orbitals
 Input files: isodata, rcsf.inp, optional rwfn file
 Output file: rwfn.inp

 Default settings ?
>>y
 Loading CSF file ... Header only
 There are/is            9  relativistic subshells;
 The following subshell radial wavefunctions remain to be estimated:
 1s 2s 2p- 2p 3s 3p- 3p 3d- 3d

 Read subshell radial wavefunctions. Choose one below
     1 -- GRASP2K File
     2 -- Thomas-Fermi
     3 -- Screened Hydrogenic
>>2
 Enter the list of relativistic subshells:
>>*
 All required subshell radial wavefunctions  have been estimated:
Shell      e          p0        gamma        P(2)       Q(2)      MTP  SRC
```

```
  1s   0.3098D+03  0.2951D+03  0.1000D+01  0.1164D-05 -0.7545D-11  328  T-F
  2s   0.6428D+02  0.1015D+03  0.1000D+01  0.4004D-06 -0.2598D-11  346  T-F
  2p-  0.6284D+02  0.7742D+00  0.1000D+01  0.2262D-12  0.3219D-07  346  T-F
  2p   0.6217D+02  0.6940D+03  0.2000D+01  0.1080D-13 -0.7003D-19  346  T-F
  3s   0.2358D+02  0.5151D+02  0.1000D+01  0.2031D-06 -0.1318D-11  358  T-F
  3p-  0.2295D+02  0.4216D+00  0.1000D+01  0.1232D-12  0.1753D-07  358  T-F
  3p   0.2278D+02  0.3794D+03  0.2000D+01  0.5901D-14 -0.3829D-19  358  T-F
  3d-  0.2170D+02  0.5564D+00  0.2000D+01  0.1282D-20  0.1825D-15  358  T-F
  3d   0.2165D+02  0.6259D+03  0.3000D+01  0.3839D-22 -0.2491D-27  358  T-F

 RWFNESTIMATE: Execution complete.


*******************************************************************************
*         RUN RMCDHF TO OBTAIN SELF CONSISTENT SOLUTIONS                      *
*         INPUT FILES: isodata, rcsf.inp, rwfn.inp, mcp.30, mcp.31,...        *
*         OUTPUT FILES: rwfn.out, rmix.out, rmcdhf.sum, rmcdhf.log            *
*                                                                             *
*         NOTE: ORBITALS BUILDING REFERENCE STATES ARE REQUIRED TO HAVE       *
*         THE CORRECT NUMBER OF NODES. THEY ARE REFERRED TO AS SPECTROSCOPIC  *
*         ORBITALS. IN THIS RUN WE VARY 1s,2s,2p,3s,3p,3d AND THEY ARE ALL    *
*         SPECTROSCOPIC. WE CAN USE WILD CARDS FOR SPECIFYING ORBITALS        *
*                                                                             *
*         NOTE: INSTEAD OF SAYING THAT WE SHOULD OPTIMIZE ON, FOR EXAMPLE,    *
*         THE STATES 1,2,3,4 WE CAN WRITE 1-4 MEANING THE SAME THING          *
*                                                                             *
*******************************************************************************

>>rmcdhf

 RMCDHF
 This program determines the radial orbitals
 and the expansion coefficients of the CSFs
 in a self-onsistent field proceedure
 Input file:  isodata, rcsf.inp, rwfn.inp, mcp.30, ...
 Outputfiles: rwfn.out, rmix.out, rmcdhf.sum, rmcdhf.log

 Default settings?  (y/n)
>>y
 Loading CSF file ... Header only
 There are/is           9  relativistic subshells;
 Loading CSF File for ALL blocks
 There are          10  relativistic CSFs... load complete;
 Loading Radial WaveFunction File ...

 There are           4  blocks  (block   J/Parity   NCF):
  1    0+     3       2    1+     2       3    2+     4       4    3+     1

 Enter ASF serial numbers for each block
 Block           1    ncf =           3  id =    0+
>>1-3
 Block           2    ncf =           2  id =    1+
>>1-2
 Block           3    ncf =           4  id =    2+
```

```
>>1-4
 Block            4    ncf =           1  id =    3+
>>1
 level weights (1 equal;  5 standard;  9 user)
>>5
 Radial functions
 1s 2s 2p- 2p 3s 3p- 3p 3d- 3d
 Enter orbitals to be varied (Updating order)
>>*
 Which of these are spectroscopic orbitals?
>>*
 Enter the maximum number of SCF cycles:
>>100

.............

 RMCDHF: Execution complete.

******************************************************************************
*         RUN RSAVE TO SAVE OUTPUT FILES: name.c, name.w, name.m, name.sum    *
*                                          name.alog name.log                 *
******************************************************************************

>>rsave evenmr
 Created evenmr.w, evenmr.c, evenmr.m, evenmr.sum, evenmr.alog and evenmr.log

******************************************************************************
*         RUN RCSFGENERATE TO GENERATE n = 4 VALENCE-VALENCE AND             *
*         CORE-VALENCE LIST                                                  *
*         OUTPUT FILES: rcsfgenerate.log, rcsf.out                           *
******************************************************************************

>>rcsfgenerate

 RCSFGENERATE
 This program creates a list of CSFs
 Configurations should be entered in spectroscopic notation
 with occupation numbers and indications if orbitals are
 closed (c), inactive (i), active (*) or has a minimal
 occupation e.g. 1s(2,1)2s(2,*)
 Outputfiles: rcsf.out, rcsfgenerate.log

 Default, reverse, symmetry or user specified ordering? (*/r/s/u)
>>*

 Select core
        0: No core
        1: He (       1s(2)                  =  2 electrons)
        2: Ne ([He] + 2s(2)2p(6)             = 10 electrons)
        3: Ar ([Ne] + 3s(2)3p(6)             = 18 electrons)
        4: Kr ([Ar] + 3d(10)4s(2)4p(6)       = 36 electrons)
        5: Xe ([Kr] + 4d(10)5s(2)5p(6)       = 54 electrons)
        6: Rn ([Xe] + 4f(14)5d(10)6s(2)6p(6) = 86 electrons)
>>1
```

```
 Enter list of (maximum 100) configurations. End list with a blank line or an asterisk (*)

 Give configuration  1
>>2s(2,i)2p(6,5)3s(2,*)
 Give configuration  2
>>2s(2,1)2p(6,i)3s(2,*)
 Give configuration  3
>>2s(2,i)2p(6,5)3p(2,*)
 Give configuration  4
>>2s(2,1)2p(6,i)3p(2,*)
 Give configuration  5
>>2s(2,i)2p(6,5)3s(1,*)3d(1,*)
 Give configuration  6
>>2s(2,1)2p(6,i)3s(1,*)3d(1,*)
 Give configuration  7
>>
 Give set of active orbitals, as defined by the highest principal quantum number
 per l-symmetry, in a comma delimited list in s,p,d etc order, e.g. 5s,4p,3d
>>4s,4p,4d,4f
 Resulting 2*J-number? lower, higher (J=1 -> 2*J=2 etc.)
>>0,6
 Number of excitations (if negative number e.g. -2, correlation
 orbitals will always be doubly occupied)
>>2
 Generate more lists ? (y/n)
>>n
        .........

 4  blocks were found

block  J/P            NCSF
           1    0+            556
           2    1+           1448
           3    2+           1898
           4    3+           1810

*****************************************************************************
*         COPY FILES                                                        *
*         IT IS ADVISABLE TO SAVE THE rcsfgenerate.log FILE TO HAVE A       *
*         RECORD ON HOW THE LIST OF CSFs WAS CREATED                        *
*****************************************************************************

>>cp rcsfgenerate.log even4.exc
>>cp rcsf.out rcsf.inp

*****************************************************************************
*         RUN RANGULAR_MPI USING 4 PROCESSES TO GENERATE ENERGY EXPRESSION  *
*         INPUT FILE  : rcsf.inp                                            *
*         OUTPUT FILES: rangular.alog, mcp.30, mcp.31,..IN 000, 001, 002, 003 *
*****************************************************************************

>>mpirun -np 4 rangular_mpi

 ====================================================
```

```
       RANGULAR_MPI: Execution Begins ...
 ====================================================
 Participating nodes:
   Host: atom1    ID: 000
   Host: atom1    ID: 001
   Host: atom1    ID: 002
   Host: atom1    ID: 003

 Date and Time:
   atom1:   Date: 20140812  Time: 011040.566  Zone: +0200
   atom1:   Date: 20140812  Time: 011040.566  Zone: +0200
   atom1:   Date: 20140812  Time: 011040.566  Zone: +0200
   atom1:   Date: 20140812  Time: 011040.566  Zone: +0200

 Start Dir:
   atom1: /GRASP2018/grasptest/example4/script
   atom1: /GRASP2018/grasptest/example4/script
   atom1: /GRASP2018/grasptest/example4/script
   atom1: /GRASP2018/grasptest/example4/script

 Serial I/O Dir (node-0 only):
   atom1: /home/tspejo/GRASP2018/grasptest/example4/script

 Work Dir (Parallel I/O):
   atom1: /home/tspejo/GRASP2018/grasptest/example4/tmp_mpi
   atom1: /home/tspejo/GRASP2018/grasptest/example4/tmp_mpi
   atom1: /home/tspejo/GRASP2018/grasptest/example4/tmp_mpi
   atom1: /home/tspejo/GRASP2018/grasptest/example4/tmp_mpi

Full interaction?  (y/n)
>>y

.........

 mpi stopped by node-           0  from RANGULAR_MPI: Execution complete.
 mpi stopped by node-           2  from RANGULAR_MPI: Execution complete.
 mpi stopped by node-           1  from RANGULAR_MPI: Execution complete.
 mpi stopped by node-           3  from RANGULAR_MPI: Execution complete.

******************************************************************************
*         RUN RWFNESTIMATE TO GENERATE INITIAL ESTIMATES FOR RADIAL ORBITALS  *
*         INPUT FILES: isodata, rcsf.inp, previous rwfn files                 *
*         OUTPUT FILE: rwfn.inp                                               *
******************************************************************************

>>rwfnestimate

 RWFNESTIMATE
 This program estimates radial wave functions
 for orbitals
 Input files: isodata, rcsf.inp, optional rwfn file
 Output file: rwfn.inp

 Default settings ?
```

```
>>y
 Loading CSF file ... Header only
 There are/is           16  relativistic subshells;
 The following subshell radial wavefunctions remain to be estimated:
 1s 2s 2p- 2p 3s 3p- 3p 3d- 3d 4s 4p- 4p 4d- 4d 4f- 4f

 Read subshell radial wavefunctions. Choose one below
     1 -- GRASP2K File
     2 -- Thomas-Fermi
     3 -- Screened Hydrogenic
>>1
 Enter the file name (Null then "rwfn.out")

 Enter the list of relativistic subshells:
>>*
 The following subshell radial wavefunctions remain to be estimated:
 4s 4p- 4p 4d- 4d 4f- 4f

 Read subshell radial wavefunctions. Choose one below
     1 -- GRASP2K File
     2 -- Thomas-Fermi
     3 -- Screened Hydrogenic
>>2
 Enter the list of relativistic subshells:
>>*
 All required subshell radial wavefunctions  have been estimated:
Shell      e           p0        gamma        P(2)       Q(2)      MTP  SRC

  1s   0.2803D+03  0.2922D+03  0.1000D+01  0.1152D-05 -0.7471D-11  354  rwf
  2s   0.4802D+02  0.9080D+02  0.1000D+01  0.3581D-06 -0.2322D-11  358  rwf
  2p-  0.4353D+02  0.6334D+00  0.1000D+01  0.1850D-12  0.2633D-07  358  rwf
  2p   0.4305D+02  0.5671D+03  0.2000D+01  0.8821D-14 -0.5721D-19  358  rwf
  3s   0.1667D+02  0.4071D+02  0.1000D+01  0.1606D-06 -0.1041D-11  362  rwf
  3p-  0.1543D+02  0.2994D+00  0.1000D+01  0.8746D-13  0.1245D-07  363  rwf
  3p   0.1534D+02  0.2693D+03  0.2000D+01  0.4188D-14 -0.2717D-19  363  rwf
  3d-  0.1358D+02  0.2667D+00  0.2000D+01  0.6146D-21  0.8747D-16  364  rwf
  3d   0.1356D+02  0.3005D+03  0.3000D+01  0.1844D-22 -0.1196D-27  364  rwf
  4s   0.1141D+02  0.3095D+02  0.1000D+01  0.1221D-06 -0.7919D-12  366  T-F
  4p-  0.1110D+02  0.2579D+00  0.1000D+01  0.7535D-13  0.1072D-07  367  T-F
  4p   0.1104D+02  0.2325D+03  0.2000D+01  0.3616D-14 -0.2346D-19  367  T-F
  4d-  0.1053D+02  0.3856D+00  0.2000D+01  0.8888D-21  0.1265D-15  367  T-F
  4d   0.1051D+02  0.4341D+03  0.3000D+01  0.2663D-22 -0.1728D-27  367  T-F
  4f-  0.9897D+01  0.2129D+00  0.3000D+01  0.2902D-29  0.4130D-24  367  T-F
  4f   0.9890D+01  0.2598D+03  0.4000D+01  0.6286D-31 -0.4078D-36  367  T-F

 RWFNESTIMATE: Execution complete.

*******************************************************************************
*         RUN RMCDHF_MPI USING 4 PROCESSES TO OBTAIN SELF CONSISTENT SOLUTIONS*
*         INPUT FILES: isodata, rcsf.inp, rwfn.inp, mcp.30, mcp.31,...        *
*         OUTPUT FILES: rwfn.out, rmix.out, rmcdhf.sum, rmcdhf.log            *
*                                                                             *
*         NOTE: FOR CORRELATION ORBITALS THERE ARE NO RESTRICTIONS ON THE     *
*         NUMBER OF NODES, I.E. THEY ARE NOT SPECTROSCOPIC. IN THIS RUN WE    *
```

```
*        VARY THE CORRELATION ORBITALS 4s,4p,4d,4f. NONE OF THESE ARE       *
*        SPECTROSCOPIC. WE CAN USE WILD CARDS * FOR SPECIFYING ORBITALS     *
*        4* MEANS 4s, 4p-, 4p, 4d-, 4d, 4f-, 4f                             *
*****************************************************************************

>>mpirun -np 4 rmcdhf_mpi

 ====================================================
       RMCDHF_MPI: Execution Begins ...
 ====================================================
 Participating nodes:
   Host: atom1    ID: 000
   Host: atom1    ID: 001
   Host: atom1    ID: 002
   Host: atom1    ID: 003

 Date and Time:
   atom1:   Date: 20140812  Time: 011653.965  Zone: +0200
   atom1:   Date: 20140812  Time: 011653.965  Zone: +0200
   atom1:   Date: 20140812  Time: 011653.965  Zone: +0200
   atom1:   Date: 20140812  Time: 011653.965  Zone: +0200

 Start Dir:
   atom1: /GRASP2018/grasptest/example4/script
   atom1: /GRASP2018/grasptest/example4/script
   atom1: /GRASP2018/grasptest/example4/script
   atom1: /GRASP2018/grasptest/example4/script

 Serial I/O Dir (node-0 only):
   atom1: /home/tspejo/GRASP2018/grasptest/example4/script

 Work Dir (Parallel I/O):
   atom1: /home/tspejo/GRASP2018/grasptest/example4/tmp_mpi
   atom1: /home/tspejo/GRASP2018/grasptest/example4/tmp_mpi
   atom1: /home/tspejo/GRASP2018/grasptest/example4/tmp_mpi
   atom1: /home/tspejo/GRASP2018/grasptest/example4/tmp_mpi

Default settings?  (y/n)
>>y
 Loading CSF file ... Header only
 There are/is           16  relativistic subshells;
 Loading CSF File for ALL blocks
 There are         5712  relativistic CSFs... load complete;
 There are            4  blocks  (block   J/Parity   NCF):
  1    0+   556       2    1+  1448       3    2+  1898       4    3+  1810

 Enter ASF serial numbers for each block
 Block           1    ncf =         556  id =    0+
>>1-3
 Block           2    ncf =        1448  id =    1+
>>1-2
 Block           3    ncf =        1898  id =    2+
>>1-4
 Block           4    ncf =        1810  id =    3+
```

```
>>1
 level weights (1 equal;  5 standard;  9 user)
>>5
  Radial functions
 1s 2s 2p- 2p 3s 3p- 3p 3d- 3d 4s 4p- 4p 4d- 4d 4f- 4f
 Enter orbitals to be varied (Updating order)
>>4*
 Which of these are spectroscopic orbitals?
>>
 Enter the maximum number of SCF cycles:
>>100

 .....

 mpi stopped by node-           1  from RMCDHF_MPI: Execution complete.
 mpi stopped by node-           0  from RMCDHF_MPI: Execution complete.
 mpi stopped by node-           3  from RMCDHF_MPI: Execution complete.
 mpi stopped by node-           2  from RMCDHF_MPI: Execution complete.

*******************************************************************************
*         RUN RSAVE TO SAVE OUTPUT FILES: name.c, name.w, name.m, name.sum    *
*                                         name.alog, name.log                 *
*******************************************************************************

>>rsave even4
 Created even4.w, even4.c, even4.m, even4.sum, even4.alog and even4.log

*******************************************************************************
*         RUN RCI_MPI USING 4 PROCESSES TO INCLUDE BREIT AND QED EFFECTS      *
*         INPUT FILES : isodata, even4.c, even4.w                             *
*         OUTPUT FILES: even4.cm, even4.csum                                  *
*                                                                             *
*         THE TRANSVERSE PHOTON FREQUENCIES CAN BE SET TO THE LOW FREQUENCY   *
*         LIMIT. RECOMMENDED IN CASES WHERE YOU HAVE CORRELATION ORBITALS     *
*         THE SELF ENERGY CORRECTION MAY FAIL FOR CORRELATION ORBITALS WITH   *
*         HIGH N.                                                             *
*******************************************************************************

>>mpirun -np 4 rci_mpi

====================================================
        RCI_MPI: Execution Begins ...
 ====================================================
 Participating nodes:
   Host: atom1    ID: 000
   Host: atom1    ID: 001
   Host: atom1    ID: 002
   Host: atom1    ID: 003

 Date and Time:
   atom1:   Date: 20140812  Time: 012250.312  Zone: +0200
   atom1:   Date: 20140812  Time: 012250.312  Zone: +0200
   atom1:   Date: 20140812  Time: 012250.312  Zone: +0200
   atom1:   Date: 20140812  Time: 012250.312  Zone: +0200
```

```
 Start Dir:
   atom1: /GRASP2018/grasptest/example4/script
   atom1: /GRASP2018/grasptest/example4/script
   atom1: /GRASP2018/grasptest/example4/script
   atom1: /GRASP2018/grasptest/example4/script

 Serial I/O Dir (node-0 only):
   atom1: /home/tspejo/GRASP2018/grasptest/example4/script

 Work Dir (Parallel I/O):
   atom1: /home/tspejo/GRASP2018/grasptest/example4/tmp_mpi
   atom1: /home/tspejo/GRASP2018/grasptest/example4/tmp_mpi
   atom1: /home/tspejo/GRASP2018/grasptest/example4/tmp_mpi
   atom1: /home/tspejo/GRASP2018/grasptest/example4/tmp_mpi

Default settings?
>>y
Name of state:
>>even4
 Block           1 ,  ncf =         556
 Block           2 ,  ncf =        1448
 Block           3 ,  ncf =        1898
 Block           4 ,  ncf =        1810
 Loading CSF file ... Header only
 There are/is           16  relativistic subshells;
 Include contribution of H (Transverse)?
>>y
 Modify all transverse photon frequencies?
>>y
 Enter the scale factor:
>>1.d-6
 Include H (Vacuum Polarisation)?
>>y
 Include H (Normal Mass Shift)?
>>n
 Include H (Specific Mass Shift)?
>>n
 Estimate self-energy?
>>y
 Largest n quantum number for including self-energy for orbital
 n should be less or equal 8
>>4
 There are            4  blocks  (block   J/Parity   NCF):
  1    0+   556       2    1+  1448       3    2+  1898       4    3+  1810

 Enter ASF serial numbers for each block
 Block           1    ncf =         556  id =    0+
>>1-3
 Block           2    ncf =        1448  id =    1+
>>1-2
 Block           3    ncf =        1898  id =    2+
>>1-4
 Block           4    ncf =        1810  id =    3+
```

```
>>1

 .....

 mpi stopped by node-           0  from RCI_MPI: Execution complete.
 mpi stopped by node-           3  from RCI_MPI: Execution complete.
 mpi stopped by node-           1  from RCI_MPI: Execution complete.
 mpi stopped by node-           2  from RCI_MPI: Execution complete.

*******************************************************************************
*         RUN JJ2LSJ TO GET THE LSJ-COMPOSITION                               *
*         INPUT FILE: even4.c, even4.cm                                       *
*         OUTPUT FILE: even4.lsj.lbl, even4.uni.lsj.lbl                       *
*******************************************************************************

>>jj2lsj

 jj2lsj: Transformation of ASFs from a jj-coupled CSF basis
         into an LSJ-coupled CSF basis  (Fortran 95 version)
         (C) Copyright by   G. Gaigalas and Ch. F. Fischer,
         (2017).
 Input files: name.c, name.(c)m
 Output files: name.lsj.lbl
   (optional)  name.lsj.c, name.lsj.j,
               name.uni.lsj.lbl, name.uni.lsj.sum

 Name of state
>>even4
 Loading Configuration Symmetry List File ...
 There are 16 relativistic subshells;
 There are 5712 relativistic CSFs;
  ... load complete;

 Mixing coefficients from a CI calc.?
>>y
 Do you need a unique labeling? (y/n)
>>y
    nelec  =          12
    ncftot =        5712
    nw     =          16
    nblock =           4

   block     ncf     nev    2j+1  parity
       1     556       3       1       1
       2    1448       2       3       1
       3    1898       4       5       1
       4    1810       1       7       1
 Default settings?  (y/n)
>>y

 jj2lsj: Execution complete.

*******************************************************************************
*         RUN RCSFGENERATE TO GENERATE LIST OF CSFs FOR                       *
```

```
*         CONFIGURATIONS 3s3p, 3p3d                                        *
*         OUTPUT FILES: rcsfgenerate.log, rcsf.out                         *
****************************************************************************

>>rcsfgenerate

 RCSFGENERATE
 This program creates a list of CSFs
 Configurations should be entered in spectroscopic notation
 with occupation numbers and indications if orbitals are
 closed (c), inactive (i), active (*) or has a minimal
 occupation e.g. 1s(2,1)2s(2,*)
 Outputfiles: rcsf.out, rcsfgenerate.log

 Default, reverse, symmetry or user specified ordering? (*/r/s/u)
>>*

 Select core
        0: No core
        1: He (       1s(2)                  =  2 electrons)
        2: Ne ([He] + 2s(2)2p(6)             = 10 electrons)
        3: Ar ([Ne] + 3s(2)3p(6)             = 18 electrons)
        4: Kr ([Ar] + 3d(10)4s(2)4p(6)       = 36 electrons)
        5: Xe ([Kr] + 4d(10)5s(2)5p(6)       = 54 electrons)
        6: Rn ([Xe] + 4f(14)5d(10)6s(2)6p(6) = 86 electrons)
>>1
 Enter list of (maximum 100) configurations. End list with a blank line or an asterisk (*)

 Give configuration  1
>>2s(2,i)2p(6,i)3s(1,i)3p(1,i)
 Give configuration  2
>>2s(2,i)2p(6,i)3p(1,i)3d(1,i)
 Give configuration  3
>>
 Give set of active orbitals, as defined by the highest principal quantum number
 per l-symmetry, in a comma delimited list in s,p,d etc order, e.g. 5s,4p,3d
>>3s,3p,3d
 Resulting 2*J-number? lower, higher (J=1 -> 2*J=2 etc.)
>>0,8
 Number of excitations (if negative number e.g. -2, correlation
 orbitals will always be doubly occupied)
>>0
 Generate more lists ? (y/n)
>>n

        .........

  5  blocks were created

      block  J/P            NCSF
          1    0-              2
          2    1-              5
          3    2-              5
          4    3-              3
```

```
           5    4-              1


*******************************************************************************
*         COPY FILES                                                          *
*         IT IS ADVISABLE TO SAVE THE rcsfgenerate.log FILE TO HAVE A         *
*         RECORD ON HOW THE LIST OF CSFs WAS CREATED                          *
*******************************************************************************

>>cp rcsfgenerate.log oddmr.exc
>>cp rcsf.out rcsf.inp

*******************************************************************************
*         RUN RANGULAR TO GENERATE ENERGY EXPRESSION                          *
*         INPUT FILE  : rcsf.inp                                              *
*         OUTPUT FILES: rangular.alog, mcp.30, mcp.31,....                    *
*******************************************************************************

>>rangular

 RANGULAR
 This program performs angular integration
 Input file:  rcsf.inp
 Outputfiles: mcp.30, mcp.31, ....
              rangular.log

 Full interaction?  (y/n)
>>y

  .....

 RANGULAR: Execution complete.

*******************************************************************************
*         RUN RWFNESTIMATE TO GENERATE INITIAL ESTIMATES FOR RADIAL ORBITALS  *
*         INPUT FILES: isodata, rcsf.inp, previous rwfn files                 *
*         OUTPUT FILE: rwfn.inp, rwfnestimate.log                             *
*******************************************************************************

>>rwfnestimate

 RWFNESTIMATE
 This program estimates radial wave functions
 for orbitals
 Input files: isodata, rcsf.inp, optional rwfn file
 Output file: rwfn.inp

 Default settings ?
>>y
 Loading CSF file ... Header only
 There are/is            9  relativistic subshells;
 The following subshell radial wavefunctions remain to be estimated:
 1s 2s 2p- 2p 3s 3p- 3p 3d- 3d
```

```
 Read subshell radial wavefunctions. Choose one below
     1 -- GRASP2K File
     2 -- Thomas-Fermi
     3 -- Screened Hydrogenic
>>2
 Enter the list of relativistic subshells:
>>*
 All required subshell radial wavefunctions  have been estimated:
Shell      e           p0        gamma        P(2)       Q(2)      MTP  SRC

  1s   0.3098D+03  0.2951D+03  0.1000D+01  0.1164D-05 -0.7545D-11  328  T-F
  2s   0.6428D+02  0.1015D+03  0.1000D+01  0.4004D-06 -0.2598D-11  346  T-F
  2p-  0.6284D+02  0.7742D+00  0.1000D+01  0.2262D-12  0.3219D-07  346  T-F
  2p   0.6217D+02  0.6940D+03  0.2000D+01  0.1080D-13 -0.7003D-19  346  T-F
  3s   0.2358D+02  0.5151D+02  0.1000D+01  0.2031D-06 -0.1318D-11  358  T-F
  3p-  0.2295D+02  0.4216D+00  0.1000D+01  0.1232D-12  0.1753D-07  358  T-F
  3p   0.2278D+02  0.3794D+03  0.2000D+01  0.5901D-14 -0.3829D-19  358  T-F
  3d-  0.2170D+02  0.5564D+00  0.2000D+01  0.1282D-20  0.1825D-15  358  T-F
  3d   0.2165D+02  0.6259D+03  0.3000D+01  0.3839D-22 -0.2491D-27  358  T-F

 RWFNESTIMATE: Execution complete.

******************************************************************************
*         RUN RMCDHF TO OBTAIN SELF CONSISTENT SOLUTIONS                     *
*         INPUT FILES: isodata, rcsf.inp, rwfn.inp, mcp.30, mcp.31,...       *
*         OUTPUT FILES: rwfn.out, rmix.out, rmcdhf.sum, rmcdhf.log           *
*                                                                            *
*         NOTE: ORBITALS BUILDING REFERENCE STATES ARE REQUIRED TO HAVE      *
*         THE CORRECT NUMBER OF NODES. THEY ARE REFERRED TO AS SPECTROSCOPIC *
*         ORBITALS. IN THIS RUN WE VARY 1s,2s,2p,3s,3p,3d AND THEY ARE ALL   *
*         SPECTROSCOPIC. WE CAN USE WILD CARDS * FOR SPECIFYING ORBITALS     *
******************************************************************************

>>rmcdhf

 RMCDHF
 This program determines the radial orbitals
 and the expansion coefficients of the CSFs
 in a self-onsistent field proceedure
 Input file:  isodata, rcsf.inp, rwfn.inp, mcp.30, ...
 Outputfiles: rwfn.out, rmix.out, rmcdhf.sum, rmcdhf.log

 Default settings?  (y/n)
>>y
 Loading CSF file ... Header only
 There are/is            9  relativistic subshells;
 Loading CSF File for ALL blocks
 There are           16  relativistic CSFs... load complete;
 Loading Radial WaveFunction File ...

 There are            5  blocks  (block   J/Parity   NCF):
  1    0-     2       2    1-     5       3    2-     5       4    3-     3
  5    4-     1
```

```
 Enter ASF serial numbers for each block
 Block            1    ncf =           2  id =    0-
>>1-2
 Block            2    ncf =           5  id =    1-
>>1-5
 Block            3    ncf =           5  id =    2-
>>1-5
 Block            4    ncf =           3  id =    3-
>>1-3
 Block            5    ncf =           1  id =    4-
>>1
 level weights (1 equal;  5 standard;  9 user)
>>5
 Radial functions
 1s 2s 2p- 2p 3s 3p- 3p 3d- 3d
 Enter orbitals to be varied (Updating order)
>>*
 Which of these are spectroscopic orbitals?
>>*
 Enter the maximum number of SCF cycles:
>>100


..............

 RMCDHF: Execution complete.

*******************************************************************************
*         RUN RSAVE TO SAVE OUTPUT FILES: name.c, name.w, name.m, name.sum    *
*                                         name.alog, name.log                 *
*******************************************************************************

>>rsave oddmr
 Created oddmr.w, oddmr.c, oddmr.m, oddmr.sum, oddmr.alog and oddmr.log

*******************************************************************************
*         RUN RCSFGENERATE TO GENERATE n = 4 VALENCE-VALENCE AND              *
*         CORE-VALENCE LIST                                                   *
*         OUTPUT FILES: rcsfgenerate.log, rcsf.out                            *
*******************************************************************************

>>rcsfgenerate

 RCSFGENERATE
 This program creates a list of CSFs
 Configurations should be entered in spectroscopic notation
 with occupation numbers and indications if orbitals are
 closed (c), inactive (i), active (*) or has a minimal
 occupation e.g. 1s(2,1)2s(2,*)
 Outputfiles: rcsf.out, rcsfgenerate.log

 Default, reverse, symmetry or user specified ordering? (*/r/s/u)
>>*
```

```
 Select core
        0: No core
        1: He (       1s(2)                  =  2 electrons)
        2: Ne ([He] + 2s(2)2p(6)             = 10 electrons)
        3: Ar ([Ne] + 3s(2)3p(6)             = 18 electrons)
        4: Kr ([Ar] + 3d(10)4s(2)4p(6)       = 36 electrons)
        5: Xe ([Kr] + 4d(10)5s(2)5p(6)       = 54 electrons)
        6: Rn ([Xe] + 4f(14)5d(10)6s(2)6p(6) = 86 electrons)
>>1
 Enter list of (maximum 100) configurations. End list with a blank line or an asterisk (*)

 Give configuration  1
>>2s(2,i)2p(6,5)3s(1,*)3p(1,*)
 Give configuration  2
>>2s(2,1)2p(6,i)3s(1,*)3p(1,*)
 Give configuration  3
>>2s(2,i)2p(6,5)3p(1,*)3d(1,*)
 Give configuration  4
>>2s(2,1)2p(6,i)3p(1,*)3d(1,*)
 Give configuration  5
>>
 Give set of active orbitals, as defined by the highest principal quantum number
 per l-symmetry, in a comma delimited list in s,p,d etc order, e.g. 5s,4p,3d
>>4s,4p,4d,4f
 Resulting 2*J-number? lower, higher (J=1 -> 2*J=2 etc.)
>>0,8
 Number of excitations (if negative number e.g. -2, correlation
 orbitals will always be doubly occupied)
>>2
 Generate more lists ? (y/n)
>>n
        .........

  5  blocks were created

      block  J/P           NCSF
          1    0-            546
          2    1-           1456
          3    2-           1891
          4    3-           1814
          5    4-           1393


******************************************************************************
*         COPY FILES                                                         *
*         IT IS ADVISABLE TO SAVE THE rcsfgenerate.log FILE TO HAVE A        *
*         RECORD ON HOW THE LIST OF CSFs WAS CREATED                         *
******************************************************************************

>>cp rcsfgenerate.log odd4.exc
>>cp rcsf.out rcsf.inp

******************************************************************************
*         RUN RANGULAR_MPI USING 4 PROCESSES TO GENERATE ENERGY EXPRESSION   *
```

```
*        INPUT FILE  : rcsf.inp                                            *
*        OUTPUT FILES: rangular.log, mcp.30, mcp.31,...IN 000, 001, 002, 003 *
***************************************************************************

>>mpirun -np 4 rangular_mpi

 ====================================================
       RANGULAR_MPI: Execution Begins ...
 ====================================================
 Participating nodes:
   Host: atom1    ID: 000
   Host: atom1    ID: 001
   Host: atom1    ID: 002
   Host: atom1    ID: 003

 Date and Time:
   atom1:   Date: 20140812  Time: 011040.566  Zone: +0200
   atom1:   Date: 20140812  Time: 011040.566  Zone: +0200
   atom1:   Date: 20140812  Time: 011040.566  Zone: +0200
   atom1:   Date: 20140812  Time: 011040.566  Zone: +0200

 Start Dir:
   atom1: /GRASP2018/grasptest/example4/script
   atom1: /GRASP2018/grasptest/example4/script
   atom1: /GRASP2018/grasptest/example4/script
   atom1: /GRASP2018/grasptest/example4/script

 Serial I/O Dir (node-0 only):
   atom1: /home/tspejo/GRASP2018/grasptest/example4/script

 Work Dir (Parallel I/O):
   atom1: /home/tspejo/GRASP2018/grasptest/example4/tmp_mpi
   atom1: /home/tspejo/GRASP2018/grasptest/example4/tmp_mpi
   atom1: /home/tspejo/GRASP2018/grasptest/example4/tmp_mpi
   atom1: /home/tspejo/GRASP2018/grasptest/example4/tmp_mpi

Full interaction?  (y/n)
>>y

.........

 mpi stopped by node-           0  from RANGULAR_MPI: Execution complete.
 mpi stopped by node-           2  from RANGULAR_MPI: Execution complete.
 mpi stopped by node-           1  from RANGULAR_MPI: Execution complete.
 mpi stopped by node-           3  from RANGULAR_MPI: Execution complete.

***************************************************************************
*        RUN RWFNESTIMATE TO GENERATE INITIAL ESTIMATES FOR RADIAL ORBITALS  *
*        INPUT FILES: isodata, rcsf.inp, previous rwfn files               *
*        OUTPUT FILE: rwfn.inp                                             *
***************************************************************************

>>rwfnestimate
```

```
 RWFNESTIMATE
 This program estimates radial wave functions
 for orbitals
 Input files: isodata, rcsf.inp, optional rwfn file
 Output file: rwfn.inp

 Default settings ?
>>y
 Loading CSF file ... Header only
 There are/is           16  relativistic subshells;
 The following subshell radial wavefunctions remain to be estimated:
 1s 2s 2p- 2p 3s 3p- 3p 3d- 3d 4s 4p- 4p 4d- 4d 4f- 4f

 Read subshell radial wavefunctions. Choose one below
     1 -- GRASP2K File
     2 -- Thomas-Fermi
     3 -- Screened Hydrogenic
>>1
 Enter the file name (Null then "rwfn.out")
>>
 Enter the list of relativistic subshells:
>>*
 The following subshell radial wavefunctions remain to be estimated:
 4s 4p- 4p 4d- 4d 4f- 4f

 Read subshell radial wavefunctions. Choose one below
     1 -- GRASP2K File
     2 -- Thomas-Fermi
     3 -- Screened Hydrogenic
>>2
 Enter the list of relativistic subshells:
>>*
 All required subshell radial wavefunctions  have been estimated:
Shell      e           p0        gamma        P(2)       Q(2)      MTP  SRC

  1s   0.2803D+03  0.2922D+03  0.1000D+01  0.1152D-05 -0.7471D-11  351  rwf
  2s   0.4797D+02  0.9076D+02  0.1000D+01  0.3579D-06 -0.2321D-11  357  rwf
  2p-  0.4349D+02  0.6335D+00  0.1000D+01  0.1851D-12  0.2634D-07  358  rwf
  2p   0.4301D+02  0.5672D+03  0.2000D+01  0.8823D-14 -0.5722D-19  358  rwf
  3s   0.1680D+02  0.4085D+02  0.1000D+01  0.1611D-06 -0.1045D-11  362  rwf
  3p-  0.1542D+02  0.2993D+00  0.1000D+01  0.8742D-13  0.1244D-07  363  rwf
  3p   0.1533D+02  0.2692D+03  0.2000D+01  0.4187D-14 -0.2715D-19  363  rwf
  3d-  0.1358D+02  0.2660D+00  0.2000D+01  0.6129D-21  0.8723D-16  364  rwf
  3d   0.1357D+02  0.2997D+03  0.3000D+01  0.1838D-22 -0.1192D-27  364  rwf
  4s   0.1141D+02  0.3095D+02  0.1000D+01  0.1221D-06 -0.7919D-12  366  T-F
  4p-  0.1110D+02  0.2579D+00  0.1000D+01  0.7535D-13  0.1072D-07  367  T-F
  4p   0.1104D+02  0.2325D+03  0.2000D+01  0.3616D-14 -0.2346D-19  367  T-F
  4d-  0.1053D+02  0.3856D+00  0.2000D+01  0.8888D-21  0.1265D-15  367  T-F
  4d   0.1051D+02  0.4341D+03  0.3000D+01  0.2663D-22 -0.1728D-27  367  T-F
  4f-  0.9897D+01  0.2129D+00  0.3000D+01  0.2902D-29  0.4130D-24  367  T-F
  4f   0.9890D+01  0.2598D+03  0.4000D+01  0.6286D-31 -0.4078D-36  367  T-F

 RWFNESTIMATE: Execution complete.
```

```
****************************************************************************
*         RUN RMCDHF_MPI USING 4 PROCESSES TO OBTAIN SELF CONSISTENT SOLUTIONS*
*         INPUT FILES: isodata, rcsf.inp, rwfn.inp, mcp.30, mcp.31,...       *
*         OUTPUT FILES: rwfn.out, rmix.out, rmcdhf.sum, rmcdhf.log           *
*                                                                            *
*         NOTE: FOR CORRELATION ORBITALS THERE ARE NO RESTRICTIONS ON THE    *
*         NUMBER OF NODES, I.E. THEY ARE NOT SPECTROSCOPIC. IN THIS RUN WE   *
*         VARY THE CORRELATION ORBITALS 4s,4p,4d,4f. NONE OF THESE ARE       *
*         SPECTROSCOPIC. WE CAN USE WILD CARDS * FOR SPECIFYING ORBITALS     *
*         4* MEANS 4s, 4p-, 4p, 4d-, 4d, 4f-, 4f                             *
****************************************************************************

>>mpirun -np 4 rmcdhf_mpi

 ====================================================
       RMCDHF_MPI: Execution Begins ...
 ====================================================
 Participating nodes:
   Host: atom1    ID: 000
   Host: atom1    ID: 001
   Host: atom1    ID: 002
   Host: atom1    ID: 003

 Date and Time:
   atom1:   Date: 20140812  Time: 020654.423  Zone: +0200
   atom1:   Date: 20140812  Time: 020654.422  Zone: +0200
   atom1:   Date: 20140812  Time: 020654.422  Zone: +0200
   atom1:   Date: 20140812  Time: 020654.422  Zone: +0200

 Start Dir:
   atom1: /GRASP2018/grasptest/example4/script
   atom1: /GRASP2018/grasptest/example4/script
   atom1: /GRASP2018/grasptest/example4/script
   atom1: /GRASP2018/grasptest/example4/script

 Serial I/O Dir (node-0 only):
   atom1: /home/tspejo/GRASP2018/grasptest/example4/script

 Work Dir (Parallel I/O):
   atom1: /home/tspejo/GRASP2018/grasptest/example4/tmp_mpi
   atom1: /home/tspejo/GRASP2018/grasptest/example4/tmp_mpi
   atom1: /home/tspejo/GRASP2018/grasptest/example4/tmp_mpi
   atom1: /home/tspejo/GRASP2018/grasptest/example4/tmp_mpi

Default settings?  (y/n)
>>y
 Loading CSF file ... Header only
 There are/is           16  relativistic subshells;
 Loading CSF File for ALL blocks
 There are         7100  relativistic CSFs... load complete;

 There are            5  blocks  (block   J/Parity   NCF):
  1    0-   546       2    1-  1456       3    2-  1891       4    3-  1814
  5    4-  1393
```

```
 Enter ASF serial numbers for each block
 Block           1    ncf =         546  id =    0-
>>1-2
 Block           2    ncf =        1456  id =    1-
>>1-5
 Block           3    ncf =        1891  id =    2-
>>1-5
 Block           4    ncf =        1814  id =    3-
>>1-3
 Block           5    ncf =        1393  id =    4-
>>1
 level weights (1 equal;  5 standard;  9 user)
>>5
  Radial functions
 1s 2s 2p- 2p 3s 3p- 3p 3d- 3d 4s 4p- 4p 4d- 4d 4f- 4f
 Enter orbitals to be varied (Updating order)
>>4*
 Which of these are spectroscopic orbitals?
>>
 Enter the maximum number of SCF cycles:
>>100

...........

 mpi stopped by node-           0  from RMCDHF_MPI: Execution complete.
 mpi stopped by node-           2  from RMCDHF_MPI: Execution complete.
 mpi stopped by node-           1  from RMCDHF_MPI: Execution complete.
 mpi stopped by node-           3  from RMCDHF_MPI: Execution complete.

*******************************************************************************
*         RUN RSAVE TO SAVE OUTPUT FILES: name.c, name.w, name.m, name.sum    *
*                                         name.alog, name.log                 *
*******************************************************************************

>>rsave odd4
 Created odd4.w, odd4.c, odd4.m, odd4.sum, odd4.alog and odd4.log

*******************************************************************************
*         RUN RCI_MPI USING 4 PROCESSES TO INCLUDE BREIT AND QED EFFECTS      *
*         INPUT FILES : isodata, odd4.c, odd4.w                               *
*         OUTPUT FILES: odd4.cm, odd4.csum                                    *
*                                                                             *
*         THE TRANSVERSE PHOTON FREQUENCIES CAN BE SET TO THE LOW FREQUENCY   *
*         LIMIT. RECOMMENDED IN CASES WHERE YOU HAVE CORRELATION ORBITALS     *
*         THE SELF ENERGY CORRECTION MAY FAIL FOR CORRELATION ORBITALS WITH   *
*         HIGH N.                                                             *
*******************************************************************************

>>mpirun -np 4 rci_mpi

 ===================================================
       RCI_MPI: Execution Begins ...
 ===================================================
```

```
 Participating nodes:
   Host: atom1    ID: 000
   Host: atom1    ID: 001
   Host: atom1    ID: 002
   Host: atom1    ID: 003

 Date and Time:
   atom1:   Date: 20140812  Time: 021251.038  Zone: +0200
   atom1:   Date: 20140812  Time: 021251.038  Zone: +0200
   atom1:   Date: 20140812  Time: 021251.038  Zone: +0200
   atom1:   Date: 20140812  Time: 021251.038  Zone: +0200

 Start Dir:
   atom1: /GRASP2018/grasptest/example4/script
   atom1: /GRASP2018/grasptest/example4/script
   atom1: /GRASP2018/grasptest/example4/script
   atom1: /GRASP2018/grasptest/example4/script

 Serial I/O Dir (node-0 only):
   atom1: /home/tspejo/GRASP2018/grasptest/example4/script

 Work Dir (Parallel I/O):
   atom1: /home/tspejo/GRASP2018/grasptest/example4/tmp_mpi
   atom1: /home/tspejo/GRASP2018/grasptest/example4/tmp_mpi
   atom1: /home/tspejo/GRASP2018/grasptest/example4/tmp_mpi
   atom1: /home/tspejo/GRASP2018/grasptest/example4/tmp_mpi

Default settings?
>>y
Name of state:
>>odd4
 Block           1 ,  ncf =         546
 Block           2 ,  ncf =        1456
 Block           3 ,  ncf =        1891
 Block           4 ,  ncf =        1814
 Block           5 ,  ncf =        1393
 Loading CSF file ... Header only
 There are/is          16  relativistic subshells;
 Include contribution of H (Transverse)?
>>y
 Modify all transverse photon frequencies?
>>y
 Enter the scale factor:
>>1.d-6
 Include H (Vacuum Polarisation)?
>>y
 Include H (Normal Mass Shift)?
>>n
 Include H (Specific Mass Shift)?
>>n
 Estimate self-energy?
>>y
 Largest n quantum number for including self-energy for orbital
 n should be less or equal 8
```

```
>>4
 There are           5  blocks  (block   J/Parity   NCF):
  1    0-   546       2    1-  1456       3    2-  1891       4    3-  1814
  5    4-  1393

 Enter ASF serial numbers for each block
 Block           1    ncf =         546  id =    0-
>>1-2
 Block           2    ncf =        1456  id =    1-
>>1-5
 Block           3    ncf =        1891  id =    2-
>>1-5
 Block           4    ncf =        1814  id =    3-
>>1-3
 Block           5    ncf =        1393  id =    4-
>>1

....

 mpi stopped by node-           0  from RCI_MPI: Execution complete.
 mpi stopped by node-           2  from RCI_MPI: Execution complete.
 mpi stopped by node-           3  from RCI_MPI: Execution complete.
 mpi stopped by node-           1  from RCI_MPI: Execution complete.

*******************************************************************************
*         RUN JJ2LSJ TO GET THE LSJ-COMPOSITION                               *
*         INPUT FILE: odd4.c, odd4.cm                                         *
*         OUTPUT FILE: odd4.lsj.lbl, odd4.uni.lsj.lbl                         *
*******************************************************************************

>>jj2lsj

 jj2lsj: Transformation of ASFs from a jj-coupled CSF basis
         into an LSJ-coupled CSF basis  (Fortran 95 version)
         (C) Copyright by   G. Gaigalas and Ch. F. Fischer,
         (2017).
 Input files: name.c, name.(c)m
 Output files: name.lsj.lbl
   (optional)  name.lsj.c, name.lsj.j,
               name.uni.lsj.lbl, name.uni.lsj.sum

 Name of state
>>odd4
 Loading Configuration Symmetry List File ...
 There are 16 relativistic subshells;
 There are 7100 relativistic CSFs;
  ... load complete;

 Mixing coefficients from a CI calc.?
>>y
 Do you need a unique labeling? (y/n)
>>y
    nelec  =           12
    ncftot =         7100
```

```
   nw     =           16
   nblock =            5

  block     ncf     nev    2j+1  parity
      1     546       2       1      -1
      2    1456       5       3      -1
      3    1891       5       5      -1
      4    1814       3       7      -1
      5    1393       1       9      -1
 Default settings?  (y/n)
>>y

 jj2lsj: Execution complete.

*****************************************************************************
*         RUN RLEVELS TO VIEW ENERGIES AND ENERGY SEPARATIONS               *
*         NOTE: SINCE LSJ-INFORMATION NOW IS AVAILABLE OUTPUT LABELS        *
*         WILL BE IN LSJ-COUPLING                                           *
*****************************************************************************

>>rlevels even4.cm odd4.cm

 nblock =           4   ncftot =         5712   nw =           16   nelec =           12
 nblock =           5   ncftot =         7100   nw =           16   nelec =           12

 Energy levels for ...
 Rydberg constant is   109737.31569
 Splitting is the energy difference with the lower neighbor
--------------------------------------------------------------------------------------
 No Pos  J Parity Energy Total    Levels     Splitting     Configuration
                     (a.u.)      (cm^-1)     (cm^-1)
--------------------------------------------------------------------------------------
  1  1   0  +   -1182.4117318       0.00        0.00  2s(2).2p(6).3s(2)_1S
  2  1   0  -   -1181.3458964  233923.84   233923.84  2s(2).2p(6).3s_2S.3p_3P
  3  1   1  -   -1181.3192507  239771.90     5848.05  2s(2).2p(6).3s_2S.3p_3P
  4  1   2  -   -1181.2547649  253924.89    14152.99  2s(2).2p(6).3s_2S.3p_3P
  5  2   1  -   -1180.8024570  353194.99    99270.10  2s(2).2p(6).3s_2S.3p_1P
  6  2   0  +   -1179.8827380  555049.98   201854.99  2s(2).2p(6).3p(2)3P2_3P
  7  1   2  +   -1179.8591476  560227.47     5177.49  2s(2).2p(6).3p(2)1D2_1D
  8  1   1  +   -1179.8373877  565003.22     4775.75  2s(2).2p(6).3p(2)3P2_3P
  9  2   2  +   -1179.7587033  582272.45    17269.23  2s(2).2p(6).3p(2)3P2_3P
 10  3   0  +   -1179.3935084  662423.47    80151.01  2s(2).2p(6).3p(2)1S0_1S
 11  2   1  +   -1179.3157358  679492.58    17069.12  2s(2).2p(6).3s_2S.3d_3D
 12  3   2  +   -1179.3110726  680516.04     1023.45  2s(2).2p(6).3s_2S.3d_3D
 13  1   3  +   -1179.3037683  682119.14     1603.10  2s(2).2p(6).3s_2S.3d_3D
 14  4   2  +   -1178.9200935  766326.03    84206.89  2s(2).2p(6).3s_2S.3d_1D
 15  2   2  -   -1178.1773269  929344.46   163018.43  2s(2).2p(6).3p_2P.3d_3F
 16  1   3  -   -1178.1320708  939277.02     9932.56  2s(2).2p(6).3p_2P.3d_3F
 17  3   2  -   -1178.0860233  949383.27    10106.25  2s(2).2p(6).3p_2P.3d_1D
 18  1   4  -   -1178.0797003  950771.02     1387.75  2s(2).2p(6).3p_2P.3d_3F
 19  3   1  -   -1177.9272497  984230.05    33459.02  2s(2).2p(6).3p_2P.3d_3D
 20  4   2  -   -1177.9243601  984864.26      634.21  2s(2).2p(6).3p_2P.3d_3P
 21  2   3  -   -1177.8729498  996147.50    11283.24  2s(2).2p(6).3p_2P.3d_3D
 22  2   0  -   -1177.8671333  997424.08     1276.58  2s(2).2p(6).3p_2P.3d_3P
```

```
 23  4   1  -   -1177.8658076   997715.04      290.96  2s(2).2p(6).3p_2P.3d_3P
 24  5   2  -   -1177.8644860   998005.10      290.07  2s(2).2p(6).3p_2P.3d_3D
 25  3   3  -   -1177.5442550  1068287.68    70282.57  2s(2).2p(6).3p_2P.3d_1F
 26  5   1  -   -1177.4836853  1081581.20    13293.52  2s(2).2p(6).3p_2P.3d_1P
-----------------------------------------------------------------------------------------
```

We compare with the energy levels given in the NIST database.

```
------------------------------------------------------------
Configuration        | Term    |  J  |             Level      |
---------------------|---------|-----|-----------------------|
                     |         |     |                        |
2p6.3s2              | 1S      |  0  |                 0      |
                     |         |     |                        |
3s.3p                | 3P*     |  0  |            233842      |
                     |         |  1  |            239660      |
                     |         |  2  |            253820      |
                     |         |     |                        |
3s.3p                | 1P*     |  1  |            351911      |
                     |         |     |                        |
3p2                  | 3P      |  0  |            554524      |
                     |         |     |                        |
3p2                  | 1D      |  2  |            559600      |
                     |         |     |                        |
3p2                  | 3P      |  1  |            564602      |
                     |         |  2  |            581803      |
                     |         |     |                        |
3p2                  | 1S      |  0  |            659627      |
                     |         |     |                        |
3s.3d                | 3D      |  1  |            678772      |
                     |         |  2  |            679785      |
                     |         |  3  |            681416      |
                     |         |     |                        |
3s.3d                | 1D      |  2  |            762093      |
                     |         |     |                        |
3p.3d                | 3F*     |  2  |            928241      |
                     |         |  3  |            938126      |
                     |         |     |                        |
3p.3d                | 1D*     |  2  |            948513      |
                     |         |     |                        |
3p.3d                | 3F*     |  4  |            949658      |
                     |         |     |                        |
3p.3d                | 3D*     |  1  |            982868      |
                     |         |     |                        |
3p.3d                | 3P*     |  2  |            983514      |
                     |         |     |                        |
3p.3d                | 3D*     |  3  |            994852      |
                     |         |     |                        |
3p.3d                | 3P*     |  0  |            995889      |
                     |         |  1  |            996243      |
                     |         |     |                        |
3p.3d                | 3D*     |  2  |            996623      |
                     |         |     |                        |
3p.3d                | 1F*     |  3  |           1062515      |
```

```
                      |         |      |                        |
3p.3d                 | 1P*     |    1 |           1074887      |
-----------------------------------------------------------------
```

Note that this is just a very small example calculation. The agreement between theory and experiment is improved when the active set is extended. If desired the we may display the energy separations in eV by running `rlevelseV`.

```
**************************************************************************
*         RUN RLEVELSEV TO VIEW ENERGIES AND ENERGY SEPARATIONS IN EV        *
*         NOTE: SINCE LSJ-INFORMATION NOW IS AVAILABLE OUTPUT LABELS         *
*         WILL BE IN LSJ-COUPLING                                            *
**************************************************************************

>>rlevelseV even4.cm odd4.cm

 nblock =            4   ncftot =         5712   nw =           16   nelec =           12
 nblock =            5   ncftot =         7100   nw =           16   nelec =           12

 Energy levels for ...
 Rydberg constant is   109737.31569
 Splitting is the energy difference with the lower neighbor
-------------------------------------------------------------------------------------------
 No Pos  J Parity Energy Total    Levels     Splitting     Configuration
                     (a.u.)        (eV)         (eV)
-------------------------------------------------------------------------------------------
  1  1   0  +   -1182.4117318        0.00000        0.00000  2s(2).2p(6).3s(2)_1S
  2  1   0  -   -1181.3458964       29.00286       29.00286  2s(2).2p(6).3s_2S.3p_3P
  3  1   1  -   -1181.3192507       29.72793        0.72507  2s(2).2p(6).3s_2S.3p_3P
  4  1   2  -   -1181.2547649       31.48267        1.75475  2s(2).2p(6).3s_2S.3p_3P
  5  2   1  -   -1180.8024570       43.79060       12.30792  2s(2).2p(6).3s_2S.3p_1P
  6  2   0  +   -1179.8827380       68.81743       25.02683  2s(2).2p(6).3p(2)3P2_3P
  7  1   2  +   -1179.8591476       69.45935        0.64193  2s(2).2p(6).3p(2)1D2_1D
  8  1   1  +   -1179.8373877       70.05147        0.59212  2s(2).2p(6).3p(2)3P2_3P
  9  2   2  +   -1179.7587033       72.19258        2.14111  2s(2).2p(6).3p(2)3P2_3P
 10  3   0  +   -1179.3935084       82.13004        9.93746  2s(2).2p(6).3p(2)1S0_1S
 11  2   1  +   -1179.3157358       84.24634        2.11630  2s(2).2p(6).3s_2S.3d_3D
 12  3   2  +   -1179.3110726       84.37323        0.12689  2s(2).2p(6).3s_2S.3d_3D
 13  1   3  +   -1179.3037683       84.57199        0.19876  2s(2).2p(6).3s_2S.3d_3D
 14  4   2  +   -1178.9200935       95.01232       10.44032  2s(2).2p(6).3s_2S.3d_1D
 15  2   2  -   -1178.1773269      115.22403       20.21171  2s(2).2p(6).3p_2P.3d_3F
 16  1   3  -   -1178.1320708      116.45551        1.23148  2s(2).2p(6).3p_2P.3d_3F
 17  3   2  -   -1178.0860233      117.70852        1.25302  2s(2).2p(6).3p_2P.3d_1D
 18  1   4  -   -1178.0797003      117.88058        0.17206  2s(2).2p(6).3p_2P.3d_3F
 19  3   1  -   -1177.9272497      122.02897        4.14839  2s(2).2p(6).3p_2P.3d_3D
 20  4   2  -   -1177.9243601      122.10760        0.07863  2s(2).2p(6).3p_2P.3d_3P
 21  2   3  -   -1177.8729498      123.50655        1.39894  2s(2).2p(6).3p_2P.3d_3D
 22  2   0  -   -1177.8671333      123.66482        0.15828  2s(2).2p(6).3p_2P.3d_3P
 23  4   1  -   -1177.8658076      123.70090        0.03607  2s(2).2p(6).3p_2P.3d_3P
 24  5   2  -   -1177.8644860      123.73686        0.03596  2s(2).2p(6).3p_2P.3d_3D
 25  3   3  -   -1177.5442550      132.45079        8.71393  2s(2).2p(6).3p_2P.3d_1F
 26  5   1  -   -1177.4836853      134.09898        1.64819  2s(2).2p(6).3p_2P.3d_1P
-------------------------------------------------------------------------------------------

**************************************************************************
```

```
*         RUN RBIOTRANSFORM_MPI USING 4 PROCESSES                              *
*         INPUT FILES: even4.c, even4.w, even4.cm                              *
*                      odd4.c, odd4.w, odd4.cm, isodata                        *
*         OUPUT FILES: even4.bw, even4.cbm, odd4.bw, odd4.cbm                  *
*******************************************************************************

>>mpirun -np 4 rbiotransform_mpi

 ====================================================
        RBIOTRANSFORM_MPI: Execution Begins ...
 ====================================================
 Participating nodes:
   Host: per-vaio    ID: 000
   Host: per-vaio    ID: 001
   Host: per-vaio    ID: 002
   Host: per-vaio    ID: 003

 Date and Time:
   per-vaio:   Date: 20141120  Time: 002539.729  Zone: +0100
   per-vaio:   Date: 20141120  Time: 002539.729  Zone: +0100
   per-vaio:   Date: 20141120  Time: 002539.729  Zone: +0100
   per-vaio:   Date: 20141120  Time: 002539.729  Zone: +0100

 Start Dir:
   per-vaio: /home/per/graspruns/FeXIII
   per-vaio: /home/per/graspruns/FeXIII
   per-vaio: /home/per/graspruns/FeXIII
   per-vaio: /home/per/graspruns/FeXIII

 Serial I/O Dir (node-0 only):
   per-vaio: /home/per/graspruns/FeXIII

 Work Dir (Parallel I/O):
   per-vaio: /home/per/tmp_mpi
   per-vaio: /home/per/tmp_mpi
   per-vaio: /home/per/tmp_mpi
   per-vaio: /home/per/tmp_mpi

 Default settings?
>>y

 Input from a CI calculation?
>>y

  Name of the Initial state
>>even4
  Name of the Final state
>>odd4
  Transformation of all J symmetries?
>>y

    .....

 mpi stopped by node-           0  from RBIOTRANSFORM_MPI: Execution complete.
```

```
 mpi stopped by node-           1  from RBIOTRANSFORM_MPI: Execution complete.
 mpi stopped by node-           2  from RBIOTRANSFORM_MPI: Execution complete.
 mpi stopped by node-           3  from RBIOTRANSFORM_MPI: Execution complete.

******************************************************************************
*         RUN RTRANSITION_MPI USING 4 PROCESSES                              *
*         INPUT FILES: even4.c, even4.bw, even4.cbm                          *
*                      odd4.c, odd4.bw, odd4.cbm, isodata                    *
*         OUPUT FILES: even4.odd4.ct                                         *
******************************************************************************

>>mpirun -np 4 rtranstion_mpi

 ====================================================
        RTRANSITION_MPI: Execution Begins ...
 ====================================================
 Participating nodes:
   Host: per-vaio    ID: 000
   Host: per-vaio    ID: 001
   Host: per-vaio    ID: 002
   Host: per-vaio    ID: 003

 Date and Time:
   per-vaio:   Date: 20141120  Time: 003050.621  Zone: +0100
   per-vaio:   Date: 20141120  Time: 003050.621  Zone: +0100
   per-vaio:   Date: 20141120  Time: 003050.621  Zone: +0100
   per-vaio:   Date: 20141120  Time: 003050.621  Zone: +0100

 Start Dir:
   per-vaio: /home/per/graspruns/FeXIII
   per-vaio: /home/per/graspruns/FeXIII
   per-vaio: /home/per/graspruns/FeXIII
   per-vaio: /home/per/graspruns/FeXIII

 Serial I/O Dir (node-0 only):
   per-vaio: /home/per/graspruns/FeXIII

  Default settings?
>>y
  Input from a CI calculation?
>>y

  Name of the Initial state
>>even4
  Name of the Final state
>>odd4

 Enter the list of transition specifications
  e.g.,  E1,M2  or  E1 M2  or  E1;M2 :
 >>E1,M2

     .....

 mpi stopped by node-           0  from RTRANSITION_MPI: Execution complete.
```

```
mpi stopped by node-           3  from RTRANSITION_MPI: Execution complete.
mpi stopped by node-           1  from RTRANSITION_MPI: Execution complete.
mpi stopped by node-           2  from RTRANSITION_MPI: Execution complete.
```

Comment: it does not matter in which order the files `even4` and `odd4` are specified.

7.5 Fifth example: the study of energy spectra for Ni XIV, obtaining unique labels

A wave function or a corresponding energy level is often designated by the label of the CSF with the largest expansion coefficient. This example presents a study of energy spectra for Ni XIV in which a few levels have the same identification. To get the energy spectra with unique labels we use the unique option in the `jj2lsj` program. The program uses the algorithm described in [31, 32, 20]: for a given set of wave functions for the same J and parity, the CSF with largest expansion coefficient is used as the label for the function containing this largest component. Once a label is assigned, the corresponding CSF is removed from consideration in the determination of the next label. The last remaining label for a wave function may be based on a contribution that is exceedingly small.

Overview

1. Define nuclear data.
2. Obtain common spectroscopic orbitals for the MR set.
   (a) Generate list of CSFs describing the even states belonging to the $3s3p^4$, $3s^23p^23d$ configurations and the odd states belonging to the $3s^23p^3$ configuration.
   (b) Perform angular integration.
   (c) Generate initial estimates of radial orbitals.
   (d) Perform self-consistent field calculation on the weighted average of all states belonging to $3s3p^4$, $3s^23p^23d$, and $3s^23p^3$.
   (e) Save output to `Ni_mr`.
3. Improve even states
   (a) Generate CSF list from SD-excitations from $3s3p^4$ and $3s^23p^23d$ to $n = 4$.
   (b) Run `rcsfinteract` to extract CSFs that interact with CSFs belonging to $3s3p^4$ and $3s^23p^23d$.
   (c) Perform angular integration.
   (d) Generate initial estimates of radial orbitals.
   (e) Perform self-consistent field calculation on the weighted average of all states belonging to $3s3p^4$ and $3s^23p^23d$.
   (f) Save output to `Ni_even_n4`.
   (g) Perform RCI calculation in which the transverse photon interaction (Breit) and vacuum polarization and self-energy (QED) corrections are added.
4. Transform from jj- to LSJ-coupling
5. Improve odd states
   (a) Generate CSF list from SD-excitations from $3s^23p^4$ to $n = 4$.

(b) Run `rcsfinteract` to extract CSFs that interact with CSFs belonging to $3s^2 3p^4$.

(c) Perform angular integration.

(d) Generate initial estimates of radial orbitals.

(e) Perform self-consistent field calculation on the weighted average of all states belonging to $3s^2 3p^4$.

(f) Save output to `Ni_odd_n4`.

(g) Perform RCI calculation in which the transverse photon interaction (Breit) and vacuum polarization and self-energy (QED) corrections are added.

6. Transform from jj- to LSJ-coupling using the unique label option.

7. Run `rlevels` to view energy separations (several states have the same label).

8. Copy files so that `rlevels` will display unique labels.

9. Run `rlevels` to view energy separations for levels now with unique labels.

10. Compute transition rates from the RCI wave functions. Computation in two steps: biorthonormal transformation and then evaluation of transition matrix elements using standard Racah algebra methods.

Program input

In the test-runs input marked by `>>` or `>>3`, for example, indicates that the user should input 3 and then strike the return key. When `>>` is followed by blanks just strike the return key.

```
*******************************************************************************
* RUN RNUCLEUS TO GENERATE NUCLEAR DATA AND DEFINE RADIAL GRID                *
* OUTPUT FILE: isodata                                                        *
*******************************************************************************

>>rnucleus

 RNUCLEUS
 This program defines nuclear data and the radial grid
 Outputfile: isodata

 Enter the atomic number:
>>28
 Enter the mass number (0 if the nucleus is to be modelled as a point source:
>>61
 The default root mean squared radius is    3.8609116450734162       fm;
 the default nuclear skin thickness is    2.2999999999999998       fm;
 Revise these values?
>>n
 Enter the mass of the neutral atom (in amu) (0 if the nucleus is to be static):
>>58.6934
 Enter the nuclear spin quantum number (I) (in units of h / 2 pi):
>>1
 Enter the nuclear dipole moment (in nuclear magnetons):
>>1
 Enter the nuclear quadrupole moment (in barns):
>>1
```

```
******************************************************************************
*        RUN RCSFGENERATE TO GENERATE LIST FOR ALL                           *
*        STATES OF 3s3p(4), 3s(2)3p(2)3d and 3s(2)3p(3)                      *
*        OUTPUT FILES: rcsfgenerate.log, rcsf.out                            *
******************************************************************************

>>rcsfgenerate

 RCSFGENERATE
 This program generates a list of CSFs
 Configurations should be entered in spectroscopic notation
 with occupation numbers and indications if orbitals are
 closed (c), inactive (i), active (*) or has a minimal
 occupation e.g. 1s(2,1)2s(2,*)
 OUTPUT FILES: rcsf.out, rcsfgenerate.log

 Default, reverse, symmetry or user specified ordering? (*/r/s/u)
>>*

 Select core
        0: No core
        1: He (       1s(2)                  =  2 electrons)
        2: Ne ([He] + 2s(2)2p(6)             = 10 electrons)
        3: Ar ([Ne] + 3s(2)3p(6)             = 18 electrons)
        4: Kr ([Ar] + 3d(10)4s(2)4p(6)       = 36 electrons)
        5: Xe ([Kr] + 4d(10)5s(2)5p(6)       = 54 electrons)
        6: Rn ([Xe] + 4f(14)5d(10)6s(2)6p(6) = 86 electrons)
>>1
 Enter list of (maximum 100) configurations. End list with a blank line or an asterisk (*)

 Give configuration  1
>>2s(2,i)2p(6,i)3s(1,i)3p(4,i)
 Give configuration  2
>>2s(2,i)2p(6,i)3s(2,i)3p(2,i)3d(1,i)
 Give configuration  3
>>
 Give set of active orbitals, as defined by the highest principal quantum number
 per l-symmetry, in a comma delimited list in s,p,d etc order, e.g. 5s,4p,3d
>>3s,3p,3d
 Resulting 2*J-number? lower, higher (J=1 -> 2*J=2 etc.)
>>1,9
 Number of excitations (if negative number e.g. -2, correlation
 orbitals will always be doubly occupied)
>>0
 Generate more lists ? (y/n)
>>y
 Enter list of (maximum 100) configurations. End list with a blank line or an asterisk (*).

 Give configuration  1
>>2s(2,i)2p(6,i)3s(2,i)3p(3,i)
 Give configuration  2
>>
 Give set of active orbitals in a comma delimited list ordered by l-symmetry, e.g. 5s,4p,3d
>>3s,3p
```

```
 Resulting 2*J-number? lower, higher (J=1 -> 2*J=2 etc.)
>>1,5
 Number of excitations (if negative number e.g. -2, correlation
 orbitals will always be doubly occupied)
>>0
 Generate more lists ? (y/n)
>>n

        .........

 8 blocks were created

      block  J/P           NCSF
          1  1/2+             8
          2  1/2-             1
          3  3/2+            11
          4  3/2-             3
          5  5/2+            10
          6  5/2-             1
          7  7/2+             5
          8  9/2+             2

******************************************************************************
*         COPY FILES                                                         *
*         IT IS ADVISABLE TO SAVE THE rcsfgenerate.log FILE TO HAVE A        *
*         RECORD ON HOW THE LIST OF CSFs WAS CREATED                         *
******************************************************************************

>>cp rcsfgenerate.log Ni_mr.exc
>>cp rcsf.out rcsf.inp


******************************************************************************
*         RUN RANGULAR TO GENERATE ENERGY EXPRESSION                         *
*         INPUT FILE  : rcsf.inp                                             *
*         OUTPUT FILES: rangular.log, mcp.30, mcp.31,....                    *
******************************************************************************

>>rangular

 RANGULAR
 This program performs angular integration
 Input file:  rcsf.inp
 Outputfiles: mcp.30, mcp.31, ....
              rangular.log

 Full interaction?  (y/n)
>>y

  .....

 RANGULAR: Execution complete.

******************************************************************************
```

```
*         RUN RWFNESTIMATE TO GENERATE INITIAL ESTIMATES FOR RADIAL ORBITALS  *
*         INPUT FILES: isodata, rcsf.inp, previous rwfn files                 *
*         OUTPUT FILE: rwfn.inp, rwfnestimate.log                             *
*******************************************************************************

>>rwfnestimate

 RWFNESTIMATE
 This program estimates radial wave functions
 for orbitals
 Input files: isodata, rcsf.inp, optional rwfn file
 Output file: rwfn.inp

 Default settings ?
>>y
 Loading CSF file ... Header only
 There are/is            9  relativistic subshells;
 The following subshell radial wavefunctions remain to be estimated:
 1s 2s 2p- 2p 3s 3p- 3p 3d- 3d

 Read subshell radial wavefunctions. Choose one below
     1 -- GRASP2K File
     2 -- Thomas-Fermi
     3 -- Screened Hydrogenic
>>2
 Enter the list of relativistic subshells:
>>*
Shell      e           p0        gamma        P(2)       Q(2)      MTP  SRC

  1s   0.3531D+03  0.3347D+03  0.1000D+01  0.1226D-05 -0.7733D-11  329  T-F
  2s   0.7017D+02  0.1144D+03  0.1000D+01  0.4190D-06 -0.2645D-11  346  T-F
  2p-  0.6820D+02  0.1007D+01  0.1000D+01  0.2459D-12  0.3609D-07  346  T-F
  2p   0.6732D+02  0.8290D+03  0.2000D+01  0.1112D-13 -0.7019D-19  346  T-F
  3s   0.2444D+02  0.5705D+02  0.1000D+01  0.2089D-06 -0.1319D-11  358  T-F
  3p-  0.2358D+02  0.5369D+00  0.1000D+01  0.1312D-12  0.1925D-07  359  T-F
  3p   0.2336D+02  0.4440D+03  0.2000D+01  0.5955D-14 -0.3759D-19  359  T-F
  3d-  0.2191D+02  0.7312D+00  0.2000D+01  0.1308D-20  0.1920D-15  359  T-F
  3d   0.2185D+02  0.7606D+03  0.3000D+01  0.3736D-22 -0.2358D-27  359  T-F

RWFNESTIMATE: Execution complete.

*******************************************************************************
*         RUN RMCDHF TO OBTAIN SELF CONSISTENT SOLUTIONS                      *
*         INPUT FILES: isodata, rcsf.inp, rwfn.inp, mcp.30, mcp.31,...        *
*         OUTPUT FILES: rwfn.out, rmix.out, rmcdhf.sum, rmcdhf.log            *
*                                                                             *
*         NOTE: ORBITALS BUILDING REFERENCE STATES ARE REQUIRED TO HAVE       *
*         THE CORRECT NUMBER OF NODES. THEY ARE REFERRED TO AS SPECTROSCOPIC  *
*         ORBITALS. IN THIS RUN WE VARY 1s,2s,2p,3s,3p,3d AND THEY ARE ALL    *
*         SPECTROSCOPIC. WE CAN USE WILD CARDS FOR SPECIFYING ORBITALS        *
*                                                                             *
*         NOTE: INSTEAD OF SAYING THAT WE SHOULD OPTIMIZE ON, FOR EXAMPLE,    *
*         THE STATES 1,2,3,4 WE CAN WRITE 1-4 MEANING THE SAME THING          *
*                                                                             *
```

```
*******************************************************************************

>>rmcdhf

 RMCDHF
 This program determines the radial orbitals
 and the expansion coefficients of the CSFs
 in a self-onsistent field proceedure
 Input file:  isodata, rcsf.inp, rwfn.inp, mcp.30, ...
 Outputfiles: rwfn.out, rmix.out, rmcdhf.sum, rmcdhf.log

 Default settings?  (y/n)
>>y
 Loading CSF file ... Header only
 There are/is            9  relativistic subshells;
 Loading CSF File for ALL blocks
 There are           41  relativistic CSFs... load complete;
 Loading Radial WaveFunction File ...
 There are            8  blocks  (block   J/Parity   NCF):
  1  1/2+     8       2  1/2-     1       3  3/2+    11       4  3/2-     3
  5  5/2+    10       6  5/2-     1       7  7/2+     5       8  9/2+     2

 Enter ASF serial numbers for each block
 Block           1    ncf =           8  id =  1/2+
>>1-8
 Block           2    ncf =           1  id =  1/2-
>>1
 Block           3    ncf =          11  id =  3/2+
>>1-11
 Block           4    ncf =           3  id =  3/2-
>>1-3
 Block           5    ncf =          10  id =  5/2+
>>1-10
 Block           6    ncf =           1  id =  5/2-
>>1
 Block           7    ncf =           5  id =  7/2+
>>1-5
 Block           8    ncf =           2  id =  9/2+
>>1-2
 level weights (1 equal;  5 standard;  9 user)
>>5
 Radial functions
 1s 2s 2p- 2p 3s 3p- 3p 3d- 3d
 Enter orbitals to be varied (Updating order)
>>*
 Which of these are spectroscopic orbitals?
>>*
 Enter the maximum number of SCF cycles:
>>999

..............

 RMCDHF: Execution complete.
```

```
******************************************************************************
*         RUN RSAVE TO SAVE OUTPUT FILES: name.c, name.w, name.m, name.sum    *
*                                         name.alog, name.log                 *
******************************************************************************

>>rsave Ni_mr
 Created Ni_mr.w, Ni_mr.c, Ni_mr.m, Ni_mr.sum, Ni_mr.alog and Ni_mr.log

******************************************************************************
*         RUN RCSFGENERATE TO GENERATE LIST FOR ALL                           *
*         STATES OF 3s3p(4), 3s(2)3p(2)3d                                     *
*         OUTPUT FILES: rcsfgenerate.log, rcsf.out                            *
******************************************************************************

>>rcsfgenerate

 RCSFGENERATE
 This program generates a list of CSFs
 Configurations should be entered in spectroscopic notation
 with occupation numbers and indications if orbitals are
 closed (c), inactive (i), active (*) or has a minimal
 occupation e.g. 1s(2,1)2s(2,*)
 OUTPUT FILES: rcsf.out, rcsfgenerate.log

 Default, reverse, symmetry or user specified ordering? (*/r/s/u)
>>*

 Select core
        0: No core
        1: He (       1s(2)                  =  2 electrons)
        2: Ne ([He] + 2s(2)2p(6)             = 10 electrons)
        3: Ar ([Ne] + 3s(2)3p(6)             = 18 electrons)
        4: Kr ([Ar] + 3d(10)4s(2)4p(6)       = 36 electrons)
        5: Xe ([Kr] + 4d(10)5s(2)5p(6)       = 54 electrons)
        6: Rn ([Xe] + 4f(14)5d(10)6s(2)6p(6) = 86 electrons)
>>1
 Enter list of (maximum 100) configurations. End list with a blank line or an asterisk (*)

 Give configuration  1
>>2s(2,i)2p(6,i)3s(1,i)3p(4,i)
 Give configuration  2
>>2s(2,i)2p(6,i)3s(2,i)3p(2,i)3d(1,i)
 Give configuration  3
>>
 Give set of active orbitals, as defined by the highest principal quantum number
 per l-symmetry, in a comma delimited list in s,p,d etc order, e.g. 5s,4p,3d
>>3s,3p,3d
 Resulting 2*J-number? lower, higher (J=1 -> 2*J=2 etc.)
>>1,9
 Number of excitations (if negative number e.g. -2, correlation
 orbitals will always be doubly occupied)
>>0
 Generate more lists ? (y/n)
>>n
```

```
        .........

  5 blocks were created

      block  J/P           NCSF
          1  1/2+             8
          2  3/2+            11
          3  5/2+            10
          4  7/2+             5
          5  9/2+             2

*******************************************************************************
*         COPY FILES                                                          *
*         NOTE THAT WE COPY THE FILE TO RCSFMR.INP FOR USE                    *
*         TOGETHER WITH RCSFINTERACT                                          *
*******************************************************************************

>>cp rcsf.out rcsfmr.inp

*******************************************************************************
*         RUN RCSFGENERATE TO GENERATE LIST OBTAINED BY                       *
*         SD-EXCITATIONS FROM 3s3p(4) and 3s(2)3p(2)3d TO n = 4               *
*         OUTPUT FILES: rcsfgenerate.log, rcsf.out                            *
*******************************************************************************

>>rcsfgenerate

 RCSFGENERATE
 This program generates a list of CSFs
 Configurations should be entered in spectroscopic notation
 with occupation numbers and indications if orbitals are
 closed (c), inactive (i), active (*) or has a minimal
 occupation e.g. 1s(2,1)2s(2,*)
 OUTPUT FILES: rcsf.out, rcsfgenerate.log

 Default, reverse, symmetry or user specified ordering? (*/r/s/u)
>>*

 Select core
        0: No core
        1: He (       1s(2)                  =  2 electrons)
        2: Ne ([He] + 2s(2)2p(6)             = 10 electrons)
        3: Ar ([Ne] + 3s(2)3p(6)             = 18 electrons)
        4: Kr ([Ar] + 3d(10)4s(2)4p(6)       = 36 electrons)
        5: Xe ([Kr] + 4d(10)5s(2)5p(6)       = 54 electrons)
        6: Rn ([Xe] + 4f(14)5d(10)6s(2)6p(6) = 86 electrons)
>>1
 Enter list of (maximum 100) configurations. End list with a blank line or an asterisk (*)

 Give configuration  1
>>2s(2,i)2p(6,i)3s(1,*)3p(4,*)
 Give configuration  2
>>2s(2,i)2p(6,i)3s(2,*)3p(2,*)3d(1,*)
```

```
 Give configuration  3
>>
 Give set of active orbitals, as defined by the highest principal quantum number
 per l-symmetry, in a comma delimited list in s,p,d etc order, e.g. 5s,4p,3d
>>4s,4p,4d,4f
 Resulting 2*J-number? lower, higher (J=1 -> 2*J=2 etc.)
>>1,9
 Number of excitations (if negative number e.g. -2, correlation
 orbitals will always be doubly occupied)
>>2
 Generate more lists ? (y/n)
>>n

        .........

  5 blocks were created

      block  J/P            NCSF
          1  1/2+           1664
          2  3/2+           2837
          3  5/2+           3271
          4  7/2+           2972
          5  9/2+           2264

******************************************************************************
*         COPY FILES                                                          *
******************************************************************************

>>cp rcsf.out rcsf.inp

******************************************************************************
*         RUN RCSFINTERACT PROGRAM TO DETERMINE WHICH OF THE CSFs IN THE      *
*         rcsf.inp LIST INTERACTS WITH THE CSFs IN rcsfmr.inp                 *
*         THE INTERACTING CSFs ARE WRITTEN TO rcsf.out                        *
*         INPUT FILES: rcsfmr.inp, rcsf.inp                                   *
*         OUTPUT FILE: rcsf.out                                               *
******************************************************************************

>>rcsfinteract

 RCSFinteract: Determines all the CSFs (rcsf.inp) that interact
               with the CSFs in the multireference (rcsfmr.inp)
               (C)  Copyright by G. Gaigalas and Ch. F. Fischer
               (Fortran 95 version)               NIST  (2017).
               Input files: rcsfmr.inp, rcsf.inp
               Output file: rcsf.out

 Reduction based on Dirac-Coulomb (1) or
 Dirac-Coulomb-Breit (2) Hamiltonian?
>>2

   ....

 There are 16 relativistic subshells;
```

```
  Block    MR NCSF   Befor NCSF   After NCSF
    1          8         1664         1047
    2         11         2837         1862
    3         10         3271         2112
    4          5         2972         1537
    5          2         2264          801

 RCSFINTERACT: Execution complete

*******************************************************************************
*         COPY FILES                                                          *
*******************************************************************************

>>cp rcsf.out rcsf.inp

*******************************************************************************
*         RUN RANGULAR TO GENERATE ENERGY EXPRESSION                          *
*         INPUT FILE  : rcsf.inp                                              *
*         OUTPUT FILES: rangular.log, mcp.30, mcp.31,....                     *
*******************************************************************************

>>rangular

 RANGULAR
 This program performs angular integration
 Input file:  rcsf.inp
 Outputfiles: mcp.30, mcp.31, ....
              rangular.log

 Full interaction?  (y/n)
>>y

  .....

 RANGULAR: Execution complete.

*******************************************************************************
*         RUN RWFNESTIMATE TO GENERATE INITIAL ESTIMATES FOR RADIAL ORBITALS  *
*         INPUT FILES: isodata, rcsf.inp, previous rwfn files                 *
*         OUTPUT FILE: rwfn.inp, rwfnestimate.log                             *
*******************************************************************************

>>rwfnestimate

 RWFNESTIMATE
 This program estimates radial wave functions
 for orbitals
 Input files: isodata, rcsf.inp, optional rwfn file
 Output file: rwfn.inp

 Default settings ?
>>y
 Loading CSF file ... Header only
 There are/is           16  relativistic subshells;
```

```
 The following subshell radial wavefunctions remain to be estimated:
 1s 2s 2p- 2p 3s 3p- 3p 3d- 3d 4s 4p- 4p 4d- 4d 4f- 4f

 Read subshell radial wavefunctions. Choose one below
     1 -- GRASP2K File
     2 -- Thomas-Fermi
     3 -- Screened Hydrogenic
>>1
 Enter the file name (Null then "rwfn.out")
>>Ni_mr.w
 Enter the list of relativistic subshells:
>>*
 The following subshell radial wavefunctions remain to be estimated:
 4s 4p- 4p 4d- 4d 4f- 4f

 Read subshell radial wavefunctions. Choose one below
     1 -- GRASP2K File
     2 -- Thomas-Fermi
     3 -- Screened Hydrogenic
>>2
 Enter the list of relativistic subshells:
>>*
Shell      e           p0        gamma        P(2)       Q(2)      MTP  SRC

  1s   0.3241D+03  0.3324D+03  0.1000D+01  0.1217D-05 -0.7677D-11  357  Ni_
  2s   0.5340D+02  0.1042D+03  0.1000D+01  0.3818D-06 -0.2409D-11  359  Ni_
  2p-  0.4833D+02  0.8505D+00  0.1000D+01  0.2077D-12  0.3049D-07  359  Ni_
  2p   0.4767D+02  0.6995D+03  0.2000D+01  0.9382D-14 -0.5921D-19  360  Ni_
  3s   0.1702D+02  0.4579D+02  0.1000D+01  0.1677D-06 -0.1058D-11  364  Ni_
  3p-  0.1558D+02  0.3892D+00  0.1000D+01  0.9504D-13  0.1395D-07  364  Ni_
  3p   0.1545D+02  0.3217D+03  0.2000D+01  0.4314D-14 -0.2723D-19  364  Ni_
  3d-  0.1321D+02  0.3699D+00  0.2000D+01  0.6617D-21  0.9712D-16  366  Ni_
  3d   0.1319D+02  0.3854D+03  0.3000D+01  0.1893D-22 -0.1195D-27  366  Ni_
  4s   0.1130D+02  0.3353D+02  0.1000D+01  0.1228D-06 -0.7753D-12  368  T-F
  4p-  0.1090D+02  0.3204D+00  0.1000D+01  0.7828D-13  0.1149D-07  368  T-F
  4p   0.1083D+02  0.2655D+03  0.2000D+01  0.3561D-14 -0.2248D-19  368  T-F
  4d-  0.1015D+02  0.4906D+00  0.2000D+01  0.8779D-21  0.1288D-15  369  T-F
  4d   0.1013D+02  0.5108D+03  0.3000D+01  0.2509D-22 -0.1584D-27  369  T-F
  4f-  0.9324D+01  0.2669D+00  0.3000D+01  0.2623D-29  0.3849D-24  369  T-F
  4f   0.9316D+01  0.3018D+03  0.4000D+01  0.5429D-31 -0.3427D-36  369  T-F

RWFNESTIMATE: Execution complete.


********************************************************************************
*         RUN RMCDHF TO OBTAIN SELF CONSISTENT SOLUTIONS                       *
*         INPUT FILES: isodata, rcsf.inp, rwfn.inp, mcp.30, mcp.31,...         *
*         OUTPUT FILES: rwfn.out, rmix.out, rmcdhf.sum, rmcdhf.log             *
*                                                                              *
*         NOTE: FOR CORRELATION ORBITALS THERE ARE NO RESTRICTIONS ON THE      *
*         NUMBER OF NODES, I.E. THEY ARE NOT SPECTROSCOPIC. IN THIS RUN WE     *
*         VARY THE CORRELATION ORBITALS 4s, 4p, 4d, 4f. NONE OF THESE ARE      *
*         SPECTROSCOPIC. WE CAN USE WILD CARDS * FOR SPECIFYING ORBITALS       *
*                                                                              *
```

```
*         NOTE: INSTEAD OF SAYING THAT WE SHOULD OPTIMIZE ON, FOR EXAMPLE,    *
*         THE STATES 1,2,3,4 WE CAN WRITE 1-4 MEANING THE SAME THING          *
*                                                                             *
*******************************************************************************

>>rmcdhf

 RMCDHF
 This program determines the radial orbitals
 and the expansion coefficients of the CSFs
 in a self-onsistent field proceedure
 Input file:  isodata, rcsf.inp, rwfn.inp, mcp.30, ...
 Outputfiles: rwfn.out, rmix.out, rmcdhf.sum, rmcdhf.log

 Default settings?  (y/n)
>>y
 Loading CSF file ... Header only
 There are/is           16  relativistic subshells;
 Loading CSF File for ALL blocks
 There are         7359  relativistic CSFs... load complete;
 Loading Radial WaveFunction File ...
 There are            5  blocks  (block   J/Parity   NCF):
  1  1/2+  1047       2  3/2+  1862       3  5/2+  2112       4  7/2+  1537
  5  9/2+   801

 Enter ASF serial numbers for each block
 Block           1    ncf =        1047  id =  1/2+
>>1-8
 Block           2    ncf =        1862  id =  3/2+
>>1-11
 Block           3    ncf =        2112  id =  5/2+
>>1-10
 Block           4    ncf =        1537  id =  7/2+
>>1-5
 Block           5    ncf =         801  id =  9/2+
>>1-2
 level weights (1 equal;  5 standard;  9 user)
>>5
 Radial functions
 1s 2s 2p- 2p 3s 3p- 3p 3d- 3d 4s 4p- 4p 4d- 4d 4f- 4f
 Enter orbitals to be varied (Updating order)
>>4*
 Which of these are spectroscopic orbitals?
>>
 Enter the maximum number of SCF cycles:
>>999

.............

 RMCDHF: Execution complete.

*******************************************************************************
*         RUN RSAVE TO SAVE OUTPUT FILES: name.c, name.w, name.m, name.sum    *
*                                         name.alog, name.log                 *
```

```
******************************************************************************

>>rsave Ni_even_n4
 Created Ni_even_n4.w, Ni_even_n4.c, Ni_even_n4.m, Ni_even_n4.sum, Ni_even_n4.alog
  and Ni_even_n4.log

******************************************************************************
*         RUN RCI TO INCLUDE TRANSVERSE PHOTON INTERACTION AND QED EFFECTS    *
*         INPUT FILES : isodata, Ni_even_n4.c, Ni_even_n4.w                   *
*         OUTPUT FILES: Ni_even_n4.cm, Ni_even_n4.csum, Ni_even_n4.clog,      *
*         rci.res                                                             *
*                                                                             *
*         THE TRANSVERSE PHOTON FREQUENCIES CAN BE SET TO THE LOW FREQUENCY   *
*         LIMIT. RECOMMENDED IN CASES WHERE YOU HAVE CORRELATION ORBITALS     *
*         THE SELF ENERGY CORRECTION MAY FAIL FOR CORRELATION ORBITALS WITH   *
*         HIGH N.                                                             *
******************************************************************************

>>rci

 RCI
 This is the configuration interaction program
 Input file:  isodata, name.c, name.w
 Outputfiles: name.cm, name.csum, name.clog, rci.res

 Default settings?
>>y
 Name of state:
>>Ni_even_n4

 Block            1 ,  ncf =         1047
 Block            2 ,  ncf =         1862
 Block            3 ,  ncf =         2112
 Block            4 ,  ncf =         1537
 Block            5 ,  ncf =          801
 Loading CSF file ... Header only
 There are/is           16  relativistic subshells;
 Include contribution of H (Transverse)?
>>y
 Modify all transverse photon frequencies?
>>n
 Include H (Vacuum Polarisation)?
>>y
 Include H (Normal Mass Shift)?
>>n
 Include H (Specific Mass Shift)?
>>n
 Estimate self-energy?
>>y
 Largest n quantum number for including self-energy for orbital
 n should be less or equal 8
>>4
 Loading Radial WaveFunction File ...
 There are            5  blocks  (block   J/Parity   NCF):
```

```
  1  1/2+  1047       2  3/2+  1862       3  5/2+  2112       4  7/2+  1537
  5  9/2+   801

 Enter ASF serial numbers for each block
 Block           1    ncf =         1047  id =  1/2+
>>1-8
 Block           2    ncf =         1862  id =  3/2+
>>1-11
 Block           3    ncf =         2112  id =  5/2+
>>1-10
 Block           4    ncf =         1537  id =  7/2+
>>1-5
 Block           5    ncf =          801  id =  9/2+
>>1-2



 ....

 RCI: Execution complete.

******************************************************************************
*         RUN JJ2LSJ TO GET THE LSJ-COMPOSITION                              *
*         INPUT FILE: Ni_even_n4.c, Ni_even_n4.cm                            *
*         OUTPUT FILE: Ni_even_n4.lsj.lbl, Ni_even_n4.uni.lsj.lbl            *
******************************************************************************

>>jj2lsj

 jj2lsj: Transformation of ASFs from a jj-coupled CSF basis
         into an LSJ-coupled CSF basis  (Fortran 95 version)
         (C) Copyright by   G. Gaigalas and Ch. F. Fischer,
         (2017).
 Input files: name.c, name.(c)m
 Output files: name.lsj.lbl
   (optional)  name.lsj.c, name.lsj.j,
               name.uni.lsj.lbl, name.uni.lsj.sum

 Name of state
>>Ni_even_n4
 Loading Configuration Symmetry List File ...
 There are 16 relativistic subshells;
 There are 7359 relativistic CSFs;
  ... load complete;

 Mixing coefficients from a CI calc.?
>>y
  Do you need a unique labeling? (y/n)
>>y
    nelec  =          15
    ncftot =        7359
    nw     =          16
    nblock =           5
```

```
   block     ncf     nev    2j+1  parity
       1    1047       8       2       1
       2    1862      11       4       1
       3    2112      10       6       1
       4    1537       5       8       1
       5     801       2      10       1
 Default settings?  (y/n)
>>y

      ..........

 jj2lsj: Execution complete.


*******************************************************************************
*         RUN RCSFGENERATE TO GENERATE LIST FOR ALL                           *
*         STATES OF 3s(2)3p(3)                                                *
*         OUTPUT FILES: rcsfgenerate.log, rcsf.out                            *
*******************************************************************************

>>rcsfgenerate

 RCSFGENERATE
 This program generates a list of CSFs
 Configurations should be entered in spectroscopic notation
 with occupation numbers and indications if orbitals are
 closed (c), inactive (i), active (*) or has a minimal
 occupation e.g. 1s(2,1)2s(2,*)
 OUTPUT FILES: rcsf.out, rcsfgenerate.log

 Default, reverse, symmetry or user specified ordering? (*/r/s/u)
>>*

 Select core
        0: No core
        1: He (       1s(2)                  =  2 electrons)
        2: Ne ([He] + 2s(2)2p(6)             = 10 electrons)
        3: Ar ([Ne] + 3s(2)3p(6)             = 18 electrons)
        4: Kr ([Ar] + 3d(10)4s(2)4p(6)       = 36 electrons)
        5: Xe ([Kr] + 4d(10)5s(2)5p(6)       = 54 electrons)
        6: Rn ([Xe] + 4f(14)5d(10)6s(2)6p(6) = 86 electrons)
>>1
 Enter list of (maximum 100) configurations. End list with a blank line or an asterisk (*)

 Give configuration  1
>>2s(2,i)2p(6,i)3s(2,i)3p(3,i)
 Give configuration  2
>>
 Give set of active orbitals, as defined by the highest principal quantum number
 per l-symmetry, in a comma delimited list in s,p,d etc order, e.g. 5s,4p,3d
>>3s,3p
 Resulting 2*J-number? lower, higher (J=1 -> 2*J=2 etc.)
>>1,5
 Number of excitations (if negative number e.g. -2, correlation
```

```
 orbitals will always be doubly occupied)
>>0
 Generate more lists ? (y/n)
>>n

        .........

 3 blocks were created

      block  J/P            NCSF
          1  1/2-              1
          2  3/2-              3
          3  5/2-              1

*****************************************************************************
*        COPY FILES                                                         *
*        NOTE THAT WE COPY THE FILE TO RCSFMR.INP FOR USE                   *
*        TOGETHER WITH RCSFINTERACT                                         *
*****************************************************************************

>>cp rcsf.out rcsfmr.inp

*****************************************************************************
*        RUN RCSFGENERATE TO GENERATE LIST OBTAINED BY                      *
*        SD-EXCITATIONS FROM 3s(2)3p(3) TO n = 4                            *
*        OUTPUT FILES: rcsfgenerate.log, rcsf.out                           *
*****************************************************************************

>>rcsfgenerate

 RCSFGENERATE
 This program generates a list of CSFs
 Configurations should be entered in spectroscopic notation
 with occupation numbers and indications if orbitals are
 closed (c), inactive (i), active (*) or has a minimal
 occupation e.g. 1s(2,1)2s(2,*)
 OUTPUT FILES: rcsf.out, rcsfgenerate.log

 Default, reverse, symmetry or user specified ordering? (*/r/s/u)
>>*

 Select core
        0: No core
        1: He (       1s(2)                  =  2 electrons)
        2: Ne ([He] + 2s(2)2p(6)             = 10 electrons)
        3: Ar ([Ne] + 3s(2)3p(6)             = 18 electrons)
        4: Kr ([Ar] + 3d(10)4s(2)4p(6)       = 36 electrons)
        5: Xe ([Kr] + 4d(10)5s(2)5p(6)       = 54 electrons)
        6: Rn ([Xe] + 4f(14)5d(10)6s(2)6p(6) = 86 electrons)
>>1
 Enter list of (maximum 100) configurations. End list with a blank line or an asterisk (*)

 Give configuration  1
>>2s(2,i)2p(6,i)3s(2,*)3p(3,*)
```

```
 Give configuration  2
>>
 Give set of active orbitals, as defined by the highest principal quantum number
 per l-symmetry, in a comma delimited list in s,p,d etc order, e.g. 5s,4p,3d
>>4s,4p,4d,4f
 Resulting 2*J-number? lower, higher (J=1 -> 2*J=2 etc.)
>>1,5
 Number of excitations (if negative number e.g. -2, correlation
 orbitals will always be doubly occupied)
>>2
 Generate more lists ? (y/n)
>>n

        .........

 3 blocks were created

      block  J/P            NCSF
          1  1/2-            481
          2  3/2-            802
          3  5/2-            868

******************************************************************************
*        COPY FILES                                                          *
******************************************************************************

>>cp rcsf.out rcsf.inp

******************************************************************************
*        RUN RCSFINTERACT PROGRAM TO DETERMINE WHICH OF THE CSFs IN THE      *
*        rcsf.inp LIST INTERACTS WITH THE CSFs IN rcsfmr.inp                 *
*        THE INTERACTING CSFs ARE WRITTEN TO rcsf.out                        *
*        INPUT FILES: rcsfmr.inp, rcsf.inp                                   *
*        OUTPUT FILE: rcsf.out                                               *
******************************************************************************

>>rcsfinteract

 RCSFinteract: Determines all the CSFs (rcsf.inp) that interact
              with the CSFs in the multireference (rcsfmr.inp)
              (C)  Copyright by G. Gaigalas and Ch. F. Fischer
              (Fortran 95 version)               NIST  (2017).
              Input files: rcsfmr.inp, rcsf.inp
              Output file: rcsf.out

 Reduction based on Dirac-Coulomb (1) or
 Dirac-Coulomb-Breit (2) Hamiltonian?
>>2

   ....

 There are 16 relativistic subshells;
  Block    MR NCSF   Befor NCSF   After NCSF
    1           1          481          237
```

```
   2           3          802          577
   3           1          868          480

 RCSFINTERACT: Execution complete

******************************************************************************
*         COPY FILES                                                         *
******************************************************************************

>>cp rcsf.out rcsf.inp

******************************************************************************
*         RUN RANGULAR TO GENERATE ENERGY EXPRESSION                         *
*         INPUT FILE  : rcsf.inp                                             *
*         OUTPUT FILES: rangular.log, mcp.30, mcp.31,....                    *
******************************************************************************

>>rangular

 RANGULAR
 This program performs angular integration
 Input file:  rcsf.inp
 Outputfiles: mcp.30, mcp.31, ....
              rangular.log

 Full interaction?  (y/n)
>>y

  .....

 RANGULAR: Execution complete.

******************************************************************************
*         RUN RWFNESTIMATE TO GENERATE INITIAL ESTIMATES FOR RADIAL ORBITALS  *
*         INPUT FILES: isodata, rcsf.inp, previous rwfn files                *
*         OUTPUT FILE: rwfn.inp, rwfnestimate.log                            *
******************************************************************************

>>rwfnestimate

 RWFNESTIMATE
 This program estimates radial wave functions
 for orbitals
 Input files: isodata, rcsf.inp, optional rwfn file
 Output file: rwfn.inp

 Default settings ?
>>y
 Loading CSF file ... Header only
 There are/is           16  relativistic subshells;
 The following subshell radial wavefunctions remain to be estimated:
 1s 2s 2p- 2p 3s 3p- 3p 3d- 3d 4s 4p- 4p 4d- 4d 4f- 4f

 Read subshell radial wavefunctions. Choose one below
```

```
     1 -- GRASP2K File
     2 -- Thomas-Fermi
     3 -- Screened Hydrogenic
>>1
 Enter the file name (Null then "rwfn.out")
>>Ni_mr.w
 Enter the list of relativistic subshells:
>>*
 The following subshell radial wavefunctions remain to be estimated:
 4s 4p- 4p 4d- 4d 4f- 4f

 Read subshell radial wavefunctions. Choose one below
     1 -- GRASP2K File
     2 -- Thomas-Fermi
     3 -- Screened Hydrogenic
>>2
 Enter the list of relativistic subshells:
>>*
Shell      e           p0        gamma        P(2)       Q(2)      MTP  SRC

  1s   0.3241D+03  0.3324D+03  0.1000D+01  0.1217D-05 -0.7677D-11  357  Ni_
  2s   0.5340D+02  0.1042D+03  0.1000D+01  0.3818D-06 -0.2409D-11  359  Ni_
  2p-  0.4833D+02  0.8505D+00  0.1000D+01  0.2077D-12  0.3049D-07  359  Ni_
  2p   0.4767D+02  0.6995D+03  0.2000D+01  0.9382D-14 -0.5921D-19  360  Ni_
  3s   0.1702D+02  0.4579D+02  0.1000D+01  0.1677D-06 -0.1058D-11  364  Ni_
  3p-  0.1558D+02  0.3892D+00  0.1000D+01  0.9504D-13  0.1395D-07  364  Ni_
  3p   0.1545D+02  0.3217D+03  0.2000D+01  0.4314D-14 -0.2723D-19  364  Ni_
  3d-  0.1321D+02  0.3699D+00  0.2000D+01  0.6617D-21  0.9712D-16  366  Ni_
  3d   0.1319D+02  0.3854D+03  0.3000D+01  0.1893D-22 -0.1195D-27  366  Ni_
  4s   0.1130D+02  0.3353D+02  0.1000D+01  0.1228D-06 -0.7753D-12  368  T-F
  4p-  0.1090D+02  0.3204D+00  0.1000D+01  0.7828D-13  0.1149D-07  368  T-F
  4p   0.1083D+02  0.2655D+03  0.2000D+01  0.3561D-14 -0.2248D-19  368  T-F
  4d-  0.1015D+02  0.4906D+00  0.2000D+01  0.8779D-21  0.1288D-15  369  T-F
  4d   0.1013D+02  0.5108D+03  0.3000D+01  0.2509D-22 -0.1584D-27  369  T-F
  4f-  0.9324D+01  0.2669D+00  0.3000D+01  0.2623D-29  0.3849D-24  369  T-F
  4f   0.9316D+01  0.3018D+03  0.4000D+01  0.5429D-31 -0.3427D-36  369  T-F

RWFNESTIMATE: Execution complete.


******************************************************************************
*         RUN RMCDHF TO OBTAIN SELF CONSISTENT SOLUTIONS                     *
*         INPUT FILES: isodata, rcsf.inp, rwfn.inp, mcp.30, mcp.31,...       *
*         OUTPUT FILES: rwfn.out, rmix.out, rmcdhf.sum, rmcdhf.log           *
*                                                                            *
*         NOTE: FOR CORRELATION ORBITALS THERE ARE NO RESTRICTIONS ON THE    *
*         NUMBER OF NODES, I.E. THEY ARE NOT SPECTROSCOPIC. IN THIS RUN WE   *
*         VARY THE CORRELATION ORBITALS 4s, 4p, 4d, 4f. NONE OF THESE ARE    *
*         SPECTROSCOPIC. WE CAN USE WILD CARDS * FOR SPECIFYING ORBITALS     *
*                                                                            *
*         NOTE: INSTEAD OF SAYING THAT WE SHOULD OPTIMIZE ON, FOR EXAMPLE,   *
*         THE STATES 1,2,3,4 WE CAN WRITE 1-4 MEANING THE SAME THING         *
*                                                                            *
******************************************************************************
```

```
>>rmcdhf

 RMCDHF
 This program determines the radial orbitals
 and the expansion coefficients of the CSFs
 in a self-onsistent field proceedure
 Input file:  isodata, rcsf.inp, rwfn.inp, mcp.30, ...
 Outputfiles: rwfn.out, rmix.out, rmcdhf.sum, rmcdhf.log

 Default settings?  (y/n)
>>y
 Loading CSF file ... Header only
 There are/is           16  relativistic subshells;
 Loading CSF File for ALL blocks
 There are         1294  relativistic CSFs... load complete;
 Loading Radial WaveFunction File ...
 There are            3  blocks  (block   J/Parity   NCF):
  1  1/2-   237       2  3/2-   577       3  5/2-   480

 Enter ASF serial numbers for each block
 Block            1    ncf =         237  id =  1/2-
>>1
 Block            2    ncf =         577  id =  3/2-
>>1-3
 Block            3    ncf =         480  id =  5/2-
>>1
 level weights (1 equal;  5 standard;  9 user)
>>5
 Radial functions
 1s 2s 2p- 2p 3s 3p- 3p 3d- 3d 4s 4p- 4p 4d- 4d 4f- 4f
 Enter orbitals to be varied (Updating order)
>>4*
 Which of these are spectroscopic orbitals?
>>
 Enter the maximum number of SCF cycles:
>>999

.............

 RMCDHF: Execution complete.

******************************************************************************
*         RUN RSAVE TO SAVE OUTPUT FILES: name.c, name.w, name.m, name.sum    *
*                                         name.log                            *
******************************************************************************

>>rsave Ni_odd_n4
 Created Ni_odd_n4.w, Ni_odd_n4.c, Ni_odd_n4.m, Ni_odd_n4.sum and Ni_odd_n4.log

******************************************************************************
*         RUN RCI TO INCLUDE TRANSVERSE PHOTON INTERACTION AND QED EFFECTS    *
*         INPUT FILES : isodata, Ni_odd_n4.c, Ni_odd_n4.w                     *
*         OUTPUT FILES: Ni_odd_n4.cm, Ni_odd_n4.csum, Ni_odd_n4.clog, rci.res *
```

```
*                                                                              *
*         THE TRANSVERSE PHOTON FREQUENCIES CAN BE SET TO THE LOW FREQUENCY    *
*         LIMIT. RECOMMENDED IN CASES WHERE YOU HAVE CORRELATION ORBITALS      *
*         THE SELF ENERGY CORRECTION MAY FAIL FOR CORRELATION ORBITALS WITH    *
*         HIGH N.                                                              *
********************************************************************************

>>rci

 RCI
 This is the configuration interaction program
 Input file:  isodata, name.c, name.w
 Outputfiles: name.cm, name.csum, name.clog, rci.res

 Default settings?
>>y
 Name of state:
>>Ni_odd_n
 Block            1 ,  ncf =          237
 Block            2 ,  ncf =          577
 Block            3 ,  ncf =          480
 Loading CSF file ... Header only
 There are/is           16  relativistic subshells;
 Include contribution of H (Transverse)?
>>y
 Modify all transverse photon frequencies?
>>n
 Include H (Vacuum Polarisation)?
>>y
 Include H (Normal Mass Shift)?
>>n
 Include H (Specific Mass Shift)?
>>n
 Estimate self-energy?
>>y
 Largest n quantum number for including self-energy for orbital
 n should be less or equal 8
>>4
 Loading Radial WaveFunction File ...
 There are            3  blocks  (block   J/Parity   NCF):
  1  1/2-   237       2  3/2-   577       3  5/2-   480

 Enter ASF serial numbers for each block
 Block            1    ncf =          237  id =  1/2-
>>1
 Block            2    ncf =          577  id =  3/2-
>>1-3
 Block            3    ncf =          480  id =  5/2-
>>1


 ....

 RCI: Execution complete.
```

```
***************************************************************************
*         RUN JJ2LSJ TO GET THE LSJ-COMPOSITION                           *
*         INPUT FILE: Ni_odd_n4.c, Ni_odd_n4.cm                           *
*         OUTPUT FILE: Ni_odd_n4.lsj.lbl, Ni_odd_n4.uni.lsj.lbl           *
***************************************************************************

>>jj2lsj

 jj2lsj: Transformation of ASFs from a jj-coupled CSF basis
         into an LSJ-coupled CSF basis  (Fortran 95 version)
         (C) Copyright by   G. Gaigalas and Ch. F. Fischer,
         (2017).
 Input files: name.c, name.(c)m
 Output files: name.lsj.lbl
   (optional)  name.lsj.c, name.lsj.j,
               name.uni.lsj.lbl, name.uni.lsj.sum

 Name of state
>>Ni_odd_n4
 Loading Configuration Symmetry List File ...
 There are 16 relativistic subshells;
 There are 1294 relativistic CSFs;
  ... load complete;

 Mixing coefficients from a CI calc.?
>>y
 Do you need a unique labeling? (y/n)
>>y
    nelec  =          15
    ncftot =        1294
    nw     =          16
    nblock =           3

   block     ncf     nev    2j+1  parity
       1     237       1       2      -1
       2     577       3       4      -1
       3     480       1       6      -1
 Default settings?  (y/n)
>>y

      ...........

 jj2lsj: Execution complete.


***************************************************************************
*         RUN RLEVELS TO VIEW ENERGIES AND ENERGY SEPARATIONS.            *
*         IF DESIRED WE CAN INSTEAD RUN RLEVELSEV TO GET THE SEPARATION IN EV *
***************************************************************************

>> rlevels Ni_even_n4.cm Ni_odd_n4.cm

 nblock =       5   ncftot =         7359   nw =           16   nelec =    15
```

```
 nblock =       3   ncftot =         1294   nw =           16   nelec =    15

 Energy levels for ...
 Rydberg constant is   109737.31534
 Splitting is the energy difference with the lower neighbor
-----------------------------------------------------------------------------------------
 No Pos  J Parity Energy Total    Levels     Splitting     Configuration
                      (a.u.)      (cm^-1)     (cm^-1)
-----------------------------------------------------------------------------------------
  1  1  3/2 -    -1443.2223116        0.00         0.00  2s(2).2p(6).3s(2).3p(3)4S3_4S
  2  2  3/2 -    -1443.0054751    47590.12     47590.12  2s(2).2p(6).3s(2).3p(3)2D3_2D
  3  1  5/2 -    -1442.9698639    55405.86      7815.74  2s(2).2p(6).3s(2).3p(3)2D3_2D
  4  1  1/2 -    -1442.8230089    87636.81     32230.95  2s(2).2p(6).3s(2).3p(3)2P1_2P
  5  3  3/2 -    -1442.7717172    98894.05     11257.24  2s(2).2p(6).3s(2).3p(3)2P1_2P
  6  1  5/2 +    -1441.7818000   316155.75    217261.70  2s(2).2p(6).3s_2S.3p(4)3P2_4P
  7  1  3/2 +    -1441.7162297   330546.77     14391.02  2s(2).2p(6).3s_2S.3p(4)3P2_4P
  8  1  1/2 +    -1441.6892182   336475.11      5928.35  2s(2).2p(6).3s_2S.3p(4)3P2_4P
  9  2  3/2 +    -1441.4302453   393313.09     56837.98  2s(2).2p(6).3s_2S.3p(4)1D2_2D
 10  2  5/2 +    -1441.4140033   396877.81      3564.72  2s(2).2p(6).3s_2S.3p(4)1D2_2D
 11  3  3/2 +    -1441.1718022   450034.79     53156.98  2s(2).2p(6).3s(2).3p(2)3P2_3P.3d_2P
 12  2  1/2 +    -1441.1456357   455777.69      5742.90  2s(2).2p(6).3s_2S.3p(4)1S0_2S
 13  3  1/2 +    -1441.0406282   478824.17     23046.48  2s(2).2p(6).3s_2S.3p(4)1S0_2S
 14  4  3/2 +    -1441.0024109   487211.89      8387.72  2s(2).2p(6).3s(2).3p(2)3P2_3P.3d_4F
 15  3  5/2 +    -1440.9761532   492974.79      5762.91  2s(2).2p(6).3s(2).3p(2)3P2_3P.3d_4F
 16  1  7/2 +    -1440.9365580   501664.94      8690.14  2s(2).2p(6).3s(2).3p(2)3P2_3P.3d_4F
 17  4  5/2 +    -1440.9107001   507340.08      5675.14  2s(2).2p(6).3s(2).3p(2)1D2_1D.3d_2F
 18  1  9/2 +    -1440.8904792   511778.07      4437.99  2s(2).2p(6).3s(2).3p(2)3P2_3P.3d_4F
 19  2  7/2 +    -1440.8807592   513911.35      2133.28  2s(2).2p(6).3s(2).3p(2)3P2_3P.3d_4D
 20  4  1/2 +    -1440.8806230   513941.26        29.91  2s(2).2p(6).3s(2).3p(2)3P2_3P.3d_4D
 21  5  3/2 +    -1440.8742942   515330.26      1389.01  2s(2).2p(6).3s(2).3p(2)3P2_3P.3d_4D
 22  5  5/2 +    -1440.8440756   521962.48      6632.21  2s(2).2p(6).3s(2).3p(2)3P2_3P.3d_4D
 23  3  7/2 +    -1440.7787492   536299.97     14337.49  2s(2).2p(6).3s(2).3p(2)3P2_3P.3d_4D
 24  4  7/2 +    -1440.6228432   570517.38     34217.42  2s(2).2p(6).3s(2).3p(2)1D2_1D.3d_2G
 25  2  9/2 +    -1440.5995839   575622.19      5104.81  2s(2).2p(6).3s(2).3p(2)1D2_1D.3d_2G
 26  6  3/2 +    -1440.5809891   579703.29      4081.10  2s(2).2p(6).3s_2S.3p(4)3P2_2P
 27  6  5/2 +    -1440.5346963   589863.37     10160.08  2s(2).2p(6).3s(2).3p(2)3P2_3P.3d_4P
 28  7  3/2 +    -1440.5082308   595671.88      5808.50  2s(2).2p(6).3s(2).3p(2)3P2_3P.3d_4P
 29  5  1/2 +    -1440.5058658   596190.95       519.07  2s(2).2p(6).3s(2).3p(2)3P2_3P.3d_4P
 30  6  1/2 +    -1440.4809830   601652.08      5461.13  2s(2).2p(6).3s(2).3p(2)3P2_3P.3d_4P
 31  8  3/2 +    -1440.4610704   606022.40      4370.32  2s(2).2p(6).3s(2).3p(2)1S0_1S.3d_2D
 32  7  5/2 +    -1440.3894396   621743.54     15721.14  2s(2).2p(6).3s(2).3p(2)1S0_1S.3d_2D
 33  9  3/2 +    -1440.3039779   640500.22     18756.68  2s(2).2p(6).3s(2).3p(2)1D2_1D.3d_2D
 34  8  5/2 +    -1440.2990681   641577.79      1077.57  2s(2).2p(6).3s(2).3p(2)1D2_1D.3d_2D
 35  7  1/2 +    -1440.2295105   656843.92     15266.12  2s(2).2p(6).3s(2).3p(2)1D2_1D.3d_2P
 36  9  5/2 +    -1440.1954728   664314.33      7470.41  2s(2).2p(6).3s(2).3p(2)3P2_3P.3d_2F
 37 10  3/2 +    -1440.1737424   669083.60      4769.27  2s(2).2p(6).3s(2).3p(2)1D2_1D.3d_2P
 38  8  1/2 +    -1440.1602120   672053.18      2969.58  2s(2).2p(6).3s(2).3p(2)1D2_1D.3d_2S
 39  5  7/2 +    -1440.1589598   672328.00       274.82  2s(2).2p(6).3s(2).3p(2)3P2_3P.3d_2F
 40 10  5/2 +    -1440.0333381   699898.78     27570.78  2s(2).2p(6).3s(2).3p(2)3P2_3P.3d_2D
 41 11  3/2 +    -1440.0283152   701001.19      1102.41  2s(2).2p(6).3s(2).3p(2)3P2_3P.3d_2D
-----------------------------------------------------------------------------------------
```

The `rlevels` program reads the `name.cm` files along with the `name.lsj.lbl` files, which display the LSJ composition as generated by `jj2lsj`. From the energy spectra we see that some even levels have the same identification. These pairs of the levels are: 12 and 13; 19 and 23; 29 and 30. If a unique label was required `jj2lsj` also outputs files `name.uni.lsj.lb` in which unique labels are determined according to the prescription given in [31, 32, 20]. To display the energies with unique labels we should copy `name.cm` to `name.uni.cm` and rerun `rlevels` with `name.uni.cm` as the input file.

```
****************************************************************************
*         COPY FILES TO HAVE EVEN PARITY LEVELS WITH UNIQUE LABELS         *
*         THAT SHOULD BE USED IN FURTHER CALCULATIONS                      *
****************************************************************************

>>cp Ni_even_n4.cm Ni_even_n4.uni.cm

****************************************************************************
*         RUN RLEVELS TO VIEW ENERGIES AND ENERGY SEPARATIONS.             *
*         ENERGY LEVELS HAVE UNIQUE LABELS                                 *
****************************************************************************

>> rlevels Ni_even_n4.uni.cm Ni_odd_n4.cm

 nblock =       5   ncftot =         7359   nw =            16   nelec =    15
 nblock =       3   ncftot =         1294   nw =            16   nelec =    15

 Energy levels for ...
 Rydberg constant is   109737.31534
 Splitting is the energy difference with the lower neighbor
-------------------------------------------------------------------------------------------
 No Pos  J Parity Energy Total    Levels     Splitting     Configuration
                      (a.u.)      (cm^-1)     (cm^-1)
-------------------------------------------------------------------------------------------
  1  1  3/2 -   -1443.2223116        0.00         0.00  2s(2).2p(6).3s(2).3p(3)4S3_4S
  2  2  3/2 -   -1443.0054751    47590.12     47590.12  2s(2).2p(6).3s(2).3p(3)2D3_2D
  3  1  5/2 -   -1442.9698639    55405.86      7815.74  2s(2).2p(6).3s(2).3p(3)2D3_2D
  4  1  1/2 -   -1442.8230089    87636.81     32230.95  2s(2).2p(6).3s(2).3p(3)2P1_2P
  5  3  3/2 -   -1442.7717172    98894.05     11257.24  2s(2).2p(6).3s(2).3p(3)2P1_2P
  6  1  5/2 +   -1441.7818000   316155.75    217261.70  2s(2).2p(6).3s_2S.3p(4)3P2_4P
  7  1  3/2 +   -1441.7162297   330546.77     14391.02  2s(2).2p(6).3s_2S.3p(4)3P2_4P
  8  1  1/2 +   -1441.6892182   336475.11      5928.35  2s(2).2p(6).3s_2S.3p(4)3P2_4P
  9  2  3/2 +   -1441.4302453   393313.09     56837.98  2s(2).2p(6).3s_2S.3p(4)1D2_2D
 10  2  5/2 +   -1441.4140033   396877.81      3564.72  2s(2).2p(6).3s_2S.3p(4)1D2_2D
 11  3  3/2 +   -1441.1718022   450034.79     53156.98  2s(2).2p(6).3s(2).3p(2)3P2_3P.3d_2P
 12  2  1/2 +   -1441.1456357   455777.69      5742.90  2s(2).2p(6).3s_2S.3p(4)3P2_2P
 13  3  1/2 +   -1441.0406282   478824.17     23046.48  2s(2).2p(6).3s_2S.3p(4)1S0_2S
 14  4  3/2 +   -1441.0024109   487211.89      8387.72  2s(2).2p(6).3s(2).3p(2)3P2_3P.3d_4F
 15  3  5/2 +   -1440.9761532   492974.79      5762.91  2s(2).2p(6).3s(2).3p(2)3P2_3P.3d_4F
 16  1  7/2 +   -1440.9365580   501664.94      8690.14  2s(2).2p(6).3s(2).3p(2)3P2_3P.3d_4F
 17  4  5/2 +   -1440.9107001   507340.08      5675.14  2s(2).2p(6).3s(2).3p(2)1D2_1D.3d_2F
 18  1  9/2 +   -1440.8904792   511778.07      4437.99  2s(2).2p(6).3s(2).3p(2)3P2_3P.3d_4F
 19  2  7/2 +   -1440.8807592   513911.35      2133.28  2s(2).2p(6).3s(2).3p(2)1D2_1D.3d_2F
 20  4  1/2 +   -1440.8806230   513941.26        29.91  2s(2).2p(6).3s(2).3p(2)3P2_3P.3d_4D
 21  5  3/2 +   -1440.8742942   515330.26      1389.01  2s(2).2p(6).3s(2).3p(2)3P2_3P.3d_4D
 22  5  5/2 +   -1440.8440756   521962.48      6632.21  2s(2).2p(6).3s(2).3p(2)3P2_3P.3d_4D
```

```
 23  3  7/2 +   -1440.7787492   536299.97    14337.49  2s(2).2p(6).3s(2).3p(2)3P2_3P.3d_4D
 24  4  7/2 +   -1440.6228432   570517.38    34217.42  2s(2).2p(6).3s(2).3p(2)1D2_1D.3d_2G
 25  2  9/2 +   -1440.5995839   575622.19     5104.81  2s(2).2p(6).3s(2).3p(2)1D2_1D.3d_2G
 26  6  3/2 +   -1440.5809891   579703.29     4081.10  2s(2).2p(6).3s_2S.3p(4)3P2_2P
 27  6  5/2 +   -1440.5346963   589863.37    10160.08  2s(2).2p(6).3s(2).3p(2)3P2_3P.3d_4P
 28  7  3/2 +   -1440.5082308   595671.88     5808.50  2s(2).2p(6).3s(2).3p(2)3P2_3P.3d_4P
 29  5  1/2 +   -1440.5058658   596190.95      519.07  2s(2).2p(6).3s(2).3p(2)3P2_3P.3d_2P
 30  6  1/2 +   -1440.4809830   601652.08     5461.13  2s(2).2p(6).3s(2).3p(2)3P2_3P.3d_4P
 31  8  3/2 +   -1440.4610704   606022.40     4370.32  2s(2).2p(6).3s(2).3p(2)1S0_1S.3d_2D
 32  7  5/2 +   -1440.3894396   621743.54    15721.14  2s(2).2p(6).3s(2).3p(2)1S0_1S.3d_2D
 33  9  3/2 +   -1440.3039779   640500.22    18756.68  2s(2).2p(6).3s(2).3p(2)1D2_1D.3d_2D
 34  8  5/2 +   -1440.2990681   641577.79     1077.57  2s(2).2p(6).3s(2).3p(2)1D2_1D.3d_2D
 35  7  1/2 +   -1440.2295105   656843.92    15266.12  2s(2).2p(6).3s(2).3p(2)1D2_1D.3d_2P
 36  9  5/2 +   -1440.1954728   664314.33     7470.41  2s(2).2p(6).3s(2).3p(2)3P2_3P.3d_2F
 37 10  3/2 +   -1440.1737424   669083.60     4769.27  2s(2).2p(6).3s(2).3p(2)1D2_1D.3d_2P
 38  8  1/2 +   -1440.1602120   672053.18     2969.58  2s(2).2p(6).3s(2).3p(2)1D2_1D.3d_2S
 39  5  7/2 +   -1440.1589598   672328.00      274.82  2s(2).2p(6).3s(2).3p(2)3P2_3P.3d_2F
 40 10  5/2 +   -1440.0333381   699898.78    27570.78  2s(2).2p(6).3s(2).3p(2)3P2_3P.3d_2D
 41 11  3/2 +   -1440.0283152   701001.19     1102.41  2s(2).2p(6).3s(2).3p(2)3P2_3P.3d_2D
-----------------------------------------------------------------------------------------

*****************************************************************************
*         COPY FILES TO HAVE LEVELS WITH UNIQUE LABELS                      *
*         THAT SHOULD BE USED IN FURTHER CALCULATIONS                       *
*****************************************************************************

>>cp Ni_even_n4.c Ni_even_n4.uni.c
>>cp Ni_even_n4.w Ni_even_n4.uni.w
```

Now we can use these files with unique labels in further calculations e.g. transition properties.

```
*****************************************************************************
*         RUN RBIOTRANSFORM FOR Ni_odd_n4 AND Ni_even_n4.uni                *
*         TO TRANSFORM WAVE FUNCTIONS                                       *
*         INPUT FILES:  isodata, Ni_odd_n4.c, Ni_odd_n4.w, Ni_odd_n4.cm,    *
*                       Ni_even_n4.uni.c, Ni_even_n4.uni.w, Ni_even_n4.uni.cm *
*         OUTPUT FILES: Ni_odd_n4.cbm, Ni_odd_n4.bw,                        *
*                       Ni_even_n4.uni.cbm, Ni_even_n4.uni.bw               *
*                       Ni_odd_n4.TB, Ni_even_n4.uni.TB (angular files)     *
*****************************************************************************

>>rbiotransform

 RBIOTRANSFORM
 This program transforms the initial and final wave
 functions so that standard tensor albegra can be
 used in evaluation of the transition parameters
 Input files:  isodata, name1.c, name1.w, name1.(c)m
               name2.c, name2.w, name2.(c)m
               name1.TB, name2.TB (optional angular files)
 Output files: name1.bw, name1.(c)bm,
               name2.bw, name2.(c)bm
```

```
              name1.TB, name2.TB (angular files)
 Default settings?
>>y
 Input from a CI calculation?
>>y
  Name of the Initial state
>>Ni_odd_n4
  Name of the Final state
>>Ni_even_n4.uni
  Transformation of all J symmetries?
>>y

   ....

 BIOTRANSFORM: Execution complete.

*******************************************************************************
*         RUN RTRANSITION FOR Ni_odd_n4 AND Ni_even_n4.uni                    *
*         TO COMPUTE TRANSITION PARAMETERS                                    *
*         INPUT FILES:  isodata, Ni_odd_n4.c, Ni_odd_n4.bw, Ni_odd_n4.cbm,    *
*                       Ni_even_n4.uni.c, Ni_even_n4.uni.bw, Ni_even_n4.uni.cbm*
*         OUTPUT FILES: Ni_odd_n4.Ni_even_n4.uni.ct                           *
*                       Ni_odd_n4.Ni_even_n4.uni.-1T (angular files)          *
*******************************************************************************

>>rtransition

 RTRANSITION
 This program computes transition parameters from
 transformed wave functions
 Input files:  isodata, name1.c, name1.bw, name1.(c)bm
               name2.c, name2.bw, name2.(c)bm
               optional, name1.lsj.lbl, name2.lsj.lbl
               name1.name2.KT (optional angular files)
 Output files: name1.name2.(c)t
               optional, name1.name2.(c)t.lsj
               name1.name2.KT (angular files)
 Here K is parity and rank of transition: -1,+1 etc

 Default settings?
>>y
 Input from a CI calculation?
>>y
 Name of the Initial state
>>Ni_odd_n4
 Name of the Final state
>>Ni_even_n4.uni

 MRGCSL: Execution begins ...
 Loading Configuration Symmetry List File ...
 There are 16 relativistic subshells;
 There are 1294 relativistic CSFs;
  ... load complete;
 Loading Configuration Symmetry List File ...
```

```
 There are 16 relativistic subshells;
 There are 7359 relativistic CSFs;
  ... load complete;
           1 s
           2 s
           2 p-
           2 p
           3 s
           3 p-
           3 p
           3 d-
           3 d
           4 s
           4 p-
           4 p
           4 d-
           4 d
           4 f-
           4 f
           3
         237          814         1294
           5
        1047         2909         5021         6558         7359
 Loading Configuration Symmetry List File ...
  there are 16 relativistic subshells;
  there are 8653 relativistic CSFs;
  ... load complete;
 Enter the list of transition specifications
  e.g.,  E1,M2  or  E1 M2  or  E1;M2 :
>>E1

   .....

 RTRANSITION: Execution complete.
```

Transition data are in `Ni_odd_n4.Ni_even_n4.uni.ct.lsj` file in which all levels have the unique labels.

An alternative way to get unique labels than the one described above is to denote the states by the LS composition. This can be done with the PERL scrip `lscomp.pl` and it is described in more detail in section 8.10.

7.6 Sixth example: the study of energy spectra for Ni XIV, extended MR

To obtain good transition energies it is often necessary to extend the MR. This is facilitated by the program `rcsfmr`. The `rcsfmr` program reads the `name.lsj.lbl` file produced by `jj2lsj` and extracts the configurations that give rise to LSJ-coupled CSFs with weights exceeding a user defined cut-off. Below is part of the `Ni_even_n4.lsj.lbl` file from the fifth example.

```
 Pos   J   Parity      Energy Total      Comp. of ASF
  1  1/2     +         -1441.689218199      99.941%
       -0.92754035    0.86033110   2s(2).2p(6).3s_2S.3p(4)3P2_4P
```

```
      -0.31645458    0.10014350   2s(2).2p(6).3s(2).3p(2)3P2_3P.3d_4P
      -0.13107331    0.01718021   2s(2).2p(6).3s_2S.3p(4)1S0_2S
      -0.06808263    0.00463524   2s(2).2p(6).3s_2S.3p(2)3P2_4P.3d(2)1S0_4P
      -0.06306078    0.00397666   2s(2).2p(6).3s_2S.3p(2)3P2_4P.3d(2)1D2_4P
      -0.06139973    0.00376993   2s(2).2p(6).3p(4)3P2_3P.3d_4P
      -0.04384457    0.00192235   2s(2).2p(6).3s_2S.3p(2)1D2_2D.3d(2)3P2_4P
       0.04315649    0.00186248   2s(2).2p(6).3s(2).3p(2)1D2_1D.3d_2S
       0.04160783    0.00173121   2s(2).2p(6).3s_2S.3p(2)3P2_4P.3d(2)3P2_4P
  2  1/2      +          -1441.145635652       99.870%
       0.55232128    0.30505880   2s(2).2p(6).3s_2S.3p(4)1S0_2S
       0.54902077    0.30142381   2s(2).2p(6).3s_2S.3p(4)3P2_2P
      -0.51853413    0.26887764   2s(2).2p(6).3s(2).3p(2)3P2_3P.3d_2P
      -0.25240762    0.06370961   2s(2).2p(6).3s(2).3p(2)1D2_1D.3d_2S
       0.14975498    0.02242655   2s(2).2p(6).3s(2).3p(2)1D2_1D.3d_2P
       0.08843968    0.00782158   2s(2).2p(6).3p(4)1D2_1D.3d_2P
      -0.07915249    0.00626512   2s(2).2p(6).3s(2).3p(2)3P2_3P.3d_4D
       0.06957005    0.00483999   2s(2).2p(6).3s_2S.3p(2)1S0_2S.3d(2)1S0_2S
      -0.06792256    0.00461347   2s(2).2p(6).3s_2S.3p(4)3P2_4P
      -0.04635869    0.00214913   2s(2).2p(6).3s_2S.3p(2)1D2_2D.3d(2)1D2_2P
      -0.04440113    0.00197146   2s(2).2p(6).3s_2S.3p(2)3P2_4P.3d(2)3P2_2P
       0.03795533    0.00144061   2s(2).2p(6).3s_2S.3p(2)1D2_2D.3d(2)3P2_2P
      -0.03449961    0.00119022   2s(2).2p(6).3s_2S.3p(2)3P2_4P.3d(2)3P2_2S
      -0.03371279    0.00113655   2s(2).2p(6).3s(2).3p(2)3P2_3P.3d_4P
      -0.03274468    0.00107221   2s(2).2p(6).3s_2S.3p(2)1D2_2D.3d(2)1D2_2S
       0.03172321    0.00100636   2s(2).2p(6).3s_2S.3p(2)1D2_2D.3d(2)3F2_2P
  3  1/2      +          -1441.040628181       99.883%
       0.69858668    0.48802335   2s(2).2p(6).3s_2S.3p(4)1S0_2S
       0.44942201    0.20198014   2s(2).2p(6).3s(2).3p(2)3P2_3P.3d_2P
      -0.37637049    0.14165475   2s(2).2p(6).3s_2S.3p(4)3P2_2P
      -0.31031417    0.09629489   2s(2).2p(6).3s(2).3p(2)1D2_1D.3d_2S
      -0.14516095    0.02107170   2s(2).2p(6).3s(2).3p(2)1D2_1D.3d_2P
       0.11019096    0.01214205   2s(2).2p(6).3s(2).3p(2)3P2_3P.3d_4D
      -0.10593148    0.01122148   2s(2).2p(6).3s_2S.3p(4)3P2_4P
       0.08895336    0.00791270   2s(2).2p(6).3s_2S.3p(2)1S0_2S.3d(2)1S0_2S
      -0.06646197    0.00441719   2s(2).2p(6).3p(4)1D2_1D.3d_2P
      -0.04537373    0.00205877   2s(2).2p(6).3s(2).3p(2)3P2_3P.3d_4P
      -0.04337124    0.00188106   2s(2).2p(6).3s_2S.3p(2)3P2_4P.3d(2)3P2_2S
      -0.04274332    0.00182699   2s(2).2p(6).3s_2S.3p(2)1D2_2D.3d(2)1D2_2S
       0.04115899    0.00169406   2s(2).2p(6).3s_2S.3p(2)1D2_2D.3d(2)1D2_2P
       0.03553786    0.00126294   2s(2).2p(6).3s_2S.3p(2)3P2_4P.3d(2)3P2_2P

      ....................

  1  9/2      +          -1440.890479157       99.130%
       0.96822869    0.93746679   2s(2).2p(6).3s(2).3p(2)3P2_3P.3d_4F
      -0.18912065    0.03576662   2s(2).2p(6).3s(2).3p(2)1D2_1D.3d_2G
       0.09063756    0.00821517   2s(2).2p(6).3s_2S.3p(2)1D2_2D.3d(2)3F2_4F
       0.04435550    0.00196741   2s(2).2p(6).3s_2S.3p(2)3P2_4P.3d(2)1G2_4F
      -0.04085292    0.00166896   2s(2).2p(6).3s_2S.3p(2)3P2_2P.3d(2)3F2_4F
      -0.03263768    0.00106522   2s(2).2p(6).3s_2S.3p(2)3P2_4P.3d(2)1D2_4F
  2  9/2      +          -1440.599583939       99.058%
       0.96757572    0.93620277   2s(2).2p(6).3s(2).3p(2)1D2_1D.3d_2G
       0.18857093    0.03555900   2s(2).2p(6).3s(2).3p(2)3P2_3P.3d_4F
      -0.08465882    0.00716712   2s(2).2p(6).3s_2S.3p(2)3P2_2P.3d(2)1G2_2G
```

```
-0.06716151    0.00451067   2s(2).2p(6).3s_2S.3p(2)3P2_2P.3d(2)3F2_2G
-0.04376137    0.00191506   2s(2).2p(6).3s(2).3d(3)2G3_2G
-0.04016044    0.00161286   2s(2).2p(6).3s_2S.3p(2)1D2_2D.3d(2)1D2_2G
```

We see that the states are strongly mixed and that it is desirable to extend the MR. The size of the extended MR is a compromise between available computational resources and the desired accuracy of computed properties. Often an exploratory approach is needed. In this example we will somewhat ad hoc determine an MR from the LSJ-coupled CSFs with weights larger than 0.03.

Overview

1. Run `rcsfmr` for `Ni_even_n4.lsj.lbl` with a cut-off 0.03.
2. Use the output from `rcsfmr` as an input to `rcsfgenerate` with no excitations. Copy to `rcsfmr.inp`
3. Use the output from `rcsfmr` as an input to `rcsfgenerate` and do SD excitation from the extended MR. Copy to `rcsf.inp`
4. Run `rcsfinteract`

Program input

```
******************************************************************************
*         RUN RCSFMR FOR Ni_even_n4                                          *
******************************************************************************

>>rcsfmr

 RCSFMR
 This program reads the name.lsj.lbl file and extracts a
 set of MR configuartions that give rise to LSJ coupled
 CSFs with absolute weights larger than a specified cut-off
 Input file: namel.lsj.lbl
 Ouput is written to screen

 Name of state
>>Ni_even_n4
 Give cut-off for weight
>>0.03

 Configurations in the MR

2s(2,*)2p(6,*)3s(1,*)3p(4,*)
2s(2,*)2p(6,*)3s(2,*)3p(2,*)3d(1,*)
2s(2,*)2p(6,*)3s(1,*)3p(2,*)3d(2,*)
2s(2,*)2p(6,*)3p(4,*)3d(1,*)
2s(2,*)2p(6,*)3s(2,*)3d(3,*)
2s(2,*)2p(6,*)3s(2,*)3p(1,*)3d(1,*)4f(1,*)
2s(2,*)2p(6,*)3s(1,*)3p(3,*)4f(1,*)

******************************************************************************
*         RUN RCSFGENERATE USING THE OBTAINED CONFIGURATIONS FROM RCSFMR     *
*         BY REQUESTING ZERO EXCITATIONS WE WILL GET THE CSFs OF THE MR      *
```

```
******************************************************************************

>>rcsfgenerate

 RCSFGENERATE
 This program generates a list of CSFs

 Configurations should be entered in spectroscopic notation
 with occupation numbers and indications if orbitals are
 closed (c), inactive (i), active (*) or has a minimal
 occupation e.g. 1s(2,1)2s(2,*)
 Outputfiles: rcsf.out, rcsfgenerate.log

 Default, reverse, symmetry or user specified ordering? (*/r/s/u)
>>*

 Select core
        0: No core
        1: He (       1s(2)                  =  2 electrons)
        2: Ne ([He] + 2s(2)2p(6)             = 10 electrons)
        3: Ar ([Ne] + 3s(2)3p(6)             = 18 electrons)
        4: Kr ([Ar] + 3d(10)4s(2)4p(6)       = 36 electrons)
        5: Xe ([Kr] + 4d(10)5s(2)5p(6)       = 54 electrons)
        6: Rn ([Xe] + 4f(14)5d(10)6s(2)6p(6) = 86 electrons)
>>1

 Enter list of (maximum 100) configurations. End list with a blank line or an asterisk (*)

 Give configuration           1
>>2s(2,i)2p(6,i)3s(1,*)3p(4,*)
 Give configuration           2
>>2s(2,i)2p(6,i)3s(2,*)3p(2,*)3d(1,*)
 Give configuration           3
>>2s(2,i)2p(6,i)3s(1,*)3p(2,*)3d(2,*)
 Give configuration           4
>>2s(2,i)2p(6,i)3p(4,*)3d(1,*)
 Give configuration           5
>>2s(2,i)2p(6,i)3s(2,*)3d(3,*)
 Give configuration           6
>>2s(2,i)2p(6,i)3s(2,*)3p(1,*)3d(1,*)4f(1,*)
 Give configuration           7
>>2s(2,i)2p(6,i)3s(1,*)3p(3,*)4f(1,*)
 Give configuration           8
>>
 Give set of active orbitals, as defined by the highest principal quantum number
 per l-symmetry, in a comma delimited list in s,p,d etc order, e.g. 5s,4p,3d
>>4s,4p,4d,4f
 Resulting 2*J-number? lower, higher (J=1 -> 2*J=2 etc.)
>>1,9
 Number of excitations (if negative number e.g. -2, correlation
 orbitals will always be doubly occupied)
>>0
 Generate more lists ? (y/n)
>>n
```

```
.................

 Group CSFs into symmetry blocks

 5 blocks were created

       block  J/P           NCSF
           1  1/2+            61
           2  3/2+           104
           3  5/2+           116
           4  7/2+            96
           5  9/2+            67

*******************************************************************************
*         COPY FILES                                                          *
*******************************************************************************

>>cp rcsf.out rcsfmr.inp

*******************************************************************************
*         RUN RCSFGENERATE USING THE OBTAINED CONFIGURATIONS FROM RCSFMR      *
*         REQUEST TWO EXCITATIONS                                             *
*******************************************************************************

>>rcsfgenerate

 RCSFGENERATE
 This program generates a list of CSFs

 Configurations should be entered in spectroscopic notation
 with occupation numbers and indications if orbitals are
 closed (c), inactive (i), active (*) or has a minimal
 occupation e.g. 1s(2,1)2s(2,*)
 Outputfiles: rcsf.out, rcsfgenerate.log

 Default, reverse, symmetry or user specified ordering? (*/r/s/u)
>>*

 Select core
        0: No core
        1: He (       1s(2)                  =  2 electrons)
        2: Ne ([He] + 2s(2)2p(6)             = 10 electrons)
        3: Ar ([Ne] + 3s(2)3p(6)             = 18 electrons)
        4: Kr ([Ar] + 3d(10)4s(2)4p(6)       = 36 electrons)
        5: Xe ([Kr] + 4d(10)5s(2)5p(6)       = 54 electrons)
        6: Rn ([Xe] + 4f(14)5d(10)6s(2)6p(6) = 86 electrons)
>>1

 Enter list of (maximum 100) configurations. End list with a blank line or an asterisk (*)

 Give configuration           1
>>2s(2,i)2p(6,i)3s(1,*)3p(4,*)
 Give configuration           2
```

```
>>2s(2,i)2p(6,i)3s(2,*)3p(2,*)3d(1,*)
 Give configuration          3
>>2s(2,i)2p(6,i)3s(1,*)3p(2,*)3d(2,*)
 Give configuration          4
>>2s(2,i)2p(6,i)3p(4,*)3d(1,*)
 Give configuration          5
>>2s(2,i)2p(6,i)3s(2,*)3d(3,*)
 Give configuration          6
>>2s(2,i)2p(6,i)3s(2,*)3p(1,*)3d(1,*)4f(1,*)
 Give configuration          7
>>2s(2,i)2p(6,i)3s(1,*)3p(3,*)4f(1,*)
 Give configuration          8
>>
 Give set of active orbitals, as defined by the highest principal quantum number
 per l-symmetry, in a comma delimited list in s,p,d etc order, e.g. 5s,4p,3d
>>4s,4p,4d,4f
 Resulting 2*J-number? lower, higher (J=1 -> 2*J=2 etc.)
>>1,9
 Number of excitations (if negative number e.g. -2, correlation
 orbitals will always be doubly occupied)
>>2
 Generate more lists ? (y/n)
>>n

.................

 Group CSFs into symmetry blocks

 5 blocks were created

      block  J/P           NCSF
          1  1/2+          5061
          2  3/2+          8907
          3  5/2+         10810
          4  7/2+         10604
          5  9/2+          8889

*****************************************************************************
*         COPY FILES                                                        *
*****************************************************************************

>>cp rcsf.out rcsf.inp

*****************************************************************************
*         RUN RCSFINTERACT                                                  *
*****************************************************************************

 RCSFinteract: Determines all the CSFs (rcsf.inp) that interact
               with the CSFs in the multireference (rcsfmr.inp)
               (C)  Copyright by G. Gaigalas and Ch. F. Fischer
               (Fortran 95 version)               NIST  (2017).
               Input files: rcsfmr.inp, rcsf.inp
               Output file: rcsf.out
```

```
 Reduction based on Dirac-Coulomb (1) or
 Dirac-Coulomb-Breit (2) Hamiltonian?
2
 Loading Configuration Symmetry List File ...
 There are 16 relativistic subshells;
  Block    MR NCSF   Before NCSF   After NCSF
    1          61         5061         3563
    2         104         8907         6375
    3         116        10810         7639
    4          96        10604         7204
    5          67         8889         5764

 Wall time:
        5 seconds

 Finish Date and Time:
   Date (Yr/Mon/Day): 2018/05/17
   Time (Hr/Min/Sec): 00/54/36.490
   Zone: +0200

 RCSFinteract: Execution complete.
```

The same procedure can be applied to `Ni_even_n4.lsj.lbl`.

7.7 Seventh example: restarting rci

Follow the fifth example up to the `rci` calculation for `Ni_even_n4`. During the `rci` calculation the Hamiltonian matrix elements, in sparse representation, are successively written to the file `rci.res`. If the calculation stalls at some point the `rci` program can be restarted. During a restart all radial integrals are recomputed, and then the computation starts with computing the matrix elements following the last matrix element that was saved to `rci.res`. In this example we assume that the `rci` calculation for `Ni_even_n4` stalled in the middle of block 3 and we show how to make a restart.

Overview

1. Run `rci` for `Ni_even_n4` (run assumed to stall in the middle of block 3)
2. Use the restart file `rci.res` to restart the `rci` run.

Program input

```
*****************************************************************************
*         RUN RCI TO INCLUDE TRANSVERSE PHOTON INTERACTION AND QED EFFECTS   *
*         INPUT FILES : isodata, Ni_even_n4.c, Ni_even_n4.w, rci.res         *
*         OUTPUT FILES: Ni_even_n4.cm, Ni_even_n4.csum, Ni_even_n4.clog,     *
*         rci.res                                                            *
*         This is a restart that reads the rci.res file                      *
*****************************************************************************
>>rci
 RCI
 This is the configuration interaction program
```

```
 Input file:  isodata, name.c, name.w
 Outputfiles: name.cm, name.csum, name.clog
              rci.res (can be used for restart)

 Default settings?
>>n
 Name of state:
>>Ni_even_n4
 Block            1 ,  ncf =         1047
 Block            2 ,  ncf =         1862
 Block            3 ,  ncf =         2112
 Block            4 ,  ncf =         1537
 Block            5 ,  ncf =          801
 Loading CSF file ... Header only
 There are/is           16  relativistic subshells;
 Restarting RCI90 ?
>>y
 Calling lodres ...
 Estimate contributions from the self-energy?
>>y
 There are            5  blocks  (block   J/Parity   NCF):
  1  1/2+    1047       2  3/2+    1862       3  5/2+    2112       4  7/2+    1537
  5  9/2+     801

 Enter ASF serial numbers for each block
 Block            1    ncf =         1047  id =  1/2+
>>1-8
 Block            2    ncf =         1862  id =  3/2+
>>1-11
 Block            3    ncf =         2112  id =  5/2+
>>1-10
 Block            4    ncf =         1537  id =  7/2+
>>1-5
 Block            5    ncf =          801  id =  9/2+
>>1-2
 Calling STRSUM...
 Calling FACTT...
 Calling GENINTRK...
 Allocating space for         6071  Rk integrals
 Calling GENINTBREIT1...
 Computing       53106  Breit integrals of type 1
 Calling GENINTBREIT2...
 Computing       26494  Breit integrals of type 2
 Calling MATRIX...
 Loading CSF File for block            1
 There are         1047  relativistic CSFs... load complete;
 Entering QED ...
        1047  (total         1047 ) rows read from .res
 LAPACK routine DSPEVX selected for eigenvalue problem.
 RCI92 MIXing coefficients File generated.
 Loading CSF File for block            2
 There are         1862  relativistic CSFs... load complete;
 Entering QED ...
        1862  (total         1862 ) rows read from .res
```

```
LAPACK routine DSPEVX selected for eigenvalue problem.
RCI92 MIXing coefficients File generated.
Loading CSF File for block           3
There are         2112  relativistic CSFs... load complete;
Entering QED ...
        739  (total         2112 ) rows read from .res
Calling setham ...
Row          800 :           283  nonzero elements;  block =            3
Row          900 :           258  nonzero elements;  block =            3
Row         1000 :           393  nonzero elements;  block =            3
Row         1100 :           224  nonzero elements;  block =            3
Row         1200 :           250  nonzero elements;  block =            3
Row         1300 :           104  nonzero elements;  block =            3
Row         1400 :           297  nonzero elements;  block =            3
Row         1500 :           375  nonzero elements;  block =            3
Row         1600 :           224  nonzero elements;  block =            3
Row         1700 :           106  nonzero elements;  block =            3
Row         1800 :           141  nonzero elements;  block =            3
Row         1900 :           128  nonzero elements;  block =            3
Row         2000 :           208  nonzero elements;  block =            3
Row         2100 :           111  nonzero elements;  block =            3
Row         2111 :            74  nonzero elements;  block =            3
Row         2112 :           113  nonzero elements;  block =            3
nelmnt =               436242
 Sparse - Memory,  iniest2
RCI92 MIXing coefficients File generated.
Loading CSF File for block           4
There are         1537  relativistic CSFs... load complete;
Entering QED ...
          0  (total         1537 ) rows read from .res
Calling setham ...
Row            1 :             1  nonzero elements;  block =            4
Row          100 :            67  nonzero elements;  block =            4
Row          200 :            62  nonzero elements;  block =            4
Row          300 :           100  nonzero elements;  block =            4
Row          400 :            83  nonzero elements;  block =            4
Row          500 :           139  nonzero elements;  block =            4
Row          600 :           223  nonzero elements;  block =            4
Row          700 :           234  nonzero elements;  block =            4
Row          800 :           321  nonzero elements;  block =            4
Row          900 :           341  nonzero elements;  block =            4
Row         1000 :           289  nonzero elements;  block =            4
Row         1100 :           337  nonzero elements;  block =            4
Row         1200 :           311  nonzero elements;  block =            4
Row         1300 :           192  nonzero elements;  block =            4
Row         1400 :           191  nonzero elements;  block =            4
Row         1500 :           282  nonzero elements;  block =            4
Row         1536 :           150  nonzero elements;  block =            4
Row         1537 :           176  nonzero elements;  block =            4
LAPACK routine DSPEVX selected for eigenvalue problem.
RCI92 MIXing coefficients File generated.
Loading CSF File for block           5
There are          801  relativistic CSFs... load complete;
Entering QED ...
```

```
          0  (total          801 ) rows read from .res
 Calling setham ...
 Row           1 :            1  nonzero elements;  block =            5
 Row         100 :           13  nonzero elements;  block =            5
 Row         200 :           81  nonzero elements;  block =            5
 Row         300 :           48  nonzero elements;  block =            5
 Row         400 :          183  nonzero elements;  block =            5
 Row         500 :          207  nonzero elements;  block =            5
 Row         600 :          118  nonzero elements;  block =            5
 Row         700 :          162  nonzero elements;  block =            5
 Row         800 :          163  nonzero elements;  block =            5
 Row         801 :          184  nonzero elements;  block =            5
 LAPACK routine DSPEVX selected for eigenvalue problem.
 RCI92 MIXing coefficients File generated.


 Finish time, Statistics


 Wall time:
       54 seconds

 Finish Date and Time:
   Date (Yr/Mon/Day): 2018/07/20
   Time (Hr/Min/Sec): 23/40/40.589
   Zone: +0200

 RCI: Execution complete.
```

During the restart all matrix elements for blocks 1 and 2 were read from `rci.res`. For block 3 matrix elements up to row 739 were read and the restarted computation carries on from this point. Transforming from jj- to LSJ coupling and displaying energies with `rlevels` shows that all the energies from the restarted `rci` calculation are identical to the ones in the fifth example.

Chapter 8

Running the tools

8.1 Splitting a list of CSFs

When using scripts, see case studies in the later chapters, it is often convenient to generate a list of CSFs based on a large active set of orbitals and then split this list into a number of lists with CSFs that can be formed from different subsets of active orbitals.

Overview

1. Generate a list of CSFs for $1s^22p\ ^2P_{1/2,3/2}$ by allowing SDT excitations (CAS expansion) to the active set $n = 5$.

2. Split into three lists with CSFs that can be formed by the active sets $n = 3$, $n = 4$, and $n = 5$.

Program input

In the test-runs input marked by >> or >>3, for example, indicates that the user should input 3 and then strike the return key. When >> is followed by blanks just strike the return key.

```
*****************************************************************************
* RUN RCSFGENERATE TO GENERATE LIST FOR 1s(2)2p                             *
* OUTPUT FILES: rcsfgenerate.log, rcsf.out                                  *
*****************************************************************************

>>rcsfgenerate

 RCSFGENERATE
 This program creates a list of CSFs
 Configurations should be entered in spectroscopic notation
 with occupation numbers and indications if orbitals are
 closed (c), inactive (i), active (*) or has a minimal
 occupation e.g. 1s(2,1)2s(2,*)
 Outputfiles: rcsf.out, rcsfgenerate.log

 Default, reverse, symmetry or user specified ordering? (*/r/s/u)
>>*

 Select core
        0: No core
        1: He (       1s(2)                  =  2 electrons)
```

```
        2: Ne ([He] + 2s(2)2p(6)             = 10 electrons)
        3: Ar ([Ne] + 3s(2)3p(6)             = 18 electrons)
        4: Kr ([Ar] + 3d(10)4s(2)4p(6)       = 36 electrons)
        5: Xe ([Kr] + 4d(10)5s(2)5p(6)       = 54 electrons)
        6: Rn ([Xe] + 4f(14)5d(10)6s(2)6p(6) = 86 electrons)
>>0
 Enter list of (maximum 100) configurations. End list with a blank line or an asterisk (*)

 Give configuration  1
>>1s(2,*)2p(1,*)
 Give configuration  2
>>
 Give set of active orbitals, as defined by the highest principal quantum number
 per l-symmetry, in a comma delimited list in s,p,d etc order, e.g. 5s,4p,3d
>>5s,5p,5d,5f,5g
 Resulting 2*J-number? lower, higher (J=1 -> 2*J=2 etc.)
>>1,3
 Number of excitations (if negative number e.g. -2, correlation
 orbitals will always be doubly occupied)
>>3
 Generate more lists ? (y/n)
>>n

  ...

 2  blocks were created

      block  J/P            NCSF
          1  1/2-           1454
          2  3/2-           2478

******************************************************************************
* COPY FILES                                                                 *
******************************************************************************

>>cp rcsf.out odd.c

******************************************************************************
* RUN RCSFSPLIT AND SPLIT INTO LISTS CORRESPONDING TO ACTIVE SETS            *
* n = 3, n = 4, n = 5                                                        *
* INPUT FILE : odd.c                                                         *
* OUTPUT FILE: odd3.c, odd4.c, odd5.c                                        *
******************************************************************************

>>rcsfsplit

 RCSFSPLIT
 Splits a list name.c of CSFs into a number of lists with CSFs that
 can be formed from different sets of active orbitals.
 Orbital sets are specified by giving the highest principal quantum number
 per l-symmetry, in a comma delimited list in s,p,d etc order, e.g. 5s,4p,3d
 Input file: name.c
 Output files: namelabel1.c, namelabel2.c, ...
```

```
 Name of state
>>odd
 Number of orbital sets
>>3
 Orbital set           1
 Give set of active orbitals, as defined by the highest principal quantum number
 per l-symmetry, in a comma delimited list in s,p,d etc order, e.g. 5s,4p,3d
>>3s,3p,3d
 Give file label
 >>3
 Orbital set           2
 Give set of active orbitals, as defined by the highest principal quantum number
 per l-symmetry, in a comma delimited list in s,p,d etc order, e.g. 5s,4p,3d
>>4s,4p,4d,4f
 Give file label
>>4
 Orbital set           3
 Give set of active orbitals, as defined by the highest principal quantum number
 per l-symmetry, in a comma delimited list in s,p,d etc order, e.g. 5s,4p,3d
>>5s,5p,5d,5f,5g
 Give file label
>>5

 List odd3.c based on orbital set           1
 contains         186 CSFs

 List odd4.c based on orbital set           2
 contains        1048 CSFs

 List odd5.c based on orbital set           3
 contains        3932 CSFs
```

8.2 Extracting and condensing

The program `rmixextract` extracts and displays, for each state in a calculation, the CSFs with absolute values of the mixing coefficients larger than a certain cut-off. The program `rmixaccumulate` works somewhat differently, it accumulates the CSFs that account for a given percentage of the wave functions in a calculation. The algorithm is described as follows:

1. Start from a calculation targeting one or more states, thus start from a number of ASFs

$$ASF_1 : \Psi(\gamma_1 PJ) = \sum_{i=1}^{N} c_i^1 \Phi(\gamma_i PJ)$$
$$\dots$$
$$ASF_M : \Psi(\gamma_M PJ) = \sum_{i=1}^{N} c_i^M \Phi(\gamma_i PJ)$$

2. For i from 1 to N compute

$$s_i = (c_i^1)^2 + (c_i^2)^2 + \dots + (c_i^M)^2.$$

3. Sort $s_1, s_2, \dots, s_N$ in descending order.

4. Accumulate until a specified fraction of the total squared weight

$$M = s_1 + \ldots + s_M = \sum_{i,j} (c_i^j)^2$$

is attained.

Overview

1. Extract and display CSFs from the RCI wave functions 2p_3 in the first example.
2. Accumulate CSFs from the RCI wave functions 2p_3 in the first example accounting for 99.99 % of the total wave function content.

Program input

In the test-runs input marked by >> or >>3, for example, indicates that the user should input 3 and then strike the return key. When >> is followed by blanks just strike the return key.

```
*****************************************************************************
*         MAKE SURE YOU ARE IN THE DIRECTORY WHERE THE FILES FROM THE FIRST   *
*         EXAMPLE ARE                                                          *
*****************************************************************************


*****************************************************************************
*         RUN RMIXEXTRACT TO EXTRACT AND PRINT MIXING COEFFICIENTS FOR 2p_3   *
*         CORRESPONDING CSFs WRITTEN to rcsf.out                              *
*         INPUT FILES: 2p_3.c, 2p_3.cm                                        *
*         OUTPUT FILE: rcsf.out                                               *
*****************************************************************************

>>rmixextract

 RMIXEXTRACT
 Extract and prints mixing coefficient above a
 cut-off. Corresponding CSFs written to screen and
 to rcsf.out
 Input files: name.c, name.(c)m
 Output file: rcsf.out

 Name of state
>>2p_3
 Mixing coefficients from CI calc. ?
>>y
 Enter the cut-off value for the coefficients [0--1]
>>0.01
 Sort extracted CSFs according to mixingcoeffcients? (y/n)
>>y

  nblock =    2    ncftot =        186    nw =    9    nelec =    3

 ===========================================================================
  nb =    1  ncfblk =        76  nevblk =    1  2J+1 =    2  parity =   -1
  nb =    1  ncfblk =        76  nevblk =    1  2J+1 =    2  parity =   -1
```

```
==========================================================================
Average Energy =    8.4326213215121353         ncf_reduced =           5

Energy =   -7.4042610340834774         Coefficients and CSF :

          1   0.998449
  1s ( 2)  2p-( 1)
              1/2
               1/2-
          2  -0.034328
  2p-( 1)  3s ( 2)
      1/2
               1/2-
          3   0.032596
  2p-( 1)  3p ( 2)
      1/2        0
               1/2-
          4   0.023063
  2p-( 1)  3p-( 2)
      1/2
               1/2-
          5   0.010751
  2s ( 1)  2p-( 1)  3s ( 1)
      1/2      1/2      1/2
                    1    1/2-
==========================================================================
 nb =    2  ncfblk =       110  nevblk =    1  2J+1 =    4  parity =   -1
 nb =    2  ncfblk =       110  nevblk =    1  2J+1 =    4  parity =   -1
==========================================================================
Average Energy =    9.6081890770738791         ncf_reduced =           4

Energy =   -7.4042597216869357         Coefficients and CSF :

          1   0.998449
  1s ( 2)  2p ( 1)
              3/2
               3/2-
          2  -0.034328
  2p ( 1)  3s ( 2)
      3/2
               3/2-
          3   0.032590
  2p ( 1)  3p ( 2)
      3/2        0
               3/2-
          4   0.023072
  2p ( 1)  3p-( 2)
      3/2
               3/2-
RMIXEXTRACT: Execution complete.


*****************************************************************************
*         RUN RMIXACCUMULATE TO ACCUMULATE CSFs CONTRIBUTING TO 99.99\%       *
```

```
*         OF THE TOTAL WAVE FUNCTION CONTENT. CORRESPONDING CSFs WRITTEN TO   *
*         rcsf.out                                                             *
*         INPUT FILES: 2p_3.c, 2p_3.cm                                         *
*         OUTPUT FILE: rcsf.out                                                *
*******************************************************************************

>>rmixaccumulate

 ***************************************************************************
 Welcome to program rmixaccumulate

 The program accumulates dominating CSFs by mixing coefficients up to
 a user defined fraction of the total wave function.
 The CSFs in the output list can be sorted by mixing coefficents
 to provide better initial estimates for the subsequent diagonalisation.
 of CI matrices.

 Input files: <state>.(c)m, <state>.c
 Output file: rcsf.out

                                              J. Ekman & P. Jonsson Feb 2016
 ***************************************************************************

 Give name of the state:
>>2p_3
 Expansion coefficients resulting from CI calculation (y/n)?
>>y
 Fraction of total wave function [0-1] to be included in reduced list:
>>0.9999
 CSFs in output file sorted by mixing coefficients (y/n)?
>>y

 Block data read from mixing file
        block        ncf         nev        2j+1          parity
          1           76           1           2          -1
          2          110           1           4          -1

 Number of CSF:s written to rcsf.out
        block        ncf
          1            7
          2            7

 WARNING! Not all peel subshells are occupied in the output CSF list:
 Remove the following peel subshells:
  3d-
```

For each symmetry block only 7 CSFs are needed to account for 99.99% of the total wave function content. The extracted list of CSFs is shown below.

```
Core subshells:

Peel subshells:
  1s   2s   2p-  2p   3s   3p-  3p   3d-  3d
CSF(s):
```

```
  1s ( 2)  2p-( 1)
               1/2
                1/2-
  2p-( 1)  3s ( 2)
      1/2
                1/2-
  2p-( 1)  3p ( 2)
      1/2        0
                1/2-
  2p-( 1)  3p-( 2)
      1/2
                1/2-
  2s ( 1)  2p-( 1)  3s ( 1)
      1/2      1/2      1/2
                    1    1/2-
  2p-( 1)  3d ( 2)
      1/2        0
                1/2-
  2s ( 1)  2p-( 1)  3s ( 1)
      1/2      1/2      1/2
                    0    1/2-
 *
  1s ( 2)  2p ( 1)
               3/2
                3/2-
  2p ( 1)  3s ( 2)
      3/2
                3/2-
  2p ( 1)  3p ( 2)
      3/2        0
                3/2-
  2p ( 1)  3p-( 2)
      3/2
                3/2-
  2s ( 1)  2p ( 1)  3s ( 1)
      1/2      3/2      1/2
                    2    3/2-
  2s ( 1)  2p ( 1)  3s ( 1)
      1/2      3/2      1/2
                    1    3/2-
  2p ( 1)  3d ( 2)
      3/2        0
                3/2-
```

Indeed we see that the $3d-$ orbital is not present in any of the extracted CSFs. If we are going to use the list of extracted CSFs for some other purposes we should remove $3d-$ from the list of peel subshells. In chapter 14 we discuss the use of `rmixaccumulate` for handling large CSF expansions.

8.3 Extracting radial orbitals for plotting

The program `rwfnplot` extracts specified orbitals from a binary radial orbital file `name.w` and generates a Matlab/GNU Octave M-file (and also an Xmgrace file for convenience) that plots the large components of the radial orbitals as functions of r or $\sqrt{r}$. The Matlab/GNU Octave M-file is easy to edit to modify the appearances of the plots. By editing the Matlab/GNU Octave M-file also the small component of the radial orbitals can be plotted. In this example we extract the $1s$, $2s$ and $2p$ orbitals from `2s_2p_DF.w` generated in the example 1.

Program input

In the test-runs input marked by >> or >>3, for example, indicates that the user should input 3 and then strike the return key. When >> is followed by blanks just strike the return key.

```
***************************************************************************
*        MAKE SURE YOU ARE IN THE DIRECTORY WHERE THE FILES FROM THE FIRST   *
*        EXAMPLE ARE                                                         *
***************************************************************************


***************************************************************************
*       RUN RWFNPLOT TO PLOT THE 1s, 2s, 2p  ORBITALS                        *
*       INPUT FILE: 2s_2p_DF.w                                               *
*       OUTPUT FILE: octave_2s_2p_DF.m  (Matlab/GNU Octave M-file)           *
***************************************************************************

>>rwfnplot

 RWFNPLOT
 Program to generate Matlab/GNU Octave and
 Xmgrace files that plot radial orbitals
 Input file:  name.w
 Output files: octave_name.m, xmgrace_name.agr

 To plot orbital: press enter
 To remove orbital: type "d" or "D" and press enter

                             Jorgen Ekman Jun 2015

 Name of state:
>>2s_2p_DF

 To have r on x-axis: type "y" otherwise "n" for sqrt(r)
>>n

   1s  =
>>
   2s  =
>>
   2p- =
>>d
   2p  =
>>
  FINISHED .....


***************************************************************************
*       START GNU OCTAVE BY TYPING octave                                    *
***************************************************************************
```

```
>>octave

******************************************************************************
*        AT THE GNU OCTAVE COMMAND LINE octave:1> INVOKE THE M-FILE          *
*        NOTE THAT ONLY THE FILE NAME AND NOT THE EXTENSION SHOULD BE GIVEN  *
******************************************************************************

octave:1> octave_2s_2p_DF
```

Executed in Matlab or GNU Octave, the `octave_2s_2p_DF.m` M-file gives the radial orbitals to the left in figure 1. To the right we have plotted converted MCHF orbitals. In figure 2 we have, for comparison, plotted the Thomas-Fermi and the screened hydrogenic initial estimates of the radial orbitals.[1] By comparing figure 1 and figure 2 one sees that Thomas-Fermi gives better initial estimates of the radial orbitals than do the screened hydrogenic estimates. Best initial estimates, however, are obtained from converted HF or MCHF orbitals.

[1]To save an Octave figure to file click on `File` and `Save As` in the upper left corner of the graphical window. By saving the figures in pdf format they can easily be imported in a LaTeX document.

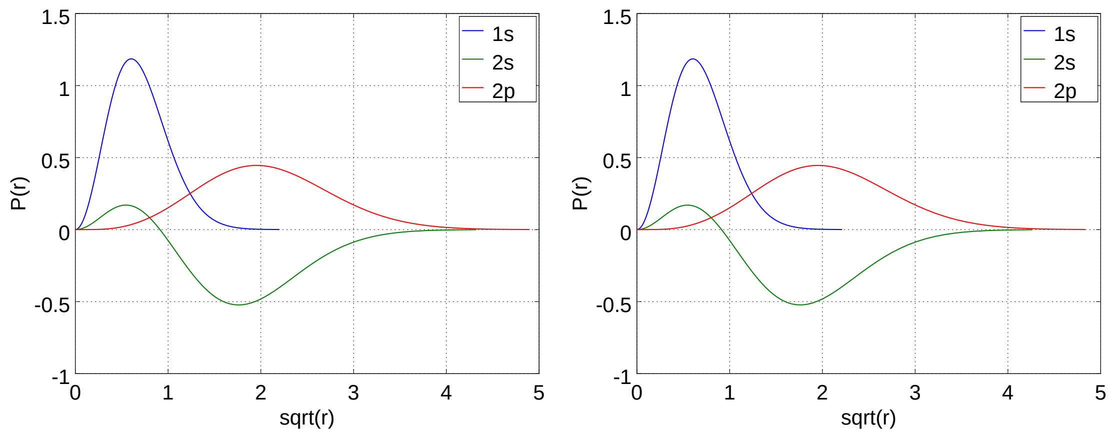

Figure 8.1: Left: plot of the large components of the $1s$, $2s$ and $2p_{3/2}$ orbitals in Li I. Right: large components from converted MCHF orbitals.

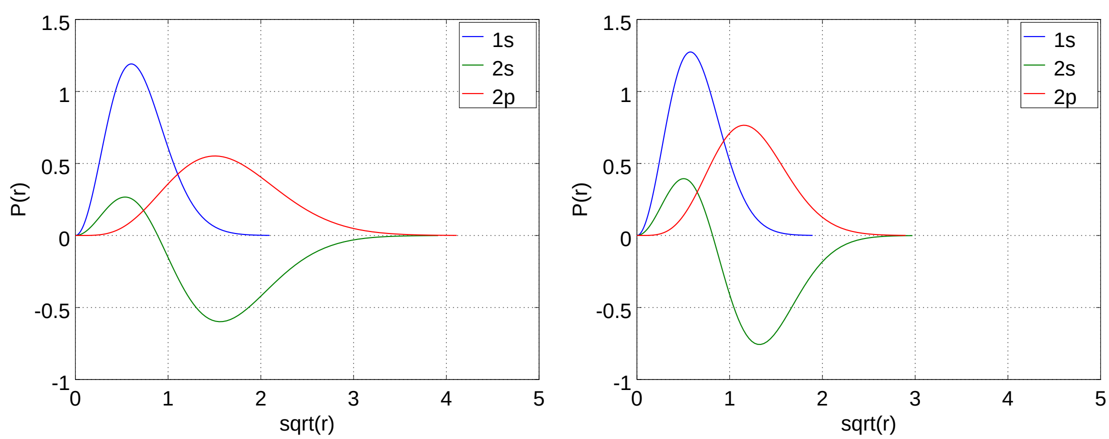

Figure 8.2: Left: plot of the large components of the $1s$, $2s$ and $2p_{3/2}$ orbitals in Li I from Thomas-Fermi estimates: Right: large components from screened hydrogenic estimates.

8.4 Output from rhfs with LSJ-labels

The program `rhfs_lsj` reads data from `name.(c)h`. Using data available from `name.lsj.lbl` output is produced with LSJ-labels.

Program input

In the test-runs input marked by >> or >>3, for example, indicates that the user should input 3 and then strike the return key. When >> is followed by blanks just strike the return key.

```
******************************************************************************
*         MAKE SURE YOU ARE IN THE DIRECTORY WHERE THE FILES FROM THE FIRST   *
*         EXAMPLE ARE                                                         *
******************************************************************************


******************************************************************************
*        RUN RHFS_LSJ TO OBTAIN OUTPUT WITH LSJ LABELS                        *
*        INPUT FILES: 2p_3.ch, 2p_3.lsj.lbl                                   *
*        OUTPUT FILE: 2p_3.chlsj                                              *
******************************************************************************

>>rhfs_lsj

 RHFS_LSJ
 This program prints output from the rhfs program
 using LSJ lables. Output can be energy sorted
 Input files: name.(c)h, name.lsj.lbl
 Output file: name.(c)hlsj

 Name of the state
>>2p_3
 Hfs data from a CI calc?
>>y
 Energy sorted output?
>>y
```

The produced file `2p_3.chlsj` is shown below.

```
 Nuclear spin                          1.500000000000000D+00 au
 Nuclear magnetic dipole moment        3.256426800000000D+00 n.m.
 Nuclear electric quadrupole moment  -4.000000000000000D-02 barns

        Energy        State    J   P       A(MHz)       B(MHz)         gJ
    -7.4042610  1s(2).2p_2P  1/2   -       4.482D+01      -0.000D+00    6.666573D-01
    -7.4042597  1s(2).2p_2P  3/2   -      -3.538D+00      -1.773D-01    1.333325D+00
```

8.5 Producing hfs tables in LaTeX

The program `rtabhfs` produces LaTeX files from output files from `rhfs_lsj`. Before running `rtabhfs` run `rhfs_lsj` on `2s_3` so that `2s_3.chlsj` is available. Now we will invoke `rtabhfs` to make a LaTeX file with hyperfine interaction constants and Landé g_J-factors for all states in the above files.

```
******************************************************************************
```

```
*        RUN RTABHFS TO PRODUCE LATEX TABLE                                  *
*        INPUT FILES: 2s_3.chlsj 2p_3.chlsj                                  *
*        OUTPUT FILE: hfs.tex                                                *
******************************************************************************

>>rtabhfs

  RTABHFS
  This program reads the output from rhfs_lsj and
  produces LaTeX tables of hfs data
  Input files: name.(c)hlsj produced by rhfs_lsj
  Output file: hfs.tex

  How many HFS files ?
>>2
  Full name of HFS file           1
>>2s_3.chlsj
  Full name of HFS file           2
>>2p_3.chlsj

  Inspect the name.(c)hlsj files and
  determine how many positions should be skipped in
  the string that determines the label. For example
  if the string is 1s(2).2s_2S.2p(2)3P2_4P and 1s(2)
  is a core then you want to skip 1s(2). i.e. 6
  positions

  How many positions should be skipped?
>>0
```

The produced LaTeX file `hfs.tex` is shown below

```
\documentclass[10pt]{article}
\usepackage{longtable}
\begin{document}
\begin{longtable}{lrrrr} \hline
State & $E$(a.u.) & $A$(MHz) & $B$(MHz) &
$g_J$ \\ \hline
 $1s^2 \,2s~^2S_{ 1/2}$ &     -7.4719740 &     3.885D+02 &    -0.000D+00 &   1.999985D+00 \\
 $1s^2 \,2p~^2P_{ 1/2}^o$ &     -7.4042610 &     4.482D+01 &    -0.000D+00 &   6.666573D-01 \\
 $1s^2 \,2p~^2P_{ 3/2}^o$ &     -7.4042597 &    -3.538D+00 &    -1.773D-01 &   1.333325D+00 \\
\hline\\
\caption{Hyperfine interaction constants}
\end{longtable}
\end{document}
```

The above LaTeX file generates the following table:

State	E(a.u.)	A(MHz)	B(MHz)	g_J
$1s^2\,2s\;^2S_{1/2}$	-7.4719740	3.885D+02	-0.000D+00	1.999985D+00
$1s^2\,2p\;^2P^o_{1/2}$	-7.4042610	4.482D+01	-0.000D+00	6.666573D-01
$1s^2\,2p\;^2P^o_{3/2}$	-7.4042597	-3.538D+00	-1.773D-01	1.333325D+00

Table 8.1: Hyperfine interaction constants

8.6 Producing energy tables in LaTeX

The program `rtablevels` produces LaTeX and ASCII files from output files from `rlevels`. As a first simple example we make a table of the energies from the calculations in example 1. It is possible to read many files from `rlevels` and make, for example, tables that show convergence of energy levels with respect to the increasing active set (see section 10.4).

```
*******************************************************************************
*         MAKE SURE YOU ARE IN THE DIRECTORY WHERE THE FILES FROM THE FIRST   *
*         EXAMPLE ARE                                                         *
*******************************************************************************

*******************************************************************************
*         RUN RLEVELS TO OBTAIN AN ENERGY FILE WITH LSJ LABELS                *
*******************************************************************************

>>rlevels 2s_3.cm 2p_3.cm > energy3

*******************************************************************************
*         RUN RTABLEVELS TO PRODUCE LATEX AND ASCII TABLES                    *
*         INPUT FILE: energy3                                                 *
*         OUTPUT FILES: energytablelatex.tex, energytableascii.txt            *
*******************************************************************************

>>rtablevels

  RTABLEVELS
  Makes LaTeX and ASCII tables of energy files produced by
  rlevels (in ljs format)
  Multiple energy files can be used as input
  Energies from file 1 fills column 1, energies from file 2
  fills column 2 etc.  Checks are done to see if the labels
  if the labels in the files are consistent
  Input file: name1, name2, ...
  Output files: energylabellatex.tex, energylabelascii.txt

  Inspect energy files and determine how many positions
  should be skipped in the string that determines the label
  e.g. if the string is 1s(2).2s_2S.2p(2)3P2_4P and 1s(2) is a core
  then you would like to skip 1s(2). i.e. 6 positions and determine
  the label from 2s_2S.2p(2)3P2_4P

  How many positions should be skipped?
>>0
 Give the number of energy files from rlevels
>>1
 Name of file 1
>>energy3
```

The produced file `energytablelatex.tex` is shown below

```
\documentclass[10pt]{article}
\usepackage{longtable}
\begin{document}
\begin{longtable}{lr} \hline
```

```
$1s^2 \,2s~^2S_{  1/2}$   &         0    \\
$1s^2 \,2p~^2P_{  1/2}^o$ &     14861    \\
$1s^2 \,2p~^2P_{  3/2}^o$ &     14861    \\
\hline\\
\caption{Energies from the files energy3,}
\end{longtable}
\end{document}
```

8.7 Producing E1 transition tables in LaTeX

The program `rtabtransE1` produces LaTeX and ASCII files from the `nam1.name2.(c)t.lsj` output file from `rtransition` (E1 transitions only). As an example we make a table of the transition data from the `2s_3.2p_3.ct.lsj` file in example 1.

```
******************************************************************************
*         MAKE SURE YOU ARE IN THE DIRECTORY WHERE THE FILES FROM THE FIRST   *
*         EXAMPLE ARE                                                          *
******************************************************************************

******************************************************************************
*         RUN RTABTRANSE1 TO PRODUCE LATEX AND ASCII TABLES                    *
*         INPUT FILE: 2s_3.2p_3.ct.lsj                                         *
*         OUTPUT FILES: transitiontable.tex, transitiontableascii.txt          *
******************************************************************************

>>rtabtransE1

  RTABTRANSE1
  Makes LaTeX tables of transition data from transition files
  name1.name2.ct.lsj
  Input file: name1.name2.ct.lsj
  Output file: transitiontable.tex

  Specify table format
  (1). Lower & Upper & Energy diff. & wavelength & S & gf & A & dT
  (2). Lower & Upper & Energy diff. & wavelength & gf & A & dT
  (3). Lower & Upper & Energy diff. & wavelength & gf & A
  (4). Lower & Upper & Energy diff. & S & gf & A & dT
  (5). Lower & Upper & Energy diff. & gf & A & dT
  (6). Lower & Upper & Energy diff. & gf & A
>>5
  Inspect the name1.name2.ct.lsj file and determine how many positions
  should be skipped in the string that determines the label
  e.g. if the string is 1s(2).2s_2S.2p(2)3P2_4P and 1s(2) is a core
  then you would like to skip 1s(2). i.e. 6 positions and determine
  the label from 2s_2S.2p(2)3P2_4P

  How many positions should be skipped?
>>0
  Name of file
>>2s_3.2p_3.ct.lsj
```

The produced file `transitiontable.tex` is shown below

```
\documentclass[10pt]{article}
\usepackage{longtable}
\begin{document}
\begin{longtable}{llrrrr}
 Lower state & Upper state & $\Delta E$ & $gf$ & $A$ & $dT$ \\ \hline
$1s^2 \,2s~~2S_{1/2}$ & $1s^2 \,2p~~2P_{1/2}$ &    14861 & 5.087D-01 & 3.747D+07 &   0.017\\
$1s^2 \,2s~~2S_{1/2}$ & $1s^2 \,2p~~2P_{3/2}$ &    14861 & 1.017D+00 & 3.747D+07 &   0.017\\
\hline\\
\caption{Transition data from the file 2s_3.2p_3.ct.lsj}
\end{longtable}
\end{document}
```

The above LaTeX file generates the following table:

Lower state	Upper state	ΔE	gf	A	dT
$1s^2\,2s\ ^2S_{1/2}$	$1s^2\,2p\ ^2P_{1/2}$	14861	5.087D−01	3.747D+07	0.017
$1s^2\,2s\ ^2S_{1/2}$	$1s^2\,2p\ ^2P_{3/2}$	14861	1.017D+00	3.747D+07	0.017

Table 8.2: Transition data from the file `2s_3.2p_3.ct.lsj`

8.8 Handling levels with the same quantum labels

For more complex systems it sometimes happens that two levels have the same dominating LSJ term. The two levels will then get the same quantum labels in the output from `rlevels` and `rtransition`. The user of the program thus needs to pay attention to this and make sure that the labels are unique. If two or more levels have the same quantum labels, open the `name.lsj.lbl` file and edit it so that all levels have unique quantum labels. After having done these changes in the label file rerun `rlevels` and `rtransition` and the changes will be reflected in the output files. Subsequent runs of `rtablevels` and `rtabtransE1` will use the new quantum labels. Alternatively, to avoid problems with quantum states having the same label use the unique label option of `jj2lsj` as described in section 7.5. If the user does not want to edit the `name.lsj.lbl` file manually to get unique quantum labels or use the unique label option of `jj2lsj` he or she can still run `rtablevels` and `rtabtrans1` (to be discussed in the next section). These two programs identify all cases with labels that are not unique and resolve them by adding indices a, b, c etc. at the end of the quantum labels so that the labels are unique.

8.9 Producing transition and lifetime tables in LaTeX

The programs `rtabtrans1` and `rtabtrans2` produce a LaTeX transition table file and a lifetime file from the `*.(c)t` output file from `rtransition`. In case of several runs for different multipoles with `rtransition` all the different `*.(c)t` output files can be concatenated before processing. For example: if we have one file `name1.name2.ct` with transition data from a run for E1 transitions between states in the two files `name1` and `name2` and one file `name1.name1.ct` from a run for M1 transitions between states in the file `name1` the transition files can be concatenated (any name can be used for the concatenated file)

```
cat name1.name2.ct name1.name1.ct > E1M1.ct
```

By processing the concatenated file `E1M1.ct` a LaTeX transition table is produced for all E1 and M1 transitions.

As an example we make a table of the transition data from the `even4.odd4.ct` file in the fourth example. In addition we produce a lifetime table. More accurate values of the lifetime, require that we also include M1, E2 transitions between states of the same parity. We neglect this for simplicity. First we create energy label data for all the levels involved in the transitions. These data are obtained by processing the `even4.cm` and `odd4.ct` files.

```
******************************************************************************
*         MAKE SURE YOU ARE IN THE DIRECTORY WHERE THE FILES FROM THE FOURTH  *
*         EXAMPLE ARE                                                         *
******************************************************************************

******************************************************************************
*         RUN RTABTRANS1 TO PRODUCE AN ENERGY LABEL FILE NEEDED FOR           *
*         FURTHER PROCESSING WITH RTABTRANS2                                  *
*         INPUT FILE: even4.cm, odd4.cm                                       *
*         OUTPUT FILES: energylabel.latex                                     *
******************************************************************************

>>rtabtrans1

  RTABTRANS1
  This program creates a file energylabel that is
  used by RTABTRANS2 to produce LaTeX tables of
  transition data
  Input files: mixing coefficient files
       name1.(c)m, name2.(c)m,.... for the wave-
       functions that are used to compute the
       transition data
  Output file: energylabel.latex(ascii)

  Type the full input file name, one for each line (NULL to terminate)

  File name ?
>>even4.cm
  File name ?
>>odd4.cm
  File name ?
>>

  Inspect the labels of the states and
  determine how many positions should be skipped in
  the string that determines the label. For example
  if all the states have a common core 1s(2) in the
  label then 6 positions should be skipped

  How many positions should be skipped?
>>12
  Output labels in LaTeX or ASCII format (0/1)?
>>0
  Energy label data written to file
  energylabel.latex(ascii)
```

The produced file `energylabel.latex` is (slightly edited to fit the page) shown below

```
 nblock =           4   ncftot =         5712   nw =           16   nelec =           12
 nblock =           5   ncftot =         7100   nw =           16   nelec =           12
```

```
 Energy levels for ...
 Rydberg constant is   109737.31534
-------------------------------------------------------------------------------------
 No - Serial number of the state; Pos - Position of the state within the
 J/P block;
--------------------------------------------------------------------------------
 No Pos  J Parity Energy Total    Levels   File          Configuration
                     (a.u.)      (cm^-1)
--------------------------------------------------------------------------------
  1  1   0  +   -1182.4117318       0.00  even4           $3s^2 ~^1S_{  0  }$
  2  1   0  -   -1181.3458964   233923.84  odd4            $3s~^2S\,3p~^3P_{  0  }^o$
  3  1   1  -   -1181.3192507   239771.90  odd4            $3s~^2S\,3p~^3P_{  1  }^o$
  4  1   2  -   -1181.2547649   253924.89  odd4            $3s~^2S\,3p~^3P_{  2  }^o$
  5  2   1  -   -1180.8024570   353194.99  odd4            $3s~^2S\,3p~^1P_{  1  }^o$
  6  2   0  +   -1179.8827380   555049.98  even4           $3p^2 (^3_2P)~^3P_{  0  }$
  7  1   2  +   -1179.8591476   560227.47  even4           $3p^2 (^1_2D)~^1D_{  2  }$
  8  1   1  +   -1179.8373877   565003.22  even4           $3p^2 (^3_2P)~^3P_{  1  }$
  9  2   2  +   -1179.7587033   582272.45  even4           $3p^2 (^3_2P)~^3P_{  2  }$
 10  3   0  +   -1179.3935084   662423.47  even4           $3p^2 (^1_0S)~^1S_{  0  }$
 11  2   1  +   -1179.3157358   679492.58  even4           $3s~^2S\,3d~^3D_{  1  }$
 12  3   2  +   -1179.3110726   680516.04  even4           $3s~^2S\,3d~^3D_{  2  }$
 13  1   3  +   -1179.3037683   682119.14  even4           $3s~^2S\,3d~^3D_{  3  }$
 14  4   2  +   -1178.9200935   766326.03  even4           $3s~^2S\,3d~^1D_{  2  }$
 15  2   2  -   -1178.1773269   929344.46  odd4            $3p~^2P\,3d~^3F_{  2  }^o$
 16  1   3  -   -1178.1320708   939277.02  odd4            $3p~^2P\,3d~^3F_{  3  }^o$
 17  3   2  -   -1178.0860233   949383.27  odd4            $3p~^2P\,3d~^1D_{  2  }^o$
 18  1   4  -   -1178.0797003   950771.02  odd4            $3p~^2P\,3d~^3F_{  4  }^o$
 19  3   1  -   -1177.9272497   984230.05  odd4            $3p~^2P\,3d~^3D_{  1  }^o$
 20  4   2  -   -1177.9243601   984864.26  odd4            $3p~^2P\,3d~^3P_{  2  }^o$
 21  2   3  -   -1177.8729498   996147.50  odd4            $3p~^2P\,3d~^3D_{  3  }^o$
 22  2   0  -   -1177.8671333   997424.08  odd4            $3p~^2P\,3d~^3P_{  0  }^o$
 23  4   1  -   -1177.8658076   997715.04  odd4            $3p~^2P\,3d~^3P_{  1  }^o$
 24  5   2  -   -1177.8644860   998005.10  odd4            $3p~^2P\,3d~^3D_{  2  }^o$
 25  3   3  -   -1177.5442550  1068287.68  odd4            $3p~^2P\,3d~^1F_{  3  }^o$
 26  5   1  -   -1177.4836853  1081581.20  odd4            $3p~^2P\,3d~^1P_{  1  }^o$
```

The LaTeX strings in the column to the right will be used as labels in the transition table and in the lifetime table. The user might want to edit the strings by, for example, removing unnecessary quantum labels. In the case above we do the global substitutions: `3s~^2S\,` → `3s` and `3p~^2P\,` → `3p`.

```
*******************************************************************************
*         RUN RTABTRANS2 TO PRODUCE THE LATEX TRANSITION FILE AND             *
*         THE LIFETIME FILE AS WELL AS A SCATTER PLOT                         *
*         INPUT FILES: even4.odd4.ct, energylabel.latex                       *
*         OUTPUT FILES: transitiontable.tex, lifetimetable.tex                *
*                       scatterplot.m                                         *
*******************************************************************************

>>rtabtrans2

  RTABTRANS2
  This program reads energy label data and transition
  data and creates transition and lifetime tables in
LaTeX or ASCII format. An Octave file with a
```

```
scatterplot of dT and 10log(A) is also produced

  Energy label data are given in the file energylabel
  created by the rtabtrans1 program
  Transition data file can be concatenated *.t or *.ct
  files.

  Input files: energylabel.latex(ascii),
  transitiondatafile
  Output files: transitiontable.tex(txt),
                lifetimetable.tex(txt),
                scatterplot.m

  Give the name of the transition data file
>>even4.odd4.ct
  Energy label file in LaTeX or ASCII format (0/1)?
>>0
  Give cut-off for printing A values
>>1e4
  Give fraction of accumulated A value for upper level
  for printing A value of a transition
>>1e-4
  Transition data wavelength sorted?
>>y
  Give number of decimals for wavelength (1,...6)
>>3

  Mean dT  5.5603142313003590E-002

  Program finished. The transition tables in latex
  have been written to file
```

Observe that there are two available criteria for selecting transitions to be printed: one criterion that is based on the value of A itself and one criterion that is related to the accumulated A value for the upper level. If a transition satisfies one of the two criteria the A value is printed. The second criterion makes it possible to select important decay channels from metastable states without having to print weak transitions from all states. A value greater than 1 for the second criterion means that printing will be based only on the first criterion, i.e. transitions will be printed if they have A values larger than the cut-off. When processing the file `transitiontable.tex` we get the transition table below. Note that we edited the file `energylabel.latex` before running `rtabtrans2`. Transition rates and gf values are given in length gauge for electric transitions. dT is a measure of the uncertainty of the electric transitions given by $dT = \dfrac{|A_l - A_v|}{\max(A_l, A_v)}$.

Upper	Lower	EM	ΔE (cm^{-1})	λ (Å)	A (s^{-1})	gf	dT
$3p3d\ ^1P^o_1$	$3s^2\ ^1S_0$	E1	1081581	92.457	1.717E+08	6.602E-04	0.112
$3p3d\ ^3P^o_1$	$3s^2\ ^1S_0$	E1	997715	100.229	6.158E+04	2.782E-07	0.299
$3s3d\ ^1D_2$	$3s3p\ ^3P^o_1$	E1	526554	189.914	3.634E+08	9.826E-03	0.003
$3p3d\ ^1P^o_1$	$3p^2(^3_2P)\ ^3P_0$	E1	526531	189.922	2.515E+08	4.080E-03	0.034
$3p3d\ ^1P^o_1$	$3p^2(^1_2D)\ ^1D_2$	E1	521353	191.808	3.915E+08	6.478E-03	0.210
$3p3d\ ^1P^o_1$	$3p^2(^3_2P)\ ^3P_1$	E1	516577	193.582	6.170E+07	1.040E-03	0.005
$3s3d\ ^1D_2$	$3s3p\ ^3P^o_2$	E1	512401	195.160	1.565E+07	4.469E-04	0.025
$3p3d\ ^1F^o_3$	$3p^2(^1_2D)\ ^1D_2$	E1	508060	196.827	1.994E+10	8.106E-01	0.063

$\vdots$

$3p3d\ ^1D_2^o$	$3s3d\ ^1D_2$	E1	183057	546.277	6.962E+08	1.557E-01	0.130
$3p3d\ ^3F_3^o$	$3s3d\ ^1D_2$	E1	172951	578.198	1.172E+07	4.112E-03	0.155
$3p3d\ ^3F_2^o$	$3s3d\ ^1D_2$	E1	163018	613.428	7.152E+07	2.017E-02	0.146

Table 8.3: Transition data

When processing the file `lifetimetable.tex` we get the table below. Note that we edited the file `energylabel` before running `rtabtrans2`. Transition rates for electric transitions entering the calculation of the lifetimes are in length and velocity gauges.

State	τ_l	τ_v
$3s3p\ ^3P_1^o$	2.3679E-08	2.2603E-08
$3s3p\ ^3P_2^o$	2.9431E-01	2.9431E-01
$3s3p\ ^1P_1^o$	4.6050E-11	4.6043E-11
$3p^2\ ^3P_0$	5.7890E-11	5.7208E-11
$3p^2\ ^1D_2$	2.0563E-10	2.0099E-10
$3p^2\ ^3P_1$	5.4422E-11	5.3844E-11
$3p^2\ ^3P_2$	5.9939E-11	5.9390E-11
$3p^2\ ^1S_0$	5.1955E-11	5.1832E-11
	$\vdots$	
$3p3d\ ^3P_2^o$	3.0579E-11	3.1347E-11
$3p3d\ ^3D_3^o$	2.5051E-11	2.5304E-11
$3p3d\ ^3P_0^o$	3.2589E-11	3.4062E-11
$3p3d\ ^3P_1^o$	3.0190E-11	3.1215E-11
$3p3d\ ^3D_2^o$	2.8117E-11	2.8766E-11
$3p3d\ ^1F_3^o$	2.3319E-11	2.3243E-11
$3p3d\ ^1P_1^o$	2.6514E-11	2.8629E-11

Table 8.4: Lifetimes in s^{-1}.

The `rtabtrans2` produces also an M-file `scatterplot.m` that, when executed under GNU Ocatve or Matlab, produces a dT and A scatterplot. The scatterplot is shown below and indicates that dT is smaller, on the average, for the stronger transitions than for the weaker ones.

8.10 Producing energy tables with LS-composition in LaTeX

The PERL script `lscomp.pl` creates a LaTeX file `lscomp.tex` which contains level information with the dominating LS-component and up to two extra LS-components if their contributions to the total wave function exceed 0.02 along with energies and, optionally, Landé g_J-factors. In addition a file `energylabel` is created which may be used together with `rtabtrans2` for the creation of a LaTeX file with transition data.

To be able to run the script, PERL has to be installed. For downloading and instructions, please visit `https://www.perl.org/get.html`

To run the script copy `lscomp.pl` to the working directory and type:

```
>>perl lscomp.pl
```

However, assuming that the script `lscomp.pl` is located in the `$HOME/GRASP2018/bin` directory, the following line may be added to the `.profile` or `.bashrc` file:

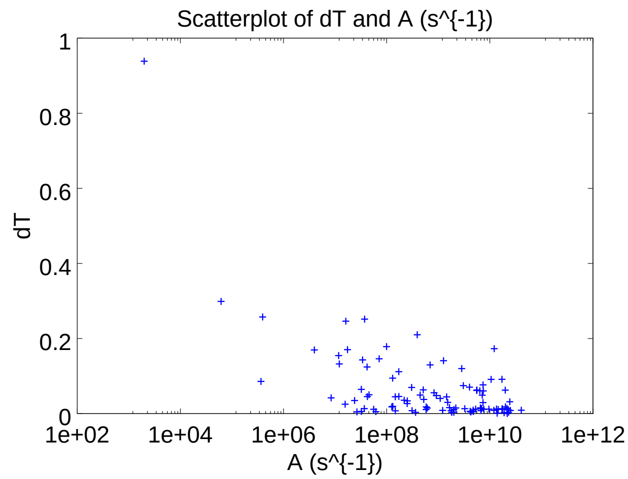

Figure 8.3: Scatterplot of dT and $A(\mathrm{s}^{-1})$.

```
alias perl_lscomp='perl $HOME/GRASP2018/bin/lscomp.pl'
```

In this case, to run the script from any working directory simply type:

```
>>perl_lscomp
```

As an example we make a table of the LS-compositions, energies and Landé g_J-factors in the fourth example. The table is obtained by processing the `even4` and `odd4` files.

```
*******************************************************************************
*         MAKE SURE YOU ARE IN THE DIRECTORY WHERE THE FILES FROM THE FOURTH  *
*         EXAMPLE ARE.                                                        *
*                                                                             *
*         BEGIN WITH RUNNING RHFS FOR EVEN4 AND ODD4 To GET THE               *
*         HFS AND LANDE FACTORS                                               *
*******************************************************************************

! For even 4

>>rhfs

 RHFS
 This is the hyperfine structure program
 Input files:  isodata, name.c, name.(c)m, name.w
 Output files: name.(c)h, name.(c)hoffd

 Default settings?
>>y

 Name of state
>>even4

 Mixing coefficients from a CI calc.?
>>y
 Loading Configuration Symmetry List File ...
 There are 16 relativistic subshells;
 There are 5712 relativistic CSFs;
```

```
  ... load complete;
 Loading Radial WaveFunction File ...
    nelec  =          12
    ncftot =        5712
    nw     =          16
    nblock =           4

   block     ncf     nev    2j+1  parity
       1     556       3       1       1
       2    1448       2       3       1
       3    1898       4       5       1
       4    1810       1       7       1
 Column 100 complete;
 Column 200 complete;
 Column 300 complete;

...................

 Column 5600 complete;
 Column 5700 complete;

 RHFS: Execution complete.

! For odd4

>>rhfs

 RHFS
 This is the hyperfine structure program
 Input files:  isodata, name.c, name.(c)m, name.w
 Output files: name.(c)h, name.(c)hoffd

 Default settings?
>>y

 Name of state
>>odd4

 Mixing coefficients from a CI calc.?
>>y
 Loading Configuration Symmetry List File ...
 There are 16 relativistic subshells;
 There are 7100 relativistic CSFs;
  ... load complete;
 Loading Radial WaveFunction File ...
    nelec  =          12
    ncftot =        7100
    nw     =          16
    nblock =           5

   block     ncf     nev    2j+1  parity
       1     546       2       1      -1
       2    1456       5       3      -1
       3    1891       5       5      -1
```

```
      4    1814       3       7      -1
      5    1393       1       9      -1
 Column 100 complete;
 Column 200 complete;
 Column 300 complete;

.......................

 Column 7000 complete;
 Column 7100 complete;

 RHFS: Execution complete.

*******************************************************************************
*         RUN LSCOMP.PL TO PRODUCE A LATEX FILE lscomp.tex WITH ENERGY LEVEL  *
*         INFORMATION WITH DOMINATING LS-COMPONENT AND UP TO TWO EXTRA        *
*         LS-COMPONENTS. IN ADDITION THE FILE energylabel.latex IS PRODUCED   *
*         WHICH CAN BE USED FOR FURTHER PROCESSING WITH RTABTRANS2            *
*         INPUT FILES: even4.lsj.lbl, even4.ch                                *
*                      odd4.lsj.lbl, odd4.ch                                  *
*         OUTPUT FILES: lscomp.tex, energylabel.latex                         *
*******************************************************************************

>>perl_lscomp

   LSCOMP.PL
   This PERL script creates files lscomp.tex and energylabel.latex

   File lscomp.tex contains energy level data with up to
   three LS components with a contribution > 0.02 of the
   total wave function.

   File energylabel.latex may be used by RTABTRANS2 to produce
   LaTeX tables of transition data.

   Input files : state1.lsj.lbl and state2.lsj.lbl
                 state1.ch and state2.ch (optional for gJ-factors)
   Output files: lscomp.tex and energylabel.latex
                                 Jorgen Ekman Sep. 2015

   State 1?
>>even4
   State 2?
>>odd4
   Necessary input file(s) exist!

   Do you want to include Lande g_J factors in the energy table? (y/n)
>>y
   Lande g_J factors from a CI calculation? (y/n)
>>y
   File(s) with g_J factors exist!

   Do you want an extra empty column for e_obs in the energy table? (y/n)
>>y
```

```
   Inspect the labels of the states and
   determine how many positions should be skipped in
   the string that determines the label. For example
   if all the states have a common core 1s(2) in the
   label then 6 positions should be skipped

   How many positions should be skipped?
>>12

   Files lscomp.tex and energylabel.latex written to disc.
```

The produced file `lscomp.tex`, slightly edited, is shown below.

```
\documentclass[12pt]{article}
\usepackage{longtable}
\usepackage[cm]{fullpage}
\thispagestyle{empty}
\begin{document}
{\scriptsize
\begin{longtable}{@{}rllrrr}
\caption{Energies.....}\\
\hline
No. & State & $LS$-composition & $E(CI) $ & $E(OBS) $  & $g_J $   \\

\hline
\endfirsthead
\caption{Continued.}\\
\hline
No. & State & $LS$-composition & $E(CI) $ & $E(OBS) $  & $g_J $   \\

\hline
\endhead
\hline
\endfoot
1   & $3s^{2}~^{1}S_{0}$               & 0.97 + 0.02~$3p^{2}(^{1}_{0}S)~^{1}S$     & 0         &  &           \\
2   & $3s~^{2}S\,3p~^{3}P_{0}^{\circ}$  & 1.00                                       & 233~924   &  &           \\
3   & $3s~^{2}S\,3p~^{3}P_{1}^{\circ}$  & 0.99                                       & 239~772   &  & 1.49513  \\
4   & $3s~^{2}S\,3p~^{3}P_{2}^{\circ}$  & 1.00                                       & 253~925   &  & 1.49886  \\
5   & $3s~^{2}S\,3p~^{1}P_{1}^{\circ}$  & 0.97 + 0.02~$3p~^{2}P\,3d~^{1}P^{\circ}$  & 353~195   &  & 1.00254  \\

           .....................

23  & $3p~^{2}P\,3d~^{3}P_{1}^{\circ}$  & 0.75 + 0.24~$3p~^{2}P\,3d~^{3}D^{\circ}$  & 997~715   &  & 1.25688  \\
24  & $3p~^{2}P\,3d~^{3}D_{2}^{\circ}$  & 0.54 + 0.45~$3p~^{2}P\,3d~^{3}P^{\circ}$  & 998~005   &  & 1.31616  \\
25  & $3p~^{2}P\,3d~^{1}F_{3}^{\circ}$  & 0.99                                       & 1~068~288 &  & 1.00120  \\
26  & $3p~^{2}P\,3d~^{1}P_{1}^{\circ}$  & 0.96 + 0.02~$3s~^{2}S\,3p~^{1}P^{\circ}$  & 1~081~581 &  & 0.9972  \\
\hline
\end{longtable}
}
\end{document}
```

After manually including observed energies from the NIST tables, the produced table looks like this

Table 8.5: Energies.....

No.	State	LS-composition	$E(CI)$	$E(OBS)$	g_J
1	$3s^2\ ^1S_0$	$0.97 + 0.02\ 3p^2(^1_0S)\ ^1S$	0		
2	$3s\ ^2S\,3p\ ^3P_0^\circ$	1.00	233 924	233 842	
3	$3s\ ^2S\,3p\ ^3P_1^\circ$	0.99	239 772	239 660	1.49513
4	$3s\ ^2S\,3p\ ^3P_2^\circ$	1.00	253 925	253 820	1.49886
5	$3s\ ^2S\,3p\ ^1P_1^\circ$	$0.97 + 0.02\ 3p\ ^2P\,3d\ ^1P^\circ$	353 195	351 911	1.00254

Table 8.5: Continued.

No.	State	LS-composition	$E(CI)$	$E(OBS)$	g_J
6	$3p^2(^3_2P)\ ^3P_0$	$0.96 + 0.03\ 3p^2(^1_0S)\ ^1S$	555 050	554 524	
7	$3p^2(^1_2D)\ ^1D_2$	$0.66 + 0.18\ 3p^2(^3_2P)\ ^3P + 0.16\ 3s\ ^2S\,3d\ ^1D$	560 227	559 600	1.09056
8	$3p^2(^3_2P)\ ^3P_1$	1.00	565 003	564 602	1.49901
9	$3p^2(^3_2P)\ ^3P_2$	$0.81 + 0.14\ 3p^2(^1_2D)\ ^1D + 0.04\ 3s\ ^2S\,3d\ ^1D$	582 272	581 803	1.40706
10	$3p^2(^1_0S)\ ^1S_0$	$0.93 + 0.03\ 3p^2(^3_2P)\ ^3P + 0.02\ 3d^2(^1_0S)\ ^1S$	662 423	659 627	
11	$3s\ ^2S\,3d\ ^3D_1$	1.00	679 493	678 772	0.49922
12	$3s\ ^2S\,3d\ ^3D_2$	1.00	680 516	679 785	1.16567
13	$3s\ ^2S\,3d\ ^3D_3$	1.00	682 119	681 416	1.33238
14	$3s\ ^2S\,3d\ ^1D_2$	$0.79 + 0.20\ 3p^2(^1_2D)\ ^1D$	766 326	762 093	0.99940
15	$3p\ ^2P\,3d\ ^3F^\circ_2$	$0.87 + 0.13\ 3p\ ^2P\,3d\ ^1D^\circ$	929 344	928 241	0.71134
16	$3p\ ^2P\,3d\ ^3F^\circ_3$	0.98	939 277	938 126	1.08503
17	$3p\ ^2P\,3d\ ^1D^\circ_2$	$0.83 + 0.12\ 3p\ ^2P\,3d\ ^3F^\circ + 0.03\ 3p\ ^2P\,3d\ ^3P^\circ$	949 383	948 513	0.97509
18	$3p\ ^2P\,3d\ ^3F^\circ_4$	1.00	950 771	949 658	1.24906
19	$3p\ ^2P\,3d\ ^3D^\circ_1$	$0.75 + 0.24\ 3p\ ^2P\,3d\ ^3P^\circ$	984 230	982 868	0.74338
20	$3p\ ^2P\,3d\ ^3P^\circ_2$	$0.51 + 0.45\ 3p\ ^2P\,3d\ ^3D^\circ + 0.03\ 3p\ ^2P\,3d\ ^1D^\circ$	984 864	983 514	1.32709
21	$3p\ ^2P\,3d\ ^3D^\circ_3$	0.98	996 147	994 852	1.32759
22	$3p\ ^2P\,3d\ ^3P^\circ_0$	1.00	997 424	995 889	
23	$3p\ ^2P\,3d\ ^3P^\circ_1$	$0.75 + 0.24\ 3p\ ^2P\,3d\ ^3D^\circ$	997 715	996 243	1.25688
24	$3p\ ^2P\,3d\ ^3D^\circ_2$	$0.54 + 0.45\ 3p\ ^2P\,3d\ ^3P^\circ$	998 005	996 623	1.31616
25	$3p\ ^2P\,3d\ ^1F^\circ_3$	0.99	1 068 288	1 062 515	1.00120
26	$3p\ ^2P\,3d\ ^1P^\circ_1$	$0.96 + 0.02\ 3s\ ^2S\,3p\ ^1P^\circ$	1 081 581	1 074 887	0.99722

Some states are strongly mixed in LS-coupling. For example, the states 20 and 24 are an almost equal mix of $^3P_2$ and $^3D_2$. The mixing is also reflected in the Landé g_J-factors which for these states are far from their pure LS values. If desired, one can apply global substitutions in the LaTeX file to get the quantum labels in the desired form.

8.11 Using rasfsplit to split files defining ASFs in symmetry blocks

The program `rasfsplit` splits the files defining a number of ASFs of different blocks (J and parity) into groups of files, one for each symmetry block. Such a splitting would make it possible to distribute computation (of transition properties, for instance) on different computer systems. For instance, calculations of transition rates for one combination of J and parity may be performed on one computer system (perhaps using MPI codes), while calculations of transition rates for another combination of J and parity may be performed on another computer system.

Overview

1. Split the ASFs defined by the files `2s22p3_2p5_3.c`, `2s22p3_2p5_3.w`, `2s22p3_2p5_3.m`, `2s22p3_2p5_3.cm` of the third example.

2. Display the energies with $J = 3/2$.

Program input

```
*******************************************************************************
*        MAKE SURE YOU ARE IN THE DIRECTORY WHERE THE FILES FROM THE THIRD    *
*        EXAMPLE ARE.                                                         *
*                                                                             *
*        RUN RASFSPLIT TO SPLIT THE 2s22p3_2p5_3.c, 2s22p3_2p5_3.w,           *
*        2s22p3_2p5_3.m, 2s22p3_2p5_3.cm FILES INTO FILES WHERE THE           *
```

```
*         EXTENSION IS odd1, odd2, odd3 ETC FOR THE DIFFERENT J BLOCKS OF     *
*         ODD PARITY                                                          *
*******************************************************************************

>>rasfsplit

 RASFSPLIT
 Splits an Atomic State Function made up by the files name.c,
 name.(c)m, name.w into the corresponding files for each
 parity and J block. If only the name.c file is available this
 file will be split
 Input files: name.c,name.(c)m, name.w
 Output files: name_even1.c, name_even1.(c)m. name_even1.w
               name_odd1.c, name_odd1.(c)m. name_odd1.w ...

 Name of the state
>>2s22p3_2p5_3

 Each of the blocks must be built from the same orbital set
 This may not be true for MR expansions, but is normally true
 for SD-MR expansions
 Is the above condition fullfilled? (y,n)
>>y

 nblock              3
 nblockodd           3
 nblockeven          0

 File 2s22p3_2p5_3.m  available

   nelec    =           7
   ncftot   =        1165
   nw       =           9
   nvectot  =           7
   nvecsize =        3212
   nblock   =           3


 Block data read from mixing file
        block        ncf         nev        2j+1          parity
          1         274           2           2           1
          2         591           4           4           1
          3         300           1           6           1
 File 2s22p3_2p5_3.cm available

   nelec    =           7
   ncftot   =        1165
   nw       =           9
   nvectot  =           7
   nvecsize =        3212
   nblock   =           3


 Block data read from mixing file
```

```
        block         ncf          nev         2j+1          parity
          1          274            2            2          -1
          2          591            4            4          -1
          3          300            1            6          -1

 Exit status of the name.w file copying for block           1 was           0
 Exit status of the name.w file copying for block           2 was           0
 Exit status of the name.w file copying for block           3 was           0
```

There are three blocks of odd parity and program has produced the files:

`2s22p3_2p5_3_odd1.c, 2s22p3_2p5_3_odd1.w, 2s22p3_2p5_3_odd1.m, 2s22p3_2p5_3_odd1.cm`
`2s22p3_2p5_3_odd2.c, 2s22p3_2p5_3_odd2.w, 2s22p3_2p5_3_odd2.m, 2s22p3_2p5_3_odd2.cm`
`2s22p3_2p5_3_odd3.c, 2s22p3_2p5_3_odd3.w, 2s22p3_2p5_3_odd3.m, 2s22p3_2p5_3_odd3.cm`

The files with the extension `odd1` define the ASFs with $J = 1/2$ and the files with the extension `odd2` define the ASFs with $J = 3/2$ etc. To see the energies of the ASFs produced by the `rci` program with $J = 3/2$ odd parity we give the command

```
rlevels  2s22p3_2p5_3_odd2.cm
```

and get the result

```
 nblock =            1   ncftot =         591   nw =            9   nelec =            7

 Energy levels for ...
 Rydberg constant is   109737.31569
 No - Serial number of the state; Pos - Position of the state within the
 J/P block; Splitting is the energy difference with the lower neighbor
-------------------------------------------------------------------------
 No Pos  J Parity Energy Total    Levels     Splitting
                     (a.u.)      (cm^-1)     (cm^-1)
-------------------------------------------------------------------------
  1  1  3/2 -    -263.2797858
  2  2  3/2 -    -262.9550573    71269.67    71269.67
  3  3  3/2 -    -262.7882760   107873.94    36604.27
  4  4  3/2 -    -259.5241195   824273.48   716399.54
-------------------------------------------------------------------------
```

Chapter 9

Interpreting the output files

In this chapter we describe in detail what information can be found in the different output files and how this information should be interpreted

9.1 Output files from the first example

The isodata file

Below is the `isodata` file for the Li example.

```
Atomic number:
   3.0000000000000000
Mass number (integer) :
   7.0000000000000000
Fermi distribution parameter a:
  0.52338755531043146
Fermi distribution parameter c:
   1.2520789669753825
Mass of nucleus (in amu):
   6.9393542602910001
Nuclear spin (I) (in units of h / 2 pi):
   1.5000000000000000
Nuclear dipole moment (in nuclear magnetons):
   3.2564267999999998
Nuclear quadrupole moment (in barns):
  -4.0000000000000001E-002
```

The calculation was for ^{7}Li with $Z = 3$ and $M = 7$. The nuclear charge distribution $\rho(r)$ was modelled as an extended Fermi distribution with

$$\rho(r) = \frac{\rho_0}{1 + e^{(r-c)/a}} \tag{9.1}$$

The parameters a and c are computed from the root mean squared radius and the skin thickness. The root mean squared radius in turn is computed from the formula $0.836A^{1/3} + 0.570$. The skin thickness is set to 2.3 fm. This gives $a = 0.52338755531043146$ fm and $c = 1.2520789669753825$ fm. On the lines following these quantities, the nuclear mass and nuclear spin I are given along with the nuclear magnetic dipole moment μ in nuclear magnetons and the nuclear quadrupole moment Q in barns.

The rcsfgenerate log-file

After each `rcsfgenerate` run there is a log-file displaying the response to the different questions. Below is the `rcsfgenerate.log` file from the $n = 3$ complete active space expansion for $1s^2 2p\ ^2P_{1/2,3/2}$.

```
           *
           0  ! Selected core
1s(2,*)2p(1,*)

3s,3p,3d
           1           3  ! Lower and higher 2*J
           3  ! Number of excitations
n
```

The log-file is a copy of the input data. By executing the command

```
rcsfgenerate < rcsfgenerate.log
```

the `rcsf.out` file will be reproduced. The `rcsfgenerate.log` file can easily be edited to give a new list of CSFs. For example

```
           *
           0  ! Selected core
1s(2,*)2p(1,*)

4s,4p,4d,4f
           1           3  ! Lower and higher 2*J
           3  ! Number of excitations
n
```

will give a file `rcsf.out` with CSFs corresponding to an active set $n = 4$.

The CSF file

The `rcsfgenerate` program produces an `rcsf.out` file. The file has a header with information about the radial orbitals and the closed shells (core shells). After this information there is a list of configuration state functions (CSFs). The configuration state functions are ordered in blocks with specified value of J. Each block is separated by a line with asterisk. Below is the file `2p_3.c`.

```
Core subshells:

Peel subshells:
  1s   2s   2p-  2p   3s   3p-  3p   3d-  3d
CSF(s):
  1s ( 2)  2p-( 1)
               1/2
                1/2-
  1s ( 2)  3p-( 1)
               1/2
                1/2-
  1s ( 1)  2s ( 1)  2p ( 1)
      1/2      1/2      3/2
                    1    1/2-

 .............
```

```
  3p-( 1)  3d-( 1)  3d ( 1)
      1/2      3/2      5/2
                    2    1/2-
  3p-( 1)  3d-( 2)
      1/2        0
               1/2-
 *
  1s ( 2)  2p ( 1)
               3/2
               3/2-
  1s ( 2)  3p ( 1)
               3/2
               3/2-
  1s ( 1)  2s ( 1)  2p ( 1)
      1/2      1/2      3/2
                    0    3/2-
............


  3p-( 1)  3d-( 1)  3d ( 1)
      1/2      3/2      5/2
                    2    3/2-
  3p-( 1)  3d-( 2)
      1/2        2
               3/2-
```

The line with core subshells is empty and in this case we have no closed core. The radial orbitals are $1s, 2s, 2p-, 2p, 3s, 3p-, 3p, 3d-, 3d$. After the radial orbitals there are lists of CSFs arranged in blocks. The first block of CSFs has $J = 1/2$. The second block has $J = 3/2$. An asterisk is separating the blocks. In the file each CSF occupies three lines. On the first line the subshells and their occupations are listed in a linear form where, for example, $1s^2$ becomes `1s ( 2)`. The second line shows the coupling of each subshell to a J quantum number and the third line shows how the J quantum numbers of each subshell are coupled from left to right to a final J.

The rangular log-file

The program `rangular` produces a log-file displaying the response to the different questions. After running `rsave` this file is saved in `name.alog`. Below is the log-file `2p_3.alog`

```
y              ! Full interaction
```

The log-file is a copy of the input data. We see that angular data were computed for all interactions. The log-file is more useful in cases where we do not have full interaction, see section 14.1. In these cases the file contains information about the zero-order space.

The rmcdhf log-file

The self-consistent field program `rmcdhf` produces a log-file displaying the response to the different questions. After running `rsave` this file is saved in `name.log`. Below is the log-file `2p_3.log` from the run on weighted average of the $1s^22p\ ^2P_{1/2,3/2}$ states.

```
y              ! Default settings
1
1
```

```
          5 ! level weights
3*

        100 ! Number of SCF cycles
```

The log-file is a copy of the input data. We see that the run was with default settings (there will be no log-file for non-default settings). It was a calculation targeting the first levels of the two blocks (ASF serial serial numbers 1 for each of the two blocks). The level weights is 5 (default option), meaning that the levels are weighted according to the statistical weight $2J+1$. `3*` means that orbitals with principal quantum numbers 3 are optimized. The blank line indicates that none of the optimized orbitals are spectroscopic. 100 SCF cycles were requested. By executing the command

```
    rmcdhf < rmcdhf.log    (or rmcdhf < name.log)
```

the `rmcdhf` run will be executed again with the settings in `rmcdhf.log`. The log-file can easily be edited and used as an input also to other runs.

The rmcdhf summary file

The self-consistent field program `rmcdhf` produces a summary file. After running `rsave` this file is saved in `name.sum`. Below is the summary file `2p_3.sum` from the run on weighted average of the $1s^22p\ ^2P_{1/2,3/2}$ states.

```
 There are 3 electrons in the cloud
  in 186 relativistic CSFs
  based on 9 relativistic subshells.

The atomic number is   3.0000000000;
 the mass of the nucleus is  1.264966898269D+04 electron masses;
  Fermi nucleus:
  c = 2.366086335030D-05 Bohr radii,
  a = 9.890591370096D-06 Bohr radii;
  there are 73 tabulation points in the nucleus.

Speed of light =  137.0359991390D+00 atomic units.

Radial grid: R(I) = RNT*(exp((I-1)*H)-1), I = 1, ..., N;

 RNT  =  6.666666666667D-07 Bohr radii;
 H    =  5.000000000000D-02 Bohr radii;
 N    =  590;
 R(1) =  0.000000000000D+00 Bohr radii;
 R(2) =  3.418073091735D-08 Bohr radii;
 R(N) =  4.110372988964D+06 Bohr radii.

 EOL calculation.
 2 levels will be optimised;
  their indices are: 1, 1.
 Each is assigned its statistical weight;

Radial wavefunction summary:

                                                             Self
Subshell      e             p0     gamma     P(2)       Q(2)   Consistency MTP
```

```
  1s   2.5177395463D+00  9.281D+00  1.00  3.172D-07 -4.202D-12  0.000D+00  355
  2s   1.9634308441D-01  1.453D+00  1.00  4.965D-08 -6.576D-13  0.000D+00  361
  2p-  1.2867248392D-01  5.116D-05  1.00  2.864D-15  1.598D-10  0.000D+00  366
  2p   1.2866992751D-01  4.265D-01  2.00  4.983D-16 -6.601D-21  0.000D+00  366
  3s   8.0600845411D+00  1.181D+01  1.00  5.027D-07 -9.543D-13  8.965D-08  360
  3p-  8.7786094243D+00  2.853D-03  1.00  2.269D-13  8.922D-09  8.940D-07  364
  3p   8.7823538492D+00  2.381D+01  2.00  2.786D-14 -1.403D-18  1.303D-06  364
  3d-  1.6298092509D+01  8.146D-03  2.00  3.117D-20  1.739D-15  6.785D-06  358
  3d   1.6306599831D+01  8.170D+01  3.00  3.262D-21 -4.324D-26  8.697D-06  358

                  -3              -1                             2              4     Generalised
Subshell     <  r  >         <  r  >        <  r  >        <  r  >        <  r  >     occupation

  1s      0.00000D+00    2.68556D+00    5.73199D-01    4.47081D-01    5.33751D-01    1.99386D+00
  2s      0.00000D+00    3.45596D-01    3.87317D+00    1.77347D+01    5.65669D+02    2.01584D-04
  2p-     0.00000D+00    2.65023D-01    4.79564D+00    2.78265D+01    1.47509D+03    3.33335D-01
  2p      5.85643D-02    2.65011D-01    4.79578D+00    2.78280D+01    1.47522D+03    6.66669D-01
  3s      0.00000D+00    3.09463D+00    8.79428D-01    1.76019D+00    3.83411D+01    2.52035D-03
  3p-     0.00000D+00    1.94750D+00    6.74894D-01    8.40900D-01    2.20221D+01    1.08534D-03
  3p      1.72790D+01    1.94712D+00    6.74750D-01    8.40046D-01    2.19775D+01    2.17087D-03
  3d-     1.01792D+01    1.82956D+00    6.31485D-01    4.57970D-01    4.14736D-01    6.48046D-05
  3d      1.01575D+01    1.82935D+00    6.31441D-01    4.57852D-01    4.14179D-01    9.72851D-05

Eigenenergies:

Level  J Parity       Hartrees              Kaysers                 eV

  1     1/2 -    -7.404577002212D+00  -1.625116808012D+06  -2.014888031072D+02
  1     3/2 -    -7.404574142855D+00  -1.625116180456D+06  -2.014887253001D+02

Weights of major contributors to ASF:

Block Level J Parity      CSF contributions

  1     1   1/2 -       0.9985     -0.0343      0.0326      0.0230      0.0107
                             1          56          58          60          30
  2     1   3/2 -       0.9985     -0.0343      0.0326      0.0230      0.0099
                             1          62          67          71          32
```

The first lines of the file tell us that the calculation was for a three electron system and that there were in total 186 CSFs built on 9 relativistic radial orbitals. After this there is information about the nucleus. In this case the nucleus has $Z = 3$ and a mass of $1.264966898269 \times 10^4$ electron masses. The nuclear charge distribution is modelled by a Fermi distribution with $c = 2.36608633503 \times 10^{-5}$ Bohr radii, and $a = 9.89059137009 \times 10^{-6}$ Bohr radii. There is information about the radial grid used in the calculation. The grid is given by

$$R(I) = RNT(\exp((I-1)H) - 1), \quad I = 1, \ldots, N$$

with $RNT = 6.666666666667 \times 10^{-7}$ Bohr radii and $H = 5 \times 10^{-2}$ Bohr radii. There are $N = 590$ grid points.

We see that it is an EOL calculation and that the calculation was on the lowest state (first eigenvalue) of each block ($J = 1/2$ and $J = 3/2$). In the optimization each state is weighted according to the statistical weight $2J+1$.

The information on optimization is followed by a radial orbital summary. Important characteristics of a radial orbital are the orbital energy eigenvalue and parameters that determine the behavior near $r=0$. The radial amplitudes

$$u(r) = \begin{pmatrix} P(r) \\ Q(r) \end{pmatrix},$$

can be expanded in power series

$$u(r) = r^{\gamma}[u_0 + u_1 r + u_2 r^2 + \ldots], \quad u_k = \begin{pmatrix} p_k \\ q_k \end{pmatrix}$$

near the origin where the index γ, p_k, and q_k are constants that depend on the nuclear potential model. In the radial orbital summary `e` is orbital energy eigenvalue, `p0` is a parameter related to the leading expansion coefficients of the radial amplitudes and `gamma` is the exponent γ, for details see I.P. Grant, Relativistic Quantum Theory of Atoms and Molecules, Springer 2007, p 272 - 273 and also the subroutine `start` in `lib92`. `P(2)` and `Q(2)` are the values of the radial amplitudes at the first grid point `R(2)` away from zero. Then the self consistency (weighted change of an orbital during an iteration) is given for each orbital. In this case the orbitals $1s, 2s, 2p-, 2p$ were keep frozen and they thus have a self consistency of zero. The orbitals $3s, 3p-, 3p, 3d-, 3d$ were optimized and the self consistency is between 8.699×10^{-6} for $3d$ and 8.965×10^{-8} for $3s$. Finally, the value `MTP` gives the number of the outermost grid point used for representing the radial amplitudes of the orbital. At remaining grid points the radial amplitudes of the orbital are set to zero. Around 360 of the available 590 grid points are utilized.

Different radial expectation values

$$\langle r^k \rangle = \langle nlj | r^k | nlj \rangle$$

of the orbitals are given along with the generalized occupation numbers. The generalized occupation number $\overline{q}(nlj)$ of an orbital nlj is defined as

$$\overline{q}(nlj) = \sum_{r=1}^{NCSF} d_r^2 q_r(nlj),$$

where $q_r(nlj)$ is the number of electrons in subshell nlj in CSF r and d_r^2 is the generalized weight

$$d_r^2 = \frac{\sum_{i=1}^{n_L}(2J_i+1)c_{ri}^2}{\sum_{i=1}^{n_L}(2J_i+1)}.$$

In the expression for the generalized weight the sum is over all levels in the EOL calculation. $c_{ri}, \quad r = 1, \ldots, NCSF$ are the mixing coefficients of level i in the basis of the CSFs. An orbital with a small generalized occupation number is associated with CSFs that have small expansion coefficients.

At the end of the summary file the eigenenergies for the states are displayed in different energy units. The weights of the major CSF contributors are also given. Note that the CSFs in this case are counted block wise.

The rci log-file

The relativistic configuration interaction program `rci` produces a log-file displaying the response to the different questions. This file is saved in `name.clog`. Below is the log-file `2p_3.clog` from the run of the $1s^22p\ ^2P_{1/2,3/2}$ states.

```
y            ! Default settings
2p_3
y            ! Contribution of H Transverse?
y            ! Modify photon frequencies?
   9.9999999999999995E-007 ! Scale factor
y            ! Vacuum polarization?
n            ! Normal mass shift?
n            ! Specific mass shift?
y            ! Self energy?
           3 ! Max n for including self energy
1
1
```

The log-file is a copy of the input data. The name of the state is 2p_3 and full interaction was included. Contributions from the transverse interaction (Breit) were added where the photon frequencies were multiplied with a factor 10^{-6}. Vacuum polarization as well as self-energy corrections were added. The self-energy corrections are based on estimations for the orbitals. For correlation orbitals with high principal quantum numbers these estimations may fail. In this case self-energy corrections were based on orbitals with principal quantum numbers smaller than 3. Finally we see that the calculation determined the first levels of the two blocks (ASF serial serial numbers 1 for each of the two blocks). By executing the command

```
rci < 2p_3.clog
```

the `rci` run will be executed again with the settings in `2p_3.clog`. The log-file can easily be edited and used as an input also to other runs.

The rci summary file

The self-consistent field program `rci` produces a summary file `name.csum`. Below is the summary file `2p_3.csum` from the run on weighted average of the $1s^2 2p\ ^2P_{1/2,3/2}$ states

```
 There are 3 electrons in the cloud
  in 186 relativistic CSFs
  based on 9 relativistic subshells.

The atomic number is   3.0000000000;
 the mass of the nucleus is  1.264966898269D+04 electron masses;
  Fermi nucleus:
  c = 2.366086335030D-05 Bohr radii,
  a = 9.890591370096D-06 Bohr radii;
  there are 73 tabulation points in the nucleus.

Speed of light =  1.370359991390D+02 atomic units.

 To H (Dirac Coulomb) is added
  H (Transverse) --- factor multiplying the photon frequency:  1.00000000D-06;
  H (Vacuum Polarisation);
  the total will be diagonalised.
 Diagonal contributions from H (Self Energy) will be estimated
  from a screened hydrogenic approximation.

Radial grid: R(I) = RNT*(exp((I-1)*H)-1), I = 1, ..., N;

 RNT  =  6.666666666667D-07 Bohr radii;
 H    =  5.000000000000D-02 Bohr radii;
```

```
 N    =  590;
 R(1) =  0.000000000000D+00 Bohr radii;
 R(2) =  3.418073091735D-08 Bohr radii;
 R(N) =  4.110372988964D+06 Bohr radii.

 Subshell radial wavefunction summary:

Subshell      e             p0     gamma     P(2)       Q(2)    MTP

  1s   2.5177395463D+00  9.281D+00  1.00  3.172D-07 -4.202D-12  355
  2s   1.9634308441D-01  1.453D+00  1.00  4.965D-08 -6.576D-13  361
  2p-  1.2867248392D-01  5.116D-05  1.00  2.864D-15  1.598D-10  366
  2p   1.2866992751D-01  4.265D-01  2.00  4.983D-16 -6.601D-21  366
  3s   8.0600845411D+00  1.181D+01  1.00  5.027D-07 -9.543D-13  360
  3p-  8.7786094243D+00  2.853D-03  1.00  2.269D-13  8.922D-09  364
  3p   8.7823538492D+00  2.381D+01  2.00  2.786D-14 -1.403D-18  364
  3d-  1.6298092509D+01  8.146D-03  2.00  3.117D-20  1.739D-15  358
  3d   1.6306599831D+01  8.170D+01  3.00  3.262D-21 -4.324D-26  358

 ...........

Information about number of radial integrals, density of the
Hamiltonian matrix etc, the energies and the leading CSFs for each
level etc.
```

From the summary file we again see what operators were included in the Hamiltonian. Information about the grid and the orbitals, same as in the `name.sum` file is also available.

The hyperfine structure files

The `rhfs` program computes hyperfine structure data. In addition the Landé g_J-factor is computed. Below is the output file `2p_3.ch` from the `rhfs` run for the RCI wave function, given in the `2p_3.c`, `2p_3.w` and `2p_3.cm` files, of the $1s^2 2p\ ^2P_{1/2,3/2}$ states.

```
Nuclear spin                         1.500000000000000D+00 au
Nuclear magnetic dipole moment       3.256426800000000D+00 n.m.
Nuclear electric quadrupole moment  -4.000000000000000D-02 barns

 Interaction constants:

 Level1  J Parity         A (MHz)              B (MHz)              g_J                delta g_J

   1      1/2 -      4.4822178831D+01   -0.0000000000D+00    6.6665728979D-01   -7.7333332997D-04
   1      3/2 -     -3.5381700218D+00   -1.7729096288D-01    1.3333253717D+00    7.7333333479D-04
```

At the top the nuclear spin and moments are displayed. Then, for each level, the A and B hyperfine interaction constants are given in MHz along with the Landé g_J-factor.

The `rhfs` program gives another file `2p_3.choffd` which contains off-diagonal hyperfine data

```
Nuclear spin                         1.500000000000000D+00 au
Nuclear magnetic dipole moment       3.256426800000000D+00 n.m.
Nuclear electric quadrupole moment  -4.000000000000000D-02 barns


 Interaction constants:
```

```
Level1  J Parity  Level2  J Parity        A (MHz)             B (MHz)

  1      1/2 -       1      1/2 -      4.4822178831D+01   -0.0000000000D+00
  1      3/2 -       1      1/2 -      1.1768857887D+01   -3.8388220012D-02
  1      3/2 -       1      3/2 -     -3.5381700218D+00   -1.7729096288D-01


Matrix elements:

Level1  J Parity  Level2  J Parity    F     Matrix element (a.u.)

  1      1/2 -       1      1/2 -      1       -8.5152606438D-09
  1      1/2 -       1      1/2 -      2        5.1091563863D-09


Matrix elements:

Level1  J Parity  Level2  J Parity    F     Matrix element (a.u.)

  1      3/2 -       1      1/2 -      1        4.0297075799D-09
  1      3/2 -       1      1/2 -      2        5.3525245724D-09


Matrix elements:

Level1  J Parity  Level2  J Parity    F     Matrix element (a.u.)

  1      3/2 -       1      3/2 -      0        1.9828496377D-09
  1      3/2 -       1      3/2 -      1        1.4720532074D-09
  1      3/2 -       1      3/2 -      2        4.2351513722D-10
  1      3/2 -       1      3/2 -      3       -1.2166549923D-09
```

Given are diagonal and off-diagonal hyperfine interaction constants A and B in MHz and the F dependent hyperfine matrix elements in atomic units. The above quantities are defined in [8], equations (13-17) and Eq (7-8).

The transition file

The `rtransition` program computes transition data. Below is the output file `2s_3.2p_3.ct` from the `rtransition` electric dipole E1 run for RCI wave functions given in the `2s_3.c`, `2s_3.w`, `2s_3.cm` and `2p_3.c`, `2p_3.w` and `2p_3.cm` files.

```
 Transition between files:
 f1 = 2s_3
 f2 = 2p_3


 Electric 2**( 1)-pole transitions
 =================================

    Upper          Lower
File Lev J  P File Lev J  P    E (Kays)         A (s-1)         gf           S
 f2  1  1/2 -  f1  1  1/2 +     14861.28 C  3.81311D+07  5.17671D-01  1.14676D+01
                                         B  3.74756D+07  5.08773D-01  1.12705D+01
```

```
f2  1  3/2 -  f1  1  1/2 +     14861.57 C  3.81334D+07  1.03537D+00  2.29353D+01
                                        B  3.74782D+07  1.01758D+00  2.25413D+01
```

The first lines of the file give the name of the files defining the wave functions. Then data are given for the electric dipole transition E1. The first transition is from the upper level 1 with $J = 1/2$ and negative parity in file f2, i.e. $1s^22p\ ^2P_{1/2}$ to the lower level 1 with $J = 1/2$ and positive parity in file f1, i.e. $1s^22s\ ^2S_{1/2}$. The second transition is from the upper level 1 with $J = 3/2$ and negative parity in file f2, i.e. $1s^22p\ ^2P_{3/2}$ to the lower level 1 with $J = 1/2$ and positive parity in file f1, i.e. $1s^22s\ ^2S_{1/2}$. For each transition the transition energy E is given in Kays (cm^{-1}). Also the transition rate A in emission (Einstein A-coefficient), the weighted oscillator strength gf and the line strength S are given in Coulomb (velocity) and Babushkin (length) gauge.

If the `2s_3.lsj.lbl` and `2p_3.lsj.lbl` files produced by `jj2lsj` are available at the run of `rtransition` an additional output file `2s_3.2p_3.ct.lsj` is produced. This file is shown below

```
Transition between files:
2s_3
2p_3


  1   -7.47197402  1s(2).2s_2S
  1   -7.40426103  1s(2).2p_2P
  14861.28 CM-1      6728.89 ANGS(VAC)      6728.20 ANGS(AIR)
E1  S =  1.12705D+01   GF =  5.08773D-01   AKI =  3.74756D+07   dT =  0.01719
         1.14676D+01         5.17671D-01          3.81311D+07


  1   -7.47197402  1s(2).2s_2S
  3   -7.40425972  1s(2).2p_2P
  14861.57 CM-1      6728.76 ANGS(VAC)      6728.06 ANGS(AIR)
E1  S =  2.25413D+01   GF =  1.01758D+00   AKI =  3.74782D+07   dT =  0.01718
         2.29353D+01         1.03537D+00          3.81334D+07
```

The `2s_3.2p_3.ct.lsj` file has a different format. Here the labels of the upper and lower states in the transition are in LSJ-notation. The J quantum number (multiplied by 2) is written to the left. The transition energy is given in cm^{-1} and the wave lengths (vacuum and air) in Angstrom (ANGS) where 1 ANGS $= 10^{-10}$ m. The line strength S, the weighted oscillator strength gf and the transition rates in emission A (AKI) are given on two lines where the upper line corresponds to the Babushkin (length) gauge and the lower line to the Coulomb (velocity) gauge. Finally,

$$dT = \frac{|A_l - A_v|}{\max(A_l, A_v)},$$

where A_l and A_v are the transition rates in length and velocity gauge, is a parameter that can be related to the estimated uncertainty of the transition rates [33].

9.2 Output files from the third example

The third example case was calculations of the states belonging to the $1s^22s^22p^3$ and $1s^22p^5$ configurations in Si VIII.

The jj2lsj file

The `jj2lsj` program transforms from jj- to LSJ-coupling and gives the LSJ-composition of the states. Below is the output file `2s22p3_2p5_3.lsj.lbl` from the `jj2lsj` run of the RCI wave functions given in the `2s22p3_2p5_3.c`, `2s22p3_2p5_3.w`, `2s22p3_2p5_3.cm` files. For each case, the first line gives the position (number) of the eigenstate in the interaction matrix, parity, total energy and the percentage of the atomic state function (ASF) that has been transformed. Thus 99.907 % implies that 0.0 % has not been transformed.

```
Pos   J   Parity      Energy Total      Comp. of ASF
  1  1/2     -           -262.790635602      99.907%
         0.98656030    0.97330122   1s(2).2s(2).2p(3)2P1_2P
         0.15010904    0.02253272   1s(2).2p(5)_2P
        -0.03364614    0.00113206   1s(2).2s_2S.2p(3)2P1_1P.3d_2P
  2  1/2     -           -259.497941492      99.123%
         0.98251140    0.96532866   1s(2).2p(5)_2P
        -0.14989834    0.02246951   1s(2).2s(2).2p(3)2P1_2P
        -0.03674439    0.00135015   1s(2).2s_2S.2p(3)2D3_1D.3d_2P
         0.03527443    0.00124429   1s(2).2s_2S.2p(3)2P1_3P.3d_2P

  1  3/2     -           -263.279785800      99.550%
         0.99652486    0.99306180   1s(2).2s(2).2p(3)4S3_4S
         0.03703202    0.00137137   1s(2).2s(2).2p(3)2P1_2P
  2  3/2     -           -262.955057275      99.670%
         0.98954100    0.97919139   1s(2).2s(2).2p(3)2D3_2D
        -0.12139907    0.01473773   1s(2).2s(2).2p(3)2P1_2P
        -0.03690740    0.00136216   1s(2).2s_2S.2p(3)4S3_3S.3d_2D
  3  3/2     -           -262.788275959      99.912%
         0.97818840    0.95685255   1s(2).2s(2).2p(3)2P1_2P
         0.15010550    0.02253166   1s(2).2p(5)_2P
         0.12276472    0.01507118   1s(2).2s(2).2p(3)2D3_2D
        -0.03672205    0.00134851   1s(2).2s(2).2p(3)4S3_4S
        -0.03335975    0.00111287   1s(2).2s_2S.2p(3)2P1_1P.3d_2P
  4  3/2     -           -259.524119499      99.004%
         0.98234136    0.96499455   1s(2).2p(5)_2P
        -0.15102022    0.02280711   1s(2).2s(2).2p(3)2P1_2P
         0.03537700    0.00125153   1s(2).2s_2S.2p(3)2P1_3P.3d_2P

  1  5/2     -           -262.953822323      99.429%
         0.99713868    0.99428554   1s(2).2s(2).2p(3)2D3_2D
```

There is a total of seven states. For each state the file gives the LSJ-expansion. The lowest $J = 1/2$ state (pos 1) with negative parity and energy -262.790635602 a.u. has the LSJ-expansion

```
         0.98656030    0.97330122   1s(2).2s(2).2p(3)2P1_2P
         0.15010904    0.02253272   1s(2).2p(5)_2P
        -0.03364614    0.00113206   1s(2).2s_2S.2p(3)2P1_1P.3d_2P
```

The second lowest $J = 1/2$ state (pos 2) with negative parity and energy -259.497941492 a.u. has the LSJ-expansion

```
         0.98251140    0.96532866   1s(2).2p(5)_2P
        -0.14989834    0.02246951   1s(2).2s(2).2p(3)2P1_2P
        -0.03674439    0.00135015   1s(2).2s_2S.2p(3)2D3_1D.3d_2P
         0.03527443    0.00124429   1s(2).2s_2S.2p(3)2P1_3P.3d_2P
```

We see that the states are close to pure LSJ-coupling and the file provides meaningful labels that match labels given in, for example, the NIST data tables. The second column in the table gives the LSJ-composition, i.e. the squared expansion coefficients.

Finally, a few words about how to interpret the notation in the composition of the ASF. Each subshell in the configuration is given with occupation, LS term designation and seniority. When the subshell is singly or fully occupied the term designation and seniority are not written out. The LS term for each subshell is the coupled from left to right. Intermediate couplings are given after the underscore sign `_`.

In Table 1 below there is a list of possible terms and their seniority for commonly occurring subshells
Table 1: List of possible terms and their seniority for commonly occurring subshells.

```
------------------------------------------------------------------------------
Subshell TERMS (2S+1, L, seniority)

 s(1)   2S1
 s(2)   1S0

 p(1)   2P1
 p(2)   1S0 1D2 3P2
 p(3)   2P1 2D3 4S3

 d(1)   2D1
 d(2)   1S0 1D2 1G2 3P2 3F2
 d(3)   2D1 2P3 2D3 2F3 2G3 2H3 4P3 4F3
 d(4)   1S0 1D2 1G2 3P2 3F2 1S4 1D4 1F4 1G4 1I4 3P4 3D4 3F4 3G4 3H4 5D4
 d(5)   2D1 2P3 2D3 2F3 2G3 2H3 4P3 4F3 2S5 2D5 2F5 2G5 2I5 4D5 4G5 6S5

 f(1)   2F1
 f(2)   1S0 1D2 1G2 1I2 3P2 3F2 3H2
--------------------------------------------------------------------------------
```

As a concrete example of how to interpret the notation we look at

```
1s(2).2s_2S.2p(3)2P1_3P.3d_2P
```

The first subshell `1s(2)` is fully occupied and have only one LS term `1S0` that is not written out explicitly. The second subshell `2s` is singly occupied and has only one LS term `2S1` that is not written out explicitly. The third subshell `2p(3)` is coupled to an LS term `2P1`. The fourth subshell `3d` is singly occupied and has only one LS term `2D1` that is not written out explicitly. Coupling `1S0` and `2S1` of subshells one and two leads to an intermediate term `_2S`. Coupling `_2S` with `2P1` of the third subshell leads to the intermediate term `_3P`. Finally coupling `_3P` with `2D1` of the fourth subshells gives the final LS term `2P`.

The programs `rtablevels` and `rtabtransE1` implement a LaTeX translation of the ASCII notation. In the LaTeX translation the LS term and seniority of a subshell are given in parenthesis just after the subshell. For the intermediate terms the underscore of the ASCII notation has been replaced by a space. In the LaTeX translation the user also has a choice to omit the closed core. Translating the above example to LaTeX and omitting the `1s(2)` we get

$$2s\ ^2S\, 2p^3(^2_1P)\ ^3P\, 3d\ ^2P.$$

Note how the seniority enters as a subscript.

The transition file in LSJ-coupling

The `rtransition` program computes transition data. Below is the output file `2s22p3_2p5_3.2s22p3_2p5_3.ct` from the `rtransition` magnetic dipole M1 run of the RCI wave functions given in the `2s22p3_2p5_3.c`, `2s22p3_2p5_3.w`, `2s22p3_2p5_3.cm` files giving the states belonging to the $1s^22s^22p^3$ and $1s^22p^5$ configurations

```
Transition in file:
f = 2s22p3_2p5_3


Magnetic 2**( 1)-pole transitions
=================================

Upper       Lower
Lev   J P   Lev   J P       E (Kays)         A (s-1)          gf            S
f   2  1/2 -  f   1  1/2 -      722662.83 M  6.00621D-05  3.44839D-16  1.18001D-11
f   3  3/2 -  f   1  1/2 -         517.88 M  1.22716D-03  2.74383D-08  1.31018D+00
f   4  3/2 -  f   1  1/2 -      716917.42 M  4.29992D+00  5.01695D-11  1.73052D-06
f   1  1/2 -  f   1  3/2 -      107356.06 M  3.11300D+01  8.09867D-09  1.86549D-03
f   2  1/2 -  f   1  3/2 -      830018.89 M  4.56677D+00  1.98756D-11  5.92158D-07
f   1  1/2 -  f   2  3/2 -       36086.39 M  1.27613D+01  2.93830D-08  2.01353D-02
f   2  1/2 -  f   2  3/2 -      758749.21 M  4.24418D+00  2.21047D-11  7.20430D-07
f   2  1/2 -  f   3  3/2 -      722144.94 M  1.09837D+01  6.31521D-11  2.16256D-06
f   2  1/2 -  f   4  3/2 -        5745.41 M  3.40764D+00  3.09528D-07  1.33224D+00
f   2  3/2 -  f   1  3/2 -       71269.67 M  1.41118D+00  1.66606D-09  5.78084D-04
f   3  3/2 -  f   1  3/2 -      107873.94 M  7.52312D+01  3.87688D-08  8.88733D-03
f   4  3/2 -  f   1  3/2 -      824273.48 M  1.25260D+01  1.10557D-10  3.31682D-06
f   3  3/2 -  f   2  3/2 -       36604.27 M  2.11594D+01  9.47019D-08  6.39782D-02
f   4  3/2 -  f   2  3/2 -      753003.80 M  9.62931D+00  1.01840D-10  3.34446D-06
f   4  3/2 -  f   3  3/2 -      716399.54 M  7.21346D-02  8.42851D-13  2.90938D-08
f   1  5/2 -  f   1  3/2 -       71540.71 M  1.99970D-02  3.51453D-11  1.21484D-05
f   1  5/2 -  f   2  3/2 -         271.04 M  2.11526D-04  2.59002D-08  2.36306D+00
f   3  3/2 -  f   1  5/2 -       36333.23 M  1.17593D+01  5.34186D-08  3.63575D-02
f   4  3/2 -  f   1  5/2 -      752732.76 M  5.41422D+00  5.73023D-11  1.88251D-06
```

If the information of LSJ-coupling is available from a `jj2lsj` run `rtransition` also produces a file `2s22p3_2p5_3.2s22p3_2p5_3.ct.lsj`

```
Transition between files:
2s22p3_2p5_3
2s22p3_2p5_3


  1 -262.79063560  1s(2).2s(2).2p(3)2P1_2P
  1 -259.49794149  1s(2).2p(5)_2P
 722662.83 CM-1       138.38 ANGS(VAC)       138.38 ANGS(AIR)
M1  S =  1.18001D-11   GF =  3.44839D-16   AKI =  6.00621D-05


  1 -262.79063560  1s(2).2s(2).2p(3)2P1_2P
  3 -262.78827596  1s(2).2s(2).2p(3)2P1_2P
    517.88 CM-1    193094.26 ANGS(VAC)    193074.30 ANGS(AIR)
M1  S =  1.31018D+00   GF =  2.74383D-08   AKI =  1.22716D-03


  1 -262.79063560  1s(2).2s(2).2p(3)2P1_2P
  3 -259.52411950  1s(2).2p(5)_2P
 716917.42 CM-1       139.49 ANGS(VAC)       139.49 ANGS(AIR)
M1  S =  1.73052D-06   GF =  5.01695D-11   AKI =  4.29992D+00
```

```
  3 -263.27978580  1s(2).2s(2).2p(3)4S3_4S
  1 -262.79063560  1s(2).2s(2).2p(3)2P1_2P
 107356.06 CM-1       931.48 ANGS(VAC)       931.48 ANGS(AIR)
M1  S =  1.86549D-03   GF =  8.09867D-09   AKI =  3.11300D+01


  3 -263.27978580  1s(2).2s(2).2p(3)4S3_4S
  1 -259.49794149  1s(2).2p(5)_2P
 830018.89 CM-1       120.48 ANGS(VAC)       120.48 ANGS(AIR)
M1  S =  5.92158D-07   GF =  1.98756D-11   AKI =  4.56677D+00


  3 -262.95505728  1s(2).2s(2).2p(3)2D3_2D
  1 -262.79063560  1s(2).2s(2).2p(3)2P1_2P
  36086.39 CM-1      2771.13 ANGS(VAC)      2770.83 ANGS(AIR)
M1  S =  2.01353D-02   GF =  2.93830D-08   AKI =  1.27613D+01


  3 -262.95505728  1s(2).2s(2).2p(3)2D3_2D
  1 -259.49794149  1s(2).2p(5)_2P
 758749.21 CM-1       131.80 ANGS(VAC)       131.80 ANGS(AIR)
M1  S =  7.20430D-07   GF =  2.21047D-11   AKI =  4.24418D+00


  3 -262.78827596  1s(2).2s(2).2p(3)2P1_2P
  1 -259.49794149  1s(2).2p(5)_2P
 722144.94 CM-1       138.48 ANGS(VAC)       138.48 ANGS(AIR)
M1  S =  2.16256D-06   GF =  6.31521D-11   AKI =  1.09837D+01


  3 -259.52411950  1s(2).2p(5)_2P
  1 -259.49794149  1s(2).2p(5)_2P
   5745.41 CM-1     17405.20 ANGS(VAC)     17403.40 ANGS(AIR)
M1  S =  1.33224D+00   GF =  3.09528D-07   AKI =  3.40764D+00


  3 -263.27978580  1s(2).2s(2).2p(3)4S3_4S
  3 -262.95505728  1s(2).2s(2).2p(3)2D3_2D
  71269.67 CM-1      1403.12 ANGS(VAC)      1403.12 ANGS(AIR)
M1  S =  5.78084D-04   GF =  1.66606D-09   AKI =  1.41118D+00


  3 -263.27978580  1s(2).2s(2).2p(3)4S3_4S
  3 -262.78827596  1s(2).2s(2).2p(3)2P1_2P
 107873.94 CM-1       927.01 ANGS(VAC)       927.01 ANGS(AIR)
M1  S =  8.88733D-03   GF =  3.87688D-08   AKI =  7.52312D+01


  3 -263.27978580  1s(2).2s(2).2p(3)4S3_4S
  3 -259.52411950  1s(2).2p(5)_2P
 824273.48 CM-1       121.32 ANGS(VAC)       121.32 ANGS(AIR)
M1  S =  3.31682D-06   GF =  1.10557D-10   AKI =  1.25260D+01
```

```
  3 -262.95505728  1s(2).2s(2).2p(3)2D3_2D
  3 -262.78827596  1s(2).2s(2).2p(3)2P1_2P
  36604.27 CM-1      2731.92 ANGS(VAC)      2731.63 ANGS(AIR)
M1  S =  6.39782D-02   GF =  9.47019D-08   AKI =  2.11594D+01


  3 -262.95505728  1s(2).2s(2).2p(3)2D3_2D
  3 -259.52411950  1s(2).2p(5)_2P
 753003.80 CM-1       132.80 ANGS(VAC)       132.80 ANGS(AIR)
M1  S =  3.34446D-06   GF =  1.01840D-10   AKI =  9.62931D+00


  3 -262.78827596  1s(2).2s(2).2p(3)2P1_2P
  3 -259.52411950  1s(2).2p(5)_2P
 716399.54 CM-1       139.59 ANGS(VAC)       139.59 ANGS(AIR)
M1  S =  2.90938D-08   GF =  8.42851D-13   AKI =  7.21346D-02


  3 -263.27978580  1s(2).2s(2).2p(3)4S3_4S
  5 -262.95382232  1s(2).2s(2).2p(3)2D3_2D
  71540.71 CM-1      1397.81 ANGS(VAC)      1397.81 ANGS(AIR)
M1  S =  1.21484D-05   GF =  3.51453D-11   AKI =  1.99970D-02


  3 -262.95505728  1s(2).2s(2).2p(3)2D3_2D
  5 -262.95382232  1s(2).2s(2).2p(3)2D3_2D
    271.04 CM-1    368948.37 ANGS(VAC)    368910.23 ANGS(AIR)
M1  S =  2.36306D+00   GF =  2.59002D-08   AKI =  2.11526D-04


  5 -262.95382232  1s(2).2s(2).2p(3)2D3_2D
  3 -262.78827596  1s(2).2s(2).2p(3)2P1_2P
  36333.23 CM-1      2752.30 ANGS(VAC)      2752.01 ANGS(AIR)
M1  S =  3.63575D-02   GF =  5.34186D-08   AKI =  1.17593D+01


  5 -262.95382232  1s(2).2s(2).2p(3)2D3_2D
  3 -259.52411950  1s(2).2p(5)_2P
 752732.76 CM-1       132.85 ANGS(VAC)       132.85 ANGS(AIR)
M1  S =  1.88251D-06   GF =  5.73023D-11   AKI =  5.41422D+00
```

Here labels of the upper and lower states in the transition are in LSJ-notation. In addition to transition energies in cm^{-1} also the wave lengths (vacuum and air) are given in Angstrom (ANGS). On the next line the line strength S, the weighted oscillator strength gf and the transition rate A (AKI) are given. The format is the same as the one produced by the transition program of ATSP2K [1]

9.3 Output files from the fifth example

The fifth example was the study of energy spectra for Ni XIV giving the unique labels.

The uniquelabel summary file

Using the unique option of the `jj2lsj` program produces a summary `name.uni.lsj.sum`. Below is the summary file `Ni_even_n4.uni.lsj.sum`. In `name.uni.lsj.sum` information for the levels is given: Pos, composition of the level, serial number of composition, and the label of the level. From the `Ni_even_n4.uni.lsj.sum` file we see that the level with $J = 1/2$ and Pos $= 2$ has serial No of composition $= 2$ and the level with $J = 1/2$ and Pos $= 5$ has serial No of composition $= 4$. These levels were thus re-identified.

```
          Composition  Serial No.         Coupling
                       of compos.
 J =   1/2
--------------------------------------------------
Pos   4   0.941860160    1   2s(2).2p(6).3s(2).3p(2)3P2_3P.3d_4D
Pos   1   0.860331100    1   2s(2).2p(6).3s_2S.3p(4)3P2_4P
Pos   8   0.664848070    1   2s(2).2p(6).3s(2).3p(2)1D2_1D.3d_2S
Pos   7   0.554194670    1   2s(2).2p(6).3s(2).3p(2)1D2_1D.3d_2P
Pos   6   0.549911900    1   2s(2).2p(6).3s(2).3p(2)3P2_3P.3d_4P
Pos   3   0.488023350    1   2s(2).2p(6).3s_2S.3p(4)1S0_2S
Pos   2   0.301423810    2   2s(2).2p(6).3s_2S.3p(4)3P2_2P
Pos   5   0.112705240    4   2s(2).2p(6).3s(2).3p(2)3P2_3P.3d_2P
--------------------------------------------------


 ...........



          Composition  Serial No.         Coupling
                       of compos.
 J =   9/2
--------------------------------------------------
Pos   1   0.937466790    1   2s(2).2p(6).3s(2).3p(2)3P2_3P.3d_4F
Pos   2   0.936202770    1   2s(2).2p(6).3s(2).3p(2)1D2_1D.3d_2G
--------------------------------------------------
```

The unique label composition file

The `jj2lsj` program produces also `name.uni.lsj.lbl`. Below is the output file `Ni_even_n4.uni.lsj.lbl` from `jj2lsj` with the unique option. This file has the same format as `Ni_even_n4.lsj.lbl`, except that the levels with the same labels were re-identified. As seen from the output file, for the level with $J = 1/2$ and Pos $= 2$ the largest expansion coefficient does not appear on the first line. This level was re-identified. The users should use `name.lsj.uni.lbl` file in further calculations (`rtransition`, `rhfs`, etc.) to obtain output with unique labels.

```
 Pos   J   Parity      Energy Total      Comp. of ASF
  1  1/2     +         -1441.689218199      99.941%
       -0.92754035     0.86033110   2s(2).2p(6).3s_2S.3p(4)3P2_4P
       -0.31645458     0.10014350   2s(2).2p(6).3s(2).3p(2)3P2_3P.3d_4P
       -0.13107331     0.01718021   2s(2).2p(6).3s_2S.3p(4)1S0_2S
       -0.06808263     0.00463524   2s(2).2p(6).3s_2S.3p(2)3P2_4P.3d(2)1S0_4P
       -0.06306078     0.00397666   2s(2).2p(6).3s_2S.3p(2)3P2_4P.3d(2)1D2_4P
       -0.06139973     0.00376993   2s(2).2p(6).3p(4)3P2_3P.3d_4P
       -0.04384457     0.00192235   2s(2).2p(6).3s_2S.3p(2)1D2_2D.3d(2)3P2_4P
```

```
         0.04315649     0.00186248   2s(2).2p(6).3s(2).3p(2)1D2_1D.3d_2S
         0.04160783     0.00173121   2s(2).2p(6).3s_2S.3p(2)3P2_4P.3d(2)3P2_4P
  2  1/2      +         -1441.145635652      99.870%
         0.54902077     0.30142381   2s(2).2p(6).3s_2S.3p(4)3P2_2P
         0.55232128     0.30505880   2s(2).2p(6).3s_2S.3p(4)1S0_2S
        -0.51853413     0.26887764   2s(2).2p(6).3s(2).3p(2)3P2_3P.3d_2P
        -0.25240762     0.06370961   2s(2).2p(6).3s(2).3p(2)1D2_1D.3d_2S
         0.14975498     0.02242655   2s(2).2p(6).3s(2).3p(2)1D2_1D.3d_2P
         0.08843968     0.00782158   2s(2).2p(6).3p(4)1D2_1D.3d_2P
        -0.07915249     0.00626512   2s(2).2p(6).3s(2).3p(2)3P2_3P.3d_4D
         0.06957005     0.00483999   2s(2).2p(6).3s_2S.3p(2)1S0_2S.3d(2)1S0_2S
        -0.06792256     0.00461347   2s(2).2p(6).3s_2S.3p(4)3P2_4P
        -0.04635869     0.00214913   2s(2).2p(6).3s_2S.3p(2)1D2_2D.3d(2)1D2_2P
        -0.04440113     0.00197146   2s(2).2p(6).3s_2S.3p(2)3P2_4P.3d(2)3P2_2P
         0.03795533     0.00144061   2s(2).2p(6).3s_2S.3p(2)1D2_2D.3d(2)3P2_2P
        -0.03449961     0.00119022   2s(2).2p(6).3s_2S.3p(2)3P2_4P.3d(2)3P2_2S
        -0.03371279     0.00113655   2s(2).2p(6).3s(2).3p(2)3P2_3P.3d_4P
        -0.03274468     0.00107221   2s(2).2p(6).3s_2S.3p(2)1D2_2D.3d(2)1D2_2S
         0.03172321     0.00100636   2s(2).2p(6).3s_2S.3p(2)1D2_2D.3d(2)3F2_2P
  3  1/2      +         -1441.040628181      99.883%
         0.69858668     0.48802335   2s(2).2p(6).3s_2S.3p(4)1S0_2S
         0.44942201     0.20198014   2s(2).2p(6).3s(2).3p(2)3P2_3P.3d_2P
        -0.37637049     0.14165475   2s(2).2p(6).3s_2S.3p(4)3P2_2P
        -0.31031417     0.09629489   2s(2).2p(6).3s(2).3p(2)1D2_1D.3d_2S
        -0.14516095     0.02107170   2s(2).2p(6).3s(2).3p(2)1D2_1D.3d_2P
         0.11019096     0.01214205   2s(2).2p(6).3s(2).3p(2)3P2_3P.3d_4D
        -0.10593148     0.01122148   2s(2).2p(6).3s_2S.3p(4)3P2_4P
         0.08895336     0.00791270   2s(2).2p(6).3s_2S.3p(2)1S0_2S.3d(2)1S0_2S
        -0.06646197     0.00441719   2s(2).2p(6).3p(4)1D2_1D.3d_2P
        -0.04537373     0.00205877   2s(2).2p(6).3s(2).3p(2)3P2_3P.3d_4P
        -0.04337124     0.00188106   2s(2).2p(6).3s_2S.3p(2)3P2_4P.3d(2)3P2_2S
        -0.04274332     0.00182699   2s(2).2p(6).3s_2S.3p(2)1D2_2D.3d(2)1D2_2S
         0.04115899     0.00169406   2s(2).2p(6).3s_2S.3p(2)1D2_2D.3d(2)1D2_2P
         0.03553786     0.00126294   2s(2).2p(6).3s_2S.3p(2)3P2_4P.3d(2)3P2_2P

...........
```

Chapter 10

Case study I: $2s^22p$, $2s2p^2$ in Mo XXXVIII using scripts

In this case study we use script files to perform systematic calculations for all the states of the $2s^22p$ and $2s2p^2$ configurations in Mo XXXVIII. The 10 states are as follows:

odd: $2s^22p\ {}^2P_{1/2,3/2}$

even: $2s2p^2\ {}^4P_{1/2,3/2,5/2}$, $2s2p^2\ {}^2D_{3/2,5/2}$, $2s2p^2\ {}^2P_{1/2,3/2}$, $2s2p^2\ {}^2S_{1/2}$

The script files can be found in `grasptest/case1/script`.

In a real application a correlation model should be defined, i.e. some rule to generate the CSFs from an orbital set. The convergence of computed properties is then monitored as the orbital set is increased. For the odd state a reasonable correlation model is to start from the $\{1s^22s^22p, 1s^22p^3\}$ multireference and then generate all CSFs that can be obtained by single- and double- excitations from the multireference to an active set of orbitals. The active set of orbitals is then systematically increased. Following the normal conventions the orbital set is denoted by the highest principal quantum number. For example $n = 3$ means the orbital set $\{1s, 2s, 2p, 3s, 3p, 3d\}$. In this study we increase the active set of orbitals up to $n = 6$. For the even states we start from the $1s^22s2p^2$ reference and generate all CSFs that can be obtained by single- and double- excitations from the multireference to the active sets of orbitals. The correlation model can be easily extended by adding CSFs to the multireference.

10.1 Running script files

To automate the calculations we use script files. For convenience we have a main script that calls subscripts to perform different tasks. The construction of the scripts is greatly simplified if the names of the files are chosen in a simple and systematic way. In the case study we use the names `odd2, odd3, odd4, odd5, odd6` and `even2, even3, even4, even5, even6` to denote files for the odd and even parity states, respectively. The digit indicates which orbital set has been used to generate the expansion.

Before starting please make sure that the GRASP2018 executables are on the path.

The main script `sh_case1` is shown below. This script controls the computational flow and calls several subscripts.

```
#!/bin/sh

set -x
```

```
#    Main script for 2s(2)2p and 2s2p(2)

# 1.   Generate the expansions
        ./sh_files_c

# 2.   Get the nuclear data
        ./sh_nuc

# 3.   Get screened hydrogenic orbitals as initial estimates
        ./sh_initial

# 4.   Perform scf calculations and a final rci calculation that
#      includes the Breit correction and QED. All calculations
#      are transformed to LSJ-coupling. Files are created that
#      support creation of energy tables
        ./sh_scf

# 5.   Perform a transition calculation
        ./sh_tr
```

Each of the subscripts are given below together with some comments.

If all script files are available with execute permission (use the command `chmod +x`) we start the computation by typing the name of the main script

```
./sh_case1
```

Please note that these calculations will take several hours!

1. Generate expansions

The expansions are generated by the script `sh_files_c`. This is by far the most complicated script. It is simplified by generating lists for large active sets and then using `rcsfsplit`.

```
#!/bin/sh

set -x

#  1.  Generate CSF expansions
#      1.1 MR for 2s(2)2p, 2p(3)

rcsfgenerate <<EOF1
*
0
1s(2,i)2s(2,i)2p(1,i)
1s(2,i)2p(3,i)

2s,2p
1,3
0
n
EOF1

cp rcsf.out odd2.c

#        1.2 SD-MR for n=6
```

```
rcsfgenerate <<EOF3
*
0
1s(2,*)2s(2,*)2p(1,*)
1s(2,*)2p(3,*)

6s,6p,6d,6f,6g,6h
1,3
2
n
EOF3

cp rcsf.out odd.c

#       Split into odd3.c, odd4.c, odd5.c, odd6.c

rcsfsplit <<EOF5
odd
4
3s,3p,3d
3
4s,4p,4d,4f
4
5s,5p,5d,5f,5g
5
6s,6p,6d,6f,6g,6h
6
EOF5

##########################################

#  2.  Generate CSF expansions
#      2.1 for 2s2p(2)

rcsfgenerate <<EOF1
*
0
1s(2,i)2s(1,i)2p(2,i)

2s,2p
1,5
0
n
EOF1

cp rcsf.out even2.c

#        2.2 SD for n=6

rcsfgenerate <<EOF3
*
0
1s(2,*)2s(1,*)2p(2,*)
```

```
6s,6p,6d,6f,6g,6h
1,5
2
n
EOF3

cp rcsf.out even.c

#       Split into even3.c, even4.c, even5.c, even6.c

rcsfsplit <<EOF5
even
4
3s,3p,3d
3
4s,4p,4d,4f
4
5s,5p,5d,5f,5g
5
6s,6p,6d,6f,6g,6h
6
EOF5
```

2. Get nuclear data

Nuclear data are defined by the script `sh_nuc`. Since we are not interested in hyperfine structure the nuclear spin and moments have all been set to 1.

```
#!/bin/sh
set -x

#  2.  Get nucleardata
rnucleus <<S1
42
96
n
96
1
1
1
S1

cat isodata
```

3. Get initial estimates

The script `sh_initial` performs angular integration, gets initial estimates and performs scf calculations for the odd and even reference states (odd2 and even2). As initial estimates we use screened hydrogenic functions. For the reference states all orbitals are required to be spectroscopic, i.e. they should have the correct number of nodes.

```
#!/bin/sh
set -x

# 3. For n=2, Get initial estimates for odd.
```

```
cp odd2.c rcsf.inp
rangular  <<S4
y
S4

#  Get initial estimates of wave functions
rwfnestimate <<S5
y
3
*
S5

# Perform self-consistent field calculations
rmcdhf > outodd_rmcdhf_initial <<S6
y
1
1
5
*
*
100
S6

#  Save the result to odd2
rsave odd2

# 3. For n=2, Get initial estimates for even

cp even2.c rcsf.inp
rangular  <<S4
y
S4

#  Get initial estimates of wave functions
rwfnestimate <<S5
y
3
*
S5

# Perform self-consistent field calculations
rmcdhf > outeven_rmcdhf_initial <<S6
y
1,2,3
1,2,3
1,2
5
*
*
100
S6

#  Save the result to even2
```

```
rsave even2
```

4. rmcdhf and rci calculations

The script `sh_scf` performs angular integration, estimates the new radial functions and performs self-consistent field calculations (rmcdhf) for the odd and even states up to $n = 6$. At the end RCI calculations are performed for the largest expansions. The RCI calculations include Breit interaction and QED corrections. All results are transformed to LSJ-coupling. Please note how we loop in the script over the digit n that indicates the size of the orbital set.

```
#!/bin/sh
set -x

#   4.  Get results for odd n=3,4,5,6

for n in 3 4 5 6
do
   (cp odd${n}.c rcsf.inp

#  Get angular data
rangular <<S4
y
S4

# Get initial estimates of wave functions
m=`expr $n - 1`
echo m=$m n=$n
rwfnestimate <<S5
y
1
odd${m}.w
*
3
*
S5

# Perform self-consistent field calculations
rmcdhf > outodd_rmcdhf_${n} <<S6
y
1
1
5
${n}*

100
S6

rsave odd${n}

# transform to LSJ-coupling

jj2lsj  <<S1
odd${n}
n
```

```
y
y
S1

   echo)
done

#  Perform Breit-correction using RCI for n=6. First copy to other file names

n=6
cp odd${n}.c oddCI${n}.c
cp odd${n}.w oddCI${n}.w

rci > outodd_rci <<S7
y
oddCI${n}
y
y
1.d-6
y
n
n
y
4
1
1
S7

# transform to LSJ-coupling

jj2lsj  <<S1
oddCI${n}
y
y
y
S1


#   4.  Get results for even n=3,4,5,6

for n in 3 4 5 6
do
   (cp even${n}.c rcsf.inp

#  Get angular data
rangular <<S4
y
S4

# Get initial estimates of wave functions
m=`expr $n - 1`
echo m=$m n=$n
rwfnestimate <<S5
y
```

```
1
even${m}.w
*
3
*
S5

# Perform self-consistent field calculations
rmcdhf > outeven_rmcdhf_${n} <<S6
y
1,2,3
1,2,3
1,2
5
${n}*

100
S6

rsave even${n}

# transform to LSJ-coupling

jj2lsj  <<S1
even${n}
n
y
y
S1


   echo)
done

#  Perform Breit-correction using RCI for n=6

n=6
cp even${n}.c evenCI${n}.c
cp even${n}.w evenCI${n}.w

rci > outeven_rci <<S7
y
evenCI${n}
y
y
1.d-6
y
n
n
y
4
1,2,3
1,2,3
1,2
```

```
S7

# transform to LSJ-coupling

jj2lsj  <<S1
evenCI${n}
y
y
y
S1
```

5. Transition calculation

The script `sh_tr` computes the E1 transition rates between the odd and even states. First we perform a biorthonormal transformation and then we perform the transition calculation itself.

```
#!/bin/sh
set -x

#  6. Perform transition calculation for the n=6 CI results

n=6

#  First the biorthonormal transformations

rbiotransform > out_rbiotransform <<EOF
y
y
oddCI$n
evenCI$n
y
EOF

# Then the transition calculations

rtransition > out_transition <<EOF
y
y
oddCI$n
evenCI$n
E1
EOF
```

10.2 Comparison with experiment

To display the computed energies we give the command

```
rlevels oddCI6.cm evenCI6.cm
```

The computer returns the energies together with labels in LSJ-coupling for all the states.

```
 nblock =            2   ncftot =        20641   nw =           36   nelec =             5
 nblock =            3   ncftot =        36290   nw =           36   nelec =             5

 Energy levels for ...
```

```
Rydberg constant is   109737.31569
Splitting is the energy difference with the lower neighbor
-----------------------------------------------------------------------------------------
No Pos  J Parity Energy Total    Levels     Splitting     Configuration
                    (a.u.)      (cm^-1)     (cm^-1)
-----------------------------------------------------------------------------------------
  1  1  1/2 -   -2386.3197033        0.00        0.00  1s(2).2s(2).2p_2P
  2  1  1/2 +   -2382.2388943   895634.04   895634.04  1s(2).2s_2S.2p(2)3P2_4P
  3  1  3/2 -   -2381.9238877   964770.00    69135.96  1s(2).2s(2).2p_2P
  4  1  3/2 +   -2378.9983289  1606855.95   642085.94  1s(2).2s_2S.2p(2)3P2_4P
  5  1  5/2 +   -2378.1473251  1793629.70   186773.75  1s(2).2s_2S.2p(2)3P2_4P
  6  2  3/2 +   -2376.7266916  2105422.71   311793.02  1s(2).2s_2S.2p(2)1D2_2D
  7  2  1/2 +   -2376.5309659  2148379.52    42956.81  1s(2).2s_2S.2p(2)3P2_2P
  8  2  5/2 +   -2373.9010567  2725577.88   577198.36  1s(2).2s_2S.2p(2)1D2_2D
  9  3  1/2 +   -2371.9002512  3164703.93   439126.05  1s(2).2s_2S.2p(2)1S0_2S
 10  3  3/2 +   -2371.8558069  3174458.33     9754.40  1s(2).2s_2S.2p(2)3P2_2P
-----------------------------------------------------------------------------------------
```

The Mo XXXVIII transitions have been observed in the JET Tokamak, Myrnäs *et al.* [34] In the table below the experimental transition energies are compared with the calculated energies. Please note that the quantum labels for the $2s2p^2\ ^2P_{1/2}$ and $2s2p^2\ ^2S_{1/2}$ seem to have been swapped in the experimental paper, i.e. the highest $J = 1/2$ state should be $2s2p^2\ ^2S_{1/2}$ and not $2s2p^2\ ^2P_{1/2}$. We see that the odd states are somewhat too high. This is due to an imbalance in the multireference. As discussed in the beginning of the chapter the correlation model can be refined by extending the multireference. Adopting the multireferences $\{2s^22p, 2p^3, 2s2p3d, 2p^23d\}$ and $\{2s2p^2, 2p^23d, 2s^23d, 2s3d^2\}$ for, respectively, the odd and even parity states improves the energy separations considerably [35]. A careful investigation of the effects of increasing the multireference is part of any systematic calculation (see section 7.6).

State (exp. label)	Exp. (Myrnäs et al., [34])	Calc. (Rynkun et al. [35])
$2s^22p\ ^2P_{1/2}$	0	0
$2s2p^2\ ^4P_{1/2}$	$894\ 050 \pm 400$	895 848
$2s^22p\ ^2P_{3/2}$	$964\ 050 \pm 90$	964 715
$2s2p^2\ ^4P_{3/2}$		1 606 863
$2s2p^2\ ^4P_{5/2}$	$1\ 790\ 130 \pm 200$	1 793 682
$2s2p^2\ ^2D_{3/2}$	$2\ 102\ 900 \pm 900$	2 106 354
$2s2p^2\ ^2S_{1/2}$	$2\ 147\ 300 \pm 900$	2 149 456
$2s2p^2\ ^2D_{5/2}$		2 725 586
$2s2p^2\ ^2P_{1/2}$	$3\ 164\ 770 \pm 1500$	3 166 168
$2s2p^2\ ^2P_{3/2}$	$3\ 171\ 300 \pm 1500$	3 175 559

10.3 Transition rates

Below are the transition parameters as given in the file `oddCI6.evenCI6.ct.lsj`. The agreement between calculated values in the two gauges (the first line gives values in length gauge and the second line gives values in velocity gauge) is quite good, especially for the strong transitions. Expansions based on a larger multireference set will further improve the agreement. The quantity dT is defined as

$$dT = \frac{|A_l - A_v|}{\max(A_l, A_v)},$$

where A_l and A_v are the transition rates in length and velocity gauge. dT is a measure of the uncertainty of the computed transition rates [33].

```
Transition between files:
oddCI6
evenCI6


  1-2386.31970329  1s(2).2s(2).2p_2P
  1-2382.23889432  1s(2).2s_2S.2p(2)3P2_4P
 895634.04 CM-1       111.65 ANGS(VAC)       111.65 ANGS(AIR)
E1  S =  4.42135D-03   GF =  1.20285D-02   AKI =  3.21798D+09   dT =  0.05152
         4.66152D-03         1.26818D-02          3.39278D+09


  1-2386.31970329  1s(2).2s(2).2p_2P
  1-2376.53096594  1s(2).2s_2S.2p(2)3P2_2P
2148379.52 CM-1        46.55 ANGS(VAC)        46.55 ANGS(AIR)
E1  S =  1.87910D-02   GF =  1.22627D-01   AKI =  1.88764D+11   dT =  0.00415
         1.88692D-02         1.23137D-01          1.89550D+11


  1-2386.31970329  1s(2).2s(2).2p_2P
  1-2371.90025121  1s(2).2s_2S.2p(2)1S0_2S
3164703.93 CM-1        31.60 ANGS(VAC)        31.60 ANGS(AIR)
E1  S =  1.51569D-05   GF =  1.45703D-04   AKI =  4.86683D+08   dT =  0.06893
         1.62791D-05         1.56490D-04          5.22716D+08


  1-2386.31970329  1s(2).2s(2).2p_2P
  3-2378.99832890  1s(2).2s_2S.2p(2)3P2_4P
1606855.95 CM-1        62.23 ANGS(VAC)        62.23 ANGS(AIR)
E1  S =  1.11563D-04   GF =  5.44532D-04   AKI =  2.34455D+08   dT =  0.01690
         1.13481D-04         5.53891D-04          2.38485D+08


  1-2386.31970329  1s(2).2s(2).2p_2P
  3-2376.72669156  1s(2).2s_2S.2p(2)1D2_2D
2105422.71 CM-1        47.50 ANGS(VAC)        47.50 ANGS(AIR)
E1  S =  2.45759D-02   GF =  1.57171D-01   AKI =  1.16181D+11   dT =  0.00644
         2.47352D-02         1.58190D-01          1.16934D+11


  1-2386.31970329  1s(2).2s(2).2p_2P
  3-2371.85580688  1s(2).2s_2S.2p(2)3P2_2P
3174458.33 CM-1        31.50 ANGS(VAC)        31.50 ANGS(AIR)
E1  S =  6.34294D-04   GF =  6.11624D-03   AKI =  1.02779D+10   dT =  0.00529
         6.30940D-04         6.08390D-03          1.02236D+10


  1-2382.23889432  1s(2).2s_2S.2p(2)3P2_4P
  3-2381.92388771  1s(2).2s(2).2p_2P
  69135.96 CM-1      1446.43 ANGS(VAC)      1446.43 ANGS(AIR)
E1  S =  5.06267D-04   GF =  1.06318D-04   AKI =  8.47420D+04   dT =  0.18842
         4.10877D-04         8.62859D-05          6.87749D+04
```

```
  3-2381.92388771  1s(2).2s(2).2p_2P
  1-2376.53096594  1s(2).2s_2S.2p(2)3P2_2P
1183609.52 CM-1        84.49 ANGS(VAC)        84.49 ANGS(AIR)
E1  S =  2.36463D-03   GF =  8.50153D-03   AKI =  3.97215D+09   dT =  0.05227
         2.49505D-03         8.97042D-03          4.19123D+09


  3-2381.92388771  1s(2).2s(2).2p_2P
  1-2371.90025121  1s(2).2s_2S.2p(2)1S0_2S
2199933.92 CM-1        45.46 ANGS(VAC)        45.46 ANGS(AIR)
E1  S =  1.48021D-02   GF =  9.89138D-02   AKI =  1.59657D+11   dT =  0.00260
         1.47636D-02         9.86564D-02          1.59242D+11


  3-2381.92388771  1s(2).2s(2).2p_2P
  3-2378.99832890  1s(2).2s_2S.2p(2)3P2_4P
 642085.94 CM-1       155.74 ANGS(VAC)       155.74 ANGS(AIR)
E1  S =  1.00858D-03   GF =  1.96711D-03   AKI =  1.35237D+08   dT =  0.11531
         1.14004D-03         2.22351D-03          1.52865D+08


  3-2381.92388771  1s(2).2s(2).2p_2P
  3-2376.72669156  1s(2).2s_2S.2p(2)1D2_2D
1140652.71 CM-1        87.67 ANGS(VAC)        87.67 ANGS(AIR)
E1  S =  2.73257D-03   GF =  9.46780D-03   AKI =  2.05418D+09   dT =  0.04984
         2.87591D-03         9.96446D-03          2.16194D+09


  3-2381.92388771  1s(2).2s(2).2p_2P
  3-2371.85580688  1s(2).2s_2S.2p(2)3P2_2P
2209688.33 CM-1        45.26 ANGS(VAC)        45.26 ANGS(AIR)
E1  S =  4.49452D-02   GF =  3.01675D-01   AKI =  2.45631D+11   dT =  0.00389
         4.51208D-02         3.02853D-01          2.46590D+11


  3-2381.92388771  1s(2).2s(2).2p_2P
  5-2378.14732507  1s(2).2s_2S.2p(2)3P2_4P
 828859.69 CM-1       120.65 ANGS(VAC)       120.65 ANGS(AIR)
E1  S =  9.97377D-03   GF =  2.51110D-02   AKI =  1.91786D+09   dT =  0.08160
         1.08599D-02         2.73420D-02          2.08825D+09


  3-2381.92388771  1s(2).2s(2).2p_2P
  5-2373.90105671  1s(2).2s_2S.2p(2)1D2_2D
1760807.88 CM-1        56.79 ANGS(VAC)        56.79 ANGS(AIR)
E1  S =  1.52580D-02   GF =  8.16081D-02   AKI =  2.81286D+10   dT =  0.01679
         1.55185D-02         8.30015D-02          2.86089D+10
```

10.4 LaTeX table with energies as functions of the active set

The script `sh_scf` is written in such a way that all results are transformed to LSJ-coupling using `jj2lsj`. The output from `rlevels` will then contain quantum labels of the states in LSJ-coupling. By saving the output from `rlevels` we can generate a LaTeX table showing the convergence of the energies as the active set is increased. If we include also the output from the final RCI calculation we can see the effect of the Breit interaction and QED.

Issuing the commands below saves the output from `rlevels` corresponding to the increasing active set of orbitals in files `energy3`, `energy4`, `energy5`, `energy6`, `energyCI6`

```
>>rlevels even3.m odd3.m > energy3
>>rlevels even4.m odd4.m > energy4
>>rlevels even5.m odd5.m > energy5
>>rlevels even6.m odd6.m > energy6
>>rlevels evenCI6.cm oddCI6.cm > energyCI6
```

We now call `rtablevels` to produce a LaTeX table

```
>>rtablevels

 RTABLEVELS
 Makes LaTeX and ASCII tables of energy files produced by
 rlevels (in ljs format)
 Multiple energy files can be used as input
 Energies from file 1 fills column 1, energies from file 2
 fills column 2 etc.  Checks are done to see if the labels
 if the labels in the files are consistent
 Input file: name1, name2, ...
 Output files: energylabellatex.tex, energylabelascii.txt

 Inspect energy files and determine how many positions
 should be skipped in the string that determines the label
 e.g. if the string is 1s(2).2s_2S.2p(2)3P2_4P and 1s(2) is a core
 then you would like to skip 1s(2). i.e. 6 positions and determine
 the label from 2s_2S.2p(2)3P2_4P

 How many positions should be skipped?
>>6
 Give the number of energy files from rlevels
>>5
 Name of file 1
>>energy3
 Name of file 2
>>energy4
 Name of file 3
>>energy5
 Name of file 4
>>energy6
 Name of file 5
>>energyCI6
```

The generated LaTeX table is named `energytablelatex.tex`. The processed table is shown below.

$2s^2\,2p\;^2P^o_{1/2}$	0	0	0	0	0
$2s\;^2S\,2p^2(^3_2P)\;^4P_{1/2}$	894244	894014	893851	893916	895634
$2s^2\,2p\;^2P^o_{3/2}$	982556	982795	982807	982793	964770
$2s\;^2S\,2p^2(^3_2P)\;^4P_{3/2}$	1620236	1621103	1621263	1621536	1606855
$2s\;^2S\,2p^2(^3_2P)\;^4P_{5/2}$	1823228	1823048	1822924	1823030	1793629

$2s\ ^2S\,2p^2(^1_2D)\ ^2D_{3/2}$	2130518	2128605	2127999	2127788	2105422
$2s\ ^2S\,2p^2(^3_2P)\ ^2P_{1/2}$	2163546	2161637	2160994	2160701	2148379
$2s\ ^2S\,2p^2(^1_2D)\ ^2D_{5/2}$	2764806	2764399	2764213	2764257	2725577
$2s\ ^2S\,2p^2(^1_0S)\ ^2S_{1/2}$	3193691	3191333	3190533	3190059	3164703
$2s\ ^2S\,2p^2(^3_2P)\ ^2P_{3/2}$	3213574	3211487	3210801	3210466	3174458

Table 10.1: Energies from the files energy3, energy4, energy5, energy6, energyCI6,

From the table we see that the energies seem to be reasonably converged when the active orbital set has been increased to $n = 6$. We also see that the Breit interaction and QED as included in the final RCI calculation changes the energies substantially.

At this point it is appropriate to comment the labels. The LS term for electrons with occupation one is not written out. The LS terms for equivalent electrons with occupation two or more are written in parenthesis together with the seniority number. The angular momenta are coupled from left to right and written in a linear fashion. As an example we look at

$$2s\ ^2S\,2p^2(^3_2P)\ ^4P_{5/2}$$

The $2s$ electron has the LS term ^2S, but this is not written out explicitly. The $2p^2$ has the LS term ^3P with seniority 2. This is written as $2p^2(^3_2P)$. Coupling the ^1S term of the $1s^2$ core (not included in the label) with the ^2S term of $2s$ results in ^2S. The ^2S term in turn is coupled with the ^3P term to yield ^4P. The final L and S are then coupled to $J = 5/2$. See also section 9.2 for a more detailed account of the labels.

10.5 LaTeX table with transition data

There is also a program `rtabtransE1` that produces a transition file. There are different options for the table. Below we choose to display gf and A in the length gauge together with dT

```
>>rtabtransE1

 RTABTRANSE1
 Makes LaTeX tables of transition data from transition files
 name1.name2.ct.lsj
 Input file: name1.name2.ct.lsj
 Output file: transitiontable.tex

 Specify table format
 (1). Lower & Upper & Energy diff. & wavelength & S & gf & A & dT
 (2). Lower & Upper & Energy diff. & wavelength & gf & A & dT
 (3). Lower & Upper & Energy diff. & wavelength & gf & A
 (4). Lower & Upper & Energy diff. & S & gf & A & dT
 (5). Lower & Upper & Energy diff. & gf & A & dT
 (6). Lower & Upper & Energy diff. & gf & A
>>5
 Inspect the name1.name2.ct.lsj file and determine how many positions
 should be skipped in the string that determines the label
 e.g. if the string is 1s(2).2s_2S.2p(2)3P2_4P and 1s(2) is a core
 then you would like to skip 1s(2). i.e. 6 positions and determine
 the label from 2s_2S.2p(2)3P2_4P
```

```
 How many positions should be skipped?
>>6
Name of file
>>oddCI6.evenCI6.ct.lsj
```

The generated LaTeX table is named `transitiontable.tex`. The processed table is shown below.

Lower state	Upper state	ΔE (cm^{-1})	gf	A (s^{-1})	dT
$2s^2\,2p\;^2P_{1/2}$	$2s\;^2S\,2p^2(^3_2P)\;^4P_{1/2}$	895634	1.202D-02	3.217D+09	0.051
$2s^2\,2p\;^2P_{1/2}$	$2s\;^2S\,2p^2(^3_2P)\;^2P_{1/2}$	2148379	1.226D-01	1.887D+11	0.004
$2s^2\,2p\;^2P_{1/2}$	$2s\;^2S\,2p^2(^1_0S)\;^2S_{1/2}$	3164703	1.457D-04	4.866D+08	0.068
$2s^2\,2p\;^2P_{1/2}$	$2s\;^2S\,2p^2(^3_2P)\;^4P_{3/2}$	1606855	5.445D-04	2.344D+08	0.016
$2s^2\,2p\;^2P_{1/2}$	$2s\;^2S\,2p^2(^1_2D)\;^2D_{3/2}$	2105422	1.571D-01	1.161D+11	0.006
$2s^2\,2p\;^2P_{1/2}$	$2s\;^2S\,2p^2(^3_2P)\;^2P_{3/2}$	3174458	6.116D-03	1.027D+10	0.005
$2s\;^2S\,2p^2(^3_2P)\;^4P_{1/2}$	$2s^2\,2p\;^2P_{3/2}$	69135	1.063D-04	8.474D+04	0.188
$2s^2\,2p\;^2P_{3/2}$	$2s\;^2S\,2p^2(^3_2P)\;^2P_{1/2}$	1183609	8.501D-03	3.972D+09	0.052
$2s^2\,2p\;^2P_{3/2}$	$2s\;^2S\,2p^2(^1_0S)\;^2S_{1/2}$	2199933	9.891D-02	1.596D+11	0.002
$2s^2\,2p\;^2P_{3/2}$	$2s\;^2S\,2p^2(^3_2P)\;^4P_{3/2}$	642085	1.967D-03	1.352D+08	0.115
$2s^2\,2p\;^2P_{3/2}$	$2s\;^2S\,2p^2(^1_2D)\;^2D_{3/2}$	1140652	9.467D-03	2.054D+09	0.049
$2s^2\,2p\;^2P_{3/2}$	$2s\;^2S\,2p^2(^3_2P)\;^2P_{3/2}$	2209688	3.016D-01	2.456D+11	0.003
$2s^2\,2p\;^2P_{3/2}$	$2s\;^2S\,2p^2(^3_2P)\;^4P_{5/2}$	828859	2.511D-02	1.917D+09	0.081
$2s^2\,2p\;^2P_{3/2}$	$2s\;^2S\,2p^2(^1_2D)\;^2D_{5/2}$	1760807	8.160D-02	2.812D+10	0.016

Table 10.2: Transition data from the file oddCI6.evenCI6.ct.lsj

10.6 Editing the LaTeX table

The table programs translate the LSJ-notation from `jj2lsj` (see section 9.2) to LaTeX notation. If the user wants to simplify or change the LaTeX notation this is easily done. For example the global substitution `2s~^2S\,` $\rightarrow$ `2s` in the LaTeX file produces the following table.

Lower state	Upper state	ΔE (cm^{-1})	gf	A (s^{-1})	dT
$2s^2\,2p\;^2P_{1/2}$	$2s2p^2(^3_2P)\;^4P_{1/2}$	895634	1.202D-02	3.217D+09	0.051
$2s^2\,2p\;^2P_{1/2}$	$2s2p^2(^3_2P)\;^2P_{1/2}$	2148379	1.226D-01	1.887D+11	0.004
$2s^2\,2p\;^2P_{1/2}$	$2s2p^2(^1_0S)\;^2S_{1/2}$	3164703	1.457D-04	4.866D+08	0.068
$2s^2\,2p\;^2P_{1/2}$	$2s2p^2(^3_2P)\;^4P_{3/2}$	1606855	5.445D-04	2.344D+08	0.016
$2s^2\,2p\;^2P_{1/2}$	$2s2p^2(^1_2D)\;^2D_{3/2}$	2105422	1.571D-01	1.161D+11	0.006
$2s^2\,2p\;^2P_{1/2}$	$2s2p^2(^3_2P)\;^2P_{3/2}$	3174458	6.116D-03	1.027D+10	0.005
$2s\;^2S\,2p^2(^3_2P)\;^4P_{1/2}$	$2s^2\,2p\;^2P_{3/2}$	69135	1.063D-04	8.474D+04	0.188
$2s^2\,2p\;^2P_{3/2}$	$2s2p^2(^3_2P)\;^2P_{1/2}$	1183609	8.501D-03	3.972D+09	0.052
$2s^2\,2p\;^2P_{3/2}$	$2s2p^2(^1_0S)\;^2S_{1/2}$	2199933	9.891D-02	1.596D+11	0.002
$2s^2\,2p\;^2P_{3/2}$	$2s2p^2(^3_2P)\;^4P_{3/2}$	642085	1.967D-03	1.352D+08	0.115
$2s^2\,2p\;^2P_{3/2}$	$2s2p^2(^1_2D)\;^2D_{3/2}$	1140652	9.467D-03	2.054D+09	0.049
$2s^2\,2p\;^2P_{3/2}$	$2s2p^2(^3_2P)\;^2P_{3/2}$	2209688	3.016D-01	2.456D+11	0.003
$2s^2\,2p\;^2P_{3/2}$	$2s2p^2(^3_2P)\;^4P_{5/2}$	828859	2.511D-02	1.917D+09	0.081
$2s^2\,2p\;^2P_{3/2}$	$2s2p^2(^1_2D)\;^2D_{5/2}$	1760807	8.160D-02	2.812D+10	0.016

Table 10.3: Transition data from the file oddCI6.evenCI6.ct.lsj

10.7 Scripts for MPI codes

The scripts above can, with very small modifications, also be used for performing the runs using the MPI codes. Before running the scripts for the MPI code the file `disks` with paths to the working directory and to the directory containing temporary data must be created (see section 7.4.). In the different scripts the calls to the MPI programs amount to changes of the type

```
rangular   -->  mpirun -np 8 rangular_mpi

rmcdhf     -->  mpirun -np 8 rmcdhf_mpi

rci        -->  mpirun -np 8 rci_mpi
```

etc. Scripts for the MPI cases are included under `case1_mpi` in the test data set (see section 2.11). Consult the `README` file in the working directory for more details on setting up the file `disks`.

Chapter 11

Case study II: the Li iso-electronic sequence using scripts

In this case study we use script files to perform systematic calculations for the $1s^22s\ ^2S_{1/2}$ ground state and the $1s^22p\ ^2P_{1/2,3/2}$ excited states in the Li iso-electronic sequence. Computing data for an iso-electronic sequence angular data can be reused and need not be recomputed for each member of the sequence. The script files can be found in `grasptest/case2/script`.

We start with a single calculation of the three reference states. After that separate calculations are done for the two parities. Correlation is then included by allowing single-, double-, and triple (SDT) excitations from the reference to active sets up to $n = 5$ (complete active space calculations). Calculations including hyperfine structures and transition rates are performed from $Z = 6$ to $Z = 12$.

It is convenient to save the results for the different ions in directories `Z6`, `Z7`, `Z8`, ..., `Z12`.

11.1 Running script files

The main script `sh_case2` is shown below. This script controls the computational flow and calls several subscripts.

```
#!/bin/sh

set -x

#    Main script for iso-electronic sequence

# 1.   Generate directories Z6, Z7, .. for the elements
#      Define nuclear data for each element

        ./sh_nuc_seq

# 2.   Generate lists of CSFs in main directory

        ./sh_files_c

# 3.   Start by performing rmcdhf calculations for the 1s(2)2s, 1s(2)2p
#      reference states

       ./sh_DF
```

```
# 4.    Perform rmcdhf calculations for all the even expansions
#       Angular data computed only once and then moved to different directories

        ./sh_even

# 5.    Perform rmcdhf calculations for all the odd expansions
#       Angular data computed only once and then moved to different directories

        ./sh_odd

# 6.    Perform rci calculations for the even5 and odd5 expansions
#       Perform rhfs and transition calculations.
#       Angular data computed only once and then moved to different directories

        ./sh_even_odd
```

Each of the subscripts are given below together with some comments.

If all script files are available with execute permission (use the command `chmod +x`) we start the computation by typing the name of the main script

```
./sh_case2
```

Please note that these calculations will take several hours!

1. Generate directories and define nuclear data

The script `sh_nuc_seq` produces nuclear data for $Z = 6, 7, \ldots, 12$ in the directories `Z6`, `Z7` ,..., `Z12`. By modifying the script we can produce nuclear data for any sequence of charges.

```
#!/bin/sh

set -x

# Full loop over all Z
#   for z in 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 \
#             26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 \
#             48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 \
#             70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 \
#             92 93 94 95 96 97 98 99 100 101 102 103 104 105 106 107 108 109   \
#             110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 \
#             126 127 128 129 130 131 132 133 134 135 136 137 138

# We select Z from 6 to 12
   for z in  6 7 8 9 10 11 12
   do

# Data from Jefferson Lab (http://education.jlab.org/itselemental)
case $z in
   1) m=0; MM=1.0794;;    #  Need to use point nucleus
   2) m=4; MM=4.002602;;
   3) m=7; MM=6.941;;
   4) m=9; MM=9.012182;;
   5) m=11; MM=10.811;;
```

```
 6) m=12; MM=12.0107;;
 7) m=14; MM=14.0067;;
 8) m=16; MM=15.9994;;
 9) m=19; MM=18.9984032;;
10) m=20; MM=20.1797;;
11) m=23; MM=22.98976928;;
12) m=24; MM=24.3050;;
13) m=27; MM=26.9815386;;
14) m=28; MM=29.0855;;
15) m=31; MM=30.973762;;
16) m=32; MM=32.065;;
17) m=35; MM=35.453;;
18) m=40; MM=39.948;;
19) m=39; MM=39.0938;;
20) m=40; MM=40.078;;
21) m=45; MM=44.955912;;
22) m=48; MM=47.867;;
23) m=51; MM=50.9415;;
24) m=52; MM=51.9961;;
25) m=55; MM=54.938045;;
26) m=56; MM=55.845;;
27) m=59; MM=58.933195;;
28) m=59; MM=58.6934;;
29) m=64; MM=63.546;;
30) m=65; MM=65.409;;
31) m=70; MM=69.723;;
32) m=73; MM=72.64;;
33) m=75; MM=74.92160;;
34) m=79; MM=78.96;;
35) m=80; MM=79.904;;
36) m=84; MM=83.798;;
37) m=85; MM=85.4678;;
38) m=88; MM=87.62;;
39) m=89; MM=88.90585;;
40) m=91; MM=91.224;;
41) m=93; MM=92.90638;;
42) m=96; MM=95.94;;
43) m=98; MM=98;;
44) m=101; MM=10.07;;
45) m=103; MM=102.90550;;
46) m=106; MM=106.42;;
47) m=108; MM=107.8682;;
48) m=112; MM=112.411;;
49) m=115; MM=114.818;;
50) m=119; MM=118.710;;
51) m=122; MM=121.760;;
52) m=128; MM=127.60;;
53) m=127; MM=126.90447;;
54) m=131; MM=131.293;;
55) m=133; MM=132.9054519;;
56) m=137; MM=137.327;;
57) m=139; MM=138.90547;;
58) m=140; MM=140.116;;
59) m=141; MM=140.90765;;
```

```
 60) m=144; MM=144.242;;
 61) m=145; MM=145;;
 62) m=150; MM=150.36;;
 63) m=152; MM=151.964;;
 64) m=157; MM=157.25;;
 65) m=159; MM=158.92535;;
 66) m=163; MM=162.5;;
 67) m=165; MM=164.93032;;
 68) m=167; MM=167.259;;
 69) m=169; MM=168.93421;;
 70) m=173; MM=173.04;;
 71) m=175; MM=174.967;;
 72) m=178; MM=178.49;;
 73) m=181; MM=180.94788;;
 74) m=184; MM=183.84;;
 75) m=186; MM=186.207;;
 76) m=190; MM=190.23;;
 77) m=192; MM=192.217;;
 78) m=195; MM=195.084;;
 79) m=197; MM=196.966569;;
 80) m=201; MM=200.59;;
 81) m=204; MM=204.3833;;
 82) m=207; MM=207.2;;
 83) m=209; MM=208.9804;;
 84) m=209; MM=209;;
 85) m=210; MM=210;;
 86) m=222; MM=222;;
 87) m=223; MM=223;;
 88) m=226; MM=226;;
 89) m=227; MM=227;;
 90) m=232; MM=232.03806;;
 91) m=231; MM=231.03588;;
 92) m=238; MM=238.02891;;
 93) m=237; MM=237;;
 94) m=244; MM=244;;
 95) m=243; MM=243;;
 96) m=247; MM=247;;
 97) m=247; MM=247;;
 98) m=251; MM=251;;
 99) m=252; MM=252;;
100) m=257; MM=257;;
101) m=258; MM=258;;
102) m=259; MM=259;;
103) m=262; MM=262;;
104) m=267; MM=267;;
105) m=268; MM=268;;
106) m=271; MM=271;;
107) m=272; MM=272;;
108) m=277; MM=277;;
109) m=276; MM=276;;
110) m=281; MM=281;;
111) m=280; MM=280;;
112) m=285; MM=285;;
113) m=284; MM=284;;
```

```
 114) m=289; MM=289;;
 115) m=288; MM=288;;
 116) m=291; MM=291;;
 117) m=293; MM=293;;   #Estimated
 118) m=294; MM=294;;
 119) m=316; MM=316;;
 120) m=318; MM=318;;
 121) m=322; MM=322;;
 122) m=324; MM=324;;
 123) m=326; MM=326;;
 124) m=330; MM=330;;
 125) m=332; MM=332;;
 126) m=334; MM=334;;
 127) m=338; MM=338;;
 128) m=340; MM=340;;
 129) m=342; MM=342;;
 130) m=346; MM=346;;
 131) m=348; MM=348;;
 132) m=350; MM=350;;
 133) m=354; MM=354;;
 134) m=356; MM=356;;
 135) m=358; MM=358;;
 136) m=362; MM=362;;
 137) m=364; MM=364;;
 138) m=366; MM=366;;
esac

echo "Starting: Z::"${z}, "ZZ::"$ZZ, "mass::"${m}, "Weight::"${MM}

rm -r Z$z
mkdir Z$z
cd Z$z

rnucleus <<EOF
$z
$m
n
$MM
1
1
1
EOF

cd ..

done
```

2. Generate expansions

The expansions are generated by the script `sh_files_c`.

```
rcsfgenerate << EOF
*
0
```

```
1s(2,i)2s(1,i)

2s,2p
1,1
0
y
1s(2,i)2p(1,i)

2s,2p
1,3
0
n
EOF

cp rcsf.out DF.c

#######################################

rcsfgenerate << EOF
*
0
1s(2,*)2s(1,*)

5s,5p,5d,5f,5g
1,1
3
n
EOF

cp rcsf.out even.c

rcsfsplit << EOF
even
3
3s,3p,3d
3
4s,4p,4d,4f
4
5s,5p,5d,5f,5g
5
EOF

################################

rcsfgenerate << EOF
*
0
1s(2,*)2p(1,*)

5s,5p,5d,5f,5g
1,3
3
n
EOF
```

```
cp rcsf.out odd.c

rcsfsplit << EOF
odd
3
3s,3p,3d
3
4s,4p,4d,4f
4
5s,5p,5d,5f,5g
5
EOF
```

3. Ground and excited reference states

The script `sh_DF` performs angular integration, gets initial estimates and performs scf calculations for the $1s^2 2s$, $1s^2 2p$ reference states. The angular integration is done only once and the `mcp.30`, `mcp.31` ... files are moved between the directories.

```
for z in 6 7 8 9 10 11 12
do
   (if test $z -lt 7
    then
cd Z${z}
cp ../DF.c rcsf.inp
#  Get angular data
rangular <<S4
y
S4

#Get initial estimates of wave functions
rwfnestimate <<S5
y
2
*
S5

# Perform self-consistent field calculations
rmcdhf > out_rmcdhf <<S6
y
1
1
1
5
*
*
100
S6

rsave DF
cp DF.w even2.w
cp DF.w odd2.w
```

```
cd ..
    else

cd Z${z}
cp ../DF.c rcsf.inp
#Move mcp files from previous directory
m=`expr $z - 1`
mv ../Z${m}/mcp* .

#Get initial estimates of wave functions
rwfnestimate <<S5
y
2
*
S5


# Perform self-consistent field calculations
rmcdhf > out_rmcdhf <<S6
y
1
1
1
5
*
*
100
S6

rsave DF
cp DF.w even2.w
cp DF.w odd2.w

cd ..

    fi
    echo)
done
```

4. Perform calculations for the even states

The script `sh_even` performs angular integration, gets initial estimates and performs `rmcdhf` calculations for the even states. The script loops over both the active set and the atomic number Z. Angular files are reused and moved between the directories.

```
for n in 3 4 5
do
   (
for z in 6 7 8 9 10 11 12
do
   (if test $z -lt 7
    then
cd Z${z}
cp ../even${n}.c rcsf.inp
```

```
#  Get angular data
rangular <<S4
y
S4

k=`expr $n - 1`
#Get initial estimates of wave functions
rwfnestimate <<S5
y
1
even${k}.w
*
2
*
S5

# Perform self-consistent field calculations
rmcdhf > outeven_rmcdhf_${n} <<S6
y
1
${n}*

100
S6

rsave even${n}
cd ..
    else

cd Z${z}
cp ../even${n}.c rcsf.inp
#Move mcp files from previous directory
m=`expr $z - 1`
mv ../Z${m}/mcp* .

k=`expr $n - 1`
#Get initial estimates of wave functions
rwfnestimate <<S5
y
1
even${k}.w
*
2
*
S5

# Perform self-consistent field calculations
rmcdhf > outeven_rmcdhf_${n} <<S6
y
1
${n}*

100
S6
```

```
rsave even${n}

cd ..

    fi
    echo)
done
)
done
```

5. Perform calculations for the odd states

The script `sh_odd` performs angular integration, gets initial estimates and performs `rmcdhf` calculations for the odd states. The script loops over both the active set and the atomic number Z. Angular files are reused and moved between the directories.

```
for n in 3 4 5
do
   (
for z in 6 7 8 9 10 11 12
do
   (if test $z -lt 7
    then
cd Z${z}
cp ../odd${n}.c rcsf.inp
#  Get angular data
rangular <<S4
y
S4

k=`expr $n - 1`
#Get initial estimates of wave functions
rwfnestimate <<S5
y
1
odd${k}.w
*
2
*
S5

# Perform self-consistent field calculations
rmcdhf > outodd_rmcdhf_${n} <<S6
y
1
1
5
${n}*

100
S6

rsave odd${n}
```

```
cd ..
    else

cd Z${z}
cp ../odd${n}.c rcsf.inp
#Move mcp files from previous directory
m=‘expr $z - 1‘
mv ../Z${m}/mcp* .

k=‘expr $n - 1‘
#Get initial estimates of wave functions
rwfnestimate <<S5
y
1
odd${k}.w
*
2
*
S5

# Perform self-consistent field calculations
rmcdhf > out_rmcdhf_${n} <<S6
y
1
1
5
${n}*

100
S6

rsave odd${n}

cd ..

    fi
    echo)
done
)
done
```

6. Configuration interaction and transition calculations

The script `sh_even_odd` performs configuration interaction and transition calculations for `even5` and `odd5`. Angular files are reused and moved between the directories.

```
for z in 6 7 8 9 10 11 12
do
(cd Z${z}
# RCI calculations for even5
rci > outeven_rci <<S6
y
even5
y
```

```
y
1.d-6
y
n
n
y
3
1
S6

# RCI calculations for odd5
rci > outodd_rci <<S6
y
odd5
y
y
1.d-6
y
n
n
y
3
1
1
S6

    if test $z -lt 7
    then

#  Run rbiotransform ans save angular data
rbiotransform <<S4
y
y
even5
odd5
y
S4

#  Run rtransition save angular data
rtransition <<S4
y
y
even5
odd5
E1
S4

    else

#Move angular files from previous directory
m=‘expr $z - 1‘
mv ../Z${m}/even5.TB .
mv ../Z${m}/odd5.TB .
mv ../Z${m}/even5.odd5.-1T .
```

```
#  Run rbiotransform using available angular data
rbiotransform <<S4
y
y
even5
odd5
y
S4

#  Run rtransition using available angular data
rtransition <<S4
y
y
even5
odd5
E1
S4
    fi
cd ..
echo)
done
```

11.2 Comparison with experiment

To display the computed energies for $Z = 6$ we enter the `Z6` directory and we give the command

```
rlevels even5.cm odd5.cm
```

The computer returns the energies together with labels in LSJ-coupling for all the states.

```
 Energy levels for ...
 Rydberg constant is   109737.31569
 No - Serial number of the state; Pos - Position of the state within the
 J/P block; Splitting is the energy difference with the lower neighbor
-------------------------------------------------------------------------
 No Pos  J Parity Energy Total    Levels     Splitting
                      (a.u.)      (cm^-1)     (cm^-1)
-------------------------------------------------------------------------
  1  1  1/2 +     -34.7859395
  2  1  1/2 -     -34.4919396    64525.53    64525.53
  3  1  3/2 -     -34.4914500    64632.98      107.45
```

These energies should be compared with NIST that give 64484.0 cm^{-1}, 64591.7 cm^{-1}. Increasing the active set further will improve the agreement with experiment.

The transition parameters are given in `even5.odd5.ct`. There is a good agreement between length (B) and velocity (C) forms of the parameters. The gf values in the length form are in good agreement with the values 0.1895 and 0.3789 from large scale MCHF calculations [36]. Again, an increased active set will improve the agreement.

```
 Transition between files:
 f1 = even5
 f2 = odd5
```

```
 Electric 2**( 1)-pole transitions
 =================================

    Upper         Lower
File Lev J   P File Lev J  P     E (Kays)        A (s-1)        gf          S
 f2  1  1/2 -  f1  1  1/2 +      64525.53 C  2.68596D+08  1.93430D-01  9.86890D-01
                                          B  2.64473D+08  1.90461D-01  9.71741D-01
 f2  1  3/2 -  f1  1  1/2 +      64632.98 C  2.69981D+08  3.87564D-01  1.97408D+00
                                          B  2.65907D+08  3.81715D-01  1.94429D+00
```

11.3 Scripts for MPI codes

The scripts above can, with very small modifications, also be used for performing the calculations using the MPI codes. The most important change is that the user needs to prepare the files `disks6`, `disks7`, ..., `disks12` with paths to the working directory and to the directory containing temporary data. The `disks` files are copied to the `Z6`, `Z7`, .., `Z12` directories by `sh_nuc_seq`. In the different scripts the calls to the MPI programs amount to changes of the type

```
rangular   -->  mpirun -np 8 rangular_mpi

rmcdhf     -->  mpirun -np 8 rmcdhf_mpi

rci        -->  mpirun -np 8 rci_mpi
```

etc. For the MPI runs the saved angular files reside in the `tmp_mpi` directory and thus they need not be copied from `Z6` to `Z7` etc. Scripts for the MPI cases are included under `case2_mpi` in the test data set (see section 2.11). Consult the `README` file in the working directory for more details on setting up the file `disks`.

Chapter 12

Case study III: graphical analysis of the Mg iso-electronic sequence

In this case study we use script files to perform systematic calculations for states belonging to the $3s^2, 3p^2, 3d^2, 3s3d$ even configurations and to the $3s3p, 3p3d$ odd configurations in the Mg iso-electronic sequence. Angular data are reused from one ion to another. The script files can be found in `grasptest/case3/script`.

Calculations are done by parity and valence-valence correlation is accounted for by allowing for SD excitations from the valence orbitals to active sets up to $n = 5$. Calculations for transition rates are performed for $Z = 26, 27, \ldots, 60$.

It is convenient to save the results for the different ions in directories named `Z26`, `Z27`, ..., `Z60`. After all calculations are finished the energies from `rlevels`, the hyperfine data and the transition data are collected from the different directories and saved in files `energy26`, `energy27`, ..., `energy60`, `hfs26`, `hfs27`, ..., `hfs60`, `trans26`, `trans27`, ..., `trans60`. These files are read by the `rseqenergy` `rseqtrans` programs to produce GNU Octave/Matlab M-files that plot computed properties as functions of the nuclear charge Z of the ions. The M-files include some fitting capabilities as well.

12.1 Iso-electronic sequences

Properties of states, as specified by parity, J quantum number and order number within the symmetry (e.g. the second eigenvalue), are smoothly varying functions of the nuclear charge Z along the iso-electronic sequence. Based on hydrogenic approximations, scaling with Z can be derived for different properties (see for example [37], chapter 19). Using spline methods or least-squares fits to scaling expressions, atomic data along a sequence can be reconstructed with high accuracy from a limited set of calculations. When reconstructing data attention must be paid to label changes. These changes are consequences of the transition from LSJ to jj-coupling, which introduces a label change between the low Z and high Z regions. In the Mg-sequence a label change occurs for the $3l3l'$, $J = 2$ even parity states. At low Z the ordering is $3p^2\ ^1D_2$, $3p^2\ ^3P_2$, $3s3d\ ^3P_2$, $3s3d\ ^1D_2$. When the spin-orbit coupling becomes dominant the ordering in jj-coupling is $(1/2, 1/2)$, $(1/2, 3/2)$, $(1/2, 5/2)$, $(3/2, 3/2)$. Since in the high-Z limit the $3s_{1/2}3d_{3/2}$ state is lower than the $3p^2_{1/2}$ state there must be a label change for some Z. A label change corresponds to an energy level anti-crossing, where two energy levels with the same J and parity will be very close to each other and there will, in the multiconfiguration approximation, be strong interactions between CSFs over a range of Z values. These interactions may result in a decrease or increase of transition probabilities due to negative or positive interference between terms in the expressions for the transition matrix element. For Mg such interference effects can be seen around $Z = 45$. In section 12.2 we will generate atomic data for the $3l3l'$ states in the Mg iso-electronic sequence. In section 12.3 we will explore the energy level anti-crossings and the interference effects using the graphical tools.

12.2 Running script files

The main script `sh_case3` is shown below. This script controls the computational flow and calls several subscripts.

```
#!/bin/sh

set -x

#    Main script for iso-electronic sequence

# 1.   Generate directories for the elements and nuclear data
        ./sh_nuc_seq

# 2.   Generate lists of CSFs in main directory

        ./sh_files_c

# 3.   Perform MCDHF calculations for the even reference states

       ./sh_DF_even

# 4.   Perform MCDHF calculations for the odd reference states

       ./sh_DF_odd

# 5.   Perform MCDHF calculations for even states

       ./sh_even

# 6.   Perform MCDHF calculations for odd states

       ./sh_odd

# 7.   Perform RCI, transition calculations

       ./sh_even_odd

# 8.   Transformation to LSJ, run rlevels and pipe to energyZ

       ./sh_rlevels

# 9.   Collect all data files and copy to the main directory

       ./sh_collect
```

1. Generate directories and define nuclear data

The script `sh_nuc_seq` produces nuclear data for $Z = 26, 27, \ldots, 60$ in the directories `Z26`, `Z27` ,..., `Z60`. By modifying the script we can produce nuclear data for any sequence of charges.

```
#!/bin/sh
```

```
set -x

# Full loop over all Z
#   for z in 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 \
#            26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 \
#            48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 \
#            70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 \
#            92 93 94 95 96 97 98 99 100 101 102 103 104 105 106 107 108 109   \
#            110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 \
#            126 127 128 129 130 131 132 133 134 135 136 137 138

# We select Z from 26 to 60
   for z in  26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 \
             49 50 51 52 53 54 55 56 57 58 59 60
   do

# Data from Jefferson Lab (http://education.jlab.org/itselemental)
case $z in
   1) m=0; MM=1.0794;;    #  Need to use point nucleus
   2) m=4; MM=4.002602;;
   3) m=7; MM=6.941;;
   4) m=9; MM=9.012182;;
   5) m=11; MM=10.811;;
   6) m=12; MM=12.0107;;
   7) m=14; MM=14.0067;;
   8) m=16; MM=15.9994;;
   9) m=19; MM=18.9984032;;
  10) m=20; MM=20.1797;;
  11) m=23; MM=22.98976928;;
  12) m=24; MM=24.3050;;
  13) m=27; MM=26.9815386;;
  14) m=28; MM=29.0855;;
  15) m=31; MM=30.973762;;
  16) m=32; MM=32.065;;
  17) m=35; MM=35.453;;
  18) m=40; MM=39.948;;
  19) m=39; MM=39.0938;;
  20) m=40; MM=40.078;;
  21) m=45; MM=44.955912;;
  22) m=48; MM=47.867;;
  23) m=51; MM=50.9415;;
  24) m=52; MM=51.9961;;
  25) m=55; MM=54.938045;;
  26) m=56; MM=55.845;;
  27) m=59; MM=58.933195;;
  28) m=59; MM=58.6934;;
  29) m=64; MM=63.546;;
  30) m=65; MM=65.409;;
  31) m=70; MM=69.723;;
  32) m=73; MM=72.64;;
  33) m=75; MM=74.92160;;
  34) m=79; MM=78.96;;
  35) m=80; MM=79.904;;
```

```
36) m=84; MM=83.798;;
37) m=85; MM=85.4678;;
38) m=88; MM=87.62;;
39) m=89; MM=88.90585;;
40) m=91; MM=91.224;;
41) m=93; MM=92.90638;;
42) m=96; MM=95.94;;
43) m=98; MM=98;;
44) m=101; MM=10.07;;
45) m=103; MM=102.90550;;
46) m=106; MM=106.42;;
47) m=108; MM=107.8682;;
48) m=112; MM=112.411;;
49) m=115; MM=114.818;;
50) m=119; MM=118.710;;
51) m=122; MM=121.760;;
52) m=128; MM=127.60;;
53) m=127; MM=126.90447;;
54) m=131; MM=131.293;;
55) m=133; MM=132.9054519;;
56) m=137; MM=137.327;;
57) m=139; MM=138.90547;;
58) m=140; MM=140.116;;
59) m=141; MM=140.90765;;
60) m=144; MM=144.242;;
61) m=145; MM=145;;
62) m=150; MM=150.36;;
63) m=152; MM=151.964;;
64) m=157; MM=157.25;;
65) m=159; MM=158.92535;;
66) m=163; MM=162.5;;
67) m=165; MM=164.93032;;
68) m=167; MM=167.259;;
69) m=169; MM=168.93421;;
70) m=173; MM=173.04;;
71) m=175; MM=174.967;;
72) m=178; MM=178.49;;
73) m=181; MM=180.94788;;
74) m=184; MM=183.84;;
75) m=186; MM=186.207;;
76) m=190; MM=190.23;;
77) m=192; MM=192.217;;
78) m=195; MM=195.084;;
79) m=197; MM=196.966569;;
80) m=201; MM=200.59;;
81) m=204; MM=204.3833;;
82) m=207; MM=207.2;;
83) m=209; MM=208.9804;;
84) m=209; MM=209;;
85) m=210; MM=210;;
86) m=222; MM=222;;
87) m=223; MM=223;;
88) m=226; MM=226;;
89) m=227; MM=227;;
```

```
  90) m=232; MM=232.03806;;
  91) m=231; MM=231.03588;;
  92) m=238; MM=238.02891;;
  93) m=237; MM=237;;
  94) m=244; MM=244;;
  95) m=243; MM=243;;
  96) m=247; MM=247;;
  97) m=247; MM=247;;
  98) m=251; MM=251;;
  99) m=252; MM=252;;
 100) m=257; MM=257;;
 101) m=258; MM=258;;
 102) m=259; MM=259;;
 103) m=262; MM=262;;
 104) m=267; MM=267;;
 105) m=268; MM=268;;
 106) m=271; MM=271;;
 107) m=272; MM=272;;
 108) m=277; MM=277;;
 109) m=276; MM=276;;
 110) m=281; MM=281;;
 111) m=280; MM=280;;
 112) m=285; MM=285;;
 113) m=284; MM=284;;
 114) m=289; MM=289;;
 115) m=288; MM=288;;
 116) m=291; MM=291;;
 117) m=293; MM=293;;   #Estimated
 118) m=294; MM=294;;
 119) m=316; MM=316;;
 120) m=318; MM=318;;
 121) m=322; MM=322;;
 122) m=324; MM=324;;
 123) m=326; MM=326;;
 124) m=330; MM=330;;
 125) m=332; MM=332;;
 126) m=334; MM=334;;
 127) m=338; MM=338;;
 128) m=340; MM=340;;
 129) m=342; MM=342;;
 130) m=346; MM=346;;
 131) m=348; MM=348;;
 132) m=350; MM=350;;
 133) m=354; MM=354;;
 134) m=356; MM=356;;
 135) m=358; MM=358;;
 136) m=362; MM=362;;
 137) m=364; MM=364;;
 138) m=366; MM=366;;
esac

echo "Starting: Z::"${z}, "ZZ::"$ZZ, "mass::"${m}, "Weight::"${MM}

rm -r Z$z
```

```
mkdir Z$z
cd Z$z

rnucleus <<EOF
$z
$m
n
$MM
1
1
1
EOF

cd ..

done
```

2. Generate expansions

The expansions are generated by the script `sh_files_c`.

```
rcsfgenerate << EOF
*
2
3s(2,i)
3p(2,i)
3d(2,i)
3s(1,i)3d(1,i)

3s,3p,3d
0,8
0
n
EOF

cp rcsf.out DFeven.c

#######################################
rcsfgenerate << EOF
*
2
3s(1,i)3p(1,i)
3p(1,i)3d(1,i)

3s,3p,3d
0,8
0
n
EOF

cp rcsf.out DFodd.c

#######################################
```

```
rcsfgenerate << EOF
*
2
3s(2,*)

5s,5p,5d,5f,5g
0,8
2
n
EOF

cp rcsf.out even.c

rcsfsplit << EOF
even
2
4s,4p,4d,4f
4
5s,5p,5d,5f,5g
5
EOF

################################

rcsfgenerate << EOF
*
2
3s(1,*)3p(1,*)

5s,5p,5d,5f,5g
0,8
2
n
EOF

cp rcsf.out odd.c

rcsfsplit << EOF
odd
2
4s,4p,4d,4f
4
5s,5p,5d,5f,5g
5
EOF
```

3. Even parity reference states

The script `sh_DF_even` performs angular integration, gets initial estimates and performs `rmcdhf` calculations for the even reference states. The angular integration is done only once and the `mcp.30`, `mcp.31` ... files are moved between the directories.

```
for z in 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 \
         49 50 51 52 53 54 55 56 57 58 59 60
```

```
do
   (if test $z -lt 27
    then
cd Z${z}
cp ../DFeven.c rcsf.inp
#  Get angular data
rangular <<S4
y
S4

#Get initial estimates of wave functions
rwfnestimate <<S5
y
2
*
S5

# Perform self-consistent field calculations
rmcdhf > outeven_rmcdhf <<S6
y
1-5
1-3
1-7
1-2
1-2
5
*
*
100
S6

rsave DFeven
cp DFeven.w even3.w

cd ..
    else

cd Z${z}
cp ../DFeven.c rcsf.inp
#Move mcp files from previous directory
m=`expr $z - 1`
mv ../Z${m}/mcp* .

#Get initial estimates of wave functions
rwfnestimate <<S5
y
2
*
S5


# Perform self-consistent field calculations
rmcdhf > outeven_rmcdhf <<S6
y
```

```
1-5
1-3
1-7
1-2
1-2
5
*
*
100
S6

rsave DFeven
cp DFeven.w even3.w

cd ..

    fi
    echo)
done
```

4. Odd parity reference states

The script `sh_DF_odd` performs angular integration, gets initial estimates and performs `rmcdhf` calculations for the odd reference states. The angular integration is done only once and the `mcp.30`, `mcp.31` ... files are moved between the directories.

```
for z in 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 \
         49 50 51 52 53 54 55 56 57 58 59 60
do
   (if test $z -lt 27
    then
cd Z${z}
cp ../DFodd.c rcsf.inp
#  Get angular data
rangular <<S4
y
S4

#Get initial estimates of wave functions
rwfnestimate <<S5
y
2
*
S5

# Perform self-consistent field calculations
rmcdhf > outodd_rmcdhf <<S6
y
1-2
1-5
1-5
1-3
1
5
```

```
*
*
100
S6

rsave DFodd
cp DFodd.w odd3.w

cd ..
    else

cd Z${z}
cp ../DFodd.c rcsf.inp
#Move mcp files from previous directory
m=`expr $z - 1`
mv ../Z${m}/mcp* .

#Get initial estimates of wave functions
rwfnestimate <<S5
y
2
*
S5


# Perform self-consistent field calculations
rmcdhf > outodd_rmcdhf <<S6
y
1-2
1-5
1-5
1-3
1
5
*
*
100
S6

rsave DFodd
cp DFodd.w odd3.w

cd ..

    fi
    echo)
done
```

5. Perform calculations for even states

The script `sh_even` performs angular integration, gets initial estimates and performs `rmcdhf` calculations for the odd states. The script loops over both the active set and the atomic number Z. Angular files are reused and moved between the directories.

```
for n in 4 5
do
   (
for z in 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 \
         49 50 51 52 53 54 55 56 57 58 59 60
do
   (if test $z -lt 27
    then
cd Z${z}
cp ../even${n}.c rcsf.inp
#  Get angular data
rangular <<S4
y
S4

k=`expr $n - 1`
#Get initial estimates of wave functions
rwfnestimate <<S5
y
1
even${k}.w
*
2
*
S5

# Perform self-consistent field calculations
rmcdhf > outeven_rmcdhf_${n} <<S6
y
1-5
1-3
1-7
1-2
1-2
5
${n}*

100
S6

rsave even${n}
cd ..
    else

cd Z${z}
cp ../even${n}.c rcsf.inp
#Move mcp files from previous directory
m=`expr $z - 1`
mv ../Z${m}/mcp* .

k=`expr $n - 1`
#Get initial estimates of wave functions
rwfnestimate <<S5
y
```

```
1
even${k}.w
*
2
*
S5

# Perform self-consistent field calculations
rmcdhf > outeven_rmcdhf_${n} <<S6
y
1-5
1-3
1-7
1-2
1-2
5
${n}*

100
S6

rsave even${n}

cd ..

    fi
    echo)
done
)
done
```

6. Perform calculations for odd states

The script `sh_odd` performs angular integration, gets initial estimates and performs `rmcdhf` calculations for the odd states. The script loops over both the active set and the atomic number Z. Angular files are reused and moved between the directories.

```
for n in 4 5
do
   (
for z in 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 \
         49 50 51 52 53 54 55 56 57 58 59 60
do
   (if test $z -lt 27
    then
cd Z${z}
cp ../odd${n}.c rcsf.inp
#  Get angular data
rangular <<S4
y
S4

k=`expr $n - 1`
#Get initial estimates of wave functions
```

```
rwfnestimate <<S5
y
1
odd${k}.w
*
2
*
S5

# Perform self-consistent field calculations
rmcdhf > outodd_rmcdhf_${n} <<S6
y
1-2
1-5
1-5
1-3
1
5
${n}*

100
S6

rsave odd${n}
cd ..
    else

cd Z${z}
cp ../odd${n}.c rcsf.inp
#Move mcp files from previous directory
m=`expr $z - 1`
mv ../Z${m}/mcp* .

k=`expr $n - 1`
#Get initial estimates of wave functions
rwfnestimate <<S5
y
1
odd${k}.w
*
2
*
S5

# Perform self-consistent field calculations
rmcdhf > outodd_rmcdhf_${n} <<S6
y
1-2
1-5
1-5
1-3
1
5
${n}*
```

```
100
S6

rsave odd${n}

cd ..

    fi
    echo)
done
)
done
```

7. Perform RCI and transition calculations

The script `sh_even_odd` performs configuration interaction and transition calculations for `even5` and `odd5`. Angular files are reused and moved between the directories.

```
for z in 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 \
         49 50 51 52 53 54 55 56 57 58 59 60
do
(cd Z${z}
# RCI calculations for even5
rci > outeven_rci <<S6
y
even5
y
y
1.d-6
y
n
n
y
3
1-5
1-3
1-7
1-2
1-2
S6

# RCI calculations for odd5
rci > outodd_rci <<S6
y
odd5
y
y
1.d-6
y
n
n
y
3
```

```
1-2
1-5
1-5
1-3
1
S6

    if test $z -lt 27
    then

#  Run rbiotransform ans save angular data
rbiotransform <<S4
y
y
even5
odd5
y
S4

#  Run rtransition save angular data
rtransition <<S4
y
y
even5
odd5
E1
S4


    else

#Move angular files from previous directory
m=‘expr $z - 1‘
mv ../Z${m}/even5.TB .
mv ../Z${m}/odd5.TB .
mv ../Z${m}/even5.odd5.-1T .

#  Run rbiotransform using available angular data
rbiotransform <<S4
y
y
even5
odd5
y
S4

#  Run rtransition using available angular data
rtransition <<S4
y
y
even5
odd5
E1
S4
```

```
    fi
cd ..
echo)
done
```

Transformation to LSJ, run rlevels and pipe to energyZ

This script runs `jj2lsj` to transform to LSJ-coupling. The energy files `energyZ` are created by redirecting the output from `rlevels`.

```
for z in 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 \
         49 50 51 52 53 54 55 56 57 58 59 60
do
(cd Z${z}

jj2lsj <<S1
even5
y
y
y
S1

jj2lsj <<S2
odd5
y
y
y
S2

rlevels even5.cm odd5.cm > energy${z}

cd ..
echo)
done
```

Collect data to prepare for the runs of the iso-electronic plotting tools

This script collects, in one directory, all the energy, hfs and transition files that are needed to run the tools that create the iso-electronic plots.

```
for z in 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 \
         49 50 51 52 53 54 55 56 57 58 59 60
do
(cd Z${z}

cp energy${z} ../.
cp even5.odd5.ct ../trans${z}

cd ..
echo)
done
```

12.3 Generating plots of properties along the sequence

After the script sh_case3 has been executed the energy files energy26, energy27, ..., energy60, as obtained from rlevels, the hyperfine structure files hfs26, hfs27 ,..., hfs60 and the transition files trans26, trans27, ..., trans60 all reside in one directory. The energy file energy26 is shown below

```
 nblock =            5   ncftot =          327   nw =           25   nelec =            12
 nblock =            5   ncftot =          320   nw =           25   nelec =            12

 Energy levels for ...
 Rydberg constant is   109737.31569
 Splitting is the energy difference with the lower neighbor
------------------------------------------------------------------------------------------
 No Pos  J Parity Energy Total    Levels     Splitting     Configuration
                    (a.u.)      (cm^-1)     (cm^-1)
------------------------------------------------------------------------------------------
  1  1   0  +   -1182.3727090        0.00        0.00  3s(2)_1S0
  2  1   0  -   -1181.3113166   232948.71   232948.71  3s.3p_3P
  3  1   1  -   -1181.2846597   238799.23     5850.52  3s.3p_3P
  4  1   2  -   -1181.2204883   252883.23    14084.01  3s.3p_3P
  5  2   1  -   -1180.7554032   354957.60   102074.37  3s.3p_1P
  6  2   0  +   -1179.8372639   556465.90   201508.29  3p(2)_3P2
  7  1   2  +   -1179.8216886   559884.26     3418.37  3p(2)_1D2
  8  1   1  +   -1179.7920264   566394.37     6510.11  3p(2)_3P2
  9  2   2  +   -1179.7153461   583223.75    16829.37  3p(2)_3P2
 10  3   0  +   -1179.3515090   663076.77    79853.02  3p(2)_1S0
 11  2   1  +   -1179.2726052   680394.14    17317.37  3s.3d_3D
 12  3   2  +   -1179.2680609   681391.51      997.37  3s.3d_3D
 13  1   3  +   -1179.2609531   682951.48     1559.97  3s.3d_3D
 14  4   2  +   -1178.8797415   766617.76    83666.28  3s.3d_1D
 15  2   2  -   -1178.1399645   928980.06   162362.29  3p.3d_3F
 16  1   3  -   -1178.0955852   938720.18     9740.12  3p.3d_3F
 17  3   2  -   -1178.0442285   949991.68    11271.50  3p.3d_1D
 18  1   4  -   -1178.0435428   950142.16      150.48  3p.3d_3F
 19  3   1  -   -1177.8804977   985926.43    35784.26  3p.3d_3D
 20  4   2  -   -1177.8791112   986230.73      304.30  3p.3d_3P
 21  2   3  -   -1177.8254894   997999.35    11768.62  3p.3d_3D
 22  2   0  -   -1177.8237420   998382.86      383.51  3p.3d_3P
 23  4   1  -   -1177.8212951   998919.91      537.05  3p.3d_3P
 24  5   2  -   -1177.8187683   999474.46      554.55  3p.3d_3D
 25  3   3  -   -1177.5115780  1066894.94    67420.48  3p.3d_1F
 26  5   1  -   -1177.4559591  1079101.88    12206.94  3p.3d_1P
 27  5   2  +   -1176.1164738  1373084.92   293983.05  3d(2)_3F2
 28  2   3  +   -1176.1091088  1374701.36     1616.44  3d(2)_3F2
 29  1   4  +   -1176.1001073  1376676.96     1975.59  3d(2)_3F2
 30  6   2  +   -1175.9682840  1405608.82    28931.87  3d(2)_1D2
 31  4   0  +   -1175.9538651  1408773.40     3164.58  3d(2)_3P2
 32  3   1  +   -1175.9510570  1409389.72      616.31  3d(2)_3P2
 33  2   4  +   -1175.9496227  1409704.50      314.78  3d(2)_1G2
 34  7   2  +   -1175.9446189  1410802.71     1098.21  3d(2)_3P2
 35  5   0  +   -1175.5824289  1490294.23    79491.51  3d(2)_1S0
------------------------------------------------------------------------------------------
```

Each state is specified by the position within the symmetry, the J quantum number and the parity. For example, the four states $3p^2\ ^1D_2$, $3p^2\ ^3P_2$, $3s3d\ ^3P_2$, $3s3d\ ^1D_2$ with even parity and $J = 2$ are specified as `1 2 +`, `2 2 +`, `3 2 +`, `4 2 +`. These specifications remain valid over the iso-electronic sequence, although the LSJ designation may change. Thus, to follow states along the iso-electronic sequence the above specifications should be used. To generate a GNU Octave/Matlab M-file that plots the $3p^2\ ^1D_2$, $3p^2\ ^3P_2$, $3s3d\ ^3P_2$, $3s3d\ ^1D_2$ states along the sequence, the `rseqenergy` program should be run. The program looks for all energy files in a given range of Z. Then after having specified the states to be plotted there is an option to perform least squares fits to obtain analytical expressions of the trends. If no fits are done the data are instead interpolated using cubic splines. The `rseqenergy` program outputs an M-file with name `seqenergyplot.m`. The M-file contains all data needed for the plot and the file can also very easily be edited to comply with the desires of the user. The input session for `rseqenergy` is shown below. Please note that you should input $2J$ and not J and the sequence `1 2 +`, `2 2 +`, `3 2 +`, `4 2 +` above should thus be inserted in the program as `1 4 +`, `2 4 +`, `3 4 +`, `4 4 +`.

```
>>rseqenergy

 RSEQENERGY
 This program reads output from rlevels for several
 ions and produces a Matlab/Octave file that plots
 energy as a function of Z
 Input files: energyZ1, energyZ2, .., energyZn
 Output file: seqenergyplot.m

 Give the first Z and last Z of the sequence
>>26,60
 How many states do you want to plot?
>>4
 Give number within symmetry,2*J and parity (+/-)
>>1,4,+
 Give number within symmetry,2*J and parity (+/-)
>>2,4,+
 Give number within symmetry,2*J and parity (+/-)
>>3,4,+
 Give number within symmetry,2*J and parity (+/-)
>>4,4,+
 Least-squares fit (y/n) ?
>>n
```

`rseqenergy` produces the file `seqenergyplot.m`. To run this file open GNU Octave (or Matlab) and issue the command

```
octave:1>seqenergyplot
```

on the GNU Octave command line and the plot shown in figure 12.1 will appear. There is an energy level anti-crossing around $Z = 44$.

We now turn to the hyperfine structure. The hyperfine structure file `hfs26` is shown below

```
Nuclear spin                         1.000000000000000D+00 au
Nuclear magnetic dipole moment       1.000000000000000D+00 n.m.
Nuclear electric quadrupole moment   1.000000000000000D+00 barns
```

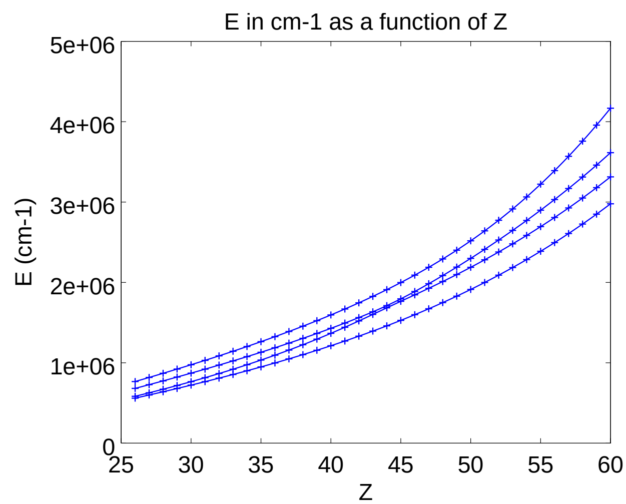

Figure 12.1: Plot of the energy of the four lowest even parity states with $J = 2$ as function of the nuclear charge Z. There is an energy level anti-crossing around $Z = 44$.

```
 Interaction constants:

 Level1  J Parity         A (MHz)             B (MHz)             g_J

   1         1 +     -3.4234387290D+02    4.4613609751D+03    1.5001883281D+00
   2         1 +     -1.9243940191D+04    6.2528207125D+02    4.9806452455D-01
   3         1 +     -1.1351221472D+01    6.2649343730D+02    1.5002882611D+00
   1         2 +      8.7390039737D+03    1.1076010824D+04    1.0772839773D+00
   2         2 +      4.4795463061D+03   -5.4746853597D+03    1.4215360611D+00
   3         2 +      8.8227565268D+03    8.9310423785D+02    1.1660563087D+00
   4         2 +      2.4077941743D+03    5.0250914053D+03    9.9939681256D-01
   5         2 +      1.8500822031D+03    5.8565537496D+02    6.6595057455D-01
   6         2 +      1.2714709452D+03   -7.1108880021D+02    1.0473515693D+00
   7         2 +      7.3078966195D+02   -1.2063797282D+03    1.4511063532D+00
   1         3 +      1.5225231973D+04    1.7639256412D+03    1.3331477513D+00
   2         3 +      1.1924106280D+03    6.4316601614D+02    1.0826447519D+00
   1         4 +      9.0356372098D+02    8.6788621658D+02    1.2493914009D+00
   2         4 +      1.2426420188D+03    3.5362685851D+03    9.9942996698D-01
```

States are specified in the same way as in the energy file by giving position (level1) within the symmetry, the J quantum number and the parity. To plot the hyperfine interaction constants or the Landé g_J factor as a function of the nuclear charge we use the program **rseqhfs**. The input session for plotting the magnetic dipole interaction constant for the states `2 2 +` and `3 2 +` is shown below (again please note that you should input $2J$ and not J)

```
>>rseqhfs

 RSEQHFS
 This program reads output from rhfs for several
 ions and produces a Matlab/Octave file that
 plots hfs parameters as functions of Z
 Input files: hfsZ1, hfsZ2, .., hfsZn or
 Output file: seqhfsplot.m

 Give the first Z and last Z of the sequence
```

```
>>26,60
 How many states do you want to plot?
>>2
 Give number within symmetry,2*J and parity (+/-)
>>2,4,+
 Give number within symmetry,2*J and parity (+/-)
>>3,4,+
 Plot A (1), B (2) or gJ (3) ?
>>1
 Least-squares fit (y/n) ?
>>n
```

`rseqhfs` produces the file `seqhfsplot.m`. To run this file open GNU Octave (or Matlab) and issue the command

```
octave:1>seqhfsplot
```

at the GNU Octave command line and the plot in figure 12.2 will now be displayed. The strong mixing of the CSFs around the level anti-crossing at $Z = 44$ causes interference effects that have large influence on the hyperfine structure constants of the two states.

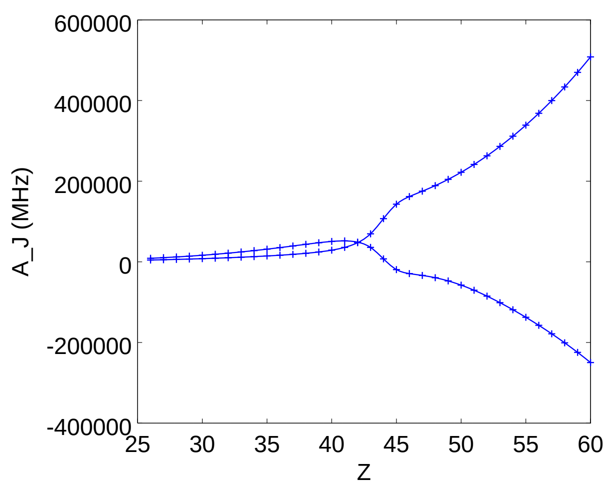

Figure 12.2: Plot of the hyperfine interaction constants A_J for the two interfering even parity states with $J = 2$ as function of the nuclear charge Z.

The transition file `trans26` is shown below. A transition is specified by giving the multipolarity along with the position (Lev) within the symmetry, the J quantum number and the parity for the upper and lower states.

```
 Transition between files:
 f1 = even5
 f2 = odd5


 Electric 2**( 1)-pole transitions
 =================================
```

```
 Upper       Lower
 Lev  J P    Lev  J P       E (Kays)         A (s-1)          gf             S
 f2  1    1 -  f1  1    0 +      238799.23 C  4.30267D+07  3.39353D-03  4.67836D-03
                                           B  4.08058D+07  3.21836D-03  4.43688D-03
 f2  2    1 -  f1  1    0 +      354957.60 C  2.33297D+10  8.32789D-01  7.72385D-01
                                           B  2.28391D+10  8.15275D-01  7.56142D-01
 f2  3    1 -  f1  1    0 +      985926.43 C  2.38994D+05  1.10580D-06  3.69239D-07
                                           B  9.80859D+04  4.53833D-07  1.51540D-07
 f2  4    1 -  f1  1    0 +      998919.91 C  6.50659D+04  2.93272D-07  9.66531D-08
                                           B  4.71470D+04  2.12506D-07  7.00353D-08
 f2  5    1 -  f1  1    0 +     1079101.88 C  3.67085D+08  1.41782D-03  4.32548D-04
                                           B  3.05390D+08  1.17953D-03  3.59850D-04
 f1  2    0 +  f2  1    1 -      317666.67 C  1.88797D+10  2.80485D-01  2.90679D-01
                                           B  1.82388D+10  2.70963D-01  2.80811D-01

                         ...............
```

To plot A, gf or S as a function of the nuclear charge we use the program `rseqtrans`. The input session for plotting the transition rate A from the states `2 2 +` and `3 2 +` to `1 1 -` is shown below. Please observe that we should input $2J$.

```
>>rseqtrans

 RSEQTRANS
 This program reads output from rtransition for several
 ions and produces a Matlab/Octave file that plots
 A, gf, or S as a function of Z
 Input files: transZ1, transZ2, .., transZn
 Output file: seqtransplot.m

 Give the first Z and last Z of the sequence
>>26,60
 Give multipolarity of transition: E1, M1, E2, M2
>>E1
 How many transitions do you want to plot?
>>2
 Give number within symmetry,2*J and parity (+/-)
 for upper and lower state
>>2,4,+,1,2,-
 Give number within symmetry,2*J and parity (+/-)
 for upper and lower state
>>3,4,+,1,2,-
 Plot A (1), gf (2) or S (3) ?
>>1
 Least-squares fit (y/n) ?
>>n
```

The `rseqtrans` program produces the file `seqtransplot.m`. To run this file open GNU Octave or Matlab and issue the command

```
octave:1>seqtransplot
```

and the plot in figure 12.3 will now be shown. The strong mixing of the CSFs around the level anti-crossing at $Z = 44$ causes interference effects that influence the rates.

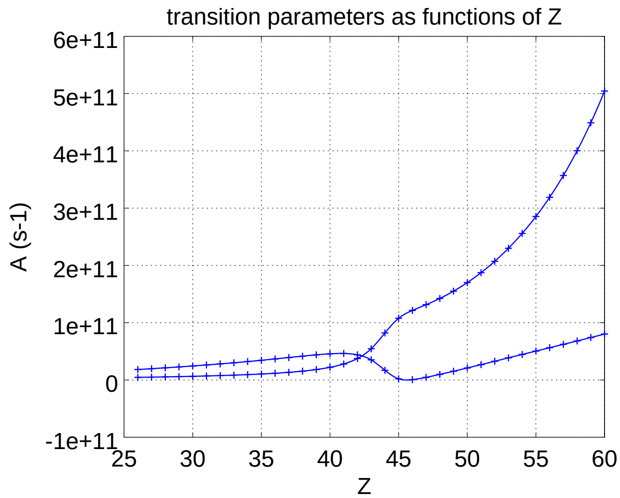

Figure 12.3: Plot of transition rates involving two interfering states.

12.4 Least-squares fits to data

If deemed important least-squares fits can be done for atomic data that are not affected by interference effects from level anti-crossings. Below we fit a polynomial to the energies for the `1 0 -`, `1 1 -`, `2 1 -`, and `1 2 -` states.

```
>>rseqenergy

 RSEQENERGY
 This program reads output from rlevels for several
 ions and produces a Matlab/Octave file that plots
 energy as a function of Z
 Input files: energyZ1, energyZ2, .., energyZn
 Output file: seqenergyplot.m

 Give the first Z and last Z of the sequence
>>26,60
 How many states do you want to plot?
>>4
 Give number within symmetry,2*J and parity (+/-)
>>1,0,-
 Give number within symmetry,2*J and parity (+/-)
>>1,2,-
 Give number within symmetry,2*J and parity (+/-)
>>2,2,-
 Give number within symmetry,2*J and parity (+/-)
>>1,4,-
 Least-squares fit (y/n) ?
>>y
 Type of fitting: a1 Z^-2 + a2 Z^-1 + ...+ a6 Z^3  (1)
                  a1 + a2 Z + a3 Z^2 + a4 Z^3      (2)
>>2
```

Starting GNU Octave (or Matlab) and giving the command

```
octave:1>seqenergyplot
```

at the GNU Octave command line gives the fitting coefficients for the four states

```
a =

  -3.8990e+00
   9.3243e-02
  -3.2068e-04
   5.5596e-06

a =

  -3.3884e+00
   5.7768e-02
   4.6734e-04
  -7.9666e-08

a =

  -3.4271e+00
   1.4637e-01
  -3.6443e-03
   4.5791e-05

a =

  -3.1674e+00
   1.3634e-01
  -3.6161e-03
   4.7051e-05
```

along with the plot in figure 12.4.

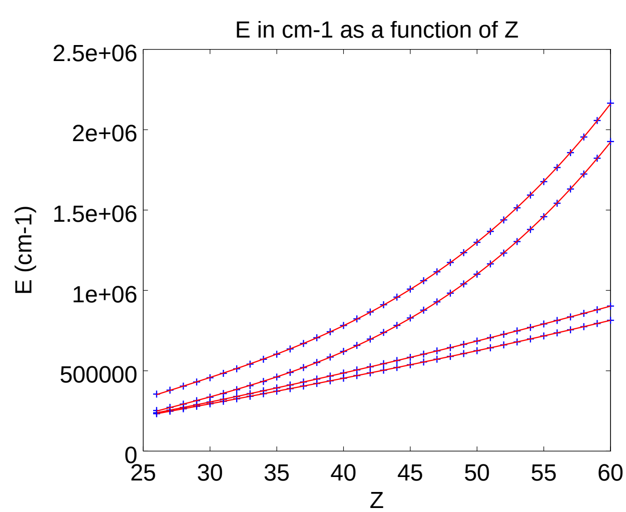

Figure 12.4: Polynomial fitted to the energies of the 1 0 -, 1 1 -, 2 1 -, and 1 2 - states

We can do fits to transition data as well. Below we fit a Laurent series to the line strength S for the transition from `1 1 -`, `2 1 -` down to `1 0 +`.

```
>>rseqtrans

 RSEQTRANS
 This program reads output from rtransition for several
 ions and produces a Matlab/Octave file that plots
 A, gf, or S as a function of Z
 Input files: transZ1, transZ2, .., transZn
 Output file: seqtransplot.m

 Give the first Z and last Z of the sequence
>>26,60
 Give multipolarity of transition: E1, M1, E2, M2
>>E1
 How many transitions do you want to plot?
>>2
 Give number within symmetry,2*J and parity (+/-)
 for upper and lower state
>>1,2,-,1,0,+
 Give number within symmetry,2*J and parity (+/-)
 for upper and lower state
>>2,2,-,1,0,+
 Plot A (1), gf (2) or S (3) ?
>>3
 Least-squares fit (y/n) ?
>>y
 Type of fitting: a1 Z^-2 + a2 Z^-1 + ...+ a6 Z^3  (1)
                  a1 + a2 Z + a3 Z^2 + a4 Z^3      (2)
>>1
```

Starting GNUOctave (or Matlab) and giving the command

```
octave:1>seqtransplot
```

at the GNU Octave command line gives the fitting coefficients

```
a =

  -2.9422e+04
   4.7527e+03
  -2.9700e+02
   8.5307e+00
  -1.1226e-01
   5.4992e-04

a =

   1.4276e+04
  -1.1873e+03
   4.8383e+01
  -1.0910e+00
   1.2034e-02
  -5.2794e-05
```

The produced plot in displayed in figure 12.5. The fitted function describes the data very well.

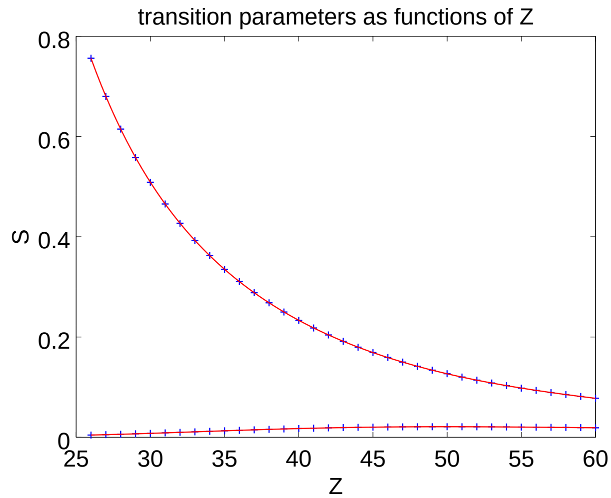

Figure 12.5: Fitted function to the line strength S for the transitions of 1 1 -, 2 1 - down to the 1 0 + groundstate.

12.5 Modifying the GNU Octave/MATLAB M-files

The M-files produced by `rseqenergy`, `rseqhfs` and `rseqtrans` are very easy to modify to include legends, change captions etc. Also other types of modifications should be considered. If, for example, calculations are done for even Z in an iso-electronic sequence then the user can easily modify `seqenergyplot.m` to output interpolated values of the energies for odd Z. Away from level anti-crossings the accuracy of the interpolated values should be quite high. In many cases data for a full iso-electronic sequence can be interpolated from a comparatively small number of ions. The M-files can be concatenated (some minor editing is needed) and it is then possible to overlay several plots.

The `seqtransplot.m` file from the last run is shown below. The data are organized in a matrix A where the first column contains the nuclear charge Z. The atomic data are stored in columns 2 and 3. Standard commands are used for plotting and least-squares fits.

```
A = [
          26   4.4368799999999998E-003  0.75614199999999998
          27   5.1946300000000004E-003  0.68008400000000002
          28   6.0161700000000004E-003  0.61473800000000001
          29   6.8959800000000003E-003  0.55813599999999997
          30   7.8269499999999992E-003  0.50873800000000002
          31   8.8007499999999995E-003  0.46534399999999998
          32   9.8050199999999994E-003  0.42698700000000001
          33   1.0828300000000001E-002  0.39289900000000000
          34   1.1857500000000000E-002  0.36245699999999997
          35   1.2878700000000000E-002  0.33515299999999998
          36   1.3878400000000001E-002  0.31056699999999998
          37   1.4843300000000000E-002  0.28835200000000000
          38   1.5761100000000000E-002  0.26822099999999999
          39   1.6621100000000000E-002  0.24992900000000001
          40   1.7414099999999998E-002  0.23326900000000000
          41   1.8133000000000000E-002  0.21806200000000001
```

```
        42   1.8772700000000000E-002  0.20415900000000001
        43   1.9330000000000000E-002  0.19142600000000001
        44   1.9803700000000000E-002  0.17974599999999999
        45   2.0194199999999999E-002  0.16901600000000000
        46   2.0503600000000000E-002  0.15914600000000001
        47   2.0734900000000001E-002  0.15005599999999999
        48   2.0892299999999999E-002  0.14167199999999999
        49   2.0980599999999999E-002  0.13392999999999999
        50   2.1004800000000001E-002  0.12677099999999999
        51   2.0970699999999998E-002  0.12014300000000000
        52   2.0883800000000001E-002  0.11399800000000000
        53   2.0749699999999999E-002  0.10829400000000000
        54   2.0573700000000000E-002  0.10299200000000000
        55   2.0360799999999998E-002   9.8056500000000005E-002
        56   2.0116100000000001E-002   9.3456800000000007E-002
        57   1.9844100000000000E-002   8.9163800000000001E-002
        58   1.9548800000000002E-002   8.5151699999999997E-002
        59   1.9234100000000001E-002   8.1397200000000003E-002
        60   1.8903300000000001E-002   7.7879100000000007E-002
 ];
  clf, hold on
zip = linspace( 26, 60);
title('transition parameters as functions of Z')
xlabel('Z')
ylabel('S')

plot(A(:,1),A(:, 2),'+')

z = A(:,1);
AD = [z.^(-2) z.^(-1) z.^0 z.^1 z.^2 z.^3];
y = A(:, 2);
m = mean(y); s = std(y);
a = AD\(y-m)/s
aiplsq = a(1)./zip.^2 + a(2)./zip + a(3) + a(4)*zip + a(5)*zip.^2 + a(6)*zip.^3;
aiplsq = s*aiplsq + m;
plot(zip,aiplsq,'r')

plot(A(:,1),A(:, 3),'+')

z = A(:,1);
AD = [z.^(-2) z.^(-1) z.^0 z.^1 z.^2 z.^3];
y = A(:, 3);
m = mean(y); s = std(y);
a = AD\(y-m)/s
aiplsq = a(1)./zip.^2 + a(2)./zip + a(3) + a(4)*zip + a(5)*zip.^2 + a(6)*zip.^3;
aiplsq = s*aiplsq + m;
plot(zip,aiplsq,'r')
```

Part IV

Issues of convergence, trouble shooting and non-default options

Chapter 13

Methods to ensure convergence

In this chapter we will try and give some practical advice on how to handle cases when the `rmcdhf` calculations for the multireference do not converge. In the calculations for the multireference the orbitals are spectroscopic and are required to have the correct number of nodes. Once the multireference is in place the remaining `rmcdhf` calculations for layers of correlation orbitals (no node counting required) are unproblematic.

13.1 Start with the core and gradually build the orbitals

An advice is to start from the inner part of the core and gradually include more and more core orbitals until simultaneous convergence of all core orbitals has been achieved. Once the core orbitals are in place, gradually start to build more and more valence orbitals. If needed keep all previous orbitals fixed the first time a layer of new orbitals is introduced. Once the new layer of orbitals is converged optimize all orbitals together. This somewhat tedious multistep procedure is in general the preferred way to achieve convergence. If not working, the methods suggested below can be used.

13.2 Using converted Hartree-Fock orbitals

It can be difficult to achieve convergence when the multireference consists of many configurations. In these cases an advice is to do a sequence of average HF calculations for the different configurations. The wave functions for each HF run are saved and at the end all wave functions are concatenated and then converted to relativistic wave functions that are used as starting estimates. As a practical example we will perform a calculation for all states belonging to $3s^2$, $3s3p$, $3s4s$, $3s3d$, $3s4p$ in Mg I using converted HF orbitals as starting estimates.

Overview

1. Define nuclear data.
2. Generate list of CSFs for the $\{3s^2, 3s3p, 3s4s, 3s3d, 3s4p\}$ MR set.
3. Perform angular integration.
4. Generate initial estimates of radial orbitals.
5. Perform self-consistent field calculation on the weighted average of the states (this will fail).
6. Perform average HF calculations for the $3s3p, 3s4s, 3s3d, 3s4p$ configurations. Save the wave functions for each run.

7. Concatenate the HF wave functions to a file `wfn.inp`.
8. Use `rwfnmchfmcdf` to convert the `wfn.inp` to `rwfn.out`.
9. Copy `rwfn.out` to `rwfn.inp` and run `rmcdhf` (this will converge).
10. Run `rsave`.

```
*****************************************************************************
* RUN RNUCLEUS TO GENERATE NUCLEAR DATA AND DEFINE RADIAL GRID              *
* OUTPUT FILE: isodata                                                      *
*****************************************************************************

>>rnucleus

 RNUCLEUS
 This program defines nuclear data and the radial grid
 Outputfile: isodata

 Enter the atomic number:
>>12
 Enter the mass number (0 if the nucleus is to be modelled as a point source:
>>24
 The default root mean squared radius is    2.9814412815539866       fm;
 the default nuclear skin thickness is    2.2999999999999998       fm;
 Revise these values?
>>n
 Enter the mass of the neutral atom (in amu) (0 if the nucleus is to be static):
>>24
 Enter the nuclear spin quantum number (I) (in units of h / 2 pi):
>>1
 Enter the nuclear dipole moment (in nuclear magnetons):
>>1
 Enter the nuclear quadrupole moment (in barns):
>>1

*****************************************************************************
*         RUN RCSFGENERATE TO GENERATE LIST OF CSFs                         *
*         FOR THE MULTIREFERNCE                                             *
*         OUTPUT FILES: rcsfgenerate.log, rcsf.out                          *
*****************************************************************************

>>rcsfgenerate

 RCSFGENERATE
 This program creates a list of CSFs
 Configurations should be entered in spectroscopic notation
 with occupation numbers and indications if orbitals are
 closed (c), inactive (i), active (*) or has a minimal
 occupation e.g. 1s(2,1)2s(2,*)
 Outputfiles: rcsf.out, rcsfgenerate.log

 Default, reverse, symmetry or user specified ordering? (*/r/s/u)
>>*
```

```
 Select core
        0: No core
        1: He (       1s(2)                  =  2 electrons)
        2: Ne ([He] + 2s(2)2p(6)             = 10 electrons)
        3: Ar ([Ne] + 3s(2)3p(6)             = 18 electrons)
        4: Kr ([Ar] + 3d(10)4s(2)4p(6)       = 36 electrons)
        5: Xe ([Kr] + 4d(10)5s(2)5p(6)       = 54 electrons)
        6: Rn ([Xe] + 4f(14)5d(10)6s(2)6p(6) = 86 electrons)
>>2
 Enter list of (maximum 100) configurations. End list with a blank line or an asterisk (*)

 Give configuration           1
>>3s(2,i)
 Give configuration           2
>>3s(1,i)3p(1,i)
 Give configuration           3
>>3s(1,i)4s(1,i)
 Give configuration           4
>>3s(1,i)3d(1,i)
 Give configuration           5
>>3s(1,i)4p(1,i)
 Give configuration           6
>>
 Give set of active orbitals, as defined by the highest principal quantum number
 per l-symmetry, in a comma delimited list in s,p,d etc order, e.g. 5s,4p,3d
>>4s,4p,3d
 Resulting 2*J-number? lower, higher (J=1 -> 2*J=2 etc.)
>>0,6
 Number of excitations (if negative number e.g. -2, correlation
 orbitals will always be doubly occupied)
>>0
 Generate more lists ? (y/n)
>>n

....

 7 blocks were created

      block  J/P            NCSF
          1    0+              2
          2    0-              2
          3    1+              2
          4    1-              4
          5    2+              2
          6    2-              2
          7    3+              1

******************************************************************************
*         COPY FILES                                                         *
*         IT IS ADVISABLE TO SAVE THE rcsfgenerate.log FILE TO HAVE A        *
*         RECORD ON HOW THE LIST OF CSFs WAS CREATED                         *
******************************************************************************

cp rcsf.out rcsf.inp
```

```
***************************************************************************
*         RUN RANGULAR TO GENERATE ENERGY EXPRESSION                      *
*         INPUT FILE  : rcsf.inp                                          *
*         OUTPUT FILES: rangular.alog, mcp.30, mcp.31,....                *
***************************************************************************

>>rangular

 RANGULAR
 This program performs angular integration
 Input file:  rcsf.inp
 Outputfiles: mcp.30, mcp.31, ....
              rangular.log

 Full interaction?  (y/n)
>>y

 RANGULAR: Execution complete.

***************************************************************************
*         RUN RWFNESTIMATE TO GENERATE INITIAL ESTIMATES FOR RADIAL ORBITALS  *
*         INPUT FILES: isodata, rcsf.inp, previous rwfn files             *
*         OUTPUT FILE: rwfn.inp, rwfnestimate.log                         *
***************************************************************************

>>rwfnestimate

 RWFNESTIMATE
 This program estimates radial wave functions
 for orbitals
 Input files: isodata, rcsf.inp, optional rwfn file
 Output file: rwfn.inp

 Default settings ?
>>y
 Loading CSF file ... Header only
 There are/is           12  relativistic subshells;
 The following subshell radial wavefunctions remain to be estimated:
 1s 2s 2p- 2p 3s 3p- 3p 3d- 3d 4s 4p- 4p

 Read subshell radial wavefunctions. Choose one below
     1 -- GRASP2K File
     2 -- Thomas-Fermi
     3 -- Screened Hydrogenic
>>2
 Enter the list of relativistic subshells:
>>*
 All required subshell radial wavefunctions  have been estimated:
Shell      e           p0        gamma        P(2)       Q(2)      MTP  SRC

  1s   0.4601D+02  0.8064D+02  0.1000D+01  0.6891D-06 -0.5886D-11  331  T-F
  2s   0.3992D+01  0.2155D+02  0.1000D+01  0.1842D-06 -0.1573D-11  355  T-F
  2p-  0.2856D+01  0.2719D-01  0.1000D+01  0.5155D-13  0.5307D-08  358  T-F
```

```
  2p   0.2845D+01  0.5573D+02  0.2000D+01  0.4069D-14 -0.3476D-19  358  T-F
  3s   0.4681D+00  0.6649D+01  0.1000D+01  0.5682D-07 -0.4854D-12  378  T-F
  3p-  0.2743D+00  0.7262D-02  0.1000D+01  0.1377D-13  0.1417D-08  383  T-F
  3p   0.2736D+00  0.1488D+02  0.2000D+01  0.1087D-14 -0.9283D-20  383  T-F
  3d-  0.8141D-01  0.4498D-03  0.2000D+01  0.1457D-22  0.1500D-17  397  T-F
  3d   0.8141D-01  0.1119D+01  0.3000D+01  0.6984D-24 -0.5966D-29  397  T-F
  4s   0.1150D+00  0.2438D+01  0.1000D+01  0.2084D-07 -0.1780D-12  394  T-F
  4p-  0.8009D-01  0.2722D-02  0.1000D+01  0.5160D-14  0.5312D-09  398  T-F
  4p   0.7998D-01  0.5583D+01  0.2000D+01  0.4077D-15 -0.3483D-20  398  T-F

 RWFNESTIMATE: Execution complete.

******************************************************************************
*         RUN RMCDHF TO OBTAIN SELF CONSISTENT SOLUTIONS                     *
*         INPUT FILES: isodata, rcsf.inp, rwfn.inp, mcp.30, mcp.31,...       *
*         OUTPUT FILES: rwfn.out, rmix.out, rmcdhf.sum, rmcdhf.log           *
*                                                                            *
*         NOTE: ORBITALS BUILDING REFERENCE STATES ARE REQUIRED TO HAVE      *
*         THE CORRECT NUMBER OF NODES. THEY ARE REFERRED TO AS SPECTROSCOPIC  *
*         ORBITALS. IN THIS RUN WE VARY 1s, 2s, 2p, 3s, 3p, 3d, 4s, 4p       *
*         AND THEY ARE ALL SPECTROSCOPIC.                                    *
******************************************************************************

>>rmcdhf

 RMCDHF
 This program determines the radial orbitals
 and the expansion coefficients of the CSFs
 in a self-onsistent field proceedure
 Input file:  isodata, rcsf.inp, rwfn.inp, mcp.30, ...
 Outputfiles: rwfn.out, rmix.out, rmcdhf.sum, rmcdhf.log

 Default settings?  (y/n)
>>y
 Loading CSF file ... Header only
 There are/is           12  relativistic subshells;
 Loading CSF File for ALL blocks
 There are           15  relativistic CSFs... load complete;
 Loading Radial WaveFunction File ...
 There are            7  blocks  (block   J/Parity   NCF):
  1    0+     2       2    0-     2       3    1+     2       4    1-     4
  5    2+     2       6    2-     2       7    3+     1

 Enter ASF serial numbers for each block
 Block           1    ncf =           2  id =    0+
>>1,2
 Block           2    ncf =           2  id =    0-
>>1,2
 Block           3    ncf =           2  id =    1+
>>1,2
 Block           4    ncf =           4  id =    1-
>>1-4
 Block           5    ncf =           2  id =    2+
>>1,2
```

```
 Block            6    ncf =           2  id =    2-
>>1,2
 Block            7    ncf =           1  id =    3+
>>1
 level weights (1 equal;  5 standard;  9 user)
>>5
 Radial functions
 1s 2s 2p- 2p 3s 3p- 3p 3d- 3d 4s 4p- 4p
 Enter orbitals to be varied (Updating order)
>>*
 Which of these are spectroscopic orbitals?
>>*
 Enter the maximum number of SCF cycles:
>>100

...

               3p    4p   -6.226030140D-03
 Method 2 unable to solve for  4s  orbital
 Iteration number: 12, limit: 12
 Present estimate of P0;  0.35970148280037D+01
 Present estimate of E(J):  0.67584059942824D-01, DELEPS: -0.56222533604713D-02
 Lower bound on energy:  0.41905738989649D-01, upper bound:  0.58877041760686D+01
 Join point:  366, Maximum tabulation point: 404
 Number of nodes counted:  3, Correct number:  3
 Sign of P at first oscillation: -1.

 Failure; equation for orbital  4s  could not be solved using method 2


 ****** Error in SUBROUTINE IMPROV ******
 Convergence not obtained

*******************************************************************************
*         RMCDHF CALCULATION DOES NOT CONVERGE. WE PERFORM HF CALCULATIONS      *
*         FOR 3s3p, 3s4s, 3s3d, 3s4p TO OBTAIN BETTER ESTIMATES OF THE        *
*         WAVE FUNCTION                                                        *
*******************************************************************************

*******************************************************************************
*         HF CALCULATION FOR 3s3p                                              *
*******************************************************************************

>>hf

                      =============================
                       H A R T R E E - F O C K . 86
                      =============================


               THE DIMENSIONS FOR THE CURRENT VERSION ARE:
                          NWF= 20        NO=220
```

```
  START OF CASE
  =============


  Enter ATOM,TERM,Z
  Examples: O,3P,8. or Oxygen,AV,8.
>>Mg,AV,12.

  List the CLOSED shells in the fields indicated (blank line if none)
  ... ... ... ... ... ... ... ... etc.
>> 1s  2s  2p     (! NOTE That shells occupy three positions and are right-justified)

  Enter electrons outside CLOSED shells (blank line if none)
  Example: 2s(1)2p(3)
>>3s(1)3p(1)

  There are   5 orbitals as follows:
    1s  2s  2p  3s  3p

  Orbitals to be varied: ALL/NONE/=i (last i)/comma delimited list/H
>>all

  Default electron parameters ? (Y/N/H)
>>y
        1s     1.00      0.000    76.282    SCREENED HYDROGENIC
        2s     3.00      0.000    22.274    SCREENED HYDROGENIC
        2p     7.00      0.000    32.977    SCREENED HYDROGENIC
        3s    10.00      0.000     3.810    SCREENED HYDROGENIC
        3p    11.00      0.000     5.111    SCREENED HYDROGENIC

  Default values for remaining parameters? (Y/N/H)
>>y

 ..................

     TOTAL ENERGY (a.u.)
     ----- ------
           Non-Relativistic     -199.52165286    Kinetic      199.52163203
           Relativistic Shift     -0.29267673    Potential   -399.04328489
           Relativistic         -199.81432960    Ratio        -2.000000104

  Additional parameters ? (Y/N/H)
>>n

  Do you wish to continue along the sequence ?
>>n


  END OF CASE
  ===========

****************************************************************************
*        COPY FILE WFN.OUT TO WFN3S3P                                      *
```

```
*****************************************************************************

>>cp wfn.out wfn3s3p

*****************************************************************************
*         HF CALCULATION FOR 3s4s COPY wfn.out TO wfn3s4s                    *
*****************************************************************************

>>hf

                      =============================
                       H A R T R E E - F O C K . 86
                      =============================


               THE DIMENSIONS FOR THE CURRENT VERSION ARE:
                          NWF= 20        NO=220


  START OF CASE
  =============


  Enter ATOM,TERM,Z
  Examples: O,3P,8. or Oxygen,AV,8.
>>Mg,AV,12.

  List the CLOSED shells in the fields indicated (blank line if none)
  ... ... ... ... ... ... ... ... etc.
>> 1s  2s  2p

  Enter electrons outside CLOSED shells (blank line if none)
  Example: 2s(1)2p(3)
>>3s(1)4s(1)

  There are   5 orbitals as follows:
    1s  2s  2p  3s  4s

  Orbitals to be varied: ALL/NONE/=i (last i)/comma delimited list/H
>>all

  Default electron parameters ? (Y/N/H)
>>y
          1s     1.00      0.000    76.282    SCREENED HYDROGENIC
          2s     3.00      0.000    22.274    SCREENED HYDROGENIC
          2p     7.00      0.000    32.977    SCREENED HYDROGENIC
          3s    10.00      0.000     3.810    SCREENED HYDROGENIC
          4s    11.00      0.000     1.625    SCREENED HYDROGENIC

  Default values for remaining parameters? (Y/N/H)
>>y

...............
```

```
     TOTAL ENERGY (a.u.)
     ----- ------
           Non-Relativistic     -199.45963917   Kinetic      199.45962102
           Relativistic Shift     -0.29285961   Potential   -398.91926019
           Relativistic         -199.75249878   Ratio        -2.000000091

  Additional parameters ? (Y/N/H)
>>n

  Do you wish to continue along the sequence ?
>>n

  END OF CASE
  ===========

*******************************************************************************
*         COPY FILES WFN.OUT TO WFN3S4S                                       *
*******************************************************************************

>>cp wfn.out wfn3s4s

*******************************************************************************
*         HF CALCULATION FOR 3s3d                                             *
*******************************************************************************

>>hf

                      =============================
                       H A R T R E E - F O C K . 86
                      =============================


               THE DIMENSIONS FOR THE CURRENT VERSION ARE:
                          NWF= 20        NO=220



  START OF CASE
  =============


  Enter ATOM,TERM,Z
  Examples: O,3P,8. or Oxygen,AV,8.
>>Mg,AV,12.

  List the CLOSED shells in the fields indicated (blank line if none)
  ... ... ... ... ... ... ... ... etc.
>> 1s  2s  2p

  Enter electrons outside CLOSED shells (blank line if none)
  Example: 2s(1)2p(3)
>>3s(1)3d(1)
```

```
  There are   5 orbitals as follows:
    1s  2s  2p  3s  3d

  Orbitals to be varied: ALL/NONE/=i (last i)/comma delimited list/H
>>all

  Default electron parameters ? (Y/N/H)
>>y
          1s     1.00      0.000    76.282    SCREENED HYDROGENIC
          2s     3.00      0.000    22.274    SCREENED HYDROGENIC
          2p     7.00      0.000    32.977    SCREENED HYDROGENIC
          3s    10.00      0.000     3.810    SCREENED HYDROGENIC
          3d    11.00      0.000     2.476    SCREENED HYDROGENIC

  Default values for remaining parameters? (Y/N/H)
>>y

.............



     TOTAL ENERGY (a.u.)
     ----- ------
           Non-Relativistic     -199.42914481    Kinetic      199.42914489
           Relativistic Shift     -0.29277506    Potential   -398.85828970
           Relativistic         -199.72191987    Ratio        -2.000000000

  Additional parameters ? (Y/N/H)
>>n

  Do you wish to continue along the sequence ?
>>n


  END OF CASE
  ===========

*******************************************************************************
*         COPY FILE WFN.OUT TO WFN3S3D                                        *
*******************************************************************************

>>cp wfn.out wfn3s3d

*******************************************************************************
*         HF CALCULATION FOR 3s4p                                             *
*******************************************************************************

>>hf

                      =============================
                       H A R T R E E - F O C K . 86
                      =============================
```

```
               THE DIMENSIONS FOR THE CURRENT VERSION ARE:
                          NWF= 20       NO=220



  START OF CASE
  =============


  Enter ATOM,TERM,Z
  Examples: O,3P,8. or Oxygen,AV,8.
>>Mg,AV,12.

  List the CLOSED shells in the fields indicated (blank line if none)
  ... ... ... ... ... ... ... ... etc.
>> 1s  2s  2p

  Enter electrons outside CLOSED shells (blank line if none)
  Example: 2s(1)2p(3)
>>3s(1)4p(1)

  There are   5 orbitals as follows:
    1s  2s  2p  3s  4p

 Orbitals to be varied: ALL/NONE/=i (last i)/comma delimited list/H
>>all

 Default electron parameters ? (Y/N/H)
>>y
         1s     1.00      0.000    76.282    SCREENED HYDROGENIC
         2s     3.00      0.000    22.274    SCREENED HYDROGENIC
         2p     7.00      0.000    32.977    SCREENED HYDROGENIC
         3s    10.00      0.000     3.810    SCREENED HYDROGENIC
         4p    11.00      0.000     3.409    SCREENED HYDROGENIC

 Default values for remaining parameters? (Y/N/H)
>>y

.........


     TOTAL ENERGY (a.u.)
     ----- ------
           Non-Relativistic     -199.43021560    Kinetic      199.43022615
           Relativistic Shift     -0.29277432    Potential   -398.86044175
           Relativistic         -199.72298992    Ratio        -1.999999947

 Additional parameters ? (Y/N/H)
>>n

 Do you wish to continue along the sequence ?
>>n
```

```
 END OF CASE
 ===========

******************************************************************************
*         COPY WFN.OUT TO WFN3S4P                                            *
******************************************************************************

>>cp wfn.out wfn3s4p

******************************************************************************
*         CONCATENATE HF WAVE FUNCTION FILES                                 *
******************************************************************************

>>cat wfn3s3p wfn3s4s wfn3s3d wfn3s4p > wfn.inp

******************************************************************************
*         RUN RWFNMCHFMCDF TO CONVERT NON-RELATIVISTIC RADIAL ORBITALS TO    *
*         RELATIVISTIC ONES                                                  *
*         INPUT FILE:  wfn.inp                                               *
*         OUTPUT FILE: rwfn.out                                              *
******************************************************************************

>>rwfnmchmcdf

 RWFNMCHFMCDF
 This program converts non-relativistic radial
 orbitals to relativistic ones in GRASP format
 Input file: wfn.inp
 Output file: rwfn.out

******************************************************************************
*         COPY FILES                                                         *
*         WE DONT NEED TO INVOKE RWFNESTIMATE SINCE ALL ORBITALS HAVE        *
*         BEEN ESTIMATED THROUGH THE MCHF MCDF CONVERSION                    *
******************************************************************************

>>cp rwfn.out rwfn.inp

******************************************************************************
*         RUN RMCDHF TO OBTAIN SELF CONSISTENT SOLUTIONS                     *
*         INPUT FILES: isodata, rcsf.inp, rwfn.inp, mcp.30, mcp.31,...       *
*         OUTPUT FILES: rwfn.out, rmix.out, rmcdhf.sum, rmcdhf.log           *
*                                                                            *
*         NOTE: ORBITALS BUILDING REFERENCE STATES ARE REQUIRED TO HAVE      *
*         THE CORRECT NUMBER OF NODES. THEY ARE REFERRED TO AS SPECTROSCOPIC *
*         ORBITALS. IN THIS RUN WE VARY 1s, 2s, 2p, 3s, 3p, 3d, 4s, 4p AND   *
*         THEY ARE ALL SPECTROSCOPIC.                                        *
******************************************************************************

>>rmcdhf

 RMCDHF
 This program determines the radial orbitals
 and the expansion coefficients of the CSFs
```

```
 in a self-onsistent field proceedure
 Input file:  isodata, rcsf.inp, rwfn.inp, mcp.30, ...
 Outputfiles: rwfn.out, rmix.out, rmcdhf.sum, rmcdhf.log

 Default settings?  (y/n)
>>y
 Loading CSF file ... Header only
 There are/is           12  relativistic subshells;
 Loading CSF File for ALL blocks
 There are           15  relativistic CSFs... load complete;
 Loading Radial WaveFunction File ...
 There are            7  blocks  (block   J/Parity   NCF):
  1    0+     2       2    0-     2       3    1+     2       4    1-     4
  5    2+     2       6    2-     2       7    3+     1

 Enter ASF serial numbers for each block
 Block           1    ncf =           2  id =    0+
>>1,2
 Block           2    ncf =           2  id =    0-
>>1,2
 Block           3    ncf =           2  id =    1+
>>1,2
 Block           4    ncf =           4  id =    1-
>>1-4
 Block           5    ncf =           2  id =    2+
>>1,2
 Block           6    ncf =           2  id =    2-
>>1,2
 Block           7    ncf =           1  id =    3+
>>1
 level weights (1 equal;  5 standard;  9 user)
>>5
 Radial functions
 1s 2s 2p- 2p 3s 3p- 3p 3d- 3d 4s 4p- 4p
 Enter orbitals to be varied (Updating order)
>>*
 Which of these are spectroscopic orbitals?
>>*
 Enter the maximum number of SCF cycles:
>>100

......

 Wall time:
      46 seconds

 Finish Date and Time:
   Date (Yr/Mon/Day): 2014/09/05
   Time (Hr/Min/Sec): 13/51/24.996
   Zone: +0200

 RMCDHF: Execution complete.

*******************************************************************************
```

```
*         THIS TIME IT CONVERGED! RUN RSAVE                                    *
********************************************************************************

>>rsave mr
 Created mr.w, mr.c, mr.m, mr.sum, mr.alog and mr.log
```

13.3 Decrease nuclear charge in small steps

Convergence can sometimes be difficult to achieve for large systems that are neutral or near neutral. In these cases one advice is to edit `isodata` and increase the nuclear charge. If the `rmcdhf` run is converged for the increased charge then copy `rwfn.out` to `rwfn.inp`, decrease the nuclear charge by a small amount and run `rmcdhf` that hopefully will converge. Repeat the procedure until you are down to the correct charge. To illustrate the technique we will perform a calculation for the ground state [Rn] $5f^{14}7s^2$ of No I ($Z = 102$).

Overview

1. Define nuclear data.
2. Generate list of CSFs.
3. Perform angular integration.
4. Generate initial estimates of radial orbitals.
5. Perform self-consistent field calculation on the weighted average of the states (this will not converge)
6. Edit `isodata` and increase nuclear charge to $Z = 105$.
7. Generate initial estimates of radial orbitals.
8. Perform self-consistent field calculation on the weighted average of the states (this will converge).
9. Copy `rwfn.out` to `rwfn.inp`. Decrease nuclear charge to $Z = 104$.
10. Perform self-consistent field calculation on the weighted average of the states (this will converge).
11. Copy `rwfn.out` to `rwfn.inp`. Decrease nuclear charge to $Z = 103$.
12. Perform self-consistent field calculation on the weighted average of the states (this will converge).
13. Copy `rwfn.out` to `rwfn.inp`. Decrease nuclear charge to $Z = 102.5$.
14. Perform self-consistent field calculation on the weighted average of the states (this will converge).
15. Copy `rwfn.out` to `rwfn.inp`. Decrease nuclear charge to $Z = 102$.
16. Perform self-consistent field calculation on the weighted average of the states (this will converge).
17. Run `rsave`.

```
******************************************************************************
* RUN RNUCLEUS TO GENERATE NUCLEAR DATA                                      *
* OUTPUT FILE: isodata                                                       *
******************************************************************************

>>rnucleus

 RNUCLEUS
 This program defines nuclear data and the radial grid
 Outputfile: isodata

 Enter the atomic number:
>>102
 Enter the mass number (0 if the nucleus is to be modelled as a point source:
>>259
 The default root mean squared radius is    5.8989240695087215       fm;
 The default nuclear skin thickness is      2.2999999999999998       fm;
 Revise these values?
>>n
 Enter the mass of the neutral atom (in amu) (0 if the nucleus is to be static):
>>259
 Enter the nuclear spin quantum number (I) (in units of h / 2 pi):
>>1
 Enter the nuclear dipole moment (in nuclear magnetons):
>>1
 Enter the nuclear quadrupole moment (in barns):
>>1

******************************************************************************
*         RUN RCSFGENERATE TO GENERATE LIST OF CSFs                          *
*         OUTPUT FILES: rcsfgenerate.log, rcsf.out                           *
******************************************************************************

>>rcsfgenerate

 RCSFGENERATE
 This program creates a list of CSFs
 Configurations should be entered in spectroscopic notation
 with occupation numbers and indications if orbitals are
 closed (c), inactive (i), active (*) or has a minimal
 occupation e.g. 1s(2,1)2s(2,*)
 Outputfiles: rcsf.out, rcsfgenerate.log

 Default, reverse, symmetry or user specified ordering? (*/r/s/u)
>>*

 Select core
       0: No core
       1: He (       1s(2)                  =  2 electrons)
       2: Ne ([He] + 2s(2)2p(6)             = 10 electrons)
       3: Ar ([Ne] + 3s(2)3p(6)             = 18 electrons)
       4: Kr ([Ar] + 3d(10)4s(2)4p(6)       = 36 electrons)
       5: Xe ([Kr] + 4d(10)5s(2)5p(6)       = 54 electrons)
       6: Rn ([Xe] + 4f(14)5d(10)6s(2)6p(6) = 86 electrons)
```

```
>>6
 Enter list of (maximum 100) configurations. End list with a blank line or an asterisk (*)

 Give configuration           1
>>5f(14,i)7s(2,i)
 Give configuration           2
>>
 Give set of active orbitals, as defined by the highest principal quantum number
 per l-symmetry, in a comma delimited list in s,p,d etc order, e.g. 5s,4p,3d
>>7s,6p,5d,5f
 Resulting 2*J-number? lower, higher (J=1 -> 2*J=2 etc.)
>>0,0
 Number of excitations (if negative number e.g. -2, correlation
 orbitals will always be doubly occupied)
>>0
 Generate more lists ? (y/n)
>>n

        .........

 1 blocks were created

      block  J/P            NCSF
          1  1/2+              1

*****************************************************************************
*         COPY FILES                                                        *
*         IT IS ADVISABLE TO SAVE THE rcsfgenerate.log FILE TO HAVE A     *
*         RECORD ON HOW THE LIST OF CSFs WAS CREATED                        *
*****************************************************************************

>>cp rcsfgenerate.log mr.exc
>>cp rcsf.out rcsf.inp

*****************************************************************************
*         RUN RANGULAR TO GENERATE ENERGY EXPRESSION                        *
*         INPUT FILE  : rcsf.inp                                            *
*         OUTPUT FILES: rangular.alog, mcp.30, mcp.31,....                  *
*****************************************************************************

>>rangular

 RANGULAR
 This program performs angular integration
 Input file:  rcsf.inp
 Outputfiles: mcp.30, mcp.31, ....
              rangular.log

 Full interaction?  (y/n)
>>y

    ......

 RANGULAR: Execution complete.
```

```
***************************************************************************
*         RUN RWFNESTIMATE TO GENERATE INITIAL ESTIMATES FOR RADIAL ORBITALS  *
*         INPUT FILES: isodata, rcsf.inp                                      *
*         OUTPUT FILE: rwfn.inp, rwfnestimate.log                             *
***************************************************************************

>>rwfnestimate

 RWFNESTIMATE
 This program estimates radial wave functions
 for orbitals
 Input files: isodata, rcsf.inp, optional rwfn file
 Output file: rwfn.inp
 Default settings ?
>>y
 Loading CSF file ... Header only
 There are/is           27  relativistic subshells;
 The following subshell radial wavefunctions remain to be estimated:
 1s 2s 2p- 2p 3s 3p- 3p 3d- 3d 4s 4p- 4p 4d- 4d 4f- 4f 5s 5p- 5p 5d- 5d 6s 6p-
 6p 5f- 5f 7s

 Read subshell radial wavefunctions. Choose one below
     1 -- GRASP92 File
     2 -- Thomas-Fermi
     3 -- Screened Hydrogenic
>>2
 Enter the list of relativistic subshells:
*
 All required subshell radial wavefunctions  have been estimated:
Shell      e           p0        gamma        P(2)       Q(2)      MTP  SRC

  1s   0.5535D+04  0.8367D+04  0.1000D+01  0.8412D-05 -0.3343D-10  328  T-F
  2s   0.1090D+04  0.3673D+04  0.1000D+01  0.3692D-05 -0.1473D-10  344  T-F
  2p-  0.1062D+04  0.5365D+03  0.1000D+01  0.5982D-11  0.1449D-05  344  T-F
  2p   0.8187D+03  0.4714D+05  0.2000D+01  0.4764D-13 -0.1901D-18  347  T-F
  3s   0.2880D+03  0.1820D+04  0.1000D+01  0.1830D-05 -0.7307D-11  358  T-F
  3p-  0.2739D+03  0.2826D+03  0.1000D+01  0.3153D-11  0.7634D-06  358  T-F
  3p   0.2159D+03  0.2629D+05  0.2000D+01  0.2657D-13 -0.1061D-18  360  T-F
  3d-  0.1922D+03  0.6033D+03  0.2000D+01  0.1354D-19  0.3277D-14  361  T-F
  3d   0.1810D+03  0.1243D+06  0.3000D+01  0.1263D-21 -0.5044D-27  362  T-F
  4s   0.7989D+02  0.9782D+03  0.1000D+01  0.9834D-06 -0.3928D-11  370  T-F
  4p-  0.7297D+02  0.1523D+03  0.1000D+01  0.1700D-11  0.4114D-06  371  T-F
  4p   0.5672D+02  0.1435D+05  0.2000D+01  0.1451D-13 -0.5794D-19  374  T-F
  4d-  0.4539D+02  0.3560D+03  0.2000D+01  0.7988D-20  0.1933D-14  375  T-F
  4d   0.4246D+02  0.7373D+05  0.3000D+01  0.7491D-22 -0.2992D-27  376  T-F
  4f-  0.2737D+02  0.4518D+03  0.3000D+01  0.1529D-28  0.3700D-23  379  T-F
  4f   0.2652D+02  0.1174D+06  0.4000D+01  0.1200D-30 -0.4791D-36  379  T-F
  5s   0.1976D+02  0.5211D+03  0.1000D+01  0.5239D-06 -0.2092D-11  384  T-F
  5p-  0.1675D+02  0.7958D+02  0.1000D+01  0.8881D-12  0.2150D-06  385  T-F
  5p   0.1243D+02  0.7413D+04  0.2000D+01  0.7492D-14 -0.2992D-19  388  T-F
  5d-  0.7880D+01  0.1754D+03  0.2000D+01  0.3936D-20  0.9526D-15  392  T-F
  5d   0.7220D+01  0.3613D+05  0.3000D+01  0.3671D-22 -0.1466D-27  393  T-F
  6s   0.3744D+01  0.2510D+03  0.1000D+01  0.2523D-06 -0.1008D-11  400  T-F
```

```
  6p-  0.2724D+01  0.3631D+02  0.1000D+01  0.4053D-12  0.9809D-07  403  T-F
  6p   0.1859D+01  0.3246D+04  0.2000D+01  0.3281D-14 -0.1310D-19  406  T-F
  5f-  0.2174D-01  0.1722D+01  0.3000D+01  0.5827D-31  0.1410D-25  457  T-F
  5f   0.2173D-01  0.4523D+03  0.4000D+01  0.4620D-33 -0.1845D-38  457  T-F
  7s   0.5319D+00  0.9661D+02  0.1000D+01  0.9712D-07 -0.3879D-12  419  T-F
 RWFNESTIMATE: Execution complete.

******************************************************************************
*         RUN RMCDHF FOR Z = 102 (WILL NOT CONVERGE)                         *
******************************************************************************

>>rmcdhf

 RMCDHF
 This program determines the radial orbitals
 and the expansion coefficients of the CSFs
 in a self-onsistent field proceedure
 Input file:  isodata, rcsf.inp, rwfn.inp, mcp.30, ...
 Outputfiles: rwfn.out, rmix.out, rmcdhf.sum, rmcdhf.log

Default settings?  (y/n)
>>y
 Loading CSF file ... Header only
 There are/is           27  relativistic subshells;
 Loading CSF File for ALL blocks
 There are            1  relativistic CSFs... load complete;
 Loading Radial WaveFunction File ...
 There are            1  blocks  (block   J/Parity   NCF):
  1    0+       1

 Enter ASF serial numbers for each block
 Block           1    ncf =           1  id =    0+
>>1
 Radial functions
 1s 2s 2p- 2p 3s 3p- 3p 3d- 3d 4s 4p- 4p 4d- 4d 4f- 4f 5s 5p- 5p 5d- 5d 6s 6p-
 6p 5f- 5f 7s
 Enter orbitals to be varied (Updating order)
>>*
 Which of these are spectroscopic orbitals?
>>*
 Enter the maximum number of SCF cycles:
>>100

..................

 Iteration number   2
 --------------------
                                         Self-           Damping
Subshell    Energy    Method   P0    consistency  Norm-1  factor  JP MTP INV NNP

  1s    5.5386641D+03  1  8.370D+03  1.36D-03  1.62D-05 0.000   274   377  0  0
  2s    1.0949842D+03  1  3.649D+03  4.51D-03 -1.88D-04 0.000   302   382  0  1
  2p-   1.0594702D+03  1  5.267D+02  4.11D-03 -2.43D-05 0.000   302   384  0  0
  2p    8.2094271D+02  1  4.636D+04  7.53D-03 -1.04D-05 0.000   306   385  0  0
```

```
  3s    2.9768023D+02  1  1.800D+03  1.17D-02 -7.39D-04 0.000   320   385  0  2
  3p-   2.8135963D+02  1  2.760D+02  1.16D-02 -6.04D-04 0.000   321   388  0  1
  3p    2.2441931D+02  1  2.576D+04  1.98D-02 -7.16D-04 0.000   324   389  0  1
  3d-   1.9934224D+02  1  5.773D+02  1.86D-02 -2.83D-04 0.000   325   411  0  0
  3d    1.8864448D+02  1  1.194D+05  2.21D-02 -2.02D-04 0.000   326   411  0  0
  4s    9.0606030D+01  1  9.551D+02  1.73D-02 -1.59D-03 0.100   335   393  0  3
  4p-   8.2942623D+01  1  1.468D+02  1.77D-02 -1.56D-03 0.100   336   404  0  2
  4p    6.7111537D+01  1  1.395D+04  2.78D-02 -2.27D-03 0.100   339   408  0  2
  4d-   5.5119356D+01  1  3.369D+02  2.90D-02 -2.26D-03 0.100   341   434  0  1
  4d    5.2313964D+01  1  7.011D+04  2.95D-02 -1.95D-03 0.100   342   434  0  1
  4f-   3.6521991D+01  1  4.126D+02  2.19D-03  1.73D-04 0.050   346   408  0  0
  4f    3.5747535D+01  1  1.084D+05  6.02D-03 -2.02D-04 0.100   346   408  0  0
  5s    3.0255721D+01  1  5.092D+02  1.83D-02  3.85D-03 0.100   348   407  0  4
  5p-   2.6881305D+01  1  7.699D+01  2.36D-02  5.21D-03 0.100   350   428  0  3
  5p    2.2514178D+01  1  7.371D+03  3.91D-02  6.74D-03 0.100   352   429  0  3
  5d-   1.7326694D+01  1  1.729D+02  5.78D-02  1.18D-02 0.100   356   440  0  2
  5d    1.6650195D+01  1  3.611D+04  6.18D-02  1.05D-02 0.100   356   440  0  2
  6s    1.2077305D+01  1  2.810D+02  4.61D-03 -1.36D-03 0.050   361   429  0  5
  6p-   1.0606432D+01  1  4.187D+01  7.79D-03 -2.88D-03 0.050   363   436  0  4
  6p    9.2631692D+00  1  3.969D+03  4.70D-02 -1.32D-02 0.190   365   438  0  4
 Method 2 unable to solve for  5f- orbital
 Iteration number: 15, limit: 15
 Present estimate of P0;  0.19118742906131D+01
 Present estimate of E(J):  0.98080405136720D+02, DELEPS: -0.60275500767866D+02
 Lower bound on energy:  0.13875636426564D+02, upper bound:  0.26234163356640D+03
 Join point:  339, Maximum tabulation point: 457
 Number of nodes counted:  2, Correct number:  1
 Sign of P at first oscillation: -1.

 Failure; equation for orbital  5f- could not be solved using method 2


 ****** Error in SUBROUTINE IMPROV ******
 Convergence not obtained

*****************************************************************************
*         Increase nuclear charge to Z = 105                                *
*****************************************************************************

>>open editor and change Z to 105 in isodata

*****************************************************************************
*         RUN RWFNESTIMATE TO GENERATE INITIAL ESTIMATES FOR RADIAL ORBITALS  *
*         INPUT FILES: isodata, rcsf.inp, previous rwfn files               *
*         OUTPUT FILE: rwfn.inp, rwfnestimate.log                           *
*****************************************************************************

>>rwfnestimate

 RWFNESTIMATE
 This program estimates radial wave functions
 for orbitals
 Input files: isodata, rcsf.inp, optional rwfn file
 Output file: rwfn.inp
```

```
 Default settings ?
>>y
 Loading CSF file ... Header only
 There are/is           27  relativistic subshells;
 The following subshell radial wavefunctions remain to be estimated:
 1s 2s 2p- 2p 3s 3p- 3p 3d- 3d 4s 4p- 4p 4d- 4d 4f- 4f 5s 5p- 5p 5d- 5d 6s 6p-
 6p 5f- 5f 7s

 Read subshell radial wavefunctions. Choose one below
     1 -- GRASP92 File
     2 -- Thomas-Fermi
     3 -- Screened Hydrogenic
>>2
 Enter the list of relativistic subshells:
>>*
 All required subshell radial wavefunctions  have been estimated:
Shell      e           p0        gamma        P(2)       Q(2)      MTP  SRC

  1s   0.5997D+04  0.9512D+04  0.1000D+01  0.9290D-05 -0.3691D-10  328  T-F
  2s   0.1211D+04  0.4259D+04  0.1000D+01  0.4160D-05 -0.1660D-10  344  T-F
  2p-  0.1185D+04  0.6741D+03  0.1000D+01  0.7086D-11  0.1718D-05  344  T-F
  2p   0.8980D+03  0.5365D+05  0.2000D+01  0.5117D-13 -0.2042D-18  347  T-F
  3s   0.3318D+03  0.2128D+04  0.1000D+01  0.2078D-05 -0.8298D-11  357  T-F
  3p-  0.3177D+03  0.3582D+03  0.1000D+01  0.3768D-11  0.9130D-06  357  T-F
  3p   0.2480D+03  0.3038D+05  0.2000D+01  0.2897D-13 -0.1157D-18  360  T-F
  3d-  0.2242D+03  0.7505D+03  0.2000D+01  0.1542D-19  0.3737D-14  360  T-F
  3d   0.2109D+03  0.1473D+06  0.3000D+01  0.1372D-21 -0.5477D-27  361  T-F
  4s   0.9893D+02  0.1162D+04  0.1000D+01  0.1135D-05 -0.4531D-11  369  T-F
  4p-  0.9179D+02  0.1966D+03  0.1000D+01  0.2068D-11  0.5011D-06  370  T-F
  4p   0.7154D+02  0.1696D+05  0.2000D+01  0.1618D-13 -0.6462D-19  372  T-F
  4d-  0.5977D+02  0.4572D+03  0.2000D+01  0.9396D-20  0.2276D-14  374  T-F
  4d   0.5611D+02  0.9032D+05  0.3000D+01  0.8412D-22 -0.3360D-27  374  T-F
  4f-  0.4043D+02  0.6235D+03  0.3000D+01  0.1877D-28  0.4547D-23  377  T-F
  4f   0.3931D+02  0.1559D+06  0.4000D+01  0.1418D-30 -0.5664D-36  377  T-F
  5s   0.2861D+02  0.6415D+03  0.1000D+01  0.6265D-06 -0.2502D-11  382  T-F
  5p-  0.2533D+02  0.1071D+03  0.1000D+01  0.1127D-11  0.2730D-06  383  T-F
  5p   0.1943D+02  0.9208D+04  0.2000D+01  0.8782D-14 -0.3508D-19  385  T-F
  5d-  0.1433D+02  0.2428D+03  0.2000D+01  0.4990D-20  0.1209D-14  388  T-F
  5d   0.1337D+02  0.4789D+05  0.3000D+01  0.4460D-22 -0.1781D-27  389  T-F
  6s   0.7815D+01  0.3387D+03  0.1000D+01  0.3307D-06 -0.1321D-11  395  T-F
  6p-  0.6526D+01  0.5491D+02  0.1000D+01  0.5778D-12  0.1400D-06  396  T-F
  6p   0.5004D+01  0.4650D+04  0.2000D+01  0.4435D-14 -0.1771D-19  399  T-F
  5f-  0.7111D+01  0.3258D+03  0.3000D+01  0.9809D-29  0.2376D-23  394  T-F
  5f   0.6883D+01  0.8130D+05  0.4000D+01  0.7395D-31 -0.2954D-36  395  T-F
  7s   0.2311D+01  0.1686D+03  0.1000D+01  0.1647D-06 -0.6577D-12  408  T-F

 RWFNESTIMATE: Execution complete.

******************************************************************************
*         RUN RMCDHF FOR Z = 105 (WILL CONVERGE)                             *
******************************************************************************

>>rmcdhf
```

```
 RMCDHF
 This program determines the radial orbitals
 and the expansion coefficients of the CSFs
 in a self-onsistent field proceedure
 Input file:  isodata, rcsf.inp, rwfn.inp, mcp.30, ...
 Outputfiles: rwfn.out, rmix.out, rmcdhf.sum, rmcdhf.log

Default settings?  (y/n)
>>y
 Loading CSF file ... Header only
 There are/is           27  relativistic subshells;
 Loading CSF File for ALL blocks
 There are            1  relativistic CSFs... load complete;
 Loading Radial WaveFunction File ...
 There are            1  blocks  (block   J/Parity   NCF):
  1    0+       1

 Enter ASF serial numbers for each block
 Block            1    ncf =           1  id =    0+
>>1
 Radial functions
 1s 2s 2p- 2p 3s 3p- 3p 3d- 3d 4s 4p- 4p 4d- 4d 4f- 4f 5s 5p- 5p 5d- 5d 6s 6p-
 6p 5f- 5f 7s
 Enter orbitals to be varied (Updating order)
>>*
 Which of these are spectroscopic orbitals?
>>*
 Enter the maximum number of SCF cycles:
>>100

..................

 Wall time:
       75 seconds

 Finish Date and Time:
   Date (Yr/Mon/Day): 2018/11/26
   Time (Hr/Min/Sec): 12/58/42.303
   Zone: +0100

 RMCDHF: Execution complete.

*******************************************************************************
*         RMCDHF IS CONVERGING                                                *
*         EDIT ISODATA AND DECREASE Z TO Z = 104 (SMALLER STEPS MAY BE NEEDED)*
*******************************************************************************

>>open editor and change Z to 104 in isodata

*******************************************************************************
*         COPY RWFN.OUT TO RWFN.INP                                           *
*         THUS WE USE THE PREVIOUS OUTPUT AS INPUT FOR THE NEW RUN            *
*******************************************************************************
```

```
>>cp rwfn.out rwfn.inp

***************************************************************************
*         RUN RMCDHF FOR Z = 104 (WILL CONVERGE)                          *
***************************************************************************

>>rmcdhf

 RMCDHF
 This program determines the radial orbitals
 and the expansion coefficients of the CSFs
 in a self-onsistent field proceedure
 Input file:  isodata, rcsf.inp, rwfn.inp, mcp.30, ...
 Outputfiles: rwfn.out, rmix.out, rmcdhf.sum, rmcdhf.log

Default settings?  (y/n)
>>y
 Loading CSF file ... Header only
 There are/is           27  relativistic subshells;
 Loading CSF File for ALL blocks
 There are            1  relativistic CSFs... load complete;
 Loading Radial WaveFunction File ...
 There are            1  blocks  (block   J/Parity   NCF):
  1    0+       1

 Enter ASF serial numbers for each block
 Block            1    ncf =           1  id =    0+
>>1
 Radial functions
 1s 2s 2p- 2p 3s 3p- 3p 3d- 3d 4s 4p- 4p 4d- 4d 4f- 4f 5s 5p- 5p 5d- 5d 6s 6p-
 6p 5f- 5f 7s
 Enter orbitals to be varied (Updating order)
>>*
 Which of these are spectroscopic orbitals?
>>*
 Enter the maximum number of SCF cycles:
>>100

..................

 Wall time:
      75 seconds

 Finish Date and Time:
   Date (Yr/Mon/Day): 2018/11/26
   Time (Hr/Min/Sec): 12/58/42.303
   Zone: +0100

 RMCDHF: Execution complete.

***************************************************************************
*         CONTINUE IN THE SAME WAY UNTIL REACHING THE ORIGINAL CHARGE Z = 102 *
*         IF THINGS DO NOT CONVERGE TRY MAKING THE CHANGE IN NUCLEAR CHARGE   *
*         SMALLER AND REDO THINGS                                         *
```

```
******************************************************************************
```

13.4 Using non-default options

If combining HF estimates and decreasing the nuclear charge in small steps does not ensure convergence then the remaining alternative is to use the non-default options in `rmcdhf`. The user may play around with the threshold for node counting. An oscillation in the large-component of the radial wavefunction is disregarded for the purposes of node counting if its amplitude is less than 1/20 the maximum amplitude. The user may change this value. If convergence is achieved then it is required that all the spectroscopic orbitals are plotted and inspected so that they have the correct node structure. The user may also want to set accelerating parameters `odamp` for subshell radial wavefunctions. Setting `odamp` to a value close to 1 damps the changes in the radial wave functions at each iteration. This may sometimes help.

13.5 Changing the grid

For neutral or near neutral super heavy systems it is sometimes desirable to increase the number of grid points and change the grid parameters. To install the program with the extended grid, follow the instructions in section 2.6. In this case change the `NNNP` and `NNN1` variables to, respectively, 1990 and 2000 and recompile. Below we perform a calculation for all states of the U I ground configuration $5f^3 6d7s^2$ where the grid parameters have been changed to smaller values and the number of grid points has been set to 1990.

Overview

1. Define nuclear data.
2. Generate list of CSFs.
3. Perform angular integration.
4. Generate initial estimates of radial orbitals. Override default options and change grid parameters.
5. Perform self-consistent field calculation on the weighted average of the states. Override default options and change grid parameters.

```
******************************************************************************
* RUN RNUCLEUS TO GENERATE NUCLEAR DATA                                      *
* OUTPUT FILE: isodata                                                       *
******************************************************************************

>>rnucleus

 Enter the atomic number:
>>92
 Enter the mass number (0 if the nucleus is to be modelled as a point source:
>>238
 The default root mean squared radius is    5.7508211074435813       fm;
  the default nuclear skin thickness is    2.2999999999999998       fm;
 Revise these values?
>>n
 Enter the mass of the neutral atom (in amu) (0 if the nucleus is to be static):
>>238
```

```
 Enter the nuclear spin quantum number (I) (in units of h / 2 pi):
>>1
 Enter the nuclear dipole moment (in nuclear magnetons):
>>1
 Enter the nuclear quadrupole moment (in barns):
>>1

*******************************************************************************
*         RUN RCSFGENERATE TO GENERATE LIST OF CSFs                           *
*         OUTPUT FILES: rcsfgenerate.log, rcsf.out                            *
*******************************************************************************

>>rcsfgenerate

 RCSFGENERATE
 This program generates a list of CSFs

 Configurations should be entered in spectroscopic notation
 with occupation numbers and indications if orbitals are
 closed (c), inactive (i), active (*) or has a minimal
 occupation e.g. 1s(2,1)2s(2,*)
 Outputfiles: rcsf.out, rcsfgenerate.log

 Default, reverse, symmetry or user specified ordering? (*/r/s/u)
>>*

 Select core
        0: No core
        1: He (       1s(2)                  =  2 electrons)
        2: Ne ([He] + 2s(2)2p(6)             = 10 electrons)
        3: Ar ([Ne] + 3s(2)3p(6)             = 18 electrons)
        4: Kr ([Ar] + 3d(10)4s(2)4p(6)       = 36 electrons)
        5: Xe ([Kr] + 4d(10)5s(2)5p(6)       = 54 electrons)
        6: Rn ([Xe] + 4f(14)5d(10)6s(2)6p(6) = 86 electrons)
>>6

 Enter list of (maximum 100) configurations. End list with a blank line or an asterisk (*)

 Give configuration           1
>>5f(3,*)6d(1,*)7s(2,*)
 Give configuration           2
>>
 Give set of active orbitals, as defined by the highest principal quantum number
 per l-symmetry, in a comma delimited list in s,p,d etc order, e.g. 5s,4p,3d
>>7s,6d,5f
 Resulting 2*J-number? lower, higher (J=1 -> 2*J=2 etc.)
>>0,22
 Number of excitations (if negative number e.g. -2, correlation
 orbitals will always be doubly occupied)
>>0
 Generate more lists ? (y/n)
>>n

12 blocks were created
```

```
      block  J/P           NCSF
          1    0-             13
          2    1-             35
          3    2-             51
          4    3-             61
          5    4-             61
          6    5-             54
          7    6-             44
          8    7-             31
          9    8-             19
         10    9-             11
         11   10-              5
         12   11-              1

*******************************************************************************
*         RUN RWFNESTIMATE TO GENERATE INITIAL ESTIMATES FOR RADIAL ORBITALS  *
*         NON-DEFAULT OPTIONS ARE USED TO CHANGE GRID PARAMETERS              *
*         INPUT FILES: isodata, rcsf.inp, previous rwfn files                 *
*         OUTPUT FILE: rwfn.inp, rwfnestimate.log                             *
*******************************************************************************

>>rwfnestimate

 RWFNESTIMATE
 This program estimates radial wave functions
 for orbitals
 Input files: isodata, rcsf.inp, optional rwfn file
 Output file: rwfn.inp
 Default settings ?
>>n
 Generate debug printout?
>>n
 File  erwf.sum  will be created as the ERWF SUMmary File;
 enter another file name if this is not acceptable; null otherwise:
>>
 Loading CSF file ... Header only
 There are/is           29  relativistic subshells;
 Change the default speed of light or radial grid parameters?
>>y
 The physical speed of light in atomic units is   137.03599913900001      ;
 revise this value?
>>n
 The default radial grid parameters for this case are:
  RNT =    2.1739130434782606E-008 ;
  H =    5.0000000000000003E-002 ;
  HP =    0.0000000000000000      ;
  N =        1990 ;
 revise these values?
>>y
 Enter RNT:
>>2.17d-08
 Enter H:
>>1.5d-02
```

```
 Enter HP:
>>0.0d0
 Enter N:
>>1990
 The following subshell radial wavefunctions remain to be estimated:
 1s 2s 2p- 2p 3s 3p- 3p 3d- 3d 4s 4p- 4p 4d- 4d 4f- 4f 5s 5p- 5p 5d- 5d 6s 6p-
 6p 5f- 5f 6d- 6d 7s

 Read subshell radial wavefunctions. Choose one below
     1 -- GRASP92 File
     2 -- Thomas-Fermi
     3 -- Screened Hydrogenic
>>2
 Enter the list of relativistic subshells:
>>*
 All required subshell radial wavefunctions  have been estimated:
Shell      e           p0        gamma        P(2)       Q(2)      MTP  SRC

  1s   0.4269D+04  0.5561D+04  0.1000D+01  0.1824D-05 -0.1370D-11 1111  T-F
  2s   0.8051D+03  0.2306D+04  0.1000D+01  0.7562D-06 -0.6840D-12 1166  T-F
  2p-  0.7805D+03  0.2581D+03  0.1000D+01  0.3154D-12  0.2522D-06 1166  T-F
  2p   0.6379D+03  0.3093D+05  0.2000D+01  0.3327D-14 -0.4013D-20 1172  T-F
  3s   0.2060D+03  0.1128D+04  0.1000D+01  0.3700D-06 -0.2233D-12 1211  T-F
  3p-  0.1939D+03  0.1342D+03  0.1000D+01  0.1641D-12  0.1311D-06 1213  T-F
  3p   0.1609D+03  0.1678D+05  0.2000D+01  0.1805D-14 -0.2178D-20 1219  T-F
  3d-  0.1407D+03  0.3047D+03  0.2000D+01  0.2444D-21  0.1952D-15 1222  T-F
  3d   0.1339D+03  0.7419D+05  0.3000D+01  0.2617D-23 -0.3158D-29 1223  T-F
  4s   0.5410D+02  0.5952D+03  0.1000D+01  0.1952D-06 -0.2356D-12 1255  T-F
  4p-  0.4849D+02  0.7077D+02  0.1000D+01  0.8656D-13  0.6914D-07 1258  T-F
  4p   0.3965D+02  0.8921D+04  0.2000D+01  0.9595D-15 -0.1158D-20 1265  T-F
  4d-  0.3040D+02  0.1730D+03  0.2000D+01  0.1388D-21  0.1109D-15 1272  T-F
  4d   0.2873D+02  0.4227D+05  0.3000D+01  0.1491D-23 -0.1800D-29 1274  T-F
  4f-  0.1647D+02  0.1935D+03  0.3000D+01  0.7638D-31  0.6101D-25 1289  T-F
  4f   0.1602D+02  0.5769D+05  0.4000D+01  0.6673D-33 -0.8054D-39 1290  T-F
  5s   0.1225D+02  0.3062D+03  0.1000D+01  0.1004D-06 -0.1212D-12 1304  T-F
  5p-  0.9992D+01  0.3545D+02  0.1000D+01  0.4336D-13  0.3463D-07 1309  T-F
  5p   0.7827D+01  0.4411D+04  0.2000D+01  0.4744D-15 -0.5726D-21 1317  T-F
  5d-  0.4418D+01  0.7947D+02  0.2000D+01  0.6376D-22  0.5093D-16 1334  T-F
  5d   0.4088D+01  0.1929D+05  0.3000D+01  0.6804D-24 -0.8212D-30 1337  T-F
  6s   0.2056D+01  0.1381D+03  0.1000D+01  0.4527D-07 -0.5465D-13 1362  T-F
  6p-  0.1383D+01  0.1482D+02  0.1000D+01  0.1812D-13  0.1447D-07 1374  T-F
  6p   0.1019D+01  0.1772D+04  0.2000D+01  0.1906D-15 -0.2300D-21 1384  T-F
  5f-  0.4670D+00  0.6503D+02  0.3000D+01  0.2566D-31  0.2050D-25 1405  T-F
  5f   0.4268D+00  0.1910D+05  0.4000D+01  0.2210D-33 -0.2667D-39 1408  T-F
  6d-  0.2724D+00  0.2248D+02  0.2000D+01  0.1803D-22  0.1440D-16 1428  T-F
  6d   0.2470D+00  0.5300D+04  0.3000D+01  0.1870D-24 -0.2257D-30 1432  T-F
  7s   0.2991D+00  0.4891D+02  0.1000D+01  0.1604D-07 -0.1936D-13 1428  T-F
 Revise any of these estimates?
>>n

 RWFNESTIMATE: Execution complete.

*******************************************************************************
*         RUN RMCDHF WITH NON-DEFAULT OPTIONS FOR GRID PARAMETERS             *
```

```
****************************************************************************

>>rmcdhf

 RMCDHF
 This program determines the radial orbitals
 and the expansion coefficients of the CSFs
 in a self-onsistent field proceedure
 Input file:  isodata, rcsf.inp, rwfn.inp, mcp.30, ...
 Outputfiles: rwfn.out, rmix.out, rmcdhf.sum, rmcdhf.log

 Default settings?  (y/n)
>>n
 Generate debug output?  (y/n)
>>n
 Loading CSF file ... Header only
 There are/is           29  relativistic subshells;
 Loading CSF File for ALL blocks
 There are          385  relativistic CSFs... load complete;
 Change the default speed of
 light or radial grid parameters?  (y/n)
>>y
 Speed of light =    137.03599913900001      ; revise ?
>>n
 The default radial grid parameters for this case are:
  RNT =    2.1739130434782606E-008
  H =    5.0000000000000003E-002
  HP =    0.0000000000000000
  N =         1990
  revise these values?
>>y
 Enter RNT:
>>2.17d-08
 Enter H:
>>1.5d-02
 Enter HP:
>>0.0d0
 Enter N:
>>1990
 Revised RNT =    2.1699999999999999E-008
 Revised H   =    1.4999999999999999E-002
 Revised HP  =    0.0000000000000000
 Revised N   =         1990
 Revise the default ACCY =    1.5625000000000006E-008
>>n
 Loading Radial WaveFunction File ...
 There are           11  blocks  (block   J/Parity   NCF):
  1    0-      13       2    1-      35       3    2-      51       4    3-      61
  5    4-      61       6    5-      54       7    6-      44       8    7-      31
  9    8-      19      10    9-      11      11   10-       5

 Enter ASF serial numbers for each block
 Block            1    ncf =          13  id =    0-
>>1-13
```

```
 Block            2    ncf =           35  id =    1-
>>1-35
 Block            3    ncf =           51  id =    2-
>>1-51
 Block            4    ncf =           61  id =    3-
>>1-61
 Block            5    ncf =           61  id =    4-
>>1-61
 Block            6    ncf =           54  id =    5-
>>1-54
 Block            7    ncf =           44  id =    6-
>>1-44
 Block            8    ncf =           31  id =    7-
>>1-31
 Block            9    ncf =           19  id =    8-
>>1-19
 Block           10    ncf =           11  id =    9-
>>1-11
 Block           11    ncf =            5  id =   10-
>>1-5
 level weights (1 equal;  5 standard;  9 user)
>>5
 Radial functions
 1s 2s 2p- 2p 3s 3p- 3p 3d- 3d 4s 4p- 4p 4d- 4d 4f- 4f 5s 5p- 5p 5d- 5d 6s 6p-
 6p 5f- 5f 6d- 6d 7s
 Enter orbitals to be varied (Updating order)
>>*
 Which of these are spectroscopic orbitals?
>>*
 Enter the maximum number of SCF cycles:
>>100
 Modify other defaults?  (y/n)
>>n
 Orthonomalization order?
      1 -- Update order
      2 -- Self consistency connected
>>1

    ........


 Wall time:
       98 seconds

 Finish Date and Time:
   Date (Yr/Mon/Day): 2018/11/26
   Time (Hr/Min/Sec): 23/43/02.595
   Zone: +0100

 RMCDHF: Execution complete.
```

Chapter 14

Managing large expansions

14.1 Rearrange lists of CSFs into zero- and first-order spaces

Sometimes the CSF expansions get so large that they cannot be handled by the normal SCF procedure. In these cases an approximate optimization scheme can be employed in which the CSF list is rearranged into zero- and a first-order spaces:

$$\underbrace{\Phi(\gamma_1^0 PJ), \Phi(\gamma_2^0 PJ), \ldots, \Phi(\gamma_M^0 PJ)}_{\text{zero-order space}}, \underbrace{\Phi(\gamma_1^1 PJ), \Phi(\gamma_2^1 PJ), \ldots, \Phi(\gamma_N^1 PJ)}_{\text{first-order space}}$$

where $M + N$ is the total number of CSFs in the original list. The zero-order space contains the most important CSFs while the first-order space contain less important CSFs that can be regarded as minor corrections. Normally $M \ll N$. Associated with the rearrangement of the CSFs is a decomposition of the Hamiltonian interaction matrix in submatrices

$$\begin{pmatrix} H^{00} & H^{01} \\ H^{10} & H^{11} \end{pmatrix}.$$

The energy expression, on which to optimize, is now obtained from the limited interaction matrix where the full H^{00}, H^{01}, H^{10} submatrices are included (interactions within the zero-order space and between the zero- and first-order spaces) but only the diagonal part of H^{11}. The rearrangement of the list of CSFs in zero- and first-order spaces is done by the program `rcsfzerofirst`.

As an example we use zero- and first-order spaces for a calculation of the states belonging to the $3s^2 3p^2$ configuration in Si-like iron. The calculation accounts for valence-valence and core-valence correlation and is based on a multireference of the form $\{3s^2 3p^2, 3s3p^2 3d, 3p^4\}$.

Overview

1. Define nuclear data.
2. Obtain common orbitals for the $\{3s^2 3p^2, 3s3p^2 3d, 3p^4\}$ MR set from DHF
   (a) Generate list of CSFs for MR
   (b) Perform angular integration.
   (c) Generate initial estimates of radial orbitals.
   (d) Perform self-consistent field calculation on the weighted average of the $3s^2 3p^2$ states.
   (e) Save output to `mr`.
3. Improve the states using the zero- and first-order method where only part of the interactions are retained

(a) Generate $n = 4$ valence-valence and core-valence CSF expansion from the MR

(b) Rearrange CSFs in zero- and first-order spaces using `rcsfzerofirst`

(c) Perform angular integration.

(d) Generate initial estimates of radial orbitals.

(e) Perform self-consistent field calculation on the weighted average of the $3s^2 3p^2$ states.

(f) Save output to `zerofirst_n4`

(g) Perform RCI calculations in which the transverse photon interaction (Breit) and vacuum polarization and self-energy (QED) corrections are added.

```
*****************************************************************************
* RUN RNUCLEUS TO GENERATE NUCLEAR DATA AND DEFINE RADIAL GRID              *
* OUTPUT FILE: isodata                                                      *
*****************************************************************************

>>rnucleus

 RNUCLEUS
 This program defines nuclear data and the radial grid
 Outputfile: isodata

 Enter the atomic number:
>>26
 Enter the mass number (0 if the nucleus is to be modelled as a point source:
>>56
 The default root mean squared radius is    3.7684209375954341       fm;
 The default nuclear skin thickness is      2.2999999999999998       fm;
 Revise these values?
>>n
 Enter the mass of the neutral atom (in amu) (0 if the nucleus is to be static):
>>56
 Enter the nuclear spin quantum number (I) (in units of h / 2 pi):
>>1
 Enter the nuclear dipole moment (in nuclear magnetons):
>>1
 Enter the nuclear quadrupole moment (in barns):
>>1

*****************************************************************************
*        RUN RCSFGENERATE TO GENERATE LIST OF CSFs                          *
*        FOR THE MULTIREFERNCE                                              *
*        OUTPUT FILES: rcsfgenerate.log, rcsf.out                           *
*****************************************************************************

>>rcsfgenerate

 RCSFGENERATE
 This program creates a list of CSFs
 Configurations should be entered in spectroscopic notation
 with occupation numbers and indications if orbitals are
 closed (c), inactive (i), active (*) or has a minimal
 occupation e.g. 1s(2,1)2s(2,*)
 Outputfiles: rcsf.out, rcsfgenerate.log
```

```
 Default, reverse, symmetry or user specified ordering? (*/r/s/u)
>>*

 Select core
        0: No core
        1: He (       1s(2)                  =  2 electrons)
        2: Ne ([He] + 2s(2)2p(6)             = 10 electrons)
        3: Ar ([Ne] + 3s(2)3p(6)             = 18 electrons)
        4: Kr ([Ar] + 3d(10)4s(2)4p(6)       = 36 electrons)
        5: Xe ([Kr] + 4d(10)5s(2)5p(6)       = 54 electrons)
        6: Rn ([Xe] + 4f(14)5d(10)6s(2)6p(6) = 86 electrons)
>>1
 Enter list of (maximum 100) configurations. End list with a blank line or an asterisk (*)

 Give configuration           1
>>2s(2,i)2p(6,i)3s(2,i)3p(2,i)
 Give configuration           2
>>2s(2,i)2p(6,i)3s(1,i)3p(2,i)3d(1,i)
 Give configuration           3
>>2s(2,i)2p(6,i)3p(4,i)
 Give configuration           4
>>
 Give set of active orbitals, as defined by the highest principal quantum number
 per l-symmetry, in a comma delimited list in s,p,d etc order, e.g. 5s,4p,3d
>>3s,3p,3d
 Resulting 2*J-number? lower, higher (J=1 -> 2*J=2 etc.)
>>0,4
 Number of excitations (if negative number e.g. -2, correlation
 orbitals will always be doubly occupied)
>>0
 Generate more lists ? (y/n)
>>n

        .........

       block  J/P            NCSF
           1    0+              9
           2    1+             15
           3    2+             20


******************************************************************************
*         COPY FILES                                                         *
*         IT IS ADVISABLE TO SAVE THE rcsfgenerate.log FILE TO HAVE A      *
*         RECORD ON HOW THE LIST OF CSFs WAS CREATED                         *
******************************************************************************

>>cp rcsfgenerate.log mr.exc
>>cp rcsf.out rcsf.inp


******************************************************************************
*         RUN RANGULAR TO GENERATE ENERGY EXPRESSION                         *
```

```
*         INPUT FILE  : rcsf.inp                                             *
*         OUTPUT FILES: rangular.alog, mcp.30, mcp.31,....                   *
******************************************************************************

>>rangular

 RANGULAR
 This program performs angular integration
 Input file:  rcsf.inp
 Outputfiles: mcp.30, mcp.31, ....
              rangular.log

 Full interaction?  (y/n)
>>y

  .....

 RANGULAR: Execution complete.

******************************************************************************
*         RUN RWFNESTIMATE TO GENERATE INITIAL ESTIMATES FOR RADIAL ORBITALS  *
*         INPUT FILES: isodata, rcsf.inp, previous rwfn files                *
*         OUTPUT FILE: rwfn.inp, rwfnestimate.log                            *
******************************************************************************

>>rwfnestimate

 RWFNESTIMATE
 This program estimates radial wave functions
 for orbitals
 Input files: isodata, rcsf.inp, optional rwfn file
 Output file: rwfn.inp

 Default settings ?
>>y
 Loading CSF file ... Header only
 There are/is            9  relativistic subshells;
 The following subshell radial wavefunctions remain to be estimated:
 1s 2s 2p- 2p 3s 3p- 3p 3d- 3d

 Read subshell radial wavefunctions. Choose one below
     1 -- GRASP2K File
     2 -- Thomas-Fermi
     3 -- Screened Hydrogenic
>>2
 Enter the list of relativistic subshells:
>>*
 All required subshell radial wavefunctions  have been estimated:
Shell      e           p0        gamma        P(2)       Q(2)      MTP  SRC

  1s   0.3024D+03  0.2943D+03  0.1000D+01  0.1161D-05 -0.7526D-11  329  T-F
  2s   0.5961D+02  0.1000D+03  0.1000D+01  0.3945D-06 -0.2559D-11  346  T-F
  2p-  0.5782D+02  0.7555D+00  0.1000D+01  0.2207D-12  0.3141D-07  346  T-F
  2p   0.5718D+02  0.6771D+03  0.2000D+01  0.1053D-13 -0.6832D-19  346  T-F
```

```
  3s   0.2068D+02  0.4969D+02  0.1000D+01  0.1960D-06 -0.1271D-11  359  T-F
  3p-  0.1992D+02  0.4010D+00  0.1000D+01  0.1172D-12  0.1667D-07  359  T-F
  3p   0.1976D+02  0.3606D+03  0.2000D+01  0.5610D-14 -0.3640D-19  359  T-F
  3d-  0.1847D+02  0.5063D+00  0.2000D+01  0.1167D-20  0.1660D-15  359  T-F
  3d   0.1842D+02  0.5694D+03  0.3000D+01  0.3493D-22 -0.2266D-27  359  T-F

 RWFNESTIMATE: Execution complete.

*******************************************************************************
*         RUN RMCDHF TO OBTAIN SELF CONSISTENT SOLUTIONS                      *
*         INPUT FILES: isodata, rcsf.inp, rwfn.inp, mcp.30, mcp.31,...        *
*         OUTPUT FILES: rwfn.out, rmix.out, rmcdhf.sum, rmcdhf.log            *
*                                                                             *
*         NOTE: ORBITALS BUILDING REFERENCE STATES ARE REQUIRED TO HAVE       *
*         THE CORRECT NUMBER OF NODES. THEY ARE REFERRED TO AS SPECTROSCOPIC  *
*         ORBITALS. IN THIS RUN WE VARY 1s, 2s, 2p, 3s, 3p, 3d AND THEY ARE   *
*         ALL SPECTROSCOPIC. WE CAN USE WILD CARDS FOR SPECIFYING ORBITALS    *
*                                                                             *
*         NOTE WE HAVE ASKED FOR 900 ITERATIONS                               *
*******************************************************************************

>>rmcdhf

 RMCDHF
 This program determines the radial orbitals
 and the expansion coefficients of the CSFs
 in a self-onsistent field proceedure
 Input file:  isodata, rcsf.inp, rwfn.inp, mcp.30, ...
 Outputfiles: rwfn.out, rmix.out, rmcdhf.sum, rmcdhf.log

 Default settings?  (y/n)
>>y
 Loading CSF file ... Header only
 There are/is            9  relativistic subshells;
 Loading CSF File for ALL blocks
 There are           44  relativistic CSFs... load complete;
 Loading Radial WaveFunction File ...
 There are            3  blocks  (block   J/Parity   NCF):
  1    0+     9       2    1+    15       3    2+    20

 Enter ASF serial numbers for each block
 Block           1    ncf =           9  id =    0+
>>1,2
 Block           2    ncf =          15  id =    1+
>>1
 Block           3    ncf =          20  id =    2+
>>1,2
 level weights (1 equal;  5 standard;  9 user)
>>5
 Radial functions
 1s 2s 2p- 2p 3s 3p- 3p 3d- 3d
 Enter orbitals to be varied (Updating order)
>>*
 Which of these are spectroscopic orbitals?
```

```
>>*
 Enter the maximum number of SCF cycles:
>>900

............

 RMCDHF: Execution complete.

***************************************************************************
*         RUN RSAVE TO SAVE OUTPUT FILES: name.c, name.w, name.m, name.sum    *
*                                         name.alog, name.log                 *
***************************************************************************

>>rsave mr
 Created mr.w, mr.c, mr.m, mr.sum, mr.alog and mr.log

***************************************************************************
*         RUN RCSFGENERATE TO GENERATE LIST OF CSFs                           *
*         ACCOUNTING FOR VALENCE-VALENCE AND CORE-VALENCE CORRELATION         *
*         OUTPUT FILES: rcsfgenerate.log, rcsf.out                            *
***************************************************************************

>>rcsfgenerate

 RCSFGENERATE
 This program creates a list of CSFs
 Configurations should be entered in spectroscopic notation
 with occupation numbers and indications if orbitals are
 closed (c), inactive (i), active (*) or has a minimal
 occupation e.g. 1s(2,1)2s(2,*)
 Outputfiles: rcsf.out, rcsfgenerate.log

 Default, reverse, symmetry or user specified ordering? (*/r/s/u)
>>*

 Select core
       0: No core
       1: He (       1s(2)                  =  2 electrons)
       2: Ne ([He] + 2s(2)2p(6)             = 10 electrons)
       3: Ar ([Ne] + 3s(2)3p(6)             = 18 electrons)
       4: Kr ([Ar] + 3d(10)4s(2)4p(6)       = 36 electrons)
       5: Xe ([Kr] + 4d(10)5s(2)5p(6)       = 54 electrons)
       6: Rn ([Xe] + 4f(14)5d(10)6s(2)6p(6) = 86 electrons)
>>1
 Enter list of (maximum 100) configurations. End list with a blank line or an asterisk (*)

 Give configuration           1
>>2s(2,i)2p(6,5)3s(2,*)3p(2,*)
 Give configuration           2
>>2s(2,i)2p(6,5)3s(1,*)3p(2,*)3d(1,*)
 Give configuration           3
>>2s(2,i)2p(6,5)3p(4,*)
 Give configuration           4
>>
```

```
 Give set of active orbitals, as defined by the highest principal quantum number
 per l-symmetry, in a comma delimited list in s,p,d etc order, e.g. 5s,4p,3d
>>4s,4p,4d,4f
 Resulting 2*J-number? lower, higher (J=1 -> 2*J=2 etc.)
>>0,4
 Number of excitations (if negative number e.g. -2, correlation
 orbitals will always be doubly occupied)
>>2
 Generate more lists ? (y/n)
>>n

        .........

  3 blocks were created

      block  J/P            NCSF
          1    0+            4720
          2    1+           12774
          3    2+           17554


******************************************************************************
*         COPY FILES                                                         *
******************************************************************************

>>cp rcsf.out rcsf.inp


******************************************************************************
*         RUN RCSFZEROFIRST TO ARRANGE LIST                                  *
******************************************************************************

>>rcsfzerofirst


 RCSFzerofirst: Takes a list of CSFs and partitions each symmetry
                block into a zero- and first-order CSF space from
                a zero-order list.
                (C)   Copyright by G. Gaigalas and Ch. F. Fischer
                (Fortran 95 version)                NIST  (2017).
                Input files:     list with CSFs to be partitioned
                                 list with CSFs defining
                                             the zero-order space
                Output file:     rcsf.out

 Give the full name of the list that contains the zero-order space
mr.c
 Give the full name of the list that should be partitioned
rcsf.inp
 Loading Configuration Symmetry List File ...
 There are 16 relativistic subshells;
   Block    Zero-order Space   Complete Space
    1                  9               4720
    2                 15              12774
```

```
   3                  20                17554

 RCSFzerofirst: Execution complete.

*******************************************************************************
*         COPY FILES                                                          *
*******************************************************************************

>>cp rcsf.out rcsf.inp

*******************************************************************************
*         RUN RANGULAR TO GENERATE ENERGY EXPRESSION                          *
*         INPUT FILE  : rcsf.inp                                              *
*         OUTPUT FILES: rangular.alog, mcp.30, mcp.31,...                     *
*         NOTE EXECUTION VERY FAST SINCE WE DO NOT INCLUDE ALL INTERACTIONS   *
*******************************************************************************

>>rangular

 RANGULAR
 This program performs angular integration
 Input file:  rcsf.inp
 Outputfiles: mcp.30, mcp.31, ....
              rangular.log

 Full interaction?  (y/n)
>>n

 Block            1 ,  ncf =        4720
 Block            2 ,  ncf =       12774
 Block            3 ,  ncf =       17554
 Loading CSF file ... Header only
 There are/is           16  relativistic subshells;
 The contribution of CSFs 1 -- ICCUT will be treated variationally;
 the remainder perturbatively; enter ICCUT:
 Give ICCUT for block           1
>>9
 Give ICCUT for block           2
>>15
 Give ICCUT for block           3
>>20

  .....

 RANGULAR: Execution complete.

*******************************************************************************
*         RUN RWFNESTIMATE TO GENERATE INITIAL ESTIMATES FOR RADIAL ORBITALS  *
*         INPUT FILES: isodata, rcsf.inp, previous rwfn files                 *
*         OUTPUT FILE: rwfn.inp, rwfnestimate.log                             *
*******************************************************************************

>>rwfnestimate
```

```
 RWFNESTIMATE
 This program estimates radial wave functions
 for orbitals
 Input files: isodata, rcsf.inp, optional rwfn file
 Output file: rwfn.inp

 Default settings ?
>>y
 Loading CSF file ... Header only
 There are/is          16  relativistic subshells;
 The following subshell radial wavefunctions remain to be estimated:
 1s 2s 2p- 2p 3s 3p- 3p 3d- 3d 4s 4p- 4p 4d- 4d 4f- 4f

 Read subshell radial wavefunctions. Choose one below
     1 -- GRASP2K File
     2 -- Thomas-Fermi
     3 -- Screened Hydrogenic
>>1
 Enter the file name (Null then "rwfn.out")
>>
 Enter the list of relativistic subshells:
>>*
 The following subshell radial wavefunctions remain to be estimated:
 4s 4p- 4p 4d- 4d 4f- 4f

 Read subshell radial wavefunctions. Choose one below
     1 -- GRASP2K File
     2 -- Thomas-Fermi
     3 -- Screened Hydrogenic
>>2
 Enter the list of relativistic subshells:
>>*
 All required subshell radial wavefunctions  have been estimated:
Shell      e           p0        gamma        P(2)       Q(2)      MTP  SRC

  1s   0.2768D+03  0.2922D+03  0.1000D+01  0.1152D-05 -0.7470D-11  358  rwf
  2s   0.4499D+02  0.9140D+02  0.1000D+01  0.3605D-06 -0.2338D-11  361  rwf
  2p-  0.4040D+02  0.6301D+00  0.1000D+01  0.1840D-12  0.2620D-07  359  rwf
  2p   0.3993D+02  0.5641D+03  0.2000D+01  0.8774D-14 -0.5691D-19  359  rwf
  3s   0.1448D+02  0.3728D+02  0.1000D+01  0.1470D-06 -0.9536D-12  364  rwf
  3p-  0.1327D+02  0.2856D+00  0.1000D+01  0.8343D-13  0.1187D-07  364  rwf
  3p   0.1318D+02  0.2569D+03  0.2000D+01  0.3996D-14 -0.2592D-19  364  rwf
  3d-  0.1475D+02  0.1697D+00  0.2000D+01  0.3910D-21  0.5565D-16  364  rwf
  3d   0.1477D+02  0.1889D+03  0.3000D+01  0.1159D-22 -0.7515D-28  364  rwf
  4s   0.9572D+01  0.2915D+02  0.1000D+01  0.1150D-06 -0.7458D-12  368  T-F
  4p-  0.9220D+01  0.2388D+00  0.1000D+01  0.6977D-13  0.9927D-08  368  T-F
  4p   0.9167D+01  0.2151D+03  0.2000D+01  0.3346D-14 -0.2171D-19  368  T-F
  4d-  0.8577D+01  0.3386D+00  0.2000D+01  0.7804D-21  0.1110D-15  369  T-F
  4d   0.8563D+01  0.3811D+03  0.3000D+01  0.2338D-22 -0.1517D-27  369  T-F
  4f-  0.7863D+01  0.1681D+00  0.3000D+01  0.2292D-29  0.3261D-24  369  T-F
  4f   0.7857D+01  0.2052D+03  0.4000D+01  0.4965D-31 -0.3221D-36  369  T-F

 RWFNESTIMATE: Execution complete.
```

```
***************************************************************************
*        RUN RMCDHF TO OBTAIN SELF CONSISTENT SOLUTIONS                   *
*        INPUT FILES: isodata, rcsf.inp, rwfn.inp, mcp.30, mcp.31,...     *
*        OUTPUT FILES: rwfn.out, rmix.out, rmcdhf.sum, rmcdhf.log         *
*                                                                         *
*        NOTE: ORBITALS BUILDING REFERENCE STATES ARE REQUIRED TO HAVE    *
*        THE CORRECT NUMBER OF NODES. THEY ARE REFERRED TO AS SPECTROSCOPIC  *
*        ORBITALS. IN THIS RUN WE VARY 4s,4p,4d,4f AND THEY ARE ALL       *
*        CORRELATION ORBITALS WITH NO NODE COUNTING                       *
***************************************************************************

>>rmcdhf

 RMCDHF
 This program determines the radial orbitals
 and the expansion coefficients of the CSFs
 in a self-onsistent field proceedure
 Input file:  isodata, rcsf.inp, rwfn.inp, mcp.30, ...
 Outputfiles: rwfn.out, rmix.out, rmcdhf.sum, rmcdhf.log

 Default settings?  (y/n)
>>y
 Loading CSF file ... Header only
 There are/is           16  relativistic subshells;
 Loading CSF File for ALL blocks
 There are        35048  relativistic CSFs... load complete;
 Loading Radial WaveFunction File ...
 There are            3  blocks  (block   J/Parity   NCF):
  1    0+  4720       2    1+ 12774       3    2+ 17554

 Enter ASF serial numbers for each block
 Block           1    ncf =        4720  id =    0+
>>1,2
 Block           2    ncf =       12774  id =    1+
>>1
 Block           3    ncf =       17554  id =    2+
>>1,2
 level weights (1 equal;  5 standard;  9 user)
>>5
 Radial functions
 1s 2s 2p- 2p 3s 3p- 3p 3d- 3d 4s 4p- 4p 4d- 4d 4f- 4f
 Enter orbitals to be varied (Updating order)
>>4*
 Which of these are spectroscopic orbitals?
>>
 Enter the maximum number of SCF cycles:
>>100

.......


***************************************************************************
*        RUN RSAVE TO SAVE OUTPUT FILES: name.c, name.w, name.m, name.sum    *
*                                        name.alog, name.log              *
```

```
*****************************************************************************

>>rsave zerofirst_n4
 Created zerofirst_n4.w, zerofirst_n4.c, zerofirst_n4.m, zerofirst_n4.sum
         zerofirst_n4.alog and zerofirst_n4.log

*****************************************************************************
*         RUN RCI TO INCLUDE TRANSVERSE PHOTON INTERACTION AND QED EFFECTS    *
*         OUTPUT FILE: zerofirst_n4.cm, zerofirst_n4.csum, ...., rci.res      *
*                                                                             *
*         THE TRANSVERSE PHOTON FREQUENCIES CAN BE SET TO THE LOW FREQUENCY   *
*         LIMIT. RECOMMENDED IN CASES WHERE YOU HAVE CORRELATION ORBITALS     *
*         THE SELF ENERGY CORRECTION MAY FAIL FOR CORRELATION ORBITALS WITH   *
*         HIGH N.                                                             *
*         NOTE THAT THIS IS VERY FAST SINCE WE DO NOT INCLUDE ALL INTERACTIONS*
*****************************************************************************

>>rci

 RCI
 This is the configuration interaction program
 Input file:  isodata, name.c, name.w
 Outputfiles: name.cm, name.csum, name.clog, rci.res
              rci.res (can be used for restart)

 Default settings?
>>n
 Name of state:
>>zerofirst_n4
 Block           1 ,  ncf =        4720
 Block           2 ,  ncf =       12774
 Block           3 ,  ncf =       17554
 Loading CSF file ... Header only
 There are/is           16  relativistic subshells;
 Restarting RCI ?
>>n
 Revise the physical speed of light (   137.03599913900001       in a.u.) ?
>>n
 Treat contributions of some CSFs as first-order perturbations?
>>y
 There are            3 blocks. They are:
   block     J Parity     No of CSFs
           1    0+        4720
           2    1+       12774
           3    2+       17554

 Enter iccut for each block
 Block           1    ncf =        4720  id =    0+
>>9
 Block           2    ncf =       12774  id =    1+
>>15
 Block           3    ncf =       17554  id =    2+
>>20
 Include contribution of H (Transverse)?
```

```
>>y
 Modify all transverse photon frequencies?
>>n
 Include H (Vacuum Polarisation)?
>>y
 Include H (Normal Mass Shift)?
>>n
 Include H (Specific Mass Shift)?
>>n
 Estimate self-energy?
>>y
 Largest n quantum number for including self-energy for orbital
 n should be less or equal 8
>>3
 The default radial grid parameters for this case are:
  RNT =    1.9999999999999999E-006 ;
  H =    5.0000000000000003E-002 ;
  HP =    0.0000000000000000      ;
  N =          590 ;
  revise these values?
>>n
 Loading Radial WaveFunction File ...
 There are            3  blocks  (block   J/Parity   NCF):
  1    0+    4720       2    1+   12774       3    2+   17554

 Enter ASF serial numbers for each block
 Block           1    ncf =        4720  id =    0+
>>1,2
 Block           2    ncf =       12774  id =    1+
>>1
 Block           3    ncf =       17554  id =    2+
>>1,2

.........

 RCI: Execution complete.
```

Below we compare the energies from calculations with zero- and first-order spaces and limited interactions and calculations with full interaction in different combinations. In addition we show the energies from the `rci` run of the MR and the experimental energies from NIST.

Energies from the `rci` run with zero- and first-order spaces and wave functions from an `rmcdhf` calculation with zero- and first-order spaces:

```
----------------------------------------------------------------------
 No Pos  J Parity Energy Total    Levels     Splitting
                    (a.u.)      (cm^-1)     (cm^-1)
----------------------------------------------------------------------
  1  1   0  +   -1210.0269828
  2  1   1  +   -1209.9855297    9097.89     9097.89
  3  1   2  +   -1209.9431027   18409.56     9311.67
  4  2   2  +   -1209.8061665   48463.58    30054.02
  5  2   0  +   -1209.6072762   92114.95    43651.37
----------------------------------------------------------------------
```

Energies from the `rci` run with full interaction and wave functions from an `rmcdhf` calculation with zero- and first-order spaces:

```
---------------------------------------------------------------------------
 No Pos  J Parity Energy Total    Levels     Splitting
                     (a.u.)      (cm^-1)     (cm^-1)
---------------------------------------------------------------------------
  1  1   0  +   -1210.0256252
  2  1   1  +   -1209.9837007    9201.37     9201.37
  3  1   2  +   -1209.9411507   18540.02     9338.65
  4  2   2  +   -1209.8027091   48924.45    30384.43
  5  2   0  +   -1209.5999946   93415.14    44490.69
---------------------------------------------------------------------------
```

Energies from an `rci` run with the full interaction wave functions from an `rmcdhf` calculation with the full interaction:

```
---------------------------------------------------------------------------
 No Pos  J Parity Energy Total    Levels     Splitting
                     (a.u.)      (cm^-1)     (cm^-1)
---------------------------------------------------------------------------
  1  1   0  +   -1210.0256751
  2  1   1  +   -1209.9837486    9201.80     9201.80
  3  1   2  +   -1209.9411988   18540.41     9338.61
  4  2   2  +   -1209.8027721   48921.54    30381.14
  5  2   0  +   -1209.6000408   93415.93    44494.39
---------------------------------------------------------------------------
```

Energies from an `rci` run for the MR:

```
---------------------------------------------------------------------------
 No Pos  J Parity Energy Total    Levels     Splitting
                     (a.u.)      (cm^-1)     (cm^-1)
---------------------------------------------------------------------------
  1  1   0  +   -1209.9394925
  2  1   1  +   -1209.8995764    8760.57     8760.57
  3  1   2  +   -1209.8558763   18351.63     9591.06
  4  2   2  +   -1209.7026579   51979.19    33627.56
  5  2   0  +   -1209.4545733  106427.47    54448.28
---------------------------------------------------------------------------
```

Experimental energies from NIST:

```
--------------------------------------------------------------
Configuration         | Term    |   J |            Level     |
----------------------|---------|-----|----------------------|
                      |         |     |                      |
3s2.3p2               | 3P      |   0 |               0.0    |
                      |         |   1 |            9302.5    |
                      |         |   2 |           18561.0    |
                      |         |     |                      |
3s2.3p2               | 1D      |   2 |           48068      |
                      |         |     |                      |
3s2.3p2               | 1S      |   0 |           91508      |
--------------------------------------------------------------
```

In the figure below we compare the $4s, 4p, 4d, 4f$ correlation orbitals from `rmcdhf` calculations with limited and full interactions, respectively. The differences between the orbitals are very small.

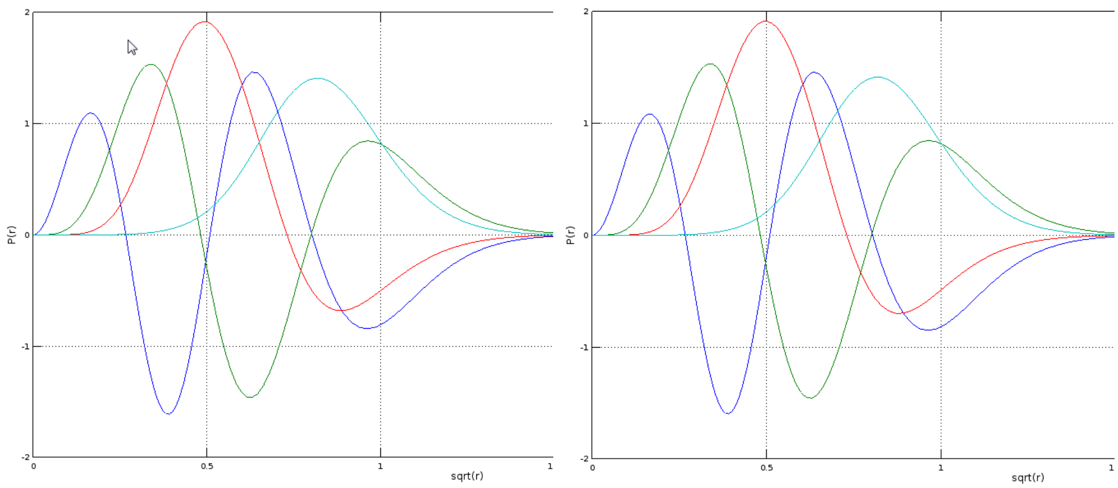

Figure 14.1: Plot of orbitals from an **rmcdhf** calculation using the full interaction matrix and an **rmcdhf** calculation with only part of the interaction.

The conclusion of all this, energy tables and shapes of radial orbitals, is that a limited interaction self-consistent calculation combined with full interaction RCI recovers almost perfectly the result of a full interaction self-consistent calculation combined with full interaction RCI.

14.2 Accumulating the wave function to a specified fraction

A very good way of selecting the zero-order space is to accumulate the wave function to a specified fraction of the squared weights. This is done by the following procedure:

1. Start from a calculation targeting one or more states, thus start from a number of ASFs

$$ASF_1 : \Psi(\gamma_1 PJ) = \sum_{i=1}^{N} c_i^1 \Phi(\gamma_i PJ)$$

$$\ldots$$

$$ASF_M : \Psi(\gamma_M PJ) = \sum_{i=1}^{N} c_i^M \Phi(\gamma_i PJ)$$

   built from a set of CSFs.

2. For i from 1 to N compute

$$s_i = (c_i^1)^2 + (c_i^2)^2 + \ldots + (c_i^M)^2.$$

3. Sort $s_1, s_2, \ldots, s_N$ in descending order.

4. Accumulate until a specified fraction of the total squared weight

$$M = s_1 + \ldots + s_M = \sum_{i,j} (c_i^j)^2$$

   is attained.

The CSFs that are associated with the accumulated fraction can then be taken as the zero order space. Alternatively, and dependent on the fraction, the method can be used to condense the list of CSFs.

Below are some different scenarios:

1. Perform some initial calculations. Use accumulation to a specified fraction to select the CSFs (configurations) in the MR. The selected CSFs can then also be used by `rcsfinteract` (see section 5.5).
2. Perform large scale calculations. To further push the calculations use accumulation to a specified fraction to select the zero order space.
3. Perform large scale calculations. Use accumulation to a specified fraction to condense the list of CSFs.

The accumulation to a specified fraction is done with the program `rmixaccumulate`

As an example we apply the accumulation to a specified fraction (0.9999 in this case) to the states defined in `zerofirst_n4`. We use the accumulated list as the zero-order space and redo the RCI calculation to see how big is the difference between the obtained energies and the energies from the full interaction calculation.

```
*******************************************************************************
*         RMIXACCUMULATE TO ACCUMULATE TO A SPECIFIED FRACTION                *
*         INPUT FILE: zerofirst_n4.c, zerofirst_n4.cm                         *
*         OUTPUT FILE: rcsf.out                                               *
*******************************************************************************


 WELCOME TO PROGRAM RMIXACCUMULATE
 Input files: <state>.(c)m, <state>.c
 Reduced CSF list is written to rcsf.out

 Give name of the state:
>>zerofirst_n4
 Expansion coefficients resulting from CI calculation (y/n)?
>>y
 Fraction of total wave function [0-1] to be included in reduced list:
>>0.9999
 CSFs in output file sorted by mixing coefficients (y/n)?
>>y

 Block data read from mixing file
        block        ncf         nev        2j+1          parity
          1         4720          2           1           1
          2        12774          1           3           1
          3        17554          2           5           1

 Number of CSF:s written to rcsf.out
        block        ncf
          1          188
          2          391
          3          712
```

For the different blocks we see that 188, 391, and 712 CSFs, respectively, contribute to 99.99 % of the total squared weight.

```
*****************************************************************************
*        COPY FILES                                                         *
*****************************************************************************

>>cp rcsf.out zero_order_0.9999.c

*****************************************************************************
*        RUN RCSFZEROFIRST TO ARRANGE LIST                                  *
*****************************************************************************

>>rcsfzerofirst

 RCSFzerofirst: Takes a list of CSFs and partitions each symmetry
               block into a zero- and first-order CSF space from
               a zero-order list.
               (C)   Copyright by G. Gaigalas and Ch. F. Fischer
               (Fortran 95 version)                NIST  (2017).
               Input files:     list with CSFs to be partitioned
                                list with CSFs defining
                                            the zero-order space
               Output file:     rcsf.out

 Give the full name of the list that contains the zero-order space
>>zero_order_0.9999.c
 Give the full name of the list that should be partitioned
>>rcsf.inp
 Loading Configuration Symmetry List File ...
 There are 16 relativistic subshells;
   Block    Zero-order Space   Complete Space
    1                188              4720
    2                391             12774
    3                712             17554

 Wall time:
       11 seconds

 Finish Date and Time:
   Date (Yr/Mon/Day): 2018/07/31
   Time (Hr/Min/Sec): 11/56/50.926
   Zone: +0200

 RCSFzerofirst: Execution complete.

 *****************************************************************************
*        COPY FILES                                                         *
*****************************************************************************

>>cp rcsf.out zerofirst_0.9999.c
>>cp zerofirst_n4.w zerofirst_0.9999.w

*****************************************************************************
*        RUN RCI TO INCLUDE TRANSVERSE PHOTON INTERACTION AND QED EFFECTS   *
*        OUTPUT FILE: zerofirst_0.9999.cm,  ...., rci.res                   *
*                                                                           *
```

```
*         THE TRANSVERSE PHOTON FREQUENCIES CAN BE SET TO THE LOW FREQUENCY   *
*         LIMIT. RECOMMENDED IN CASES WHERE YOU HAVE CORRELATION ORBITALS     *
*         THE SELF ENERGY CORRECTION MAY FAIL FOR CORRELATION ORBITALS WITH   *
*         HIGH N.                                                             *
*         NOTE THAT THIS IS VERY FAST SINCE WE DO NOT INCLUDE ALL INTERACTIONS*
*******************************************************************************

 RCI
 This is the configuration interaction program
 Input file:  isodata, name.c, name.w
 Outputfiles: name.cm, name.csum, name.clog
              rci.res (can be used for restart)

 Default settings?
>>n
 Name of state:
>>zerofirst_0.9999
 Block           1 ,  ncf =        4720
 Block           2 ,  ncf =       12774
 Block           3 ,  ncf =       17554
 Loading CSF file ... Header only
 There are/is           16  relativistic subshells;
 Restarting RCI90 ?
>>n
 Revise the physical speed of light (   137.03599913900001       in a.u.) ?
>>n
 Treat contributions of some CSFs as first-order perturbations?
>>y
 There are            3 blocks. They are:
   block     J Parity     No of CSFs
          1    0+       4720
          2    1+      12774
          3    2+      17554

 Enter iccut for each block
 Block           1    ncf =        4720  id =    0+
>>188
 Block           2    ncf =       12774  id =    1+
>>391
 Block           3    ncf =       17554  id =    2+
>>712
 Include contribution of H (Transverse)?
>>y
 Modify all transverse photon frequencies?
>>n
 Include H (Vacuum Polarisation)?
>>y
 Include H (Normal Mass Shift)?
>>n
 Include H (Specific Mass Shift)?
>>n
 Estimate self-energy?
>>y
 Largest n quantum number for including self-energy for orbital
```

```
 n should be less or equal 8
>>3
 Loading Radial WaveFunction File ...
 There are            3  blocks  (block   J/Parity   NCF):
  1    0+    4720       2    1+   12774       3    2+   17554

 Enter ASF serial numbers for each block
 Block           1    ncf =        4720  id =    0+
>>1,2
 Block           2    ncf =       12774  id =    1+
>>1
 Block           3    ncf =       17554  id =    2+
>>1,2

 ......

  Finish time, Statistics


 Wall time:
       95 seconds

 Finish Date and Time:
   Date (Yr/Mon/Day): 2018/07/31
   Time (Hr/Min/Sec): 12/03/43.197
   Zone: +0200

 RCI: Execution complete.
```

Below we display the energies from the `rci` run with the zero-order space from an accumulation to 0.9999.

```
 nblock =           3   ncftot =        35048   nw =           16   nelec =           14

 Energy levels for ...
 Rydberg constant is   109737.31569
 No - Serial number of the state; Pos - Position of the state within the
 J/P block; Splitting is the energy difference with the lower neighbor
------------------------------------------------------------------------
 No Pos  J Parity Energy Total    Levels     Splitting
                     (a.u.)      (cm^-1)     (cm^-1)
------------------------------------------------------------------------
  1  1   0  +   -1210.0256624
  2  1   1  +   -1209.9837313     9202.81     9202.81
  3  1   2  +   -1209.9411927    18538.95     9336.14
  4  2   2  +   -1209.8027664    48920.02    30381.07
  5  2   0  +   -1209.6000620    93408.50    44488.47
------------------------------------------------------------------------
```

We see that with a larger zero-order space we now have energies in very good agreement with the ones from an RCI calculation with full interaction. In this example we did two RCI calculations. The first was with a very small zero-order space in terms of the MR. We then used this calculation to accumulate to a defined fraction. By redoing the RCI with the new zero-order space we get energies that are very close to the ones from a full interaction calculation. For large expansions two calculations with limited interaction are much faster than one calculation with full interaction.

14.3 Computational strategies using zero- and first-order

Based on the experience from a number of studies we suggest the following computational strategy for large cases:

1. The MR is always generated using full interaction
2. To run `rmcdhf` for an expansion that is large:
   (a) Start by running `rmixaccumulate` with 0.99 or something similar on an expansion you have that is not too large, .e.g. an expansion based on just one or two orbital layers.
   (b) Generate your large expansion and run `rcsfinteract` to make sure you only retain CSFs that interact with the CSFs of the MR.
   (c) Run `rcsfzerofirst`
   zero-order – output from `rmixaccumulate`
   list to be partitioned – output from `rcsfgenerate` (step above)
   (d) Run `rangular` with ICCUT values for the size of the zero-order expansion from `rmixaccumulate`
   (e) Run `rmcdhf` in the usual way. Due to the fact that limited interaction is included in the angular integration the `rmcdhf` calculation will be fast.
3. Run `rci` for the large expansion with full interaction.
4. For very large expansions, consider performing the `rci` calculation with the expansion from the previous layer as a zero-order space or the expansion from the previous layer accumulated to a high fraction, say 0.99999999, as the zero-order space. Alternatively, run `rci` with a small zero-order space and accumulate to some fraction and use this list as a new zero-order space and redo the `rci` calculation.

Please remember that all strategies are dependent on the atomic system at hand, and that some explorations of the fractions used for `rmixaccumulate` are needed. See [22] for one application of the zero- and first-order strategy.

Bibliography

[1] C. Froese Fischer, G. Tachiev, G. Gaigalas, and M.R. Godefroid, Comput. Phys. Commun. 176, 559 (2007).

[2] C. Froese Fischer, M. Godefroid, T. Brage, P. Jönsson, and G. Gaigalas, Journal of Physics B 49, 182004 (2016).

[3] I.P. Grant, *Relativistic Quantum Theory of Atoms and Molecules: Theory and Computation*, Springer, New York, 2006, ISBN 0-7503-0374-3

[4] Grasp2018: C. Froese Fischer, G. Gaigalas, P. Jönsson, J. Bieroń, Comput. Phys. Commun. (in press 2018), `https://doi.org/10.1016/j.cpc.2018.10.032` (the user should update the reference when volume, issue and pages are available)

[5] Grasp2K new version: P. Jönsson, G. Gaigalas, J. Bieroń, C. Froese Fischer, I.P. Grant, Comput. Phys. Commun. 184, 2197 (2013).

[6] Grasp2K: P. Jönsson, X. He, C. Froese Fischer, and I.P. Grant, Comput. Phys. Commun. 176, 597 (2007).

[7] Grasp92: F.A. Parpia, C. Froese Fischer, I.P. Grant, Comput. Phys. Commun. 94, 249 (1996).

[8] Hfs92: P. Jönsson F.A. Parpia and C. Froese Fischer, Comput. Phys. Commun. 96, 301 (1996).

[9] Sms92: P. Jönsson and C. Froese Fischer, Comput. Phys. Commun. 94, 249 (1997).

[10] Jjgen: L. Sturesson, P. Jönsson, and C. Froese Fischer, Comput. Phys. Commun. 177, 539 (2007).

[11] Hfszeeman: M. Andersson and P. Jönsson, Comput. Phys. Commun. 178, 156 (2008).

[12] Ris4: J. Ekman, P. Jönsson, M. Godefroid, C. Nazé, G. Gaigalas, J. Bieroń, Comput. Phys. Commun. 235, 433 (2019).

[13] Ris3: C. Nazé, E. Gaidamauskas, G. Gaigalas, M. Godefroid, and P. Jönsson, Comput. Phys. Commun. 184, 2187 (2013).

[14] Dvdson: A. Stathopoulos, C. Froese Fischer, Comput. Phys. Commun. 79, 268 (1994).

[15] G. Gaigalas, Z.B. Rudzikas, and C. Froese Fischer, Journal of Physics B At. Mol. Phys. 30, 3747 (1997).

[16] G. Gaigalas, S. Fritzsche, and I.P. Grant, Comput. Phys. Commun. 139, 263 (2001).

[17] G. Gaigalas, S. Fritzsche, Z. Rudzikas, Atomic Data and Nuclear Data Tables 76, 235 (2000).

[18] G. Gaigalas, T. Žalandauskas, and Z. Rudzikas, At. Data and Nucl. Data Tables 84, 99 (2003).

[19] G. Gaigalas, T. Žalandauskas, and S. Fritzsche, Comput. Phys. Commun. 157, 239 (2004).

[20] G. Gaigalas, C. Froese Fischer, P. Rynkun, and P. Jönsson, Atoms, 5(1), 6 (2017); doi:10.3390/atoms5010006

[21] J. Olsen , M. Godefroid, P. Jönsson, P.Å. Malmqvist, and C. Froese Fischer, Phys. Rev. E 52, 4499 (1995).

[22] S. Gustafsson, P. Jönsson, C. Froese Fischer and I.P. Grant, Atoms, 5(1), 3 (2017); doi:10.3390/atoms5010003

[23] J. Grumer, T. Brage, M. Andersson, J. Li, P. Jönsson, W. Li, Y. Yang, R. Hutton, and Y. Zou, Physica Scripta 89 114002 (2014).

[24] C. Froese Fischer, T. Brage, P. Jönsson, Computational Atomic Structure - an MCHF approach, IoP, (1997).

[25] O. Zatsarinny and C. Froese Fischer, Comput. Phys. Commun. 202, 287 (2016).

[26] P. Jönsson, J. Ekman, S. Gustafsson, H. Hartman, L.B. Karlsson, R. du Rietz, G. Gaigalas, M.R. Godefroid, and C. Froese Fischer, A & A 559, A100 (2013).

[27] J. Ekman, P. Jönsson, S. Gustafsson, H. Hartman, G. Gaigalas, M.R. Godefroid, and C. Froese Fischer, A & A 564, A24 (2014).

[28] C. Froese Fischer, Atoms 2(1), 1-14 (2014). http://www.mdpi.com/2218-2004/2/1/1.

[29] S. Verdebout, P. Rynkun, P. Jönsson, G. Gaigalas, C. Froese Fischer, M. Godefroid, Journal of Physics B 46, 085003 (2013)

[30] J. Bieroń, L. Filippin, G. Gaigalas, M. Godefroid, P. Jönsson, and P. Pyykkö, Phys. Rev. A 97, 062505 (2018)

[31] C. Froese Fischer and G. Gaigalas, Phys. Rev. A **85**, 042501 (2012).

[32] C. Froese Fischer and G. Tachiev, At. Data and Nucl. Data Tables **87**, 1 (2004).

[33] J. Ekman, M. Godefroid, and H. Hartman, Atoms 2(2), 215-224 (2014), http://www.mdpi.com/2218-2004/2/2/215

[34] Myrnäs et. al. http://www.iop.org/Jet/fulltext/JETP93072.pdf, (1993).

[35] P. Rynkun, P. Jönsson, and G. Gaigalas, Atomic Data and Nuclear Data Tables **98** 481 (2012).

[36] M. Godefroid, C. Froese Fischer, and P. Jönsson, J. Phys. B: At. Mol. Opt. Phys. 34 1079 (2001).

[37] R.D. Cowan, The Theory of Atomic Structure and Spectra, University of California Press (1981).

Appendix A

Learn more about computational atomic structure

A.1 The Computational Atomic Structure Group

To meet the demands for atomic data the Computational Atomic Structure (CompAS) group has been formed. The group is involved in developing state of the art computer codes for atomic calculations in the non-relativistic scheme with relativistic corrections in the Breit-Pauli approximation ATSP2K, as well as in the fully relativistic scheme GRASP2018. The codes rely on multiconfiguration methods and the wave function for an atomic state is expanded in configuration state functions (CSFs).

In addition to the code development itself, the group includes members with expertise in the methods and is constantly developing computational techniques for the evaluation of atomic properties of the highest quality. To learn more about CompAS group and its activities go to
`http://ddwap.mah.se/tsjoek/compas/index.php`.

A.2 Suggested reading

In this section we suggest books and articles that provide the theoretical background to multiconfiguration methods, electron correlation, and the systematic computation of different atomic properties.

1. B. Swirles
   Relativistic self-consistent fields
   Proc. Roy. Soc. A **152**, 625, (1935).

   The first attempt to formulate DHF equations. This investigation was suggested in a conversation between D R Hartree and Bertha Swirles on a railway station while returning to Cambridge from a conference in Manchester.

2. B. Swirles
   Proc. Roy. Soc. A **157**, 680, (1936).

3. I. P. Grant
   Relativistic self-consistent fields.
   Proc. Roy. Soc. A **262** 555-576 (1961).

   Formulated DHF equations with the use of Racah methods to exploit the internal symmetry of Dirac central field orbitals. The mathematical tools were not available to Swirles in 1935.

4. I. P. Grant
   Relativistic self-consistent fields.
   Proc. Phys. Soc. (Lond.) **86**, 523-527 (1965)

   Sum rules for Breit interaction of a single electron with a closed subshell.

5. I. P. Grant and V. M. Burke
   The effect of relativity on atomic wavefunctions
   Proc. Phys. Soc. (Lond.) **90**, 297-314 (1967)

   First published comparison of Schrödinger and Dirac charge densities for Hg^{79+}

6. C. Froese
   Hartree-Fock Procedure for Some $nsn's\ ^1S$ Configurations
   Phys. Rev. 150.1 (1966): 1-6: doi: /10.1103/PhysRev.150.

   The $1s2s\ ^1S$ state was of considerable theoretical interest in the 1960's. This paper explores the trade-offs between orthogonal and non-orthogonal orbitals.

7. I. P. Grant
   Relativistic calculation of atomic structures
   Adv. Phys. **19**, 747-811 (1970).

   Review of progress in relativistic atomic structure calculations to 1970. This survey revealed how quickly the reformulation of DHF had been adopted after 1961 and suggested the desirability of developing MCDHF.

8. I. P. Grant
   Gauge invariance and relativistic radiative transitions
   J. Phys. B **7**, 1458-1475 (1974).

   This paper began as an attempt to reconcile two alternative relativistic expressions for electric multipole transitions, equivalent to the length and velocity forms of nonrelativistic theory. These give the same numerical result for one-electron transitions in a local potential but can be wildly different when using HF-type orbitals. The electric and longitudinal radiative transition multipole operators have the same selection rules so that the transition matrix element is an arbitrary linear combination of the two. Coulomb and Babushkin expressions correspond to different multiples of the longitudinal operator whose contribution is only null if there is local charge conservation. The difference of Coulomb and Babushkin results is now used to assess the accuracy of the wavefunctions describing initial and final states.

   Papers 9-11 formulate the reduction of Breit and transverse interaction to radial integrals used in the original GRASP

9. I. P. Grant and N. C. Pyper
   Breit interaction in multi-configuration relativistic atomic structure
   J. Phys. B **9**, 761-774 (1976).

10. I. P. Grant and B. J. McKenzie
    The transverse electron-electron interaction in atomic structure calculations
    J. Phys. B **13**, 2671-2681 (1980).

11. B. J. McKenzie, I. P. Grant and P. H. Norrington
    A program to calculate transverse Breit and QED corrections to energy levels in a MCDF environment
    Comput. Phys. Commun. **21**, 233-246 (1980); *ibid* **23**, 222 (1980).

    Papers 12-22 explore the use of MCDHF+B calculations in different contexts. The outputs were limited by the computing resources then available.

12. I. P. Grant, D. F. Mayers and N. C. Pyper
Studies in multi-configuration Dirac-Fock theory. I: The low-lying spectrum of Hf III
J. Phys. B **9**, 2777-2796 (1976).

13. S. J. Rose, N. C. Pyper and I. P. Grant
Studies in multi-configuration Dirac-Fock theory. II: The even-parity low-lying spectrum of Ba I
J. Phys. B **11**, 755-768 (1978).

14. N. C. Pyper and I. P. Grant
Studies in multi-configuration Dirac-Fock theory. III: Interpretation of the electronic structure of the neutral and ionized states of uranium
J. C. S. Faraday II **74**, 1885-1900 (1978).

15. S. J. Rose, N. C. Pyper and I. P. Grant
Studies in multi-configuration Dirac-Fock theory. IV: The low-lying spectrum of Bi I.
J. Phys. B **11**, 3499-3512 (1978).

16. S. J. Rose, N. C. Pyper and I. P. Grant
the direct and indirect effects in the relativistic modification of atomic valence orbitals
J. Phys. B **11**, 1171-1176 (1978).

17. S. J. Rose, I. P. Grant and J.-P. Connerade
Fully relativistic analysis of $5p$-excitation in atomic barium
Phil. Trans. Roy. Soc. (Lond.) A**296**, 527-544 (1980).

18. J.-P. Connerade, S. J. Rose and I. P. Grant
Two-step autoionization and the double ionization anomaly on Ba I
J. Phys. B **12**, L53-L55 (1979).

19. N. C. Pyper, S. J. Rose and I. P. Grant
Analysis of fine structure excitation energies in Dirac-Fock and perturbation theories
J. Phys. B **14**, 1319-1331 (1982).

20. I. P. Grant and N. C. Pyper
Theoretical chemistry of superheavy elements E116 and E114
Nature **265**, 715-717 (1977).

21. N. C. Pyper and I. P. Grant
On the interpretation of Hund's rules in atomic spectra
J. Phys. B **10**, 1803-1814 (1977).

22. N. Beatham, I. P. Grant, B. J. McKenzie and S. J. Rose
Spectroscopic studies with a MCDF program
Physica Scripta **21**, 423-431 (1980).

23. I. P. Grant
Many electron effects in the theory of nuclear volume isotope shift
Physica Scripta **21**, 443-447 (1980).

24. Relativistic effects in atoms, molecules and solids (ed. G. L. Malli) 1983 (Plenum: NATO ASI Series B, Vol. 87)

contain IPG's lectures:
Incidence of relativistic effects in atoms pp. 55-72
Formulation of the relativistic N-electron problem. pp. 73-88
Techniques for open-shell calculations for atoms. pp. 89-100
Self-consistency and numerical problems. pp. 101-114

25. K. G. Dyall and I. P. Grant
Phase conventions, quasi-spin and the $jj - LS$ transformation coefficients
J. Phys. B **15**, L371-L373 (1982).

26. J. R. Lemen, K. J. H. Phillips, R. D. Cowan, J. Hata and I. P. Grant
Inner shell transitions of Fe XXIII and Fe XXIV in the X-ray spectra of solar flares
Astron. & Astrophys. **135**, 313-324 (1984).

Program package publications: 27 was the first time that previously published programs were assembled in package; 28 upgraded it and was superseded by GRASP92.

27. I. P. Grant, B. J. McKenzie, P. H. Norrington, D. F. Mayers and N. C. Pyper
An atomic MCDF package.
Comput. Phys. Commun. **21**, 207-232 (1980)

28. K. G. Dyall, I. P. Grant, C. T. Johnson, F. A. Parpia and E. P. Plummer
GRASP – a general-purpose relativistic atomic structure package
Comput. Phys. Commun. **55**, 425-456 (1989).

29. M. Tong, P. Jönsson, and C. Froese Fischer
Convergence studies of atomic properties from variational methods: total energy, ionization energy, specific mass shift, and hyperfine parameters for Li
Physica Scripta 48.4 (1993): 446; doi:10.1088/0031-8949/48/4/009.

Discusses convergence of SDT excitations for a 3-electron system by partial waves, each expanded by n. Extrapolation procedures are discussed.

30. C. Froese Fischer
Convergence studies of MCHF calculations for Be and Li-.
Journal of Physics B: 26.5 (1993): 855-862; doi:10.1088/0953-4075/26/5/009.

Studies SDTQ excitations of 4-electron systems motivating what is now the MR-SD method. Introduces systematic methods, n-expansions, and extrapolations.

31. T. Brage and C. Froese Fischer.
Systematic calculations of correlation in complex ions.
Physica Scripta 1993.T47 (1993): 18-28; doi:10.1088/0031-8949/1993/T47/002.

Introduces a systematic approach with independent theoretical tests of atomic properties along with comparison with experiment.

32. T. Brage, C. Froese Fischer, and Per Jönsson.
Effects of core-valence and core-core correlation on the line strength of the resonance lines in Li I and Na I.
Phys. Rev. A 49.3 (1994): 2181-2184; doi:10.1103/PhysRevA.49.2181 .

Shows the importance of core-core correlation for accurate transition rates which, in time, agreed with experiment when experiment improved.

33. Z. Cai, V. M. Umar, and C. Froese Fischer. Large-scale relativistic correlation calculations: Levels of Pr^{+3}. Physical Review Letters 68.3 (1992): 297; doi:10.1103/PhysRevLett.68.297

Describes correlation studies for $4f^2$ in the lanthanides using GRASP92.

34. A. Ynnerman and C. Froese Fischer
Multiconfigurational-Dirac-Fock calculation of the $2s^2\ {}^1S_0 - 2s2p\ {}^3P_1$ spin-forbidden transition for the Be-like isoelectronic sequence.
Phys. Rev. A 51.3 (1995): 2020-2030; doi:10.1103/PhysRevA.51.2020.

Introduced optimization by layers with GRASP92 using active set expansions. Discussed length/velocity discrepancy in the transition rate and the numerical cancellation in the calculation of the transition rate.

35. J. Olsen, M. Godefroid, P. Jönsson, P.Å. Malmqvist and C. Froese Fischer
Transition probability calculations for atoms using nonorthogonal orbitals.
Physical Review E 52, 4499 (1995).

Describes the transformation method that makes it possible to compute transition rates between separately optimized initial and final states.

36. M. R. Godefroid, P. Jönsson and C. Froese Fischer
Atomic Structure Variational Calculations in Spectroscopy
Physica Scripta. Vol. T78, 33–46, 1998.

The paper gives examples on how computational atomic structure can be used in atomic spectroscopy for testing theoretical models or experimental results, predicting properties or interpreting them in terms of electron correlation. The effects inherent in the multiconfiguration Hartree-Fock method due to its variational nature are emphasized through some simple analysis of the wave function spatial distribution in correlation with the model used.

37. T. Brage, D. S. Leckrone, and C. Froese Fischer.
Core-valence and core-core correlation effects on hyperfine-structure parameters and oscillator strengths in Tl II and Tl III.
Phys. Rev. A 53.1 (1996): 192-200; doi:10.1103/PhysRevA.53.192.

Applies systematic procedures to the study of hyperfine-structure parameters, including zero-and first-order CSFs using GRASP92.

38. C. Froese Fischer.
Correlation and relativistic effects on transitions in lighter atoms.
Physica Scripta 1999.T83 (1999): 49; doi:10.1238/Physica.Topical.083a00049

Discusses Breit-Pauli and MCDHF results for the calculation of transition data with regard to spectrum calculations.

39. Yu Zou, and C. Froese Fischer.
Resonance transition energies and oscillator strengths in lutetium and lawrencium.
Physical Review Letters 88.18 (2002): 183001; doi: 10.1103/PhysRevLett.88.183001.

Describes correlation in a complex system with two unfilled shells. Though Lu and Lr are homologous systems, the configuration of the ground state changes.

40. J. Bieroń, C. Froese Fischer, P. Indelicato, P. Jönsson, and P. Pyykkö
Complete Active Space multiconfiguration Dirac-Hartree-Fock calculations of hyperfine structure constants of the gold atom
Phys. Rev. A **79**, 052502 (2009)
arXiv:physics/0902.4307

Describes calculations where correlation effects deep down in the core are accounted for.

41. C. Froese Fischer.
Relativistic Variational Calculations for Complex Atoms.
Advances in the Theory of Atomic and Molecular Systems Springer Netherlands, 2009. 115-128; doi:10.1007/978-90-481-2596-8-7

Reviews relativistic calculations for complex heavy atoms.

42. G. Gaigalas, E. Gaidamauskas, Z. R. Rudzikas, N. Magnani, R. Caciuffo
Theoretical studies of spectroscopic properties of Cm^{4+} and Am^{3+}
Physical Review A, Atomic, molecular, and optical physics. ISSN 1050-2947. 2009, Vol. 79, iss. 2, p. 022511-1-8.

43. G. Gaigalas, E. Gaidamauskas, Z.R. Rudzikas, N. Magnani, R. Cacciuffo
Correlation, relativistic, and quantum electrodynamics effects on the atomic structure of eka-thorium
Physical Review A, Atomic, molecular, and optical physics. ISSN 1050-2947. 2010, Vol. 81, Issue 2, p. 022508.

44. G. Gaigalas, Z. Rudzikas, E. Gaidamauskas, P. Rynkun, A. Alkauskas
Peculiarities of spectroscopic properties of W^{24+}
Physical Review A, Atomic, molecular, and optical physics. ISSN 1050-2947. 2010, Vol. 82, issue 1, p. 014502-4.

45. C. Froese Fischer and G. Gaigalas
Multiconfiguration Dirac-Hartree-Fock energy levels and transition probabilities for W XXXVIII
Physical Review A, Atomic, molecular, and optical physics. ISSN 1050-2947. 2012, Vol. 85, p. 042501.

46. S. Verdebout, P. Rynkun, P. Jönsson, G. Gaigalas, C. Froese Fischer, M. Godefroid
A Partitioned Correlation Function Interaction approach for describing electron correlation in atoms
Journal of Physics B 46, 085003 (2013).

This paper includes a discussion about electron correlation effects and the limitations of current methodologies for describing these effects. The paper gives an outline of how improved methods can be implemented based on a biorthonormal orbital transformation.

47. P. Jönsson, P. Bengtsson, J. Ekman, S. Gustafsson, L.B. Karlsson, G. Gaigalas, C. Froese Fischer, D. Kato, I. Murakami, H.A. Sakaue, H. Hara, T. Watanabe, N. Nakamura, and N. Yamamoto
Relativistic CI calculations of spectroscopic data for the $2p^6$ and $2p^5 3l$ configurations in Ne-like ions between Mg III and Kr XXVII.
Atomic Data and Nuclear Data Tables 100, 1 (2014).

Discusses accurate spectrum calculations giving energies and transition rates for hundreds of levels.

48. J. Ekman, P. Jönsson, S. Gustafsson, H. Hartman, G. Gaigalas, M.R. Godefroid, and C. Froese Fischer
Calculations with spectroscopic accuracy: energies, Landé g_J-factors, and transition rates in the carbon isoelectronic sequence from Ar XIII to Zn XXV
Astronomy & Astrophysics 564, A24 (2014).

Discusses accurate spectrum calculations giving energies and transition rates for hundreds of levels.

49. S. Verdebout, C. Nazé, P. Jönsson, P. Rynkun, M. Godefroid, G. Gaigalas
Hyperfine structures and Landé: g_J-factors for $n = 2$ states in beryllium-, boron-, carbon-, and nitrogen-like ions from relativistic configuration interaction calculations
Atomic Data and Nuclear Data Tables 100, 1111 (2014).

This paper discusses hyperfine structure and the computation of Landé g_J-factors.

50. C. Nazé, S. Verdebout, P. Rynkun, G. Gaigalas, M. Godefroid, P. Jönsson
Isotope Shifts in Beryllium-, Boron-, Carbon-, and Nitrogen-like Ions from Relativistic Configuration Interaction Calculations
Atomic Data and Nuclear Data Tables 100, 1197 (2014).

The paper gives examples of accurate IS calculations including relativistic corrections.

51. J. Ekman, M. R. Godefroid and H. Hartman
Validation and Implementation of Uncertainty Estimates of Calculated Transition Rates
Atoms 2014, 2(2), 215-224; doi:10.3390/atoms2020215.

The paper discusses how and if uncertainties can be estimated for calculated transition rates in length and velocity gauge.

52. L. Radžiūtė, D. Kato, G. Gaigalas, P. Jönsson, P. Rynkun, V. Jonauskas and S. Kucas
Energy level structure of Er^{3+}
Journal of Quantitative Spectroscopy and Radiative Transfer, ISSN 0022-4073, 2015, Vol. 152, pp. 94-106.

Describes how calculations can be done for atoms with very complex shells structures.

53. G. Gaigalas, P. Rynkun, Ch. Froese Fischer
Lifetimes of $4p^5 4d$ levels in highly ionized atoms
Physical Review A, Atomic, molecular, and optical physics. ISSN 1050-2947. 2015, Vol. 91, p. 022509-10.

Shows how, for levels (identified by their label) that cross along an iso-electronic sequence, when viewed instead by J, parity, and position, their composition changes smoothly along an isoelectronic sequence and transition matrix elements can be approximated by formulas in Z.

54. L. Radžiūtė, J. Ekman, P. Jönsson, and G. Gaigalas
Extended calculations of level and transition properties in the nitrogen isoelectronic sequence: Cr XVIII, Fe XX, Ni XXII, and Zn XXIV
Astronomy & Astrophysics 2015, 582, A61

Discusses accurate spectrum calculations giving energies and transition rates for hundreds of levels.

55. S. Gustafsson, P. Jönsson, C. Froese Fischer and I.P. Grant
Combining multiconfiguration and perturbation methods: perturbative estimates of core-core electron correlation contributions to excitation energies in Mg-like iron
Atoms, 5(1), 3 (2017); `doi:10.3390/atoms5010003`

Summarizes how perturbative techniques (zero- and first-order partitions) can be used to include electron correlation.

56. P. Jönsson, G. Gaigalas, P. Rynkun, L. Radžiūtė, J. Ekman, S. Gustafsson, H. Hartman, K. Wang, M. Godefroid, C. Froese Fischer, I. Grant, T. Brage, G. Del Zanna
Multiconfiguration Dirac-Hartree-Fock calculations with spectroscopic accuracy: applications to astrophysics
Atoms 5(2) , 16 (2017); `doi:10.3390/atoms5020016`

Summarizes a number of very accurate calculations on astrophysically important ions attaining spectroscopic accuracy, i.e. accuracy high enough to directly support line identifications in spectra.